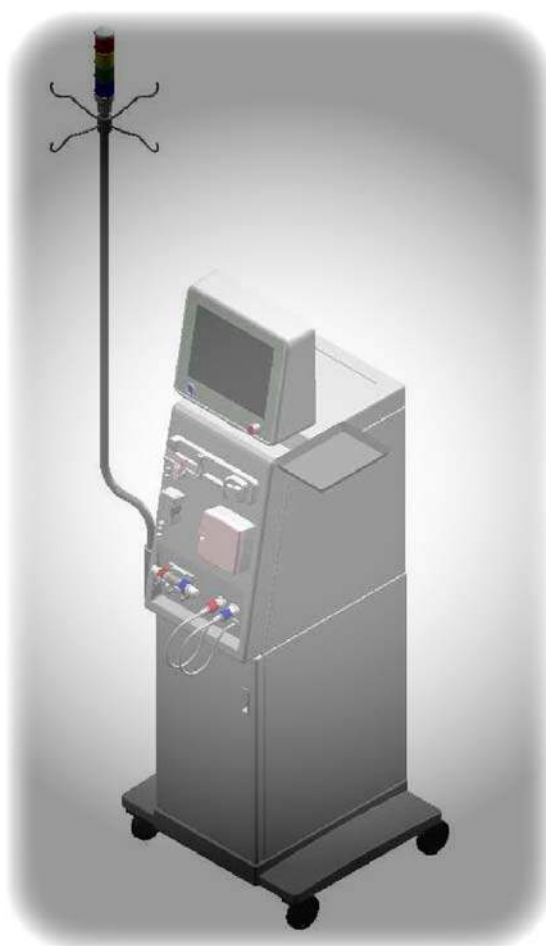


Single patient dialysis machine

SURDIAL 55Plus

Service Manual (For Type 2)



Contents

1. Trouble shooting
2. Service manual (Machine parts)
3. Service manual (Electric parts)
4. Adjustment procedure

Appendix

Maintenance tool list
Flowchart

1. Trouble shooting

Trouble Shooting-List

ID number	Error code	Alarm name	Option
CXTS00-E01	E01	RECOVER FROM EMERGENCY	
CXTS00-E03	E03	CPU ERROR (Not display)	
CXTS00-E05	E05	DIALYSATE PRESSURE MAX ALARM	
CXTS00-E06	E06	DIALYSATE PRESSURE MIN ALARM	
CXTS00-E07	E07	VENOUS PRESSURE MAX ALARM	
CXTS00-E08	E08	VENOUS PRESSURE MIN ALARM	
BHTS01-E09	E09	TMP ALARM (for Type 1)	
CXTS00-E09	E09	TMP ALARM (for Type 2)	
CXTS00-E10	E10	TMP OFFSET ALARM	
CXTS00-E11	E11	UFRP RATE OF CHANGE ALARM	
CXTS00-E12	E12	DIALYSATE TEMP MAX ALARM	
CXTS00-E13	E13	DIALYSATE TEMP MIN ALARM	
CXTS00-E14	E14	HEATER OVER HEAT ALARM	
CXTS00-E15	E15	BLOOD LEAK ALARM	
CXTS00-E17(E18)	E17	AIR BUBBLES ALARM. BP IS OPERATED AFTER REMOVAL	
	E18	AIR BUBBLES ARE DETECTED	
CXTS00-E19(E20)	E19	TUBE NON-EQUIPPED OR LARGE BUBBLE	
	E20	TUBE NON-EQUIPPED OR LARGE BUBBLE	
CXTS00-E21	E21	SINGLE SWITCH TIME ALARM	
CXTS00-E22	E22	T3&T4 DIFFERENCE ALARM	
CXTS00-E23	E23	ARTERIAL PRESSURE MAX ALARM	
CXTS00-E24	E24	ARTERIAL PRESSURE MIN ALARM	
CXTS00-E25	E25	UF COMPENSATION ERROR	
BHTS01-E27	E27	DIALYSATE CONDUCTIVITY ALARM (for Type 1)	
CXTS00-E27	E27	DIALYSATE CONDUCTIVITY ALARM (for Type 2)	
CXTS00-E28	E28	AIR BUBBLE SENSOR ERROR (SELF-TEST)	
CXTS00-E29	E29	CPU1 TRANSFER ERROR	
CXTS00-E30	E30	VENOUS PRESSURE MAX LIMIT	
CXTS00-E31	E31	VENOUS PRESSURE MIN LIMIT	
BHTS01-E32	E32	DIALYSATE PRESSURE MAX LIMIT (for Type 1)	
CXTS00-E32	E32	DIALYSATE PRESSURE MAX LIMIT (for Type 2)	
BHTS01-E33	E33	DIALYSATE PRESSURE MIN LIMIT (for Type 1)	
CXTS00-E33	E33	DIALYSATE PRESSURE MIN LIMIT (for Type 2)	
CXTS00-E34	E34	T1 TEMPERATURE SENSOR ERROR	
CXTS00-E35	E35	T2 TEMPERATURE SENSOR ERROR	
CXTS00-E36	E36	T3 TEMPERATURE SENSOR ERROR	
CXTS00-E37	E37	T4 TEMPERATURE SENSOR ERROR	
CXTS00-E38	E38	HP ROTATION ERROR	
CXTS00-E39	E39	UNINSTALLED HP ALARM	

ID number	Error code	Alarm name	Option
CXTS00-E40	E40	+5V (SUB) POWER ERROR	
CXTS00-E41	E41	Vref (SUB) POWER ERROR	
CXTS00-E42	E42	+24V POWER ERROR	
CXTS00-E43	E43	+5V POWER ERROR	
CXTS00-E44	E44	Vref POWER ERROR	
CXTS00-E45	E45	+12V POWER ERROR	
CXTS00-E46	E46	-12V POWER ERROR	
CXTS00-E47	E47	BP COVER SENSOR ERROR	
CXTS00-E48	E48	CLV FAN STOP ALARM	
BHTS01-E49	E49	UFP ERROR (for Type 1)	
BHTS01-E50	E50	SELF-CHECK ALARM (for Type 1)	
CXTS00-E50	E50	SELF-CHECK ALARM (for Type 2)	
CXTS00-E51	E51	AIR BUBBLE SENSOR ERROR	
CXTS00-E52	E52	BLOOD LEAK SENSOR ERROR	
CXTS00-E53	E53	BLOOD PUMP STOP ERROR	
CXTS00-E54	E54	CONCENTRATION SENSOR ERROR	
BHTS01-E55	E55	WATER SUPPLY STOP (for Type 1)	
CXTS00-E55	E55	WATER SUPPLY STOP (for Type 2)	
BHTS01-E56	E56	SHORT WATER SUPPLY (for Type 1)	
CXTS00-E56	E56	SHORT WATER SUPPLY (for Type 2)	
BHTS01-E57	E57	SHORT DIALYSATE FLOW (for Type 1)	
CXTS00-E57	E57	SHORT DIALYSATE FLOW (for Type 2)	
BHTS01-E58	E58	DIALYSATE FLOW STOP (for Type 1)	
BHTS01-E59	E59	AIR ELIMINATION TANK ERROR (for Type 1)	
CXTS00-E59	E59	AIR ELIMINATION TANK ERROR (for Type 2)	
BHTS01-E60	E60	CLOSED CIRCUIT LEAK ERROR (for Type 1)	
CXTS00-E60	E60	CLOSED CIRCUIT LEAK ERROR (for Type 2)	
CXTS00-E61	E61	UFP ROTATION ERROR	
BHTS01-E62	E62	UFP SPEED ERROR (for Type 1)	
CXTS00-E62	E62	UFP SPEED ERROR (for Type 2)	
CXTS00-E63	E63	SYRINGE INFUSION COMPLETION	
CXTS00-E64	E64	CLV POSITION ERROR	
CXTS00-E65	E65	UFP STARTING POINT ERROR	
CXTS00-E66	E66	BATTERY VOLTAGE TOO LOW	
CXTS00-E67	E67	ARTERIAL PRESSURE MAX LIMIT	
CXTS00-E68	E68	ARTERIAL PRESSURE MIN LIMIT	
BHTS01-E69	E69	DIALYSATE FLOW SENSOR ERROR (for Type 1)	
CXTS00-E70	E70	AIR BUBBLE SENSOR SETTING ERROR	
CXTS00-E71	E71	HP DRIVE ERROR (PH7)	
CXTS00-E72	E72	HP DRIVE ERROR (PH8)	
CXTS00-E73	E73	MAIN CPU (HMI) ERROR	
CXTS00-E74	E74	BLOOD PUMP ERROR	

ID number	Error code	Alarm name	Option
CXTS00-E75	E75	BLOOD PUMP FAULT LOAD ALARM	
CXTS00-E76	E76	MAIN CPU (CONTROL) ERROR	
CXTS00-E77	E77	SUB CPU ALARM	
CXTS00-E78	E78	PANEL CPU ALARM	
CXTS00-E80	E80	BPM RAM ERROR	○
CXTS00-E81	E81	BPM ROM ERROR	○
CXTS00-E82	E82	BPM ERROR	○
CXTS00-E83	E83	BPM COMUNICATION ERROR	○
CXTS00-E84	E84	BPM PRESSURIZATION ERROR	○
CXTS00-E85	E85	HIGHEST-BLOOD-PRESSURE MAX ERROR	○
CXTS00-E86	E86	HIGHEST-BLOOD-PRESSURE MIN ERROR	○
CXTS00-E87	E87	LOWEST-BLOOD-PRESSURE MAX ERROR	○
CXTS00-E88	E88	LOWEST-BLOOD-PRESSURE MIN ERROR	○
CXTS00-E89	E89	PULSE-RATE MAX ERROR	○
CXTS00-E90	E90	PULSE-RATE MIN ERROR	○
BHTS01-E97	E97	SHORT CHEMICAL SOLUTION ALARM (for Type 1)	
CXTS00-E97	E97	SHORT CHEMICAL SOLUTION ALARM (for Type 2)	
CXTS00-E99	E99	BATTERY ERROR	
CXTS00-E100	E100	INTERNAL DATA ERROR	
BHTS01-E101	E101	SOLUTION B CONDUCTIVITY ALARM (for Type 1)	
CXTS00-E101	E101	SOLUTION B CONDUCTIVITY ALARM (for Type 2)	
CXTS00-E102	E102	T5 TEMPERATURE SENSOR ERROR	
CXTS00-E103	E103	I/O SWITCH ERROR	
CXTS00-E104	E104	BACK LIGHT ERROR	
CXTS00-E105	E105	CPU CHECK ERROR	
CXTS00-E106	E106	FPGA2 POWER OF I/O ERROR	
CXTS00-E107	E107	FPGA2 POWER OF INT ERROR	
CXTS00-E108	E108	UFP STOP ERROR	
BHTS01-E110	E110	CONC CONNECTOR TO RINSING PORT (for Type 1)	
CXTS00-E110	E110	CONC CONNECTOR TO RINSING PORT (for Type 2)	
CXTS00-E111	E111	T6 TEMPERATURE SENSOR ERROR	
CXTS00-E112	E112	T7 TEMPERATURE SENSOR ERROR	
BHTS01-E113	E113	T5&T6 DIFFERENCE ALARM (for Type 1)	
BHTS01-E114	E114	TEMPARATURE CANNOT REACH (for Type 1)	
CXTS00-E114	E114	TARGET TEMPARATURE NOT REACHED (for Type 2)	
CXTS00-E115	E115	B CARTRIDGE HOLDER OPEN	○
BHTS01-E117	E117	B CONDUCTIVITY CANNOT REACH (for Type 1)	○
BHTS01-E118	E118	T2 OVER HEAT (for Type 1)	○
CXTS00-E118	E118	T2 OVER HEATED (for Type 2)	
BHTS01-E119	E119	T7 TEMP LOW (for Type 1)	○
CXTS00-E119	E119	TEMPERATURE IS LOW (for Type 2)	
BHTS01-E120	E120	CIRCURATION FLOW STOP (for Type 1)	○

ID number	Error code	Alarm name	Option
CXTS00-E120	E120	CIRCULATION FLOW STOP (for Type 2)	
BHTS01-E125	E125	DIALYSIS FLOW FAILURE (for Type 1)	○
CXTS00-E125	E125	DIALYSIS FLOW FAILURE (for Type 2)	
BHTS01-E126	E126	CD2&CD3 DIFFERENCE ALARM (for Type 1)	○
BHTS01-E127	E127	DIALYSIS FLOW CONTROL FAILURE (for Type 1)	○
CXTS00-E127	E127	DIALYSIS FLOW CONTROL ERROR (for Type 2)	
BHTS01-E128	E128	T2&T5 DIFFERENCE ALARM (for Type 1)	○
CXTS00-E128	E128	T2&T5 DIFFERENCE ALARM (for Type 2)	
CXTS00-E129	E129	T4&T7 DIFFERENCE ALARM	
CXTS00-E137	E137	FAN3 STOP ALARM	
BHTS01-E138	E138	DIALYSIS FLOW SENSOR ALARM (for Type 1)	○
BHTS01-E139	E139	VALVE CHECK ERROR (for Type 1)	
CXTS00-E140	E140	AVE. STROKE VOLUME MIN ALARM	
BHTS01-E141	E141	CLOSED CIRCUIT LEAK ERROR (for Type 1)	○
CXTS00-E143	E143	BLOOD DETECTED	
CXTS00-E144	E144	BLOOD NOT DETECTED	
CXTS00-E145	E145	Central system valve leak error (for Type 2)	○
CXTS00-E147	E147	HP BOLUS OPERATION ERROR	
CXTS00-E148	E148	VALVE ERROR (DISINFECTION VALVE) (for Type 2)	
CXTS00-E149	E149	FS1 SENSOR ALARM (for Type 2)	
CXTS00-E150	E150	FS2 SENSOR ALARM (for Type 2)	
CXTS00-E151	E151	UFP SPEED ERROR (1 CYCLE) (for Type 2)	
CXTS00-E152	E152	DISINFECTANT TEMPERATURE ALARM (for Type 2)	
CXTS00-E153	E153	VALVE ERROR (CHAMBER PERIPHERY VALVE) (for Type 2)	
CXTS00-E154	E154	Dialysate pressure gauge check error (for Type 2)	
CXTS00-E155	E155	Cleaning valve error (for Type 2)	
CXTS00-M01	M01	COUPLER SW IS ON	
CXTS00-M02	M02	COUPLER SW IS OFF	
CXTS00-M03	M03	UF SPEED SETTING EXCEEDED MAX LIMIT	
CXTS00-M04	M04	COMPLETION OF HP INFUSION	
CXTS00-M05	M05	PERFORMED TARGET UF	
CXTS00-M06	M06	PRIMING IS COMPLETE	
CXTS00-M07	M07	UFR IS SET TO ZERO	
CXTS00-M08	M08	HP SPEED IS SET TO ZERO	
CXTS00-M09	M09	UF GOAL IS SET TO ZERO	
CXTS00-M10	M10	TIMER COMPLETION	
CXTS00-M11	M11	CONSUMPTION EXCHANGE TIME. REFER TO MAINTENANCE	
CXTS00-M12	M12	AFTER BUBBLE REMOVAL. BP IS OPERATED	
CXTS00-M13	M13	AFTER SETTING TUBE. BLOOD PUMP IS OPERATED	
CXTS00-M14	M14	PERFORMED TARGET DIALYSES	
CXTS00-M15	M15	WATER RINSE NOT SUFFICIENT	

ID number	Error code	Alarm name	Option
CXTS00-M16	M16	RETRANS COMPLETE	
CXTS00-M20	M20	STEP TIME: LESS THAN 5 MINUTES	
CXTS00-M21	M21	F-RAM DATA WAS OVERWRITTEN	
CXTS00-M23	M23	SOLUTION PREP MODE IS INCORRECT. CHECK THE MAINTENANCE 5 SCREEN	
CXTS00-M24	M24	BLOOD DETECTED, please ensure venous line is connected to patient and press the DIALYSE Switch to start treatment	
CXTS00-M50	M50	CHECK THE CUFF AND HOSES	○
CXTS00-M51	M51	CHECK THE CONNECTION OF CUFF	○
CXTS00-M52	M52	IT CANNOT MEASURE BECAUSE OF PATIENT MOVEMENT	○
CXTS00-M53	M53	CUFF PRESSURE IS INSUFFICIENT, IT CANNOT MEASURE	○
CXTS00-M54	M54	IT CANNOT MEASURE BECAUSE OF MOVEMENT OR ARRHYTHMIC	○
CXTS00-M55	M55	IT CANNOT MEASURE BECAUSE OF MOVEMENT OR ARRHYTHMIC	○
CXTS00-M56	M56	CHECK THE CUFF AND THE BODY MOTION OF PATIENT	○
CXTS00-M57	M57	CHECK THE CUFF AND THE BODY MOTION OF PATIENT	○
CXTS00-M58	M58	CHECK THE CUFF AND THE BODY MOTION OF PATIENT	○
CXTS00-M59	M59	PULSE IS SMALL, IT CANNOT MEASURE	○
CXTS00-M60	M60	CHECK THE CUFF AND HOSES	○
CXTS00-M61	M61	BLOOD PRESSURE IS CHANGED	○
CXTS00-M65	M65	PLEASE CONFIRM THE SUPPLY CONDITION OF CONCENTRATE	

Device abnormality without the error code

ID number	Malfunction contents
CXTS00-001	LCD screen is not displayed.
CXTS00-002	Touch panel switch cannot operate.
CXTS00-003	Heparin pump does not operate.
CXTS00-004	Blood pump error.
CXTS00-005	Fluid still flows out from coupler after bypassed.
CXTS00-006	Venous chamber shows fluctuations in the level.
CXTS00-007	Air gets mixed in dialysate.
CXTS00-008	Dialysate flow rate shows instability (for Type 2).
CXTS00-009	Normal fluid drain is not achieved even by pressing the DRAIN switch (for Type 2).
CXTS00-010	Error in the UF volume exceeds the specified limit.
CXTS00-011	Pump does not operate.

E01 RECOVER FROM EMERGENCY

Detection condition	After power loss, the power returns again while the device is in operation						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse	Continues with the process immediately before failure.	○ (3 times)	○	-	-	-	-
Standby		○ (3 times)	○	-	-	-	-
Preparation		○ (3 times)	○	-	-	-	-
Dialysis		○ (3 times)	○	-	-	-	-
Retrans		○ (3 times)	○	-	-	-	-
Reset method	Press the RESET switch.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The power supply circuit breaker is intentionally set to ON or OFF.	Normal	
The equipment power comes back ON.	Normal	
The power supply plug is inserted or removed	Normal	

E03 CPU ERROR (Not display)

Detection condition	When the main CPU detects the abnormal temperature of main CPU						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
· Rinse Wait	- / - / -	○	○	-	-	-	-
· Rinse	○ / - / -	○	○	-	-	-	-
· Dia. Wait	- / - / -	○	○	○	-	-	-
· Standby	○ / - / -	○	○	○	-	-	-
· Preparation	○ / - / -	○	○	○	-	-	-
· Dialysis	○ / ○ / -	○	○	○	-	-	-
· Retrans	○ / - / -	○	○	○	-	-	-
Reset method	Recovers by "I/O" switch OFF/ON. (Restart the Device.)		Remark	The alarm message is not displayed by error condition.			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty main CPU PCB	C : Replace the main CPU PCB.	BHSM**-100

E05 DIALYSATE PRESSURE MAX ALARM



Detection condition	When the dialysate pressure exceeds the maximum set value for the dialysate pressure alarm.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dia. Wait	- / - / -	○	○	2 sec.	2 sec.	2 sec.	○
Standby	- / - / -	○	○	2 sec.	2 sec.	2 sec.	○
Preparation	- / - / -	○	○	2 sec.	2 sec.	2 sec.	○
Dialysis	- / ○ / -	○	○	2 sec.	2 sec.	2 sec.	-
Retrans	- / - / -	○	○	2 sec.	2 sec.	2 sec.	○
Power failure	- / - / -	○	-	2 sec.	2 sec.	2 sec.	○
Reset method	Press the RESET switch			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
This alarm is accompanied with the maximum venous pressure alarm	A :For detail, refer to information in the “E07 VENOUS PRESSURE MAX ALARM” section.	CXTS**-E07
	C :For detail, refer to information in the “E07 VENOUS PRESSURE MAX ALARM” section.	CXTS**-E07
Rise in fluid pressure due to faulty “PR1”	C :Make adjustment of the “PR1”. If this adjustment cannot be made, replace the “PR1”.	CXAJ**-008 CXSM**-014
Leaks from the “V10”	C :Overhaul or replace the “V10”.	CXSM**-001
Rise in fluid pressure due to the stop of UF operation	C :Remove the causes of the stop of UF operation.	
Leaks from the “V5a”	A :Clamp the OUT side of the “V5a” with forceps while the circuit on the chamber A side is being closed to check whether the leaks are remedied. C :Disassemble and clean, or replace the “V5a”.	CXSM**-001
Leaks from the “V7a”	A :Clamp the OUT side of the “V7a” with forceps while the circuit on the chamber B side is being closed to check whether the leaks are remedied. C :Disassemble and clean, or replace the “V7a”.	CXSM**-001

E06 DIALYSATE PRESSURE MIN ALARM



Detection condition	When the dialysate pressure falls below the minimum set value for the dialysate pressure alarm						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dia. Wait	- / - / -	○	○	2 sec.	2 sec.	2 sec.	○
Standby	- / - / -	○	○	2 sec.	2 sec.	2 sec.	○
Preparation	- / - / -	○	○	2 sec.	2 sec.	2 sec.	○
Dialysis	- / ○ / -	○	○	2 sec.	2 sec.	2 sec.	-
Retrans	- / - / -	○	○	2 sec.	2 sec.	2 sec.	○
Power failure	- / - / -	○	-	2 sec.	2 sec.	2 sec.	○
Reset method	Press the RESET switch.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
This alarm is accompanied with the minimum venous pressure alarm	A: For detail, refer to information in the “E08 VENOUS PRESSURE MIN ALARM” section.	CXTS**-E08
	C: For detail, refer to information in the “E08 VENOUS PRESSURE MIN ALARM” section.	CXTS**-E08
Clogging of “RV2”	C: Make adjustment of the closed circuit.	CXAJ**-010
	If this adjustment cannot be made, disassemble and clean, or replace the “RV2”.	CXSM**-005
The “V4” opens due to a malfunction of the “FSW”	C: Disassemble and clean, or replace the “AS2”.	CXSM**-015
Leaks from the “V4”	A: Clamp the “V4” line with forceps to check whether the leaks are remedied.	CXSM**-001
	C: Disassemble and clean, or replace the “V4”.	
Leaks from the “V6b”	A: Clamp the OUT side of the “V6b” with forceps while the circuit on the chamber A side is being closed to check whether the leaks are remedied.	CXSM**-001
	C: Disassemble and clean, or replace the “V6b”.	
Leaks from the “V8b”	A: Clamp the OUT side of the “V8b” with forceps while the circuit on the chamber B side is being closed to check whether the leaks are remedied.	CXSM**-001
	C: Disassemble and clean, or replace the “V8b”.	
Excessive clogging of dialyzer	C: Replace the dialyzer.	

E07 VENOUS PRESSURE MAX ALARM

Detection condition	When the venous pressure exceeds the maximum set value for the venous pressure alarm						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dia. Wait	- / - / -	○	○	○	○	-	○
Standby	- / - / -	○	○	○	○	-	○
Preparation	- / - / -	○	○	○	○	-	○
Dialysis	- / ○ / -	○	○	○	○	-	-
Retrans	- / - / -	○	○	○	○	-	○
Power failure	- / - / -	○	-	○	○	-	○
Reset method	Press the RESET switch.			Remark	This alarm is not detected during the startup test in the standby mode.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Clogging of venous pressure line in the blood circuit	A: Check the shunt part for any clogging and the blood circuit for any bended parts. C: If any clogging or bended part is observed, remedy it.	
Fluctuations in venous pressure due to the movement of a patient	C: If the venous pressure shows substantial fluctuations, slightly reduce the maximum set value for the venous pressure alarm at a physician's discretion.	
Substantial fluctuations in the blood pump discharge pressure	A: Check to be sure that the blood pump operates at a right discharge rate and at a right shutoff pressure, and cause no uneven revolutions. C: If the shutoff pressure is low, replace the blood pump. If the blood pump is normal, replace the power supply PCB, I/O PCB, and/or CPU PCB.	BHAJ**-014 BHSM**-013 BHSM**-100 CXSM**-101 BHSM**-105

E08 VENOUS PRESSURE MIN ALARM



Detection condition	When the venous pressure drops below the minimum set value for the venous pressure alarm						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dia. Wait	- / - / -	○	○	○	○	-	○
Standby	- / - / -	○	○	○	○	-	○
Preparation	- / - / -	○	○	○	○	-	○
Dialysis	- / ○ / -	○	○	○	○	-	-
Retrans	- / - / -	○	○	○	○	-	○
Power failure	- / - / -	○	-	○	○	-	○
Reset method	Press the RESET switch.			Remark	This alarm is not detected during the startup test in the standby mode.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Drop in patient's blood pressure	A: Stop the UF operation, and then check the condition of the patient. If the minimum set value for the venous pressure alarm is improper, make resetting of the value.	
Disconnection of the blood circuit from the venous pressure receiving port	A: Check to be sure that blood does not enter the venous pressure sensor line, and then connect the blood circuit to the line C: If blood enters the venous pressure sensor line (from the pressure receiving port to the PV/PA PCB), replace the line.	
Clogging of arterial pressure line in the blood circuit	A: Check the shunt part for any clogging and the blood circuit for any bended parts. C: Remedy the clogged parts.	
Fluctuations in venous pressure due to the movement of a patient	C: If the venous pressure shows substantial fluctuations, slightly reduce the minimum set value for the venous pressure alarm at a physician's discretion.	
Substantial fluctuations in the blood pump discharge pressure	A: Check to be sure that the blood pump operates at a right discharge rate and at a right shutoff pressure, and cause no uneven revolutions C: If the shutoff pressure is low, replace the blood pump. If the blood pump is normal, replace the blood pump, power supply PCB, and/or CPU PCB.	BHAJ**-014 BHSM**-013 BHSM**-100 CXSM**-101 BHSM**-105

E09 TMP ALARM (for Type 1)

Detection condition	When the TMP value exceeds the maximum or minimum set value for TMP alarm for a period of consecutive five seconds						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dialysis	- / ○ / -	○	○	○	-	-	-
Reset method	Press the RESET switch.			Remark	This alarm is not detected while the bypass operation is in progress. This alarm is not detected while the B powder cartridge exchange operation is in progress. Only type 1 of fluid circuit.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Excessive clogging of the dialyser	C: Replace the dialyser at a physician's discretion.	
Too low set value for the TMP alarm	C: Set it to a proper value.	
This alarm is accompanied with "E05 DIALYSATE PRESSURE MAX ALARM", "E06 DIALYSATE PRESSURE MIN ALARM", "E07 VENOUS PRESSURE MAX ALARM", or "E08 VENOUS PRESSURE MIN ALARM".	C: For detail, refer to information in the "E05 DIALYSATE PRESSURE MAX ALARM", "E06 DIALYSATE PRESSURE MIN ALARM", "E07 VENOUS PRESSURE MAX ALARM", or "E08 VENOUS PRESSURE MIN ALARM" section.	CXTS**-E05 CXTS**-E06 CXTS**-E07 CXTS**-E08
This alarm is accompanied with "E32 DIALYSATE PRESSURE MAX LIMIT", "E33 DIALYSATE PRESSURE MIN LIMIT", "E30 VENOUS PRESSURE MAX LIMIT" or "E31 VENOUS PRESSURE MIN LIMIT".	C: For detail, refer to information in the "E32 DIALYSATE PRESSURE MAX LIMIT", "E33 DIALYSATE PRESSURE MIN LIMIT", "E30 VENOUS PRESSURE MAX LIMIT" or "E31 VENOUS PRESSURE MIN LIMIT" section.	BHTS**-E32 BHTS**-E33 CXTS**-E30 CXTS**-E31

E09 TMP ALARM (for Type 2)

Detection condition	TMP value over the TMP maximum alarm value is detected when switching the round. TMP value below the TPM minimum alarm value is detected when switching the round						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dialysis	- / - / ○	○	○	○	○	-	-
Reset method	Press the RESET switch.			Remark	This alarm is not detected while the bypass operation is in progress. Only type 2 of fluid circuit.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Excessive clogging of the dialyser	C: Replace the dialyser at a physician's discretion.	
Too low set value for the TMP alarm	C: Set it to a proper value.	
This alarm is accompanied with "E05 DIALYSATE PRESSURE MAX ALARM", "E06 DIALYSATE PRESSURE MIN ALARM", "E07 VENOUS PRESSURE MAX ALARM", or "E08 VENOUS PRESSURE MIN ALARM".	C: For detail, refer to information in the "E05 DIALYSATE PRESSURE MAX ALARM", "E06 DIALYSATE PRESSURE MIN ALARM", "E07 VENOUS PRESSURE MAX ALARM", or "E08 VENOUS PRESSURE MIN ALARM" section.	CXTS**-E05 CXTS**-E06 CXTS**-E07 CXTS**-E08
This alarm is accompanied with "E32 DIALYSATE PRESSURE MAX LIMIT", "E33 DIALYSATE PRESSURE MIN LIMIT", "E30 VENOUS PRESSURE MAX LIMIT" or "E31 VENOUS PRESSURE MIN LIMIT".	C: For detail, refer to information in the "E32 DIALYSATE PRESSURE MAX LIMIT", "E33 DIALYSATE PRESSURE MIN LIMIT", "E30 VENOUS PRESSURE MAX LIMIT" or "E31 VENOUS PRESSURE MIN LIMIT" section.	BHTS**-E32 CXTS**-E32 BHTS**-E33 CXTS**-E33 CXTS**-E30 CXTS**-E31

E10 TMP OFFSET ALARM

Detection condition	When the TMP offset value drops below -200 mmHg or exceeds $+150$ mmHg						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dialysis	- / - / -	○	○	-	-	-	○
Reset method	Press the RESET switch.		Remark	This alarm is detected in the first round (Type 2 is after 5 minutes) after the Device shifts to the dialysis process and in a round initiated after a lapse of one hour.			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Erroneous venous pressure is displayed	A: When the "VP DISPLAY OFFSET" (venous pressure display offset) is set to "0" on the Maintenance Mode 6 screen and then the venous pressure receiving port is open to the atmosphere or when the venous pressure of 0 ± 2 mmHg is displayed and external pressure of $+300$ mmHg is added to the said pressure, check to be sure that the venous pressure of $+300\pm 5$ mmHg is displayed.	
○ Deviation from the calibration of the venous pressure sensor PCB	C: Perform calibration of the PV/PA PCB.	BHAJ**-001
○ Faulty venous pressure sensor PCB	C: Replace calibration of the PV/PA PCB.	BHSM**-104
Erroneous fluid pressure is displayed	A: When the "DP DISPLAY OFFSET" (fluid pressure display offset) is set to "0" on the Maintenance Mode 6 screen and then the tube on the outlet side of the fluid pressure receiving port is disconnected to open to the atmosphere or when the fluid pressure of 0 ± 2 mmHg is displayed and external pressure of $+300$ mmHg is added to the said pressure, check to be sure that the fluid pressure of $+300\pm 5$ mmHg is displayed.	
○ Deviation from the calibration of the fluid pressure sensor circuit	C: Perform calibration of the fluid pressure sensor.	BHAJ**-002
○ Faulty fluid pressure sensor or faulty fluid pressure sensor PCB	C: Replace the fluid pressure sensor assembly.	BHSM**-019
The blood circuit is disconnected from the venous pressure receiving port	A: Check to be sure that blood does not enter the venous pressure sensor line, and then connect the blood circuit to the line. C: If blood enters the venous pressure sensor line (from the pressure receiving port to the PV/PA PCB), replace the line.	

E11 UFRP RATE OF CHANGE ALARM



Detection condition	When the rate of change between the UFRP value of 20 rounds before and the current UFPR value exceeds the alarm set value						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dialysis	- / - / -	○	○	-	-	-	○
Reset method	Press the RESET switch.			Remark	This alarm is not detected when single needle is executed.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
TMP changes result from the clogging or leaks of the dialyzer	C: Replace the dialyzer.	
TMP changes result from broken blood circuit	C: Remedy the broken blood circuit.	
TMP changes result from leaks of closed systems	C: Remedy the leaks of the fluid circuit.	
TMP shows a proper rate of change, but the set value for UFRP rate of change alarm is improper	C: Increase the set value for UFRP rate of change alarm to a proper value.	
Erroneous venous pressure is displayed	A: When the "VP DISPLAY OFFSET" (venous pressure display offset) is set to "0" on the Maintenance Mode 6 screen. And the venous pressure transducer is open to atmosphere, a venous pressure of 0±2 mmHg must be displayed. When an external pressure of +300 mmHg is added to the VP transducer, a pressure of +300±5 mmHg must be displayed.	
○ Deviation from the calibration of the venous pressure sensor PCB	C: Perform calibration of the PV/PA PCB.	BHAJ**-001
○ Faulty venous pressure sensor PCB	C: Replace the PV/PA PCB.	BHSM**-104
Erroneous fluid pressure is displayed	A: When the "DP DISPLAY OFFSET" (fluid pressure display offset) is set to "0" on the Maintenance Mode 6 screen. Then open the tube on the outlet side of the fluid pressure transducer to atmosphere, a pressure of 0±2 mmHg must be displayed. When an external pressure of +300 mmHg is added to the DP transducer, a pressure of +300±5 mmHg must be displayed.	
○ Deviation from the calibration of the fluid pressure sensor circuit	C: Perform calibration of the fluid pressure sensor.	BHAJ**-002
○ Faulty fluid pressure sensor or faulty fluid pressure sensor PCB	C: Replace the fluid pressure sensor assembly.	BHSM**-019

E12 DIALYSATE TEMP MAX ALARM



Detection condition	When the T4 temperature exceeds the maximum set value for the dialysate temperature alarm						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Standby	- / - / ○ (Heater turned off)	○	○	-	-	-	○
Preparation	- / - / ○ (Heater turned off)	○	○	-	-	-	○
Dialysis	- / ○ / ○ (Heater turned off)	○	○	-	-	-	-
Retrans	- / - / ○ (Heater turned off)	○	○	-	-	-	○
Reset method	Press the RESET switch. If the T4 temperature deviates from the set value for a period of consecutive 15 seconds, the alarm will be automatically reset.		Remark		· While SQHD is in progress or after SQHD is reset, the alarm is not detected unless the bypass operation is reset. · While Fluid Stop is in progress or after Fluid Stop is reset, the alarm is not detected unless the bypass operation is reset. · This alarm is not detected during the startup test in the standby mode.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The actual dialysate temperature is high	A: Touch the dialysate feed line to the dialyser with hand, or if the dialyser is not connected to a patient, monitor the dialysate temperature at the coupler outlet .Use a thermometer to check whether the actual dialysate temperature is high.	
○ Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101
○ Deviation from the display temperature of the I/O PCB	C: Use a temperature calibrator to perform calibration of the “T1”, “T2”, “T3”, and “T4” circuits of the main PCB. If no calibration can be performed, replace the I/O PCB.	BHAJ**-003 BHAJ**-004 BHAJ**-022 CXSM**-101
○ The feed water temperature is high	C: Adjust the feed water temperature to the temperature specifications. 17 to 30°C (When dialysate flow is 300 to 800 mL/min) 14 to 30°C (Hot rinse option installation) 4 to 30°C (Heat exchanger option installation) The feed water temperature must be 5°C lower than the dialysate temperature set value.	
○ Low charge flow rate	C: Make adjustment of charge flow rate.	CXAJ**-009
The actual temperature is not high	A: Touch the dialysate feed line to the dialyser with hand, or if the dialyser is not connected to a patient, monitor the dialysate temperature at the coupler outlet. Use a thermometer to check the actual dialysate temperature.	

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
○ Faulty T4 temperature sensor	A: Shake the harness of the T4 temperature sensor to check any fluctuation of the T4 temperature display. Furthermore, when a temperature calibrator is connected, check the T4 temperature display for any fluctuations. C: If the analysis results show any abnormalities, replace the T4 temperature sensor.	BHSM**-008
○ Deviation from the display temperature of the I/O PCB	A: When a temperature calibrator is connected, check the T4 temperature display for any fluctuations. C: Use a temperature calibrator to perform calibration of the "T4" circuits of the main PCB. If no calibration can be performed, replace the I/O PCB.	BHAJ**-004 BHSM**-101

E13 DIALYSATE TEMP MIN ALARM



Detection condition	When the T4 temperature drops below the minimum set value for the dialysate temperature alarm						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Preparation	- / - / ○	○	○	-	-	-	○
Dialysis	- / ○ / ○	○	○	-	-	-	-
Retrans	- / - / ○	○	○	-	-	-	○
Reset method	Press the RESET switch. If the T4 temperature deviates from the set value for a period of consecutive 15 seconds, the alarm will be automatically reset.		Remark		- While SQHD is in progress or after SQHD is reset, the alarm is not detected unless the bypass operation is reset. - While Fluid Stop is in progress or after Fluid Stop is reset, the alarm is not detected unless the bypass operation is reset. - While the B powder cartridge exchange operation the alarm is not detected in progress. - Does not detect for 60 sec after being reset.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The actual dialysate temperature is low.	A: Touch the dialysate feed line to the dialyser with hand, or if the dialyser is not connected to a patient, monitor the dialysate temperature at the coupler outlet using a thermometer to check whether the actual dialysate temperature is high.	
○ Low feed water temperature	C: Adjust the feed water temperature to the temperature specifications. 17 to 30°C (When dialysate flow is 300 to 800 mL/min) 14 to 30°C (Hot rinse option installation) 4 to 30°C (Heat exchanger option installation) The feed water temperature must be 5°C lower than the dialysate temperature set value.	
○ Faulty heater	A: Turn OFF the main power supply, and then make measurement of resistance at both ends. C: If the resistance is out of the range as shown below, replace the heater. <Normal> 230V spec. :92.0 to 105.8Ω (500W×2) 110V spec. :10.5 to 12.1Ω (1000W) <Hot rinse option installation> 230V spec. :92.0 to 105.8Ω and 46.0 to 52.9Ω (500W + 1000W) 110V spec. :7.0 to 8.1Ω (1500W)	BHSM**-004
○ High charge flow rate	A: Check to be sure that the charge flow rate is corresponding to the dialysate flow rate. C: If not, make adjustment of the charge flow rate.	CXAJ**-009
The actual temperature is not low	A: Touch the dialysate feed line to the dialyser with hand, or if the dialyser is not connected to a patient, monitor the dialysate temperature at the coupler outlet .Use a thermometer to check the actual dialysate temperature.	

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
○ Faulty T4 temperature sensor	A: Shake the harness of the T4 temperature sensor to check any fluctuation of the T4 temperature display. Furthermore, when a temperature calibrator is connected, check the T4 temperature display for any fluctuations. C: If the analysis results show any abnormalities, replace the T4 temperature sensor.	BHSM**-008
○ Deviation from the display temperature of the I/O PCB	A: When a temperature calibrator is connected, check the T4 temperature display for any fluctuations. C: Use a temperature calibrator to perform calibration of the "T4" circuits of the main PCB. If no calibration can be performed, replace the I/O PCB.	BHAJ**-004 CXSM**-101

E14 HEATER OVER HEAT ALARM



Detection condition	When the “LS1a” or “LS1b” (used for 230V specification) located on the outer shell of the heater is activated						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / - (Heater turned off in all of the process)	○	○	-	-	-	○
Rinse		○	○	-	-	-	○
Dia. Wait		○	○	-	-	-	○
Standby		○	○	-	-	-	○
Preparation		○	○	-	-	-	○
Dialysis		○	○	-	-	-	-
Retrans		○	○	-	-	-	○
Reset method	Press the RESET switch.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The heater gets overheated	A: Connect a multiple meter to the heater side of the terminal block to check whether a 230V (or 110v for the 110V specification) is constantly applied to the heater. ⚠ CAUTION <u>If the heater is actually heated, its surface temperature will exceed a temperature of 60°C.(120°C at Hot Rinse spec.)</u> <u>Therefore, do not touch the heater with bare hand.</u>	
○ Faulty “T1”, “T2”, “T3”, or “T4”	C: For detail, refer to information in the “E34 T1 TEMPERATURE ERROR” “E35 T2 TEMPERATURE ERROR” “E36 T3 TEMPERATURE ERROR” “E37 T4 TEMPERATURE ERROR”	CXTS**-E34 CXTS**-E35 CXTS**-E36 CXTS**-E37
○ Faulty power supply PCB	C: Replace the power supply PCB.	BHSM**-105
○ Faulty “SSR1” or “SSR2”	C: Replace the “SSR1” or “SSR2”.	CXSM**-110
○ Faulty “relay”	C: Replace the “relay”.	CXSM**-110
The heater does not get overheated		
○ Faulty “LS1a” or “LS1b”	A: Check to be sure that the contact of the “LS1a” or “LS1b” is closed at ambient temperatures. Furthermore, where possible, check whether the contact opens when the temperature of a surface keeping in touch with the “LS1a” or “LS1b” reaches approximately 60°C. (120°C at Hot Rinse spec.) C: If the “LS1a” (or “LS1b”) malfunctions, replace it.	BHSM**-004
○ Broken wire in wirings to the “LS1a” or “LS1b” and loose connection of contact part	A: Lightly pull the lead wires of the CN111 connector to check the connector for disconnection or loose connection. C: If the connector is disconnected or loosened, remedy it.	
○ Faulty power supply PCB	C: If the “LS1a” (or “LS1b”) is normal, replace the power supply PCB.	BHSM**-105

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
○ Faulty I/O PCB	C: If the power supply PCB is normal, replace the I/O PCB.	CXSM**-101

E15 BLOOD LEAK ALARM



Detection condition	When the blood leak value exceeds the blood leak alarm set value for a period of consecutive five seconds						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dialysis	- / ○ / ○	○	○	○	○	-	-
Retrans	- / - / ○	○	○	○	○	-	○
Reset method	Press the RESET switch.			Remarks	This alarm is not detected during three rounds after the alarm is reset including a round in which it is reset.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Blood leaks	A: Collect a dialysate sample from the drain fluid or the dialyzer outlet, and check visually for any blood leaks or use a test device.	
○ Blood leaks from the dialyzer	C: After the Device shifts to the retransfusion process, press the BYPASS switch, and then replace the dialyzer.	
No blood leaks	A: Collect a dialysate sample from the drain fluid or the dialyzer outlet, and check visually for any blood leaks or use a test device.	
○ Excessive stains on the cartridge window (e.g. deposition of protein, calcium carbonate, or fingerprints)	C: Dismount the cartridge from the blood leak sensor, and then clean fingerprints or else off the window surface using a cotton-tipped swab. If the inside of the window (inside of the cartridge) has stains, clean it to remove the stains.	
○ Entry of large amount of air in the blood leak sensor	A: Check whether air enters the inside of the cartridge. Furthermore, visually check whether fine air bubbles that cannot be removed through the "AS1" or "AS2" are produced and whether the negative pressure line in the Device (e.g. the pump inlet) has leaks.	
○ Degraded or faulty LED in the blood leak sensor	C: Replace the blood leak sensor.	BHSM**-018
○ Faulty I/O PCB	C: If any abnormality occurs even after the blood leak sensor is replaced, replace the I/O PCB.	CXSM**-101

E17 AIR BUBBLES ALARM. BP IS OPERATED AFTER REMOVAL.
E18 AIR BUBBLES ARE DETECTED.



Detection condition	The air bubble sensor detects air bubbles when the air bubble sensor switch is on. (A tube must be inserted in the air bubble sensor.)						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	○	-	○
Rinse	- / - / -	○	○	-	○	-	○
Dia. Wait	- / - / -	○	○	○ SW OFF	○	-	○
Standby	- / - / -	○	○	○ SW OFF	○	-	○
Preparation	- / - / -	○	○	○ SW OFF	○	-	○
Dialysis	- / ○ / -	○	○	○ SW OFF	○	-	-
Retrans	- / - / -	○	○	○ SW OFF	○	-	○
Reset method	Clamp is released by the RESET switch. Recover BP/HP using their PUMP switches after releasing the clamp.		Remarks	When the reset type of the air bubble alarm is [standard]E17 [special]E18			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Air bubbles are produced		
○ Entry of air from the blood circuit	C: Remedy places through which air enters to return air bubbles to the drip chamber. If these places cannot be remedied, replace the blood circuit.	
No air bubbles are produced		
○ Deviation from the calibration of the air bubble sensor	C: Perform the calibration of the air bubble sensor detection circuit.	CXAJ**-006
○ Faulty air bubble sensor circuit part	C: If the circuit part cannot be adjusted, replace the I/O PCB.	CXSM**-101
○ Broken wire in the air bubble sensor harness	C: Replace the air bubble sensor harness.	
○ Malfunction due to external noises	C: Cut off a noise source or keep it at a distance.	

E19(E20) TUBE NON-EQUIPPED OR LARGE BUBBLE

Detection condition	The air bubble sensor has no tube or detects large air bubbles when the air bubble sensor switch is on.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	○	-	○
Rinse	- / - / -	○	○	-	○	-	○
Dia. Wait	- / - / -	○	○	○ SW OFF	○	-	○
Standby	- / - / -	○	○	○ SW OFF	○	-	○
Preparation	- / - / -	○	○	○ SW OFF	○	-	○
Dialysis	- / ○ / -	○	○	○ SW OFF	○	-	-
Retrans	- / - / -	○	○	○ SW OFF	○	-	○
Reset method	Clamp is released by the RESET switch.. Recover BP/HP using their PUMP switches after releasing the clamp.		Remark	When the reset type of the air bubble alarm is [standard]E19 [special]E20			

Symptom/Cause	A:Analysis / C:Countermeasure	
Air bubbles are produced		
○ Entry of air from the blood circuit	C: Remedy places through which air enters to return air bubbles to the drip chamber. If these places cannot be remedied, replace the blood circuit.	
No air bubbles are produced		
○ Tube disconnected from the air bubble sensor	C: Connect the tube again.	
○ Deviation from the calibration of the air bubble sensor	C: Perform the calibration of the air bubble sensor detection circuit	CXAJ**-006
○ Faulty air bubble sensor circuit part	C: If the calibration of the air bubble sensor detection circuit cannot be performed, replace the I/O PCB.	CXAJ**-101
○ Broken wire in the air bubble sensor harness	C: Replace the air bubble sensor harness.	
○ Malfunction due to external noises	C: Cut off a noise source or keep it at a distance.	

E21 SINGLE SWITCH TIME ALARM



Detection condition	(1) Venous pressure did not reach the upper-switch-limit pressure in 4000/(BP flow rate) or 30 seconds. (2) Venous pressure did not reach the lower-switch-limit pressure in the S/N ALARM LOW STROKE TIME.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dialysis (SN)	- / ○ / -	○	○	○	○	-	-
Reset method	Press the RESET switch.			Remarks	During SN operation When alarm is occurred, V19 turn ON. When dialysate pressure value become higher than 300mmHg within 5 seconds, V4 turn ON after V19 turn ON.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
A rise in the venous pressure is slow	A: Check to be sure that the venous pressure rises when the blood pump rotates.	
○ Dented or bended rolling tube in the blood circuit	C: Replace the blood circuit.	
○ Clamp spacing too wide	C: Make adjustment of the clamp spacing.	BHSM**-009
The clamp does not operate	A: For detail, refer to information in the "E64 CLV POSITION ERROR" section. C: For detail, refer to information in the "E64 CLV POSITION ERROR" section.	CXTS**-E64 CXTS**-E64
A rise and a drop in the venous pressure are normal	A: Check for venous pressure changes.	
○ Improper blood flow rate, single needle operation range, and single needle alarm set value	A: Make measurement of a period of time needed to shift. C: Set them to a proper value, respectively.	

E22 T3&T4 DIFFERENCE ALARM



Detection condition	When a difference between the T3 and T4 temperatures exceeds 1.2°C for a period of consecutive five seconds						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dialysis	- / ○ / ○	○	○	-	-	-	-
Reset method	Press the RESET switch.. Auto reset after release from alarm condition for 10 sec.			Remark	While SQHD is in progress or after SQHD is reset, the alarm is not detected unless the bypass operation is reset. While Fluid Stop is in progress the alarm is not detected. While the B powder cartridge exchange operation is in progress or 2rounds after this operation the alarm is not detected.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
There is an actual difference in temperatures		
○ Factors causing temperature changes present in the vicinity of the temperature sensor	A: Check for piping conditions. C: Remedy piping so as to be immune to disturbance.	
There is no actual difference in temperatures		
○ Deviation from the display temperature calibration of the I/O PCB	A: For detail, refer to information in the “E36 T3 TEMPERATURE SENSOR ERROR” For detail, refer to information in the “E37 T4 TEMPERATURE SENSOR ERROR”	CXTS**-E36 CXTS**-E37
○ Faulty T3 or T4 temperature sensor	C: For detail, refer to information in the “E36 T3 TEMPERATURE SENSOR ERROR” For detail, refer to information in the “E37 T4 TEMPERATURE SENSOR ERROR”	CXTS**-E36 CXTS**-E37

E23 ARTERIAL PRESSURE MAX ALARM



Detection condition	When the arterial pressure continuously exceeds the maximum set value for the arterial pressure alarm						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dia. Wait	- / - / -	○	○	○	-	-	○
Standby	- / - / -	○	○	○	-	-	○
Preparation	- / - / -	○	○	○	-	-	○
Dialysis	- / ○ / -	○	○	○	-	-	-
Retrans	- / - / -	○	○	○	-	-	○
Power failure	- / - / -	○	-	○	-	-	○
Reset method	Press the RESET switch.		Remark	This alarm is not detected during the startup test in the standby mode.			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Clogging of arterial pressure line in the blood circuit	A: Check the shunt part for any clogging and the blood circuit for any bended parts. C: If any clogging or bended part is observed, remedy it.	
Fluctuations in arterial pressure due to the movement of a patient	C: If the arterial pressure shows substantial fluctuations, slightly reduce the maximum set value for the arterial pressure alarm at a physician's discretion.	
Substantial fluctuations in the blood pump discharge pressure	A: Check to be sure that the blood pump operates at a right discharge rate and at a right shutoff pressure, and cause no uneven revolutions. C: If the shutoff pressure is low, replace the blood pump. If the blood pump is normal, replace the power supply PCB, I/O PCB, and/or CPU PCB.	BHAJ**-014 BHSM**-013 BHSM**-100 CXSM**-101 BHSM**-105

E24 ARTERIAL PRESSURE MIN ALARM



Detection condition	When the arterial pressure drops below the minimum set value for the arterial pressure alarm						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	○	-	-	○
Dia. Wait	- / - / -	○	○	○	-	-	○
Preparation	- / - / -	○	○	○	-	-	○
Dialysis	- / ○ / -	○	○	○	-	-	-
Retrans	- / - / -	○	○	○	-	-	○
Power failure	- / - / -	○	-	○	-	-	○
Reset method	Press the RESET switch.		Remark	This alarm is not detected during the startup test in the standby mode.			

Symptom/Cause	A:Analysis / C:Countermeasure	
Drop in patient's blood pressure	A: Stop the UF operation, and then check the patient for the condition. If the minimum set value for the arterial pressure alarm is improper, make resetting of the value.	
Disconnection of the blood circuit from the arterial pressure transducer	A: Check to be sure that blood does not enter the venous pressure sensor line, and then connect the blood circuit to the line. C: If blood enters the arterial pressure sensor line (from the pressure transducer to the PV/PA PCB), replace the line.	
Clogging of arterial pressure line in the blood circuit	A: Check the shunt part for any clogging and the blood circuit for any bended parts. C: Remedy the clogged parts.	
Fluctuations in arterial pressure due to the movement of a patient	C: If the arterial pressure shows substantial fluctuations, slightly reduce the minimum set value for the arterial pressure alarm at a physician's discretion.	
Substantial fluctuations in the blood pump discharge pressure	A: Check to be sure that the blood pump operates at a right discharge rate and at a right shutoff pressure, and cause no uneven revolutions. C: If the shutoff pressure is low, replace the blood pump. If the blood pump is normal, replace the blood pump, power supply PCB, and/or CPU PCB.	BHAJ**-014 BHSM**-013 BHSM**-100 CXSM**-101 BHSM**-105

E25 UF COMPENSATION ERROR

Detection condition	Calculated UF correction volume within the software is $\geq \pm 100 \text{g/h}$						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dialysis	- / ○ / -	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
○ Stop the treatment immediately and contact your nearest local branch or sales representative.		

E27 DIALYSATE CONDUCTIVITY ALARM (for Type 1)



Detection condition	1. When a conductivity value in excess of the maximum set value for the conductivity alarm or below the minimum set value for the conductivity alarm is detected. 2. When a conductivity value is higher than 15.6mS/cm or lower than 12.3mS/cm						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dialysis	- / ○ / ○	○	○	-	-	-	-
Retrans	- / - / ○	○	○	-	-	-	-
Reset method	Press the RESET switch. If a conductivity value falls outside of the detection condition for a period of consecutive 10 seconds, the alarm will be reset.			Remark	· While Fluid Stop is in progress or after Fluid Stop is reset unless the bypass operation is reset. · While SQHD is in progress or after SQHD is reset unless the bypass operation is reset. · While 4 rounds after the B powder cartridge exchange operation is in progress the alarm is not detected. Only type 1 of fluid circuit.		

Symptom/Cause	A: Analysis / C: Countermeasure	Work No.
Dialysate has an abnormal actual concentration		
○ Adjustment of actual concentration	A: Collect a fluid sample from the sampling port, and then make measurement of actual concentration by a specified method. C: Make adjustment of actual concentration on the Maintenance Menu 5 screen. A: Make measurement of chamber volume. C: If the chamber volume does not range from 490 to 520mL, replace the chamber assembly.	CXAJ**-021 CXSM**-006
○ Excessive charge flow rate	A: Make measurement of charge flow rate. C: If the charge flow rate does not correspond to the dialysate flow rate, make flow rate adjustment.	CXAJ**-009
○ Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101
○ Faulty “drive PCB”	C: Replace the “drive PCB”.	BHSM**-103
○ Faulty “P3” or “P4”	C: Replace the “P3” or “P4”.	CXSM**-021
○ Bended, clogged, or crushed tube in the concentrate fluid line	C: Remedy the bended, clogged, or crushed tube.	BHAJ**-009
Dialysate has a normal conductivity		
○ Stains (e.g. calcium carbonate) on the electrode of the conductivity sensor, or fluid leaks	C: Perform thorough acid rinse. If the stains cannot be removed, perform solvent rinse with a specified chemical (e.g. CHARON or SHUMMA).	CXSM**-012
○ Loose connection of the “CN218” conductivity sensor connector	A: Visually check the CN218 for any erroneous wire connections, corrosion of contact surface, or broken wires. C: Remedy erroneous wire connections, if any. If any corrosion or broken wires are observed, replace the connector.	

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
○ Broken wire in or short-circuit of the conductivity sensor	A: Disconnect the CD1 connector from the I/O PCB, and then make measurement of "CD1 - GND" resistance to check to be sure that the sensor has no short circuit, broken wires, or loose connection. C: Replace the harness of the CD1.	
○ Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101

E27 DIALYSATE CONDUCTIVITY ALARM (for Type 2)



Detection condition	1. CD1 value over the dialysate conductivity maximum alarm value is detected. Calculation is based on the maximum value in the past 7 rounds. 2. CD1 value below the dialysate conductivity minimum alarm value is detected. Calculation is based on the minimum value in the past 7 rounds.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Preparation	- / - / ○	-	-	-	-	-	-
Dialysis	- / ○ / ○	○	○	-	-	-	-
Retrans	- / - / ○	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	In Prep. complete process, this alarm can be detected only in auto-bypass. Auto reset when out of alarm condition continues consecutive 10sec. Only type 2 of fluid circuit.		

Symptom/Cause	A: Analysis / C: Countermeasure	
Dialysate has an abnormal actual concentration		
○ Adjustment of actual concentration	A: Collect a fluid sample from the sampling port, and then make measurement of actual concentration by a specified method. C: Make adjustment of actual concentration on the Maintenance Menu 5 screen. A: Make measurement of chamber volume. C: If the chamber volume does not range from 98 to 105mL, replace the chamber assembly.	CXAJ**-021 CXSM**-006
○ Excessive charge flow rate	A: Make measurement of charge flow rate. C: If the charge flow rate does not correspond to the dialysate flow rate, make flow rate adjustment.	CXAJ**-009
○ Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101
○ Faulty "drive PCB"	C: Replace the "drive PCB".	BHSM**-103
○ Faulty "P3" or "P4"	C: Replace the "P3" or "P4".	CXSM**-021
○ Bended, clogged, or crushed tube in the concentrate fluid line	C: Remedy the bended, clogged, or crushed tube.	CXAJ**-009
Dialysate has a normal conductivity		
○ Stains (e.g. calcium carbonate) on the electrode of the conductivity sensor, or fluid leaks	C: Perform thorough acid rinse. If the stains cannot be removed, perform solvent rinse with a specified chemical (e.g. CHARON or SHUMMA).	CXSM**-012
○ Loose connection of the "CN218" conductivity sensor connector	A: Visually check the CN218 for any erroneous wire connections, corrosion of contact surface, or broken wires. C: Remedy erroneous wire connections, if any. If any corrosion or broken wires are observed, replace the connector.	

Symptom/Cause	A:Analysis / C:Countermeasure	
○ Broken wire in or short-circuit of the conductivity sensor	A: Disconnect the CD1 connector from the I/O PCB, and then make measurement of "CD1 - GND" resistance to check to be sure that the sensor has no short circuit, broken wires, or loose connection. C: Replace the harness of the CD1.	
○ Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101

E28 AIR BUBBLE SENSOR ERROR (SELF-TEST)



Detection condition	· An error signal of the Bubble CPU was detected. · The error in the self-test signal of the bubble sensor was detected.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
· Rinse Wait	- / - / -	○	○	-	○	-	-
· Rinse	- / - / -	○	○	-	○	-	-
· Dia. Wait	- / - / -	○	○	○ SW OFF	○	-	-
· Standby	- / - / -	○	○	○ SW OFF	○	-	-
· Preparation	- / - / -	○	○	○ SW OFF	○	-	-
· Dialysis	- / ○ / -	○	○	○ SW OFF	○	-	-
· Retrans	- / - / -	○	○	○ SW OFF	○	-	-
Reset method	Press the RESET switch.			Remark	The air bubble ON/OFF SW switch blinking		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
○ Faulty main CPU PCB	C: Replace the main CPU PCB.	BHSM**-100
○ Faulty main I/O PCB	C: Replace the main I/O PCB.	CXSM**-101
○ Faulty air bubble sensor	C: Replace the air bubble sensor.	CXSM**-010

E29 CPU1 TRANSFER ERROR

Detection condition	Detects the abnormal condition transfer.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	○	-	-
Rinse	- / - / -	○	○	-	○	-	-
Dia. Wait	- / - / -	○	○	○	○	-	-
Standby	○ / - / -	○	○	○	○	-	-
Preparation	○ / - / -	○	○	○	○	-	-
Dialysis	○ / ○ / -	○	○	○	○	-	-
Retrans	○ / - / -	○	○	○	○	-	-
Reset method	Recovers by "I/O" switch OFF/ON. (Restart the Device.)			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
○ Faulty main CPU PCB	C: Replace the main CPU PCB.	BHSM**-100

E30 VENOUS PRESSURE MAX LIMIT



Detection condition	When the venous pressure value exceeds the maximum set value for venous pressure limiter						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	-	-	○
Rinse	- / - / -	○	○	-	-	-	○
Dia. Wait	- / - / -	○	○	○	-	-	○
Standby	- / - / -	○	○	○	-	-	○
Preparation	- / - / -	○	○	○	-	-	○
Dialysis	- / ○ / -	○	○	○	-	-	-
Retrans	- / - / -	○	○	○	-	-	○
Reset method	Press the RESET switch.			Remark	This alarm is not detected during the startup test in the standby mode.		

Symptom/Cause	A:Analysis / C:Countermeasure	
The tube in the blood circuit connecting between the venous pressure air chamber and a patient has bends, clogging, or blockage with Robert's clamp.	C: Remove the cause of the blockage, if any.	
The mesh of chamber on the venous pressure side or the connector of the shunt part has blockage due to thrombus.	C: If any blockage is present, replace the blood circuit or remove the thrombus.	
Faulty PV/PA PCB	A: Disconnect the blood circuit from the venous pressure transducer, open it to the atmosphere, and check whether the venous pressure 0 mmHg is displayed. C: If the venous pressure 0 mmHg is not displayed, perform the calibration of the PV/PA PCB. If the calibration cannot be performed, replace the PV/PA PCB.	BHAJ**-001 BHSM**-104
Faulty I/O PCB	C: If the venous pressure does not return to its normal level, even after the PV/PA PCB is replaced, replace the I/O PCB.	CXSM**-101
Faulty main CPU PCB	C: If the venous pressure does not return to its normal level, even after the I/O PCB is replaced, replace the main CPU PCB.	BHSM**-100

E31 VENOUS PRESSURE MIN LIMIT



Detection condition	When the venous pressure value drops below the minimum set value for venous pressure limiter						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	-	-	○
Rinse	- / - / -	○	○	-	-	-	○
Dia. Wait	- / - / -	○	○	○	-	-	○
Standby	- / - / -	○	○	○	-	-	○
Preparation	- / - / -	○	○	○	-	-	○
Dialysis	- / ○ / -	○	○	○	-	-	-
Retrans	- / - / -	○	○	○	-	-	○
Reset method	Press the RESET switch.		Remark	This alarm is not detected during the startup test in the standby mode.			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The DRAIN switch is pressed, while the venous pressure side of the blood circuit is blocked with the clamp and without disconnecting the coupler from the dialyzer.	A: Error in operation.	
Faulty PV/PA PCB	A: Disconnect the blood circuit from the venous pressure transducer, open it to atmosphere, and check whether the venous pressure of 0 mmHg is displayed. C: If the venous pressure of 0 mmHg is not displayed, perform the calibration of the PV/PA PCB. If the calibration cannot be performed, replace the PV/PA PCB.	BHAJ**-001 BHSM**-104
Faulty I/O PCB	C: If the venous pressure does not return to its normal level even after the PV/PA PCB is replaced, replace the I/O PCB.	CXSM**-101
Faulty main CPU PCB	C: If the venous pressure does not return to its normal level even after the I/O PCB is replaced, replace the main CPU PCB.	BHSM**-100

E32 DIALYSATE PRESSURE MAX LIMIT (for Type 1)



Detection condition	When the dialysate pressure exceeds 480 mmHg. (For 5 second in Rinse and Rinse Wait process)						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	2 sec.	2 sec.	○
Rinse	- / - / -	○	○	-	2 sec.	2 sec.	○
Dia. Wait	- / - / -	○	○	2 sec.	2 sec.	2 sec.	○
Standby	- / - / -	○	○	2 sec.	2 sec.	2 sec.	○
Preparation	- / - / -	○	○	2 sec.	2 sec.	2 sec.	○
Dialysis	- / ○ / -	○	○	2 sec.	2 sec.	2 sec.	-
Retrans	- / - / -	○	○	2 sec.	2 sec.	2 sec.	○
Reset method	Press the RESET switch.		Remark	The alarm operations of the BP/HP and clamp continue if the E05 DIALYSATE PRESSURE MAX ALARM was detected simultaneously. Only type 1 of fluid circuit.			

Symptom/Cause	A:Analysis / C:Countermeasure	
Rise in dialysate pressure due to faulty "PR1"	C: Make adjustment of the "PR1" If this adjustment cannot be made, replace the PR1	CXAJ**-008 CXSM**-014
Leaks from the "V10"	C: Overhaul or replace the "V10"	CXSM**-001
An error occurs while in gas purge process.		
○ The "PC2" is excessively closed	C: Make adjustment of the "PC2" so that the dialysate pressure will be in the range of -100 to 0 mmHg for the gas purge operation.	CXAJ**-020
○ The "R2" is clogged	C: Make adjustment of the "R2" so that the dialysate pressure will be in the range of -100 to 0 mmHg for the gas purge operation.	CXAJ**-020
○ The "V4" does not operate	A: Energize or de-energize the "V4" on the maintenance Mode 4 screen to check it for operation. If the "V4" does not operate, disconnect the connector from the "V4", check to be sure that a 24V voltage is output from the connector while the "V4" is in operation. (Acceptance range: 24V±10%) C: If a 24V voltage is output, replace the "V4" PCB. If a 24V voltage is not output, replace the I/O PCB.	CXSM**-001 CXSM**-101
Rise in dialysate pressure due to bended blood circuit	C: If the blood circuit has any bended part, remove it.	
Faulty dialysate pressure sensor assembly	A: Disconnect the hose from the dialysate pressure sensor assembly and open it to atmosphere, and check whether the dialysate pressure of 0 mmHg is displayed. C: If the dialysate pressure of 0 mmHg is not displayed, perform the calibration of the dialysate pressure sensor PCB. If the calibration cannot be performed, replace the dialysate pressure sensor assembly.	BHAJ**-002 BHSM**-019

Symptom/Cause	A:Analysis / C:Countermeasure	
Broken wire in the harness of the dialysate pressure sensor assembly or loose connection of connector	A: Lightly pull the lead wires of the sensor assembly to check them for disconnection or loose connection. C: If the lead wires have any disconnection or loose connection, remedy it.	
Clogging of drain fluid tube	A: Disconnect the drain fluid port, and then check to be sure that the tube is properly connected	
○ Bended or clogged drain fluid tube	C: If the drain fluid tube has any bend or clogging, remedy it.	
○ Drain fluid tube and drain fluid port mounted in elevated positions	C: The drain fluid tube must be installed between 0 and maximum 60 cm above the floor.	
○ Drain fluid tube too long	C: Adjust the drain fluid tube length to maximum 3m.	

E32 DIALYSATE PRESSURE MAX LIMIT (for Type 2)



Detection condition	When the dialysate pressure exceeds 630 mmHg.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
All	- / - / ○	○	○	-	-	2 sec.	○
Reset method	Press the RESET switch.			Remark	Only type 2 of fluid circuit.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Rise in dialysate pressure due to faulty "PR1"	C: Make adjustment of the "PR1" If this adjustment cannot be made, replace the PR1	CXAJ**-008 CXSM**-014
Leaks from the "V10"	C: Overhaul or replace the "V10"	CXSM**-001
An error occurs while in gas purge process.		
○ The "NV4" is excessively closed or the "NV6" is excessively opened	C: Make adjustment of the "NV4" and "NV6" so that the dialysate pressure will be in the range of 0 to +100 mmHg for the gas purge operation.	CXAJ**-020
○ RV2 and/or NV5 is clogged	C: Make adjustment of RV2 and/or NV5 so that the dialysate pressure will fall in the range of 0 to +100 mmHg for the gas purge operation.	CXAJ**-020
○ The "V4" does not operate	A: Energize or de-energize the "V4" on the maintenance Mode 4 screen to check it for operation. If the "V4" does not operate, disconnect the connector from the "V4", check to be sure that a 24V voltage is output from the connector while the "V4" is in operation. (Acceptance range: 24V±10%) C: If a 24V voltage is output, replace the "V4" PCB. If a 24V voltage is not output, replace the I/O PCB.	CXSM**-001 CXSM**-101
Rise in dialysate pressure due to bended blood circuit	C: If the blood circuit has any bended part, remove it.	
Faulty dialysate pressure sensor assembly	A: Disconnect the hose from the dialysate pressure sensor assembly and open it to atmosphere, and check whether the dialysate pressure of 0 mmHg is displayed. C: If the dialysate pressure of 0 mmHg is not displayed, perform the calibration of the dialysate pressure sensor PCB. If the calibration cannot be performed, replace the dialysate pressure sensor assembly.	BHAJ**-002 BHSM**-019
Broken wire in the harness of the dialysate pressure sensor assembly or loose connection of connector	A: Lightly pull the lead wires of the sensor assembly to check them for disconnection or loose connection. C: If the lead wires have any disconnection or loose connection, remedy it.	

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Clogging of drain fluid tube	A: Disconnect the drain fluid port, and then check to be sure that the tube is properly connected	
○ Bended or clogged drain fluid tube	C: If the drain fluid tube has any bend or clogging, remedy it.	
○ Drain fluid tube and drain fluid port mounted in elevated positions	C: The drain fluid tube must be installed between 0 and maximum 60 cm above the floor.	
○ Drain fluid tube too long	C: Adjust the drain fluid tube length to maximum 3m.	

E33 DIALYSATE PRESSURE MIN LIMIT (for Type 1)



Detection condition	When the dialysate pressure drops below -480 mmHg. (For 5 second in Rinse and Rinse Wait process)						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	2 sec	2 sec	○
Rinse	- / - / -	○	○	-	2 sec	2 sec	○
Dia. Wait	- / - / -	○	○	2 sec	2 sec	2 sec	○
Standby	- / - / -	○	○	2 sec	2 sec	2 sec	○
Preparation	- / - / -	○	○	2 sec	2 sec	2 sec	○
Dialysis	- / ○ / -	○	○	2 sec	2 sec	2 sec	-
Retrans	- / - / -	○	○	2 sec	2 sec	2 sec	○
Reset method	Press the RESET switch.		Remark	The alarm operations of the BP/HP and clamp continue if the E06 DIALYSATE PRESSURE MIN ALARM was detected simultaneously. Only type 1 of fluid circuit.			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Excessive clogging of the dialyser	C: Replace the dialyser.	
Leaks at the "V4"	A: Clamp the "V4" line with forceps to check whether the leaks are remedied. C: Disassemble and clean, or replace the "V4".	CXSM**-001
The "V4" opens due to a malfunction of the "FSW"	A: Clamp the "V4" line with forceps to check whether the leaks are remedied. C: Disassemble and clean, or replace the "AS2".	CXSM**-015
Clogging of the "RV2"	C: Make adjustment of the closed circuit. If this adjustment cannot be made, disassemble and clean or replace the "RV2".	CXAJ**-010 CXSM**-005
An error occurs while in gas purge process		
○ The "R3" is clogged	C: Make adjustment of the "R3" so that the dialysate pressure will be in the range of -100 to 0 mmHg for the gas purge operation.	
○ The "V10" does not operate	A: Energize or de-energize the "V10" on the maintenance Mode 4 screen to check it for operation. If the "V10" does not work, disconnect the connector from "V10", and then check 24V voltage at the output of the connector while the "V10" is in operation. (Acceptable range: 24V±10%) C: If a 24V voltage is output, replace the "V10". If a 24V voltage is not output, replace the I/O PCB.	CXSM**-001 CXSM**-101

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty dialysate pressure sensor assembly	A: Disconnect the hose from the dialysate pressure sensor assembly to open it to atmosphere, and check whether the dialysate pressure of 0 mmHg is displayed. C: If the dialysate pressure of 0 mmHg is not displayed, perform the calibration of the dialysate pressure sensor PCB. If the calibration cannot be performed, replace the dialysate pressure sensor assembly.	BHAJ**-002 CXSM**-007
Broken wire in the harness of the dialysate pressure sensor assembly or loose connection of connector	A: Lightly pull the lead wires of the sensor assembly to check them for disconnection or loose connection. C: If the lead wires have any disconnection or loose connection, remedy it.	

E33 DIALYSATE PRESSURE MIN LIMIT (for Type 2)



Detection condition	When the dialysate pressure drops below -630 mmHg.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
All	- / - / ○	○	○	-	-	2 sec.	○
Reset method	Press the RESET switch.			Remark	Only type 2 of fluid circuit.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Excessive clogging of the dialyser	C: Replace the dialyser.	
Leaks at the “V4”	A: Clamp the “V4” line with forceps to check whether the leaks are remedied. C: Disassemble and clean, or replace the “V4”.	CXSM**-001
The “V4” opens due to a malfunction of the “FSW”	A: Clamp the “V4” line with forceps to check whether the leaks are remedied. C: Disassemble and clean, or replace the “AS2”.	CXSM**-015
Clogging of the “RV2”	C: Make adjustment of the closed circuit. If this adjustment cannot be made, disassemble and clean or replace the “RV2”.	CXAJ**-010 BHSM**-005
An error occurs while in gas purge process		
○ The “NV4” is excessively opened or the “NV6” is excessively closed	C: Make adjustment of the “NV4” and “NV6” so that the dialysate pressure will be in the range of 0 to +100 mmHg for the gas purge operation.	
○ The “V10” does not operate	A: Energize or de-energize the “V10” on the maintenance Mode 4 screen to check it for operation. C: If the “V10” does not work, disconnect the connector from “V10”, and then check 24V voltage at the output of the connector while the “V10” is in operation. (Acceptable range: 24V±10%) C: If a 24V voltage is output, replace the “V10”. If a 24V voltage is not output, replace the I/O PCB.	CXSM**-001 CXSM**-101
Faulty dialysate pressure sensor assembly	A: Disconnect the hose from the dialysate pressure sensor assembly to open it to atmosphere, and check whether the dialysate pressure of 0 mmHg is displayed. C: If the dialysate pressure of 0 mmHg is not displayed, perform the calibration of the dialysate pressure sensor PCB. If the calibration cannot be performed, replace the dialysate pressure sensor assembly.	BHAJ**-002 CXSM**-007
Broken wire in the harness of the dialysate pressure sensor assembly or loose connection of connector	A: Lightly pull the lead wires of the sensor assembly to check them for disconnection or loose connection. C: If the lead wires have any disconnection or loose connection, remedy it.	

E34 T1 TEMPERATURE SENSOR ERROR



Detection condition	When the T1 temperature sensor detects temperatures of not less than 99.8°C for a period of consecutive 5 seconds. When the T1 temperature sensor detects temperatures of not more than 0.5°C for a period of consecutive 5 seconds.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	-	-	○
Rinse	- / - / -	○	○	-	-	-	○
Dia. Wait	- / - / -	○	○	-	-	-	○
Standby	○ / - / -	○	○	-	-	-	○
Preparation	○ / - / -	○	○	-	-	-	○
Dialysis	○ / ○ / -	○	○	-	-	-	-
Retrans	○ / - / -	○	○	-	-	-	○
Reset method	Press the RESET switch.			Remark	This alarm is not detected during the startup test in the standby mode. Heater OFF when the heater is on.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The T1 display or also another analog display is out of order	A: Check in the display of all the temperature sensors and other sensors (e.g. fluid pressure sensor) are working properly. If not, check 5.00V±0.01V at the test point on the I/O PCB. C: If the voltage does not come to 5.00V±0.01V, perform voltage calibration.	CXSM**-101 CXSM**-101
Hot water is poured from the feed water port	C: Since it is supposed that the fluid circuit part is damaged, the fluid circuit part should be overhauled.	
Loose connection of the T1 temperature sensor connector	A: Visually check the CN214 for any erroneous wire connections, corrosion of contact surface, or broken wires. C: Remedy erroneous wire connections. If any corrosion or broken wires are observed, replace the connector.	BHSM**-008
Broken wire or short circuit in the “T1 temperature sensor”	A: Disconnect the T1 temperature sensor from the CN214 on the I/O PCB. Then measure the resistance between terminals, to check that the sensor has no short circuit, broken wires, or loose connection. C: If the sensor has any short circuit, broken wire, or loose connection, replace the “T1 temperature sensor”.	BHSM**-008
Deviation from the display temperature calibration of the I/O PCB	A: Disconnect T1 temperature sensor from the PCB. Connect an another normal temperature (e.g. T2) to the connector of T1, to confirm condition of PCB. C: If the temperature display is abnormal, perform the calibration of the I/O PCB. If the calibration cannot be performed, replace the I/O PCB.	BHAJ**-022 CXSM**-101

E35 T2 TEMPERATURE SENSOR ERROR



Detection condition	When the T2 temperature sensor detects temperatures of not less than 99.8°C for a period of consecutive 5 seconds. When the T2 temperature sensor detects temperatures of not more than 0.5°C for a period of consecutive 5 seconds.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	-	-	○
Rinse	- / - / -	○	○	-	-	-	○
Dia. Wait	- / - / -	○	○	-	-	-	○
Standby	○ / - / -	○	○	-	-	-	○
Preparation	○ / - / -	○	○	-	-	-	○
Dialysis	○ / ○ / -	○	○	-	-	-	-
Retrans	○ / - / -	○	○	-	-	-	○
Reset method	Press the RESET switch.			Remark	This alarm is not detected during the startup test in the standby mode. Heater OFF when the heater is on.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The T2 display or also another analog display is out of order	A: Check in the display of all the temperature sensors and other sensors (e.g. fluid pressure sensor) are working properly. If not, check 5.00V±0.01V at the test point on the I/O PCB. C: If the voltage does not come to 5.00V±0.01V, perform voltage calibration.	CXSM**-101 CXSM**-101
Loose connection of the T2 temperature sensor connector	A: Visually check the CN215 for any erroneous wire connections, corrosion of contact surface, or broken wires. C: Remedy erroneous wire connections. If any corrosion or broken wires are observed, replace the connector.	BHSM**-008
Broken wire or short circuit in the “T2 temperature sensor”	A: Disconnect the T2 temperature sensor from the CN215 on the I/O PCB. Measure the resistance between terminals and check that the sensor has no short circuit, broken wires, or loose connection. C: If the sensor has any short circuit, broken wire, or loose connection, replace the “T2 temperature sensor”.	BHSM**-008
Deviation from the display temperature calibration of the I/O PCB	A: Disconnect T2 temperature sensor from the PCB. Connect an another normal temperature (e.g.T3) to the connector of T2, to confirm condition of PCB. C: If the temperature display is abnormal, perform the calibration of the temperature sensor module PCB. If the calibration cannot be performed, replace the I/O PCB.	BHAJ**-003 CXSM**-101

E36 T3 TEMPERATURE SENSOR ERROR



Detection condition	When the T3 temperature sensor detects temperatures of not less than 99.8°C for a period of consecutive 5 seconds. When the T3 temperature sensor detects temperatures of not more than 0.5°C for a period of consecutive 5 seconds.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	-	-	○
Rinse	- / - / -	○	○	-	-	-	○
Dia. Wait	- / - / -	○	○	-	-	-	○
Standby	○ / - / -	○	○	-	-	-	○
Preparation	○ / - / -	○	○	-	-	-	○
Dialysis	○ / ○ / -	○	○	-	-	-	-
Retrans	○ / - / -	○	○	-	-	-	○
Reset method	Press the RESET switch.			Remark	This alarm is not detected during the startup test in the standby mode. Heater OFF when the heater is on.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The T3 display or also another analog display is out of order	A: Check if the temperature sensors and other sensors (e.g. fluid pressure sensor) are working properly. If all the temperature sensors do not display the correct value. Check 5.00V±0.01V at the test point on the I/O PCB. C: If the voltage does not come to 5.00V±0.01V, perform voltage calibration.	CXSM**-101 CXSM**-101
Loose connection of the T3 temperature sensor connector	A: Visually check the CN216 for any erroneous wire connections, corrosion of contact surface, or broken wires. C: Remedy erroneous wire connections, if any. If any corrosion or broken wires are observed, replace the connector.	BHSM**-008
Broken wire or short circuit in the “T3 temperature sensor”	A: Disconnect the T3 temperature sensor from the CN216 on the I/O PC. Then measure the resistance between terminals, check if the sensor has no short circuit, broken wires, or loose connection. C: If the sensor has any short circuit, broken wire, or loose connection, replace the “T3 temperature sensor”.	BHSM**-008
Deviation from the display temperature calibration of the I/O PCB	A: Disconnect T3 temperature sensor from the PCB. Connect an another normal temperature (e.g.T4) to the connector of T3, to confirm condition of PCB. C: If the temperature display is abnormal, perform the calibration of the temperature sensor module PCB. If the calibration cannot be performed, replace the I/O PCB.	BHAJ**-003 CXSM**-101

E37 T4 TEMPERATURE SENSOR ERROR



Detection condition	When the T4 temperature sensor detects temperatures of not less than 99.8°C for a period of consecutive 5 seconds. When the T4 temperature sensor detects temperatures of not more than 0.5°C for a period of consecutive 5 seconds.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	-	-	○
Rinse	- / - / -	○	○	-	-	-	○
Dia. Wait	- / - / -	○	○	-	-	-	○
Standby	○ / - / -	○	○	-	-	-	○
Preparation	○ / - / -	○	○	-	-	-	○
Dialysis	○ / ○ / -	○	○	-	-	-	-
Retrans	○ / - / -	○	○	-	-	-	○
Reset method	Press the RESET switch.			Remark	This alarm is not detected during the startup test in the standby mode. Heater OFF when the heater is on.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The T4 display or also another analog display is out of order	A: Check if the temperature sensors and other sensors (e.g. fluid pressure sensor) are working properly. If all the temperature sensors do not display the correct value. Check 5.00V±0.01V at the test point on the I/O PCB.	CXSM**-101
	C: If the voltage does not come to 5.00V±0.01V, perform voltage calibration.	CXSM**-101
Loose connection of the T4 temperature sensor connector	A: Visually check the CN217 for any erroneous wire connections, corrosion of contact surface, or broken wires. C: Remedy erroneous wire connections, if any. If any corrosion or broken wires are observed, replace the connector.	BHSM**-008
Broken wire or short circuit in the “T4 temperature sensor”	A: Disconnect the T4 temperature sensor from the CN217 on the I/O PCB. Measure the resistance between terminals, check that the sensor has no short circuit, broken wires, or loose connection. C: If the sensor has any short circuit, broken wire, or loose connection, replace the “T4 temperature sensor”.	BHSM**-008
Deviation from the display temperature calibration of the I/O PCB	A: Disconnect T4 temperature sensor from the PCB. Connect an another normal temperature (e.g.T1) to the connector of T4, to confirm condition of PCB. C: If the temperature display is abnormal, perform the calibration of the temperature sensor module PCB (CD1). If the calibration cannot be performed, replace the I/O PCB.	BHAI**-004 CXSM**-101

E38 HP ROTATION ERROR

Detection condition	When PH8 and PH7 turn ON in the order described ideal time elapsed during starting the heparin pump. (When the heparin pump rotates backward)						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dialysis waiting	- / - / -	○	○	HP STOP	-	-	-
Standby	- / - / -	○	○	HP STOP	-	-	-
Preparation	- / - / -	○	○	HP STOP	-	-	-
Dialysis	- / - / -	○	○	HP STOP	-	-	-
Retrans	- / - / -	○	○	HP STOP	-	-	-
Reset method	Press the RESET switch.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The heparin pump motor normally rotates but stops rotating as soon as the alarm is issued	A: When starting the heparin pump, check whether the “PH7” and “PH8” turn ON in the order described on the Operation Display screen.	
○ Defective detection of the dog “PH7” or “PH8”	A: Check the “PH7” and “PH8” for any stains. C: Wipe stains off them, if any. C: If the error reoccurs even after wiping stains off them, replace the HP sensor PCB.	BHSM**-003
○ Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101
The heparin pump motor rotates backward		
○ Faulty drive PCB	C: Replace the drive PCB.	BHSM**-103
○ Faulty motor	C: Replace the motor.	BHSM**-003

E39 UNINSTALLED HP ALARM



Detection condition	When LS6 turns OFF while the heparin pump is in operation						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Preparation	- / - / -	○	○	○ HP Stop	-	-	○
Dialysis	- / - / -	○	○	○ HP Stop	-	-	-
Retrans	- / - / -	○	○	○ HP Stop	-	-	○
Power failure	- / - / -	○	-	○ HP Stop	-	-	○
Reset method	· Press the RESET switch. · Turn HP SW off. · When LS6 is ON, recover automatically.			Remark	This alarm is not detected at an infusion rate of 0.0 L/h.		

Symptom/Cause	A:Analysis / C:Countermeasure	
A syringe drops off	C: If a syringe drops off, reinstall it	
Faulty "LS6"	A: Disconnect the CN205 from the I/O PCB. Manually turn ON or OFF the "LS6". Then check for continuity between Pins 1 and 2. • Short-circuit when the "LS6" turns ON (normal state). • Open when the "LS6" turns OFF (overloaded state). C: If the "LS6" is faulty, replace it.	BHSM**-003
Broken wire in the harness of the "LS6" and loose connection of connector	A: Lightly pull the lead wires of the CN205 on the I/O PCB to check them for any disconnection or loose connection. Furthermore, check the "LS6" for soldered conditions. C: Remedy or replace the harness.	
Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101

E40 +5V (SUB) POWER ERROR

Detection condition	When the +5V (SUB) power supply voltage is out of range : +4.8 to +5.5V						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	○	-	-
Rinse	- / - / -	○	○	-	○	-	-
Dia. Wait	- / - / -	○	○	○ (SW OFF)	○	-	-
Standby	○ / - / -	○	○	○ (SW OFF)	○	-	-
Preparation	○ / - / -	○	○	○ (SW OFF)	○	-	-
Dialysis	○ / ○ / -	○	○	○ (SW OFF)	○	-	-
Retrans	○ / - / -	○	○	○ (SW OFF)	○	-	-
Power failure	- / - / -	○	-	○ (SW OFF)	○	-	-
Reset method	Press the RESET switch.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
○ Faulty I/O PCB	A: Measure a voltage between WJ1 and Pin 5 of the CN106 on the power supply PCB.	CXSM**-101
○ Faulty power supply PCB	C: If a 5V voltage is output, replace the I/O PCB. (Acceptance range: 5V±4%)	
	C: If a 5V voltage is not output, replace the power supply PCB. (Acceptance range: 5V±4%)	BHSM**-105

E41 Vref (SUB) POWER ERROR



Detection condition	When the Vref (SUB) power supply voltage is out of range : +4.85 to +5.15V						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	○	-	-
Rinse	- / - / -	○	○	-	○	-	-
Dia. Wait	- / - / -	○	○	○ (SW OFF)	○	-	-
Standby	○ / - / -	○	○	○ (SW OFF)	○	-	-
Preparation	○ / - / -	○	○	○ (SW OFF)	○	-	-
Dialysis	○ / ○ / -	○	○	○ (SW OFF)	○	-	-
Retrans	○ / - / -	○	○	○ (SW OFF)	○	-	-
Power failure	- / - / -	○	-	○ (SW OFF)	○	-	-
Reset method	Press the RESET switch.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty I/O PCB	C: Check that the Vref (SUB) power supply voltage is in a range of: +4.9 to +5.1V.	CXSM**-101
	If calibration cannot be performed, replace the I/O PCB.	CXSM**-101

E42 +24V POWER ERROR



Detection condition	When the +24 power supply voltage is out of range: +21.6 to +26.4V						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	○	-	-
Rinse	- / - / -	○	○	-	○	-	-
Dia. Wait	- / - / -	○	○	○ (SW OFF)	○	-	-
Standby	○ / - / -	○	○	○ (SW OFF)	○	-	-
Preparation	○ / - / -	○	○	○ (SW OFF)	○	-	-
Dialysis	○ / ○ / -	○	○	○ (SW OFF)	○	-	-
Retrans	○ / - / -	○	○	○ (SW OFF)	○	-	-
Reset method	Press the RESET switch.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty annunciator light	A: Disconnect the CN220 from the I/O PCB. Measure a voltage between WJ1 and Pin 1 of the CN106. C: If a 24V voltage is output, replace the annunciator light. (Acceptance range: 24V±10%)	
Faulty valve	A: Disconnect the CN211 from the I/O PCB. Measure a voltage between WJ1 and Pin 1 of the CN106. C: If a 24V voltage is output, replace any faulty part(s) out of the "V9a", "V9b", "V10", "V11", "V12", "V17", "V19", "V21", and "V22". (Acceptance range: 24V±10%) A: Disconnect the CN212 from the I/O PCB. Measure a voltage between WJ1 and Pin 1 of the CN106. C: If a 24V voltage is output, replace any faulty part(s) out of the "V1", "V4", "V5a", "V5b", "V6a", "V6b", "V7a", "V7b", "V8a" and "V8b". (Acceptance range: 24V±10%) A: Disconnect the CN237 from the I/O PCB. Measure a voltage between WJ1 and Pin 1 of the CN106. (B powder option installation) C: If a 24V voltage is output, replace any faulty part(s) out of the "V27a", "V27b", "V28a", "V28b", "V29", "V30" and "V31". (Acceptance range: 24V±10%) A: Disconnect the CN242 from the I/O PCB. Measure a voltage between WJ1 and Pin 1 of the CN106. (Hot rinse option installation) C: If a 24V voltage is output, replace any faulty part out of the "V33". (Acceptance range: 24V±10%) A: Disconnect the CN245 from the I/O PCB. Measure a voltage between WJ1 and Pin 1 of the CN106. (Hot rinse option installation) C: If a 24V voltage is output, replace any faulty part(s) out of the	CXSM**-001 CXSM**-002 CXSM**-001 CXSM**-001 BHSM**-022 CXSM**-001

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
	“V34” and “V35”. (Acceptance range: 24V±10%)	
Faulty flow meter	A: Disconnect the CN213 from the I/O PCB. Measure a voltage between WJ1 and Pin 1 of the CN106. C: If a 24V voltage is output, replace any faulty part(s) out of the “FS1”, “FS2”, and “FS3”. (Acceptance range: 24V±10%) A: Disconnect the CN241 from the I/O PCB. Measure a voltage between WJ1 and Pin 1 of the CN106. (Hot rinse option installation) C: If a 24V voltage is output, replace any faulty part out of the “FS5”. (Acceptance range: 24V±10%)	CXSM**-012
Faulty fan	A: Disconnect the CN112 from the power supply PCB. Measure a voltage between WJ1 and Pin 1 of the CN106. C: If a 24V voltage is output, replace the fan 1. (Acceptance range: 24V±10%) A: Disconnect the CN118 from the power supply PCB. Measure a voltage between WJ1 and Pin 1 of the CN106. C: If a 24V voltage is output, replace the CLV fan. (Acceptance range: 24V±10%) A: Disconnect the CN203 from the I/O PCB. Measure a voltage between WJ1 and Pin 1 of the CN106. C: If a 24V voltage is output, replace the fan 2. (Acceptance range: 24V±10%) A: Disconnect the CN248 from the I/O PCB. Measure a voltage between WJ1 and Pin 1 of the CN106. (Hot rinse option installation) C: If a 24V voltage is output, replace the fan 3a. (Acceptance range: 24V±10%) A: Disconnect the CN242 from the I/O PCB. Measure a voltage between WJ1 and Pin 1 of the CN106. (Hot rinse option installation) C: If a 24V voltage is output, replace the fan 3b. (Acceptance range: 24V±10%)	
Faulty I/O PCB	A: If the alarm is detected even after the annunciator light, valves, flow meter, and fan are replaced. Disconnect the CN106 from the power supply PCB. Measure a voltage between WJ1 and Pin 1 of the CN101. C: If a 24V voltage is output, replace the I/O PCB. (Acceptance range: 24V±10%)	CXSM**-101
Faulty heparin pump	A: Disconnect the CN1002 from the drive PCB. Measure a voltage between WJ1 and Pin 1 of the CN106. C: If a 24V voltage is output, replace the heparin pump. (Acceptance range: 24V±10%)	BHSM**-003
Faulty UF pump	A: Disconnect the CN1003 from the drive PCB. Measure a voltage between WJ1 and Pin 1 of the CN106. C: If a 24V voltage is output, replace the UF pump. (Acceptance range: 24V±10%)	

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty P3	A: Disconnect the CN1004 from the drive PCB. Measure a voltage between WJ1 and Pin 1 of the CN106. C: If a 24V voltage is output, replace the P3. (Acceptance range: 24V±10%)	CXSM**-021
Faulty P4	A: Disconnect the CN1005 from the drive PCB. Measure a voltage between WJ1 and Pin 1 of the CN106. C: If a 24V voltage is output, replace the P4. (Acceptance range: 24V±10%)	CXSM**-021
Faulty drive PCB	A: If the alarm is detected even after the heparin pump, UF pump, P3, and P4, disconnect the CN107 from the power supply PCB. Measure a voltage between WJ1 and Pin 1 of the CN101. C: If a 24V voltage is output, replace the drive PCB. (Acceptance range: 24V±10%)	BHSM**-103
Faulty blood pump	A: Disconnect the CN114 from the power supply PCB. Measure a voltage between WJ1 and Pin 1 of the CN106. C: If a 24V voltage is output, replace the blood pump. (Acceptance range: 24V±10%)	BHSM**-013
Faulty pump	A: Disconnect the CN109 from the power supply PCB. Measure a voltage between WJ1 and Pin 1 of the CN106. C: If a 24V voltage is output, replace the P1 and P2. (Acceptance range: 24V±10%)	CXSM**-007
Faulty fan for clamp	A: Disconnect the CN118 from the power supply PCB. Measure a voltage between WJ1 and Pin 1 of the CN106. C: If a 24V voltage is output, replace the fan for clamp. (Acceptance range: 24V±10%)	BHSM**-009
Faulty clamp	A: Disconnect the CN117 from the power supply PCB. Measure a voltage between WJ1 and Pin 1 of the CN106. C: If a 24V voltage is output, replace the clamp. (Acceptance range: 24V±10%)	CXSM**-112
Faulty relay	A: Disconnect the CN103 from the power supply PCB. Measure a voltage between WJ1 and Pin 1 of the CN106. C: If a 24V voltage is output, replace the relay. (Acceptance range: 24V±10%)	CXSM**-112
Faulty heater drive relay or faulty SSR	A: Disconnect the CN110 from the power supply PCB. Measure a voltage between WJ1 and Pin 1 of the CN106. C: If a 24V voltage is output, replace the relay, SSR1, and SSR2. (Acceptance range: 24V±10%)	BHSM**-105
Faulty power supply PCB	C: If the alarm is detected even after the heparin pump, UF pump, P3, and P4 are replaced, replace the power supply PCB.	

E43 +5V POWER ERROR

Detection condition	When the +5 power supply voltage is out of the range :+4.5 to +5.5V						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	○	-	-
Rinse	- / - / -	○	○	-	○	-	-
Dia. Wait	- / - / -	○	○	○ (SW OFF)	○	-	-
Standby	○ / - / -	○	○	○ (SW OFF)	○	-	-
Preparation	○ / - / -	○	○	○ (SW OFF)	○	-	-
Dialysis	○ / ○ / -	○	○	○ (SW OFF)	○	-	-
Retrans	○ / - / -	○	○	○ (SW OFF)	○	-	-
Power failure	- / - / -	○	-	○ (SW OFF)	○	-	-
Reset method	Press the RESET switch.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty I/O PCB	A: Measure a voltage between WJ1 and Pin 5 of the CN106 on the power supply PCB. C: If a 5V voltage is output, replace the I/O PCB. (Acceptance range: 5V±5%)	CXSM**-101
Faulty main CPU PCB	A: Disconnect the CN105 from the power supply PCB. Measure a voltage between WJ1 and Pin 5 of the CN106. C: If a 5V voltage is output, replace the main CPU PCB. (Acceptance range: 5V±5%)	BHSM**-100
Faulty BPC	A: Disconnect the CN113 from the power supply PCB. Measure a voltage between WJ1 and Pin 5 of the CN106. C: If a 5V voltage is output, replace the BPC. (Acceptance range: 5V±5%)	BHSM**-013
Faulty drive PCB	A: Disconnect the CN116 from the power supply PCB. Measure a voltage between WJ1 and Pin 5 of the CN106. C: If a 5V voltage is output, replace the drive PCB. (Acceptance range: 5V±5%)	BHSM**-103
Faulty panel CPU PCB	A: Disconnect the CN119 from the power supply PCB. Measure a voltage between WJ1 and Pin 5 of the CN106. C: If a 5V voltage is output, replace the panel CPU PCB. (Acceptance range: 5V±5%)	CXSM**-106

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty power supply PCB	C: If a 5V voltage is not output even after all parts are checked, replace the power supply PCB.	BHSM**-105

E44 Vref POWER ERROR

Detection condition	When the Vref power supply voltage is out of the range : +4.85 to +5.15V						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	○	-	-
Rinse	- / - / -	○	○	-	○	-	-
Dia. Wait	- / - / -	○	○	○ (SW OFF)	○	-	-
Standby	○ / - / -	○	○	○ (SW OFF)	○	-	-
Preparation	○ / - / -	○	○	○ (SW OFF)	○	-	-
Dialysis	○ / ○ / -	○	○	○ (SW OFF)	○	-	-
Retrans	○ / - / -	○	○	○ (SW OFF)	○	-	-
Power failure	- / - / -	○	-	○ (SW OFF)	○	-	-
Reset method	Press the RESET switch.		Remark				

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty I/O PCB	A: Check if the Vref power supply voltage is in a range of: +4.9 to +5.1V.	CXSM**-101
	C: If calibration cannot be performed, replace the I/O PCB.	CXSM**-101

E45 +12V POWER ERROR



Detection condition	When the +12 power supply voltage is out of the range : +10.8 to +13.2V						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	○	-	-
Rinse	- / - / -	○	○	-	○	-	-
Dia. Wait	- / - / -	○	○	○ (SW OFF)	○	-	-
Standby	○ / - / -	○	○	○ (SW OFF)	○	-	-
Preparation	○ / - / -	○	○	○ (SW OFF)	○	-	-
Dialysis	○ / ○ / -	○	○	○ (SW OFF)	○	-	-
Retrans	○ / - / -	○	○	○ (SW OFF)	○	-	-
Power failure	- / - / -	○	-	○ (SW OFF)	○	-	-
Reset method	Press the RESET switch.		Remark				

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty I/O PCB	A: Measure a voltage between WJ1 and Pin 10 of the CN106 on the power supply PCB. C: If a 12V voltage is output, replace the I/O PCB.	CXSM**-101
Faulty PV/PA PCB (on the arterial side)	A: Disconnect the CN206 from the I/O PCB. Measure a voltage between WJ1 and Pin 10 of the CN106. C: If a 12V voltage is output, replace the PA PCB. (Acceptance range: 12V±5%)	BHSM**-104
Faulty PV/PA PCB (on the venous side)	A: Disconnect the CN207 from the I/O PCB. Measure a voltage between WJ1 and Pin 10 of the CN106. C: If a 12V voltage is output, replace the PV PCB. (Acceptance range: 12V±5%)	BHSM**-104
Faulty fluid pressure sensor assembly	A: Disconnect the CN208 from the I/O PCB. Measure a voltage between WJ1 and Pin 10 of the CN106. C: If a 12V voltage is output, replace the PD assembly. (Acceptance range: 12V±5%)	BHSM**-019
Faulty blood leak sensor assembly	A: Disconnect the CN209 from the I/O PCB. Measure a voltage between WJ1 and Pin 10 of the CN106. C: If a 12V voltage is output, replace the BLD assembly. (Acceptance range: 12V±5%)	BHSM**-018
Faulty power supply PCB	C: If a 12V voltage is not output even after all parts are checked, replace the power supply PCB.	BHSM**-105

E46 -12V POWER ERROR

Detection condition	When the -12 power supply voltage is out of the range : -14.4 to -9.6V						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	○	-	-
Rinse	- / - / -	○	○	-	○	-	-
Dia. Wait	- / - / -	○	○	○ (SW OFF)	○	-	-
Standby	○ / - / -	○	○	○ (SW OFF)	○	-	-
Preparation	○ / - / -	○	○	○ (SW OFF)	○	-	-
Dialysis	○ / ○ / -	○	○	○ (SW OFF)	○	-	-
Retrans	○ / - / -	○	○	○ (SW OFF)	○	-	-
Power failure	- / - / -	○	-	○ (SW OFF)	○	-	-
Reset method	Press the RESET switch.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty I/O PCB	A: Measure a voltage between WJ1 and Pin 12 of the CN106 on the power supply PCB. C: If a -12V voltage is output, replace the I/O PCB. (Acceptance range: -12V±5%)	CXSM**-101
Faulty PV/PA PCB (on the arterial side)	A: Disconnect the CN206 from the I/O PCB. Measure a voltage between WJ1 and Pin 12 of the CN106. C: If a -12V voltage is output, replace the PA PCB. (Acceptance range: -12V±5%)	BHSM**-104
Faulty PV/PA PCB (on the venous side)	A: Disconnect the CN207 from the I/O PCB. Measure a voltage between WJ1 and Pin 12 of the CN106. C: If a -12V voltage is output, replace the PV PCB. (Acceptance range: -12V±5%)	BHSM**-104
Faulty blood leak sensor assembly	A: Disconnect the CN209 from the I/O PCB. Measure a voltage between WJ1 and Pin 12 of the CN106. C: If a -12V voltage is output, replace the BLD assembly. (Acceptance range: -12V±5%)	BHSM**-018
Faulty power supply PCB	C: If a -12V voltage is not output even after all parts are checked, replace the power supply PCB.	BHSM**-105

E47 BP COVER SENSOR ERROR



Detection condition	When the signals of BPC1 and BPC2 get out of synchronization for a period of consecutive two seconds						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	-	-	-
Rinse	- / - / -	○	○	-	-	-	-
Dia. Wait	- / - / -	○	○	○ SW OFF	-	-	-
Standby	- / - / -	○	○	○ SW OFF	-	-	-
Preparation	- / - / -	○	○	○ SW OFF	-	-	-
Dialysis	- / ○ / -	○	○	○ SW OFF	-	-	-
Retrans	- / - / -	○	○	○ SW OFF	-	-	-
Reset method	Press the RESET switch. If the signals fall outside of the detection condition, the error will be automatically reset.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The blood pump guard (cover) is open		
○ Failure to close the guard	C: Close the blood pump guard.	
○ Improper method of connecting the rolling tube to the blood pump	C: Properly connect the rolling tube to the blood pump.	
○ Guard cannot be closed due to the distortion of the blood pump guard shaft	C: Remedy or replace the hinge shaft.	
The blood pump guard is closed	A: Check the voltage between the Pins 1 and 2 of the CN113 on the power supply PCB.	
○ Faulty 5V power supply to the “BPC1” or “BPC2”	C: If the voltage there is not 5V, replace the power supply PCB. (Acceptance range: 5V±5%)	BHSM**-105
○ Faulty “BPC1” or “BPC2”	A: Check for the signals at the Pin 3 (BPC1) and the Pin 4 (BPC2) of the CN113 on the power supply PCB. - Lo state when the guard is closed. - Hi state when the guard is open. C: Replace the harness of BPC (blood pump guard).	
○ Broken wire in the wirings to the “BPC1” or “BPC2” and loose connection of connector	A: Lightly pull the lead wires of the CN113 on the power supply PCB to check them for any disconnection or loose connection. C: Replace the harness of BPC.	
○ Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
○ Faulty main CPU PCB	C: Replace the main CPU PCB.	BHSM**-100

E48 CLV FAN STOP ALARM

Detection condition	When the fan for CLV is stopped.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	○	-	-
Rinse	- / - / -	○	○	-	○	-	-
Dia. Wait	- / - / -	○	○	○ SW OFF	○	-	-
Standby	- / - / -	○	○	○ SW OFF	○	-	-
Preparation	- / - / -	○	○	○ SW OFF	○	-	-
Dialysis	- / ○ / -	○	○	○ SW OFF	○	-	-
Retrans	- / - / -	○	○	○ SW OFF	○	-	-
Reset method	Press the RESET switch. If the CLV fan falls outside of the detection condition, the alarm will be automatically reset.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Clogged CLV fan	A: Check to be sure that the CLV fan stops operation due to foreign matters caught in the clamp. C: Remove the foreign matters.	
Faulty CLV fan	A: Disconnect the CN118 from the power supply PCB. Measure a voltage between WJ1 and Pin 1 of the CN118. C: If a 24V voltage is output, replace the CLV fan. (Acceptance range: 24V±10%)	BHSM**-009

E49 UFP ERROR (for Type 1)

Detection condition	Rotation speed was 30% faster or slower than the ideal rotation speed						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dialysis	○ / ○ / -	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	Only type 1 of fluid circuit		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The UF pump normally rotates		
○ Faulty “PH5” and “PH6”, or broken wire in the harness	C: Check on the Maintenance Mode 4 screen that the “PH5” and “PH6” respond, when the dog plate shields light between the sensor slits. A: If the “PH5” and “PH6” do not respond, replace them.	BHSM**-109
○ Faulty I/O PCB	A: Replace the I/O PCB.	CXSM**-101
○ Faulty main CPU PCB	A: Replace the main CPU PCB.	BHSM**-100
The UF pump does not rotate or shows uneven rotation		
○ Inadequate motor torque	C: Lightly hold the spindle by hand with the motor in rotation or going to rotate, and then check whether the motor stops. A: Refasten the enamel lock screw.	
○ Uncoupling between the spindle and the motor shaft	A: Replace the pump assembly.	CXSM**-021
○ Wear, seizure, or sticking of coupled/joined part	C: Lightly pull the lead wires of the CN1003 on the drive PCB to check them for any disconnection or loose connection.	
○ Broken wire in the wirings to the motor and loose connection of connector	A: Remedy or replace the faulty part.	
○ Faulty drive PCB	A: Replace the drive PCB.	BHSM**-103
○ Faulty main CPU PCB	A: Replace the main CPU PCB.	BHSM**-100

E50 SELF-CHECK ALARM (for Type 1)



Detection condition	If one or more self-checks are “NG” confirmed (17 checks).						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Standby	○ / ○ / -	○	○	-	-	-	-
Reset method	Restart the self-checks by pressing the RESET switch.			Remark	Only type 1 of fluid circuit.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
[NG by self-check for MEMORY]		
○ Malfunction of CPU	A: Replace the I/O PCB.	CXSM**-101
[NG by self-check for CLV]	C: For detail, refer to information in the “E64 CLV POSITION ERROR” section.	CXTS**-E64
[NG by self-check for UF PUMP]	C: For detail, refer to information in the “E61 UFP ROTATION ERROR” section. For detail, refer to information in the “E62 UFP SPEED ERROR” section.	CXTS**-E61 BHTS**-E62
[NG by self-check for BLOOD PUMP]	C: For detail, refer to information in the “E47 BP COVER SENSOR ERROR” section. For detail, refer to information in the “E53 BLOOD PUMP STOP ERROR” section. For detail, refer to information in the “E74 BLOOD PUMP ERROR” section.	CXTS**-E47 CXTS**-E53 CXTS**-E74
[NG by self-check for T4 SENSOR]		
○ Faulty relay for false resistance switching (for T4 temperature sensor)	C: While the self-check of the T4 temperature sensor is in progress, check whether the I/O PCB “clicks”. A: If the I/O PCB does not click, replace it.	CXSM**-101
○ Faulty temperature sensor circuit	A: Replace the I/O PCB.	CXSM**-101
○ Defective adjustment of temperature sensor circuit	C: Check the condition with a temperature calibration tool. A: If the adjustment is deviated, adjust the temperature sensor circuit. If the adjustment cannot be made, replace the I/O PCB.	BHAJ**-004 CXSM**-101
[NG by self-check for T5 SENSOR]		
○ Faulty relay for false resistance switching (for T5 temperature sensor)	C: While the self-check of the T5 temperature sensor is in progress, check whether the I/O PCB “clicks”. A: If the I/O PCB does not click, replace it.	CXSM**-101

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
○ Faulty temperature sensor circuit	A: Replace the I/O PCB.	CXSM**-101
○ Defective adjustment of temperature sensor circuit	C: Check the condition with a temperature calibration tool. A: If the adjustment is deviated, adjust the temperature sensor circuit. If the adjustment cannot be made, replace the I/O PCB.	BHAJ**-004 CXSM**-101
[NG by self-check for COND. 1 SENSOR]		
○ Faulty relay for false resistance switching (for COND. 1 sensor)	C: While the self-check for the COND. 1 sensor is in progress, check whether the I/O PCB “clicks”. A: If the I/O PCB does not click, replace it.	CXSM**-101
○ Faulty conductivity sensor circuit	A: Replace the I/O PCB.	CXSM**-101
○ Defective adjustment of conductivity sensor circuit	C: Check the condition with a conductivity calibration tool. A: If the adjustment is deviated, adjust the conductivity sensor circuit. If the adjustment cannot be made, replace the I/O PCB.	CXAJ**-005 CXSM**-101
[NG by self-check for COND. 2 SENSOR]		
○ Faulty relay for false resistance switching (for COND. 2 sensor)	C: While the self-check for the COND. 2 sensor is in progress, check whether the I/O PCB “clicks”. A: If the I/O PCB does not click, replace it.	CXSM**-101
○ Faulty conductivity sensor circuit	A: Replace the I/O PCB.	CXSM**-101
○ Defective adjustment of conductivity sensor circuit	C: Check the condition with a conductivity calibration tool. A: If the adjustment is deviated, adjust the conductivity sensor circuit. If the adjustment cannot be made, replace the I/O PCB.	CXAJ**-005 CXSM**-101
[NG by self-check for COND. 3 SENSOR]	(B powder option installation)	
○ Faulty relay for false resistance switching (for COND. 3 sensor)	C: While the self-check for the COND. 3 sensor is in progress, check whether the I/O PCB “clicks”. A: If the I/O PCB does not click, replace it.	CXSM**-101
○ Faulty conductivity sensor circuit	A: Replace the I/O PCB.	CXSM**-101

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
○ Defective adjustment of conductivity sensor circuit	C: Check the conditions with a conductivity calibration tool. A: If the adjustment is deviated, adjust the temperature sensor circuit. If the adjustment cannot be made, replace the I/O PCB.	CXAJ**-005 CXSM**-101
[NG by self-check for T6 SENSOR]	(B powder option installation)	
○ Faulty relay for false resistance switching (for T6 temperature sensor)	C: While the self-check of the T6 temperature sensor is in progress, check whether the I/O PCB “clicks”. A: If the I/O PCB does not click, replace it.	CXSM**-101
○ Faulty temperature sensor circuit	A: Replace the I/O PCB.	CXSM**-101
○ Defective adjustment of temperature sensor circuit	C: Check the condition with a temperature calibration tool. A: If the adjustment is deviated, adjust the temperature sensor circuit. If the adjustment cannot be made, replace the I/O PCB.	BHAJ**-004 CXSM**-101
[NG by self-check for BLD]		
○ Broken wire in the blood leak sensor harness or loose connection of connector	C: While the self-check for the blood leak sensor is in progress, the REAL value shows no variations. Lightly pull the lead wires of the CN209 on the I/O PCB and of the connector on the blood leak sensor side to check them for any disconnection or loose connection. A: Remedy or replace the faulty part(s).	
○ Faulty blood leak sensor	C: The REAL value does not increase even after a filter is installed. Or the REAL value is too low while the self-check of the blood leak sensor is in progress. A: Perform the zero calibration of the blood leak sensor. If “NG” is still active after the completion of zero calibration, replace the blood leak sensor.	CXAJ**-007
○ Faulty I/O PCB	A: Replace the I/O PCB.	CXSM**-101
[NG by self-check for V.PRESS SENSOR]		
○ Faulty relay for false resistance switching (for venous pressure sensor)	C: While the self-check for the venous pressure sensor is in progress, check whether the I/O PCB “clicks”. A: If the I/O PCB does not click, replace it.	CXSM**-101
○ Faulty venous pressure sensor circuit	A: Replace the I/O PCB.	CXSM**-101

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
○ Defective adjustment of venous pressure sensor circuit	C: Check the displayed value when the pressure is released to the atmosphere or pressurized with a syringe. A: If the adjustment is deviated, make adjustment of the PV/PA PCB circuits. If the adjustment cannot be made, replace the I/O PCB.	BHAJ**-001 CXSM**-101
[NG by self-check for A.PRESS SENSOR]		
○ Faulty relay for false resistance switching (for artery pressure sensor)	C: While the self-check for the artery pressure sensor is in progress, check whether the I/O PCB "clicks". A: If the I/O PCB does not click, replace it.	CXSM**-101
○ Faulty artery pressure sensor circuit	A: Replace the I/O PCB.	CXSM**-101
○ Defective adjustment of artery pressure sensor circuit	C: Check the displayed value when the pressure is released to the atmosphere or pressurized with a syringe. A: If the adjustment is deviated, make adjustment of the PV/PA PCB circuits. If the adjustment cannot be made, replace the I/O PCB.	BHAJ**-001 CXSM**-101
[NG by self-check for BYPASS]		
○ Faulty FS4	C: While the self-check for the bypass operation is in progress, check to be sure that the fluid passes through the FS4 line. A: If the LED in the FS4 used for operation check does not repeat ON and OFF corresponding to the flow rate, replace the FS4.	
○ Faulty I/O PCB	A: If the LED repeats ON and OFF, replace the harness connected to the FS4 or the I/O PCB.	CXSM**-101
○ Inadequate adjustment of dialysate flow rate	C: While the self-check for the bypass operation is in progress, check to be sure that the fluid does not pass through the FS4 line. C: Check whether the NV2 dialysate flow control valve is excessively throttled. A: If the NV2 is excessively throttled, make readjustment of the dialysate flow rate to not less than 300 mL/min.	CXAJ**010
○ Faulty V9b	C: Manually operate the V9b on the Operation Display screen, to check that it is in operation. C: If the V9b is not in operation, disconnect the harness from the V9b. Then connect it to a different valve, to check this valve for operation. A: If this different valve operates, replace the V9b. A: If this different valve does not operate, replace the harness connected to the V9b or the I/O PCB.	CXSM**-001 CXSM**-101

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
○ Faulty V19	C: Check the V19V in the same procedure as that for the V9b.	
[NG by self-check for T7 SENSOR]	(Hot rinse option installation)	
○ Faulty relay for false resistance switching (for T7 temperature sensor)	C: While the self-check of the T7 temperature sensor is in progress, check whether the I/O PCB “clicks”. A: If the I/O PCB does not click, replace it.	CXSM**-101
○ Faulty temperature sensor circuit	A: Replace the I/O PCB.	CXSM**-101
○ Defective adjustment of temperature sensor circuit	C: Check the condition with a temperature calibration tool. A: If the adjustment is deviated, adjust the temperature sensor circuit. If the adjustment cannot be made, replace the I/O PCB.	BHAJ**-004 CXSM**-101
[NG by self-check for Closed Circuit Leak]	C: For detail, refer to information in the “E60 CLOSED CIRCUIT LEAK ERROR” section.	BHTS**-E60

E50 SELF-CHECK ALARM (for Type 2)



Detection condition	If one or more self-checks are “NG” confirmed (17 checks).						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Standby	○ / ○ / -	○	○	-	-	-	-
Reset method	Restart the self-checks by pressing the RESET switch.			Remark	Only type 2 of fluid circuit.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
[NG by self-check for MEMORY]		
○ Malfunction of CPU	C: Replace the I/O PCB.	CXSM**-101
[NG by self-check for CLV]	A: For detail, refer to information in the “E64 CLV POSITION ERROR” section.	CXTS**-E64
[NG by self-check for UF PUMP]	A: For detail, refer to information in the “E61 UFP ROTATION ERROR” section. For detail, refer to information in the “E62 UFP SPEED ERROR” section.	CXTS**-E61 CXTS**-E62
[NG by self-check for BLOOD PUMP]	A: For detail, refer to information in the “E47 BP COVER SENSOR ERROR” section. For detail, refer to information in the “E53 BLOOD PUMP STOP ERROR” section. For detail, refer to information in the “E74 BLOOD PUMP ERROR” section.	CXTS**-E47 CXTS**-E53 CXTS**-E74
[NG by self-check for T4 SENSOR]		
○ Faulty relay for false resistance switching (for T4 temperature sensor)	A: While the self-check of the T4 temperature sensor is in progress, check whether the I/O PCB “clicks”. C: If the I/O PCB does not click, replace it.	CXSM**-101
○ Faulty temperature sensor circuit	C: Replace the I/O PCB.	CXSM**-101
○ Defective adjustment of temperature sensor circuit	A: Check the condition with a temperature calibration tool. C: If the adjustment is deviated, adjust the temperature sensor circuit. If the adjustment cannot be made, replace the I/O PCB.	BHAJ**-004 CXSM**-101
[NG by self-check for T5 SENSOR]		
○ Faulty relay for false resistance switching (for T5 temperature sensor)	A: While the self-check of the T5 temperature sensor is in progress, check whether the I/O PCB “clicks”. C: If the I/O PCB does not click, replace it.	CXSM**-101

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
○ Faulty temperature sensor circuit	C: Replace the I/O PCB.	CXSM**-101
○ Defective adjustment of temperature sensor circuit	A: Check the condition with a temperature calibration tool. C: If the adjustment is deviated, adjust the temperature sensor circuit. If the adjustment cannot be made, replace the I/O PCB.	BHAJ**-004 CXSM**-101
[NG by self-check for COND. 1 SENSOR]		
○ Faulty relay for false resistance switching (for COND. 1 sensor)	A: While the self-check for the COND. 1 sensor is in progress, check whether the I/O PCB “clicks”. C: If the I/O PCB does not click, replace it.	CXSM**-101
○ Faulty conductivity sensor circuit	C: Replace the I/O PCB.	CXSM**-101
○ Defective adjustment of conductivity sensor circuit	A: Check the condition with a conductivity calibration tool. C: If the adjustment is deviated, adjust the conductivity sensor circuit. If the adjustment cannot be made, replace the I/O PCB.	BHAJ**-005 CXSM**-101
[NG by self-check for COND. 2 SENSOR]		
○ Faulty relay for false resistance switching (for COND. 2 sensor)	A: While the self-check for the COND. 2 sensor is in progress, check whether the I/O PCB “clicks”. C: If the I/O PCB does not click, replace it.	CXSM**-101
○ Faulty conductivity sensor circuit	C: Replace the I/O PCB.	CXSM**-101
○ Defective adjustment of conductivity sensor circuit	A: Check the condition with a conductivity calibration tool. C: If the adjustment is deviated, adjust the conductivity sensor circuit. If the adjustment cannot be made, replace the I/O PCB.	BHAJ**-005 CXSM**-101
[NG by self-check for COND. 3 SENSOR]	(B powder option installation)	
○ Faulty relay for false resistance switching (for COND. 3 sensor)	A: While the self-check for the COND. 3 sensor is in progress, check whether the I/O PCB “clicks”. C: If the I/O PCB does not click, replace it.	CXSM**-101
○ Faulty conductivity sensor circuit	C: Replace the I/O PCB.	CXSM**-101
○ Defective adjustment of conductivity sensor circuit	A: Check the conditions with a conductivity calibration tool. C: If the adjustment is deviated, adjust the temperature sensor circuit.	BHAJ**-005 CXSM**-101

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
	If the adjustment cannot be made, replace the I/O PCB.	
[NG by self-check for T6 SENSOR]	(B powder option installation)	
○ Faulty relay for false resistance switching (for T6 temperature sensor)	A: While the self-check of the T6 temperature sensor is in progress, check whether the I/O PCB “clicks”. C: If the I/O PCB does not click, replace it.	CXSM**-101
○ Faulty temperature sensor circuit	C: Replace the I/O PCB.	CXSM**-101
○ Defective adjustment of temperature sensor circuit	A: Check the condition with a temperature calibration tool. C: If the adjustment is deviated, adjust the temperature sensor circuit. If the adjustment cannot be made, replace the I/O PCB.	BHAJ**-004 CXSM**-101
[NG by self-check for BLD]		
○ Broken wire in the blood leak sensor harness or loose connection of connector	A: While the self-check for the blood leak sensor is in progress, the REAL value shows no variations. Lightly pull the lead wires of the CN209 on the I/O PCB and of the connector on the blood leak sensor side to check them for any disconnection or loose connection. C: Remedy or replace the faulty part(s).	
○ Faulty blood leak sensor	A: The REAL value does not increase even after a filter is installed. Or the REAL value is too low while the self-check of the blood leak sensor is in progress. C: Perform the zero calibration of the blood leak sensor. If “NG” is still active after the completion of zero calibration, replace the blood leak sensor.	BHAJ**-007
○ Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101
[NG by self-check for V.PRESS SENSOR]		
○ Faulty relay for false resistance switching (for venous pressure sensor)	A: While the self-check for the venous pressure sensor is in progress, check whether the I/O PCB “clicks”. C: If the I/O PCB does not click, replace it.	CXSM**-101
○ Faulty venous pressure sensor circuit	C: Replace the I/O PCB.	CXSM**-101

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
○ Defective adjustment of venous pressure sensor circuit	A: Check the displayed value when the pressure is released to the atmosphere or pressurized with a syringe. C: If the adjustment is deviated, make adjustment of the PV/PA PCB circuits. If the adjustment cannot be made, replace the I/O PCB.	BHAJ**-001 CXSM**-101
[NG by self-check for A.PRESS SENSOR]		
○ Faulty relay for false resistance switching (for artery pressure sensor)	A: While the self-check for the artery pressure sensor is in progress, check whether the I/O PCB "clicks". C: If the I/O PCB does not click, replace it.	CXSM**-101
○ Faulty artery pressure sensor circuit	C: Replace the I/O PCB.	CXSM**-101
○ Defective adjustment of artery pressure sensor circuit	A: Check the displayed value when the pressure is released to the atmosphere or pressurized with a syringe. C: If the adjustment is deviated, make adjustment of the PV/PA PCB circuits. If the adjustment cannot be made, replace the I/O PCB.	BHAJ**-001 CXSM**-101
[NG by self-check for BYPASS]		
○ Faulty FS4	A: While the self-check for the bypass operation is in progress, check to be sure that the fluid passes through the FS4 line. C: If the LED in the FS4 used for operation check does not repeat ON and OFF corresponding to the flow rate, replace the FS4.	
○ Faulty I/O PCB	C: If the LED repeats ON and OFF, replace the harness connected to the FS4 or the I/O PCB.	CXSM**-101
○ Inadequate adjustment of dialysate flow rate	A: While the self-check for the bypass operation is in progress, check to be sure that the fluid does not pass through the FS4 line. A: Check whether the NV2 dialysate flow control valve is excessively throttled. C: If the NV2 is excessively throttled, make readjustment of the dialysate flow rate to not less than 300 mL/min.	CXAJ**010
○ Faulty V9b	A: Manually operate the V9b on the Operation Display screen, to check that it is in operation. A: If the V9b is not in operation, disconnect the harness from the V9b. Then connect it to a different valve, to check this valve for operation. C: If this different valve operates, replace the V9b. C: If this different valve does not operate, replace the harness connected to the V9b or the I/O PCB.	CXSM**-001 CXSM**-101

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
○ Faulty V19	A: Check the V19V in the same procedure as that for the V9b.	
[NG by self-check for T7 SENSOR]		
○ Faulty relay for false resistance switching (for T7 temperature sensor)	A: While the self-check of the T7 temperature sensor is in progress, check whether the I/O PCB “clicks”. C: If the I/O PCB does not click, replace it.	CXSM**-101
○ Faulty temperature sensor circuit	C: Replace the I/O PCB.	CXSM**-101
○ Defective adjustment of temperature sensor circuit	A: Check the condition with a temperature calibration tool. C: If the adjustment is deviated, adjust the temperature sensor circuit. If the adjustment cannot be made, replace the I/O PCB.	BHAJ**-004 CXSM**-101
[NG BD CPU]		
○ Faulty BD sensor circuit	C: Replace the I/O PCB.	
○ Defective adjustment of BD sensor circuit	A: Check the condition of BD sensor circuit. C: If the adjustment is deviated, adjust the BD sensor circuit. If the adjustment cannot be made, replace the I/O PCB.	
[NG V12/V17]		
○ Leaks in the closed circuit	A: Check visual or with the fingertips for fluid leaks out of the closed circuit. C: Replace the part(s) from which the fluid leaks.	
○ Faulty fluid pressure sensor assembly	A: Disconnect the hose from the fluid pressure sensor assembly to open it to the atmosphere, and then check whether the fluid pressure of 0 mmHg is displayed. Then, use a syringe to pressurize it until the reference pressure gauge reads 300 mmHg. C: If the fluid pressure of 300 mmHg is not displayed, perform the calibration of the fluid pressure sensor PCB. If the calibration cannot be performed, replace the fluid pressure sensor assembly.	
○ Faulty “UFP”	C: Refer to information in the “E49 UFP ERROR” section.	
○ Malfunction of valve	A: Check each valve for operation on the Maintenance Menu 4 screen. C: If any valve does not operate, repair or replace the valve concerned.	
○ Leaks from the valve	C: Repair or replace the valve(s) from which the fluid leaks.	

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
[NG D.PRESS SENSOR]		
○ Faulty fluid pressure sensor assembly	A: Disconnect the hose from the fluid pressure sensor assembly to open it to the atmosphere, and then check whether the fluid pressure of 0 mmHg is displayed. Then, use a syringe to pressurize it until the reference pressure gauge reads 300 mmHg. C: If the fluid pressure of 300 mmHg is not displayed, perform the calibration of the fluid pressure sensor PCB. If the calibration cannot be performed, replace the fluid pressure sensor assembly.	
○ Faulty “UFP”	C: Refer to information in the “E49 UFP ERROR” section.	
[NG by self-check for Closed Circuit Leak]	A : For detail, refer to information in the “E60 CLOSED CIRCUIT LEAK ERROR” section.	CXTS**-E60

E51 AIR BUBBLE SENSOR ERROR



Detection condition	An error signal from the air bubble CPU is detected.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	○	-	-
Rinse	- / - / -	○	○	-	○	-	-
Dia. Wait	- / - / -	○	○	○ SW OFF	○	-	-
Standby	- / - / -	○	○	○ SW OFF	○	-	-
Preparation	- / - / -	○	○	○ SW OFF	○	-	-
Dialysis	- / ○ / -	○	○	○ SW OFF	○	-	-
Retrans	- / - / -	○	○	○ SW OFF	○	-	-
Reset method	Recovers by "I/O" switch OFF/ON. (Restart the Device.)			Remark	The air bubble ON/OFF SW switch blinking.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Broken wire in the air bubble sensor harness or loose connection of connector part	A: Pull the lead wires of the CN210 on the I/O PCB to check them for loose connection. Further check whether the lead wires are correctly seated in the connector. C: Replace the air bubble sensor.	CXSM**-010
Malfunction due to external noises	C: Cut off a noise source or keep it at a distance.	
Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101
Faulty main CPU PCB	C: Replace the main CPU PCB.	BHSM**-100

E52 BLOOD LEAK SENSOR ERROR



Detection condition	When the REAL blood leak value exceeds 251ppm for a period of consecutive 5 seconds						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Preparation	- / - / ○	○	○	○	-	-	○
Reset method	Press the RESET switch.		Remark	· Not detected in the period of “Blood Leak Mask” which is set on MAINT2 and started right after the release of the Bypass mode. If the BP is running (except priming time) or the DIALYSIS switch is turned on, this error is available. · While the alarm is detected, Dialysis SW is not acceptable.			

Symptom/Cause	A: Analysis / C: Countermeasure	Work No.
The dialyser causes excessive leaks	C: Replace the dialyser. ⚠ CAUTION <u>After the alarm is reset, since the inside of the Viscous chamber is in the state of leaking blood on the drain fluid side, the error may be continuously detected. In this case, keep releasing the alarm until leak blood is drained and then operate the Device.</u>	
No blood leaks	A: Collect a dialysate sample from the drain fluid port of the Device or the Dialyser outlet. Then check visual for any blood leak or use a test device.	
○ Excessive stains on the cartridge window (e.g. deposition of protein, calcium carbonate, or fingerprints)	C: Dismount the cartridge from the blood leak sensor. Clean fingerprints or else off the window surface with a cotton-tipped swab. If stains remain on the inside of the cartridge, clean it to remove the stains.	
○ Entry of large amount of air in the blood leak sensor	A: Check whether air enters the inside of the cartridge. Check visual whether fine air bubbles cannot be removed through the “AS1” or “AS2”. Check if the negative pressure line in the Device (e.g. the pump inlet) has leaks.	
○ Degraded or faulty LED in the blood leak sensor	C: Replace the blood leak sensor.	BHSM**-018
○ Faulty I/O PCB	C: If the error occurs even after the blood leak sensor is replaced, replace the I/O PCB.	CXSM**-101

E53 BLOOD PUMP STOP ERROR



Detection condition	BP stops operation due to the following reason: 1) Cover open BP stops operation for 120 seconds continuously due to the following reasons: (Message, indication lamp, alarm sound only 120 seconds after and other alarm operations immediately) 2) Flow rate of 0mL/min 3) Stop switch ON						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dialysis	- / ○ / -	○	○	○ HP Stop	-	-	-
Retrans	- / - / -	○	○	○ HP Stop	-	-	-
Reset method	Pressing the RESET switch. If the blood pump falls outside of the detection condition, the alarm will be automatically reset.			Remark	During the following process, the alarm is not detected. · Mainte2[Mask time OF BP STOP alarm (sec)] after E17 or E19 alarm is reset. · BP stop by alarm · Open Retrans screen (exclude starting BP) · After Drain operation in Retrans process		

Symptom/Cause	A:Analysis/ C:Countermeasure	Work No.
The blood pump is accidentally stopped	C: Restart the blood pump.	

E54 CONCENTRATION SENSOR ERROR

Detection condition	When CD1 detects for 2 seconds a value of 10.0 ms/cm during the last minute of the rinse process						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Water Rinse	- / - / -	○	○	-	-	-	-
Tepid Rinse	- / - / -	○	○	-	-	-	-
Reset method	Pressing the RESET switch.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Inadequate rinse	C: Check and adjust the dialysate flow rate for rinse operation. And perform the rinse operation again.	
Stains (e.g. calcium carbonate) on the electrode of the conductivity sensor, or fluid leaks	C: Perform thorough acid rinse. If the stains cannot be removed, perform resolute rinse with a specified chemical (e.g. CHARON or SHUMMA).	
Loose connection of the "CN218" conductivity sensor connector	A: Check the CN218 for any erroneous wire connection, corrosion of contact surface, or broken wires. C: Remedy erroneous wire connections. If any corrosion or broken wires are observed, replace the connector.	
Broken wire or short-circuit of the conductivity sensor	A: Disconnect the CD1 connector from the I/O PCB. Measure the "CD1 - GND" resistance, to check if the sensor has no short circuit, broken wires, or loose connection. C: If any short circuit, broken wire, or loose connection is observed, replace the harness of the CD1.	
Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101

E55 WATER SUPPLY STOP (for Type 1)



Detection condition	When the FS1 detects the absence of water flow for a period of consecutive five seconds before 10 seconds elapse after the round is switched						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse	○ / - / -	○	○	-	-	-	○
Standby	○ / - / -	○	○	-	-	-	○
Preparation	○ / - / -	○	○	-	-	-	○
Dialysis	○ / ○ / -	○	○	-	-	-	-
Retrans	○ / - / -	○	○	-	-	-	○
Reset method	· Press the RESET switch. · While in the rinse process, after five seconds elapse since the fluid circuit stops feeding water, turn ON the V1 and V10 to monitor the PD value. If the PD value is maintained at 150 mmHg for a period of consecutive five seconds, the alarm will be automatically reset.			Remark	Only type 1 of fluid circuit.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Feed water flow rate is short	A: Check for the charge flow rate.	CXAJ**-009
○ Clogged feed fluid filter	C: Clean or replace the filter.	
○ Bended or clogged feed water piping	C: Check for bended or clogged part(s).	
○ Water leak from the feed water piping line	C: Check for leak part(s).	
○ Stop of feed water	C: Operate the water feed system.	
Feed water flow rate is normal		
○ Deviation from the adjustment of or failure in "PR"	A: Check whether the fluid pressure value reaches not less than 150 mmHg when the "V1" and "V10" turn ON while in the standby-to-dialysis process. C: Make adjustment of the "PR". If this adjustment cannot be made, replace the "PR".	CXAJ**-008 CXSM**-014
○ Broken wiring in the wiring to the "FS1" or loose connection of connector part	A: Lightly pull the lead wires of the CN213 on the I/O PCB to check them for disconnection or loose connection. Furthermore, check the connector on the "FS1" side. C: If any disconnection or loose connection is observed, remedy it.	
○ Faulty "FS1"	A: Measure the power supply voltage of 24V and output voltage on "FS1" - Between "+" and "-" power supply voltage: Approx. 24V (Acceptance range: 24V±10%) - Between "OUT" and "-" of output voltage: Approx. 24V with the float located in its "up" position (Acceptance range: 24V±10%)/Approx. 0V with the float located in its "down" position (Acceptance range: Not more than 1V) C: Replace the "FS1".	CXSM**-012

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
○ Faulty I/O PCB	A: If the “FS1” is normal, check the operation display state and the display of the “FS1”. C: If the display is abnormal, replace the I/O PCB.	CXSM**-101

E55 WATER SUPPLY ERROR (for Type 2)



Detection condition	1) FS1 detects OFF status for 1 second continuously within 3 seconds after switching the round. 2) PD does not increase over 200mmHg in C0 step.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V1 & V10: ON (Depressurization)	Alarm Forced off
Rinse	○ / - / -	○	○	-	-	○	○
Standby	○ / - / -	○	○	-	-	-	○
Preparation	○ / - / -	○	○	-	-	-	○
Dialysis	○ / - / -	○	○	-	-	-	-
Retrans	○ / - / -	○	○	-	-	-	○
Reset method	· Press the RESET switch. · While in the rinse process, If the PD value is maintained at 200 mmHg after fluid stop, the alarm will be automatically reset.			Remark	Heater off. Only type 2 of fluid circuit.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Filling flow rate is short	A: Check for the filling flow rate. (filling time: 6.5 to 6.7 seconds)	CXAJ**-009
○ Clogged feed fluid filter	C: Clean or replace the filter.	
○ Bended or clogged feed water piping	C: Check for bended or clogged part(s).	
○ Water leak from the feed water piping line	C: Check for leak part(s).	
○ Stop of feed water	C: Operate the water feed system.	
Feed water flow rate is normal		
○ Deviation from the adjustment of or failure in "PR"	A: Check whether the fluid pressure value reaches not less than 150 mmHg when the "V1" and "V10" turn ON while in the standby-to-dialysis process. C: Make adjustment of the "PR". If this adjustment cannot be made, replace the "PR".	CXAJ**-008 CXSM**-014
○ Broken wiring in the wiring to the "FS1" or loose connection of connector part	A: Lightly pull the lead wires of the CN213 on the I/O PCB to check them for disconnection or loose connection. Furthermore, check the connector on the "FS1" side. C: If any disconnection or loose connection is observed, remedy it.	
○ Faulty "FS1"	A: Measure the power supply voltage of 24V and output voltage on "FS1" - Between "+" and "-" power supply voltage: Approx. 24V (Acceptance range: 24V±10%) - Between "OUT" and "-" of output voltage: Approx. 24V with the float located in its "up" position (Acceptance range: 24V±10%)/Approx. 0V with the float located in its "down" position (Acceptance range: Not more than 1V) C: Replace the "FS1".	CXSM**-012
○ Faulty I/O PCB	A: If the "FS1" is normal, check the operation display state and the display of the "FS1". C: If the display is abnormal, replace the I/O PCB.	CXSM**-101

E56 SHORT WATER SUPPLY (for Type 1)



Detection condition	When the “FS2” detects the absence of water flow for a period of consecutive five seconds before the “FS1” detects the same						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse	○ / - / -	○	○	-	-	-	○
Standby	○ / - / -	○	○	-	-	-	○
Preparation	○ / - / -	○	○	-	-	-	○
Dialysis	○ / ○ / -	○	○	-	-	-	-
Retrans	○ / - / -	○	○	-	-	-	○
Reset method	· Press the RESET switch. · While in the rinse process, after five seconds elapse since the fluid circuit stops feeding water, turn ON the V1 and V10 to monitor the PD value. If the PD value is maintained at 150 mmHg for a period of consecutive five seconds, the alarm will be automatically reset.			Remark	Only type 1 of fluid circuit.		

Symptom/Cause	A: Analysis / C: Countermeasure	Work No.
Charge flow rate too low	C: Adjust the charge flow rate so that the charge will be complete in a period of 45 to 48 seconds. (32 to 34 seconds at Hot rinse option installation.)	CXAJ**-009
Dialysate flow rate too high	C: Make adjustment of dialysate flow rate in the range of 300 to 700 mL/min. (300 to 800 mL/min at Hot rinse option installation.)	CXAJ**-010
Clogging of closed circuit	C: Overhaul or replace clogged part(s).	
Faulty “P1”	C: Overhaul or replace the pump “P1”.	CXSM**-007
Faulty “P2”	C: Overhaul or replace the pump “P2”.	CXSM**-007
Broken wiring in the wiring to the “FS1” or loose connection of connector part	A: Lightly pull the lead wires of the CN213 on the I/O PCB to check them for disconnection or loose connection. Furthermore, check for the connector on the “FS1” side. C: If any disconnection or loose connection is observed, remedy it.	
Broken wiring in the wiring to the “FS2” or loose connection of connector part	A: Lightly pull the lead wires of the CN213 on the I/O PCB to check them for disconnection or loose connection. Furthermore, check for the connector on the “FS2” side. C: If any disconnection or loose connection is observed, remedy it.	
Faulty “FS1”	A: Measure the power supply voltage of 24V and output voltage on “FS1”. - Between “+” and “-” of power supply voltage: Approx. 24V (Acceptance range: 24V±10%) - Between “OUT” and “-” of output voltage: Approx. 24V with the float located in its “up” position (Acceptance range: 24V±10%)/Approx. 0V with the float located in its “down” position (Acceptance range: Not more than 1V) C: Replace the “FS1” or the flow meter assembly.	CXSM**-012
	A: Measure the power supply voltage of 24V and output voltage on “FS2”. Between “+” and “-” of power supply voltage: Approx. 24V (Acceptance range: 24V±10%)	

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
	Between “OUT” and “-” of output voltage: Approx. 24V with the float located in its “up” position (Acceptance range: $24V \pm 10\%$)/Approx. 0V with the float located in its “down” position (Acceptance range: Not more than 1V)	
Faulty “FS2”	C: Replace the “FS2” or the flow meter assembly.	
Faulty I/O PCB	A: If the “FS1” or the “FS2” is normal, check for the operation display state and the display of the “FS1” and “FS2” C: If the display is abnormal, replace the I/O PCB.	CXSM**-101

E56 SHORT WATER SUPPLY (for Type 2)



Detection condition	FS1 detects ON status for 15 second continuously in the round.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V1 & V10: ON (Depressurization)	Alarm Forced off
Rinse	○ / - / -	○	○	-	-	○	○
Standby	○ / - / -	○	○	-	-	-	○
Preparation	○ / - / -	○	○	-	-	-	○
Dialysis	○ / - / -	○	○	-	-	-	-
Retrans	○ / - / -	○	○	-	-	-	○
Reset method	· Press the RESET switch. · While in the rinse process, If the PD value is maintained at 200 mmHg after fluid stop, the alarm will be automatically reset.			Remark	Heater OFF Only type 2 of fluid circuit.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Filling flow rate too low	C: Adjust the filling flow rate. (filling time: 6.5 to 6.7 seconds)	
Dialysate flow rate too high	C: Make adjustment of dialysate flow rate in the range of 300 to 800 mL/min.	CXAJ**-010
Clogging of closed circuit	C: Overhaul or replace clogged part(s).	
Faulty "P1"	C: Overhaul or replace the pump "P1".	CXSM**-007
Faulty "P2"	C: Overhaul or replace the pump "P2".	CXSM**-007
Broken wiring in the wiring to the "FS1" or loose connection of connector part	A: Lightly pull the lead wires of the CN213 on the I/O PCB to check them for disconnection or loose connection. Furthermore, check for the connector on the "FS1" side. C: If any disconnection or loose connection is observed, remedy it.	
Broken wiring in the wiring to the "FS2" or loose connection of connector part	A: Lightly pull the lead wires of the CN213 on the I/O PCB to check them for disconnection or loose connection. Furthermore, check for the connector on the "FS2" side. C: If any disconnection or loose connection is observed, remedy it.	
Faulty "FS1"	A: Measure the power supply voltage of 24V and output voltage on "FS1". - Between "+" and "-" of power supply voltage: Approx. 24V (Acceptance range: 24V±10%) - Between "OUT" and "-" of output voltage: Approx. 24V with the float located in its "up" position (Acceptance range: 24V±10%)/Approx. 0V with the float located in its "down" position (Acceptance range: Not more than 1V) C: Replace the "FS1" or the flow meter assembly. A: Measure the power supply voltage of 24V and output voltage on "FS2". - Between "+" and "-" of power supply voltage: Approx. 24V (Acceptance range: 24V±10%)	CXSM**-012

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
	Between “OUT” and “-” of output voltage: Approx. 24V with the float located in its “up” position (Acceptance range: $24V \pm 10\%$)/Approx. 0V with the float located in its “down” position (Acceptance range: Not more than 1V)	
Faulty “FS2”	C: Replace the “FS2” or the flow meter assembly.	
Faulty I/O PCB	A: If the “FS1” or the “FS2” is normal, check for the operation display state and the display of the “FS1” and “FS2” C: If the display is abnormal, replace the I/O PCB.	CXSM**-101

E57 SHORT DIALYSATE FLOW (for Type 1)



Detection condition	When the stop of dialysate flow is not detected even after a lapse of 127 seconds since the FS2 detects the presence of the dialysate flow after the round is switched $127 \text{ seconds} = 500 \text{ mL} \div 282 \text{ mL/min.} \times 60 \text{ sec.} \times 1.2$ Chamber capacity: 500 mL/Min. dialysate flow rate: 282 mL (A flow meter error (3% of full scale) for the minimum flow rate of 300 mL is taken into account)/Safety factor: 20%						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Standby	- / - / -	○	○	-	-	-	-
Preparation	- / - / -	○	○	-	-	-	-
Dialysis	- / - / -	○	○	-	-	-	-
Retrans	- / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	Only type 1 of fluid circuit.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Low dialysate flow rate	C: Adjust the dialysate flow rate in the range of 300 to 700 mL/min. (300 to 800 mL/min at Hot rinse option installation.)	CXAJ**-010
Clogging of closed circuit	C: Overhaul or replace clogged part(s).	
Faulty "P2"	C: Overhaul or replace the pump "P2".	CXSM**-007
Broken wiring in the wiring to the "FS2" or loose connection of connector part	A: Lightly pull the lead wires of the CN213 on the I/O PCB to check them for disconnection or loose connection. Furthermore, check for the connector on the "FS2" side. C: If any disconnection or loose connection is observed, remedy it.	
Faulty "FS2"	A: Measure the power supply voltage of 24V and output voltage on "FS2". - Between "+" and "-" of power supply voltage: Approx. 24V (Acceptance range: $24V \pm 10\%$) - Between "OUT" and "-" of output voltage: Approx. 24V with the float located in its "up" position (Acceptance range: $24V \pm 10\%$)/Approx. 0V with the float located in its "down" position (Acceptance range: Not more than 1V) C: Replace the "FS2" or the flow meter assembly.	
Faulty I/O PCB	A: If the "FS2" is normal, check the operation display state and the display of the "FS2". C: If the display is abnormal, replace the I/O PCB	CXSM**-101

E57 SHORT DIALYSATE FLOW (for Type 2)



Detection condition	FS2 detects ON status for 120 second continuously in the round.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Standby	- / - / -	○	○	-	-	-	-
Preparation	- / - / -	○	○	○	○	-	-
Dialysis	- / ○ / -	○	○	-	-	-	-
Retrans	- / - / -	○	○	○	○	-	-
Reset method	Press the RESET switch.			Remark	Only type 2 of fluid circuit.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Low dialysate flow rate	C: Adjust the dialysate flow rate. (300 to 800 mL/min)	CXAJ**-010
Clogging of closed circuit	C: Overhaul or replace clogged part(s).	
Faulty "P2"	C: Overhaul or replace the pump "P2".	CXSM**-007
Broken wiring in the wiring to the "FS2" or loose connection of connector part	A: Lightly pull the lead wires of the CN213 on the I/O PCB to check them for disconnection or loose connection. Furthermore, check for the connector on the "FS2" side. C: If any disconnection or loose connection is observed, remedy it.	
Faulty "FS2"	A: Measure the power supply voltage of 24V and output voltage on "FS2". - Between "+" and "-" of power supply voltage: Approx. 24V (Acceptance range: 24V±10%) - Between "OUT" and "-" of output voltage: Approx. 24V with the float located in its "up" position (Acceptance range: 24V±10%)/Approx. 0V with the float located in its "down" position (Acceptance range: Not more than 1V) C: Replace the "FS2" or the flow meter assembly.	
Faulty I/O PCB	A: If the "FS2" is normal, check the operation display state and the display of the "FS2". C: If the display is abnormal, replace the I/O PCB	CXSM**-101

E58 DIALYSATE FLOW STOP (for Type 1)



Detection condition	When the dialysate flow stop is detected for two seconds within a period of 20 seconds since the FS2 detects the presence of the dialysate flow after the round is switched.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse	- / - / -	○	○	-	-	-	-
Standby	○ / - / -	○	○	-	-	-	-
Preparation	○ / - / -	○	○	-	-	-	-
Retrans	○ / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	Only type 1 of fluid circuit		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Dialysate flows		
○ Dialysate flow rate too high	C: Adjust the dialysate flow rate in the range of 300 to 700 mL/min. (300 to 800 mL/min at Hot rinse option installation.)	CXAJ**-010
○ Dialysate flow rate too low	C: Adjust the dialysate flow rate in the range of 300 to 700 mL/min. (300 to 800 mL/min at Hot rinse option installation.)	CXAJ**-010
○ Broken wire in the wiring to the "FS2" or loose connection of connector part	A: Lightly pull the lead wires of the CN213 on the I/O PCB to check them for disconnection or loose connection. Furthermore, check for the connector on the "FS2" side. C: If any disconnection or loose connection is observed, remedy it.	
○ Faulty "FS2"	A: Measure a power supply voltage of 24V and output voltage on "FS2". - Between "+" and "-" of power supply voltage: Approx. 24V (Acceptance range: 24V±10%) - Between "OUT" and "-" of output voltage: Approx. 24V with the float located in the "up" position/Approx. 0V with the float located in its "down" position (Acceptance range: 24V±10%/0V±5%) C: Replace the "FS2" or the flow meter assembly.	
○ Faulty I/O PCB	A: If the "FS2" is normal, check for the operation display state and the display of the "FS2". C: If the display is abnormal, replace the I/O PCB.	CXSM**-101
Dialysate does not flow		
○ Entry of large amount of air in the closed circuit	A: Visually check for the entry of air. C: Repair part(s) in which air enters.	
○ Leaks in the closed circuit	C: Repair leak part(s).	
○ Faulty P2	C: Overhaul or replace the P2.	CXSM**-007

E59 AIR ELIMINATION TANK ERROR (for Type 1)



Detection condition	“P2” was ON, “FS2” detected flow, and “FSW” detected air continuously for 15 sec in the Dialysis process and for 30 sec in other processes.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse	- / - / -	○	○	-	-	-	○
Standby	- / - / -	○	○	-	-	-	○
Preparation	- / - / -	○	○	-	-	-	○
Dialysis	- / ○ / ○	○	○	-	-	○	-
Retrans	- / - / -	○	○	-	-	-	○
Reset method	Press the RESET switch.			Remark	Only type 1 of fluid circuit.		

Symptom/Cause	A: Analysis / C: Countermeasure	Work No.
The float is in “down” position	A: Check the “FSW” for operation on the Operation Display screen. The annunciator light turns ON to indicate that the float is in “down” position.	
○ The float absorbs water to increase the specific gravity	A: Measure the float weight. Any floats in excess of 8 g in weight are unacceptable. C: Replace the float.	CXSM**-015
○ The float stays in “down” position due to foreign matters depositing on the float or the float shaft	C: Clean the float and the float shaft.	CXSM**-015
○ Air discharge disabled due to faulty “V4”	C: Overhaul or replace the “V4”.	CXSM**-001
○ Air discharge disabled due to faulty electric circuit for driving the “V4”	A: Manually turn ON the V4 on the Operation Display screen to check whether a 24V voltage is output to the connector of the V4. (Acceptance range: 24V±10%)	
The float is in “up” position		
○ Faulty float switch due to short circuit	C: If a 24V voltage is not output, replace the I/O PCB. (Acceptance range: 24V±10%) A: Check the “FSW” for operation on the Operation Display screen. If the annunciator light constantly turns ON, the error will result from faulty electrical system.	CXSM**-101
○ Faulty input circuit of the float switch due to short circuit	A: Disconnect the connector from the float switch, and then check for continuity on the float side. C: If a short circuit occurs with the float located in its “up” position, replace the float switch. C: If the float switch is normal, replace the I/O PCB.	CXSM**-015 CXSM**-101

E59 RELEASING AIR FROM AS2 ERROR (for Type 2)



Detection condition	1) FSW ON status totals 15 seconds (Total value cleared upon detection). (Dialysis, Retrans) 2) FSW ON status is detected for 30 seconds continuously. (Standby, Preparation, all rinses)						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse	- / - / -	○	○	-	-	-	○
Standby	- / - / -	○	○	-	-	-	○
Preparation	- / - / -	○	○	-	-	-	○
Dialysis	- / ○ / ○	○	○	-	-	○ (when the reset operation)	-
Retrans	- / - / ○	○	○	-	-	○ (when the reset operation)	○
Reset method	Press the RESET switch.			Remark	Only type 2 of fluid circuit.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The float is in “down” position	A: Check the “FSW” for operation on the Operation Display screen. The annunciator light turns ON to indicate that the float is in “down” position.	
○ The float has some pin hole and fluid is into the float	A: Check the float whether pin hole is existed or not. C: If pin hole is on the float, replace the float.	
○ The float stays in “down” position due to foreign matters depositing on the float or the float shaft	C: Clean the float and the float shaft.	CXSM**-015
○ Air discharge disabled due to faulty “V4”	C: Overhaul or replace the “V4”.	CXSM**-001
○ Air discharge disabled due to faulty electric circuit for driving the “V4”	A: Manually turn ON the V4 on the Operation Display screen to check whether a 24V voltage is output to the connector of the V4. (Acceptance range: 24V±10%)	
The float is in “up” position		
○ Faulty float switch due to short circuit	C: If a 24V voltage is not output, replace the I/O PCB. (Acceptance range: 24V±10%) A: Check the “FSW” for operation on the Operation Display screen. If the annunciator light constantly turns ON, the error will result from faulty electrical system.	CXSM**-101
○ Faulty input circuit of the float switch due to short circuit	A: Disconnect the connector from the float switch, and then check for continuity on the float side. C: If a short circuit occurs with the float located in its “up” position, replace the float switch. C: If the float switch is normal, replace the I/O PCB.	CXSM**-015 CXSM**-101

E60 CLOSED CIRCUIT LEAK ERROR (for Type 1)



Detection condition	When either of low or fluctuating positive pressure of the chamber A or low or fluctuating negative pressure of the chamber B is detected * Refer to information in the “Detection condition for leaks from closed circuit using the Maintenance Menu 7 screen” section.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse (Water Rinse)	- / - / -	○	○	-	-	-	-
Standby (Startup test)	- / - / -	○	○	-	-	-	-
Reset method	The self-check is resumed by the Reset SW in the Standby process.			Remark	Only type 1 of fluid circuit.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The pressure rising check results in NG		
○ Leaks in the closed circuit	A: Check visual or with the fingertips for fluid leaks out of the closed circuit. C: Replace the part(s) from which the fluid leaks.	
○ Faulty fluid pressure sensor assembly	A: Disconnect the hose from the fluid pressure sensor assembly to open it to the atmosphere, and then check whether the fluid pressure of 0 mmHg is displayed. Then, use a syringe to pressurize it until the reference pressure gauge reads 300 mmHg. C: If the fluid pressure of 300 mmHg is not displayed, perform the calibration of the fluid pressure sensor PCB. If the calibration cannot be performed, replace the fluid pressure sensor assembly.	BHAJ**-002 BHSM**-019
○ Faulty “UFP”	C: Refer to information in the “E49 UFP ERROR” section.	BHTS**-E49
○ Malfunction of valve	A: Check each valve for operation on the Maintenance Menu 4 screen. C: If any valve does not operate, repair or replace the valve concerned.	CXSM**-001
The leak check results in NG		
○ Leaks from the coupler	C: Replace the O-ring of the coupler or the coupler by the set.	
○ Leaks from the valve	C: Repair or replace the valve(s) from which the fluid leaks.	CXSM**-001

E60 CLOSED CIRCUIT LEAK ERROR (for Type 2)



Detection condition	When either of low or fluctuating positive pressure of the chamber A or low or fluctuating negative pressure of the chamber B is detected * Refer to information in the “Detection condition for leaks from closed circuit using the Maintenance Menu 7 screen” section.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse (Water Rinse)	- / - / -	○	○	-	-	-	-
Standby (Startup test)	- / - / -	○	○	-	-	-	-
Dialysis	○ / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch. The self-check is resumed by the Reset SW in the Standby process. The check is resumed by the Reset SW in the Dialysis process.			Remark	Only type 2 of fluid circuit.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The pressure rising check results in NG		
○ Leaks in the closed circuit	A: Check visual or with the fingertips for fluid leaks out of the closed circuit. C: Replace the part(s) from which the fluid leaks.	
○ Faulty fluid pressure sensor assembly	A: Disconnect the hose from the fluid pressure sensor assembly to open it to the atmosphere, and then check whether the fluid pressure of 0 mmHg is displayed. Then, use a syringe to pressurize it until the reference pressure gauge reads 300 mmHg. C: If the fluid pressure of 300 mmHg is not displayed, perform the calibration of the fluid pressure sensor PCB. If the calibration cannot be performed, replace the fluid pressure sensor assembly.	BHAJ**-002 BHSM**-019
○ Faulty “UFP”	C: Refer to information in the “E49 UFP ERROR” section.	BHTS**-E49
○ Malfunction of valve	A: Check each valve for operation on the Maintenance Menu 4 screen. C: If any valve does not operate, repair or replace the valve concerned.	CXSM**-001
The leak check results in NG		
○ Leaks from the coupler	C: Replace the O-ring of the coupler or the coupler by the set.	
○ Leaks from the valve	C: Repair or replace the valve(s) from which the fluid leaks.	CXSM**-001

E61 UFP ROTATION ERROR

Detection condition	When two backward rotations of the UFP are detected						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dialysis	○ / ○ / -	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty "PH5" and "PH6" or broken wire in the harness	A: Check to be sure on the Maintenance Mode 4 screen that the "PH5" and "PH6" respond when the dog plate shields light between the sensor slits. C: If the "PH5" and "PH6" do not respond, replace them.	BHSM**-109
Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101
Faulty main CPU PCB	C: Replace the main CPU PCB.	BHSM**-100
Faulty drive PCB	C: Replace the drive PCB.	BHSM**-103

E62 UFP SPEED ERROR (for Type 1)

Detection condition	The number of rotation per 4 rounds was 2 times higher or lower than the ideal number of rotation						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dialysis	○ / ○ / -	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	Only type 1 of fluid circuit.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
UFP is faulty.	A: Hold the spindle lightly with a hand, while the motor is rotating or tries to rotate, and check if it stops. C: Tighten the headless set bolt for fixing securely or replace the UF pump.	BHTS**-E49
PH5 or PH6 are faulty.	A: Check that PH5 and PH6 are reactive on the maintenance 4 screen while the sensor slit is shielded by dog plate. C: Replace PH5 or PH6 if there is no response.	
The I/O board is faulty.	C: Replace the I/O board.	
The drive board is faulty	C: Replace the drive board.	

E62 UFP SPEED ERROR (for Type 2)

Detection condition	The number of rotation per 10 minutes was 3 times higher or lower than the ideal number of rotation						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dialysis	○ / ○ / -	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	Only type 2 of fluid circuit.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
UFP is faulty.	A: Hold the spindle lightly with a hand, while the motor is rotating or tries to rotate, and check if it stops. C: Tighten the headless set bolt for fixing securely or replace the UF pump.	
PH5 or PH6 are faulty.	A: Check that PH5 and PH6 are reactive on the maintenance 4 screen while the sensor slit is shielded by dog plate. C: Replace PH5 or PH6 if there is no response.	
The I/O board is faulty.	C: Replace the I/O board.	
The drive board is faulty	C: Replace the drive board.	

E63 SYRINGE INFUSION COMPLETION



Detection condition	When the LS3 detects the overload signal while the HP is in operation						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dia. Wait	- / - / -	○	○ (Yellow)	HP stop	-	-	-
Standby	- / - / -	○	○ (Yellow)	HP stop	-	-	-
Preparation	- / - / -	○	○ (Yellow)	HP stop	-	-	-
Dialysis	- / - / -	○	○ (Yellow)	HP stop	-	-	-
Retrans	- / - / -	○	○ (Yellow)	HP stop	-	-	-
Reset method	Press the RESET switch. If the LS3 falls outside of the detection condition, the alarm will be automatically reset.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The infusion of chemical solution is complete	A: Check visual if the infusion of chemical solution into the syringe is complete. C: Set the Heparin Pump Operation switch to OFF.	
The syringe bites into the line	C: Remove the causes of the bite.	
When monitoring on the Operation Display screen, the "LS3" is constantly in the ON state (displays the overloaded state)		
○ Improper positioning of the "LS3" or the screw used to turn ON the "LS3"	A: Check visual the screw used to turn ON the "LS3". C: Properly position the screw used to turn ON the "LS3"	BHSM**-003
○ Faulty "LS3"	A: Disconnect the CN205 from the I/O PCB, and then manually turn ON or OFF the "LS3" to check for the continuity between the Pins 3 and 4. • Short-circuited when the "LS3" turns ON (normal state). • Open when the "LS3" turns OFF (overloaded state). C: In case of breakdown, replace the "LS3".	BHSM**-003
○ Broken wire in the harness of the "LS3" and loose connection of connector	A: Lightly pull the lead wires of the CN205 on the I/O PCB to check them for any disconnection or loose connection. Furthermore, check the "LS3" for soldered conditions. C: If any disconnection or loose connection is observed, remedy or replace the harness.	
○ Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101

E64 CLV POSITION ERROR



Detection condition	“LS5” was turned off while the CLV was open, or “LS5” was turned on while the CLV was close						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	○	-	○
Rinse	- / - / -	○	○	-	○	-	○
Dia. Wait	- / - / -	○	○	○ SW OFF	○	-	○
Standby	- / - / -	○	○	○ SW OFF	○	-	○
Preparation	- / - / -	○	○	○ SW OFF	○	-	○
Dialysis	- / ○ / -	○	○	○ SW OFF	○	-	-
Retrans	- / - / -	○	○	○ SW OFF	○	-	○
Reset method	Press the RESET switch.			Remark	This error is not detected while the startup test is in progress.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The clamp is in operation	A: Monitor the Maintenance Mode 4 screen to check that the “CLV” turns OFF. Furthermore, check visual whether the clamp head is in “forward” position.	
○ Faulty power supply PCB	A: Measure the output voltage between the Pins 1 and 2 of the CN117 on the power supply PCB C: If a voltage of approx. 24V is not output, replace the power supply PCB. (Acceptance range: 24V±10%)	BHSM**-105
○ Faulty solenoid	C: If a voltage of approx. 24V is output, replace the solenoid. (Acceptance range: 24V±10%)	BHSM**-009
The clamp is not in operation	A: Monitor the Maintenance Mode 4 screen to check that the “CLV” turns ON and further check whether the clamp head is in “backward” position.	
○ Improper positioning of the clamp	A: Check visual the clamp head position. C: If the clamp head hits at the frame of the air bubble clamp casing, properly position the clamp head.	BHSM**-009
○ Faulty “LS5”	A: Disconnect the CN205 from the I/O PCB, and then manually turn ON or OFF the “LS5” to check for the continuity between the pins. • Short-circuited when the “LS5” turns ON. • Open when the “LS3” turns OFF. C: If a continuity error is detected, replace the “LS5” or the harness.	BHSM**-009

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
○ Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101
○ Faulty main CPU PCB	C: Replace the main CPU PCB.	BHSM**-100

E65 UFP STARTING POINT ERROR



Detection condition	When the PH5 or the PH6 remains OFF or ON for a period of consecutive two seconds while the home position setting of the UFP is in progress						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse	- / - / -	○	○	-	-	-	-
Standby	○ / - / -	○	○	-	-	-	-
Dialysis	- / - / - (End of UF)	○	○	-	-	-	-
Retrans	- / - / - (When Retrans switch is pressed)	○	○	-	-	-	-
Reset method	Press the RESET switch.		Remark	Homing operation is performed at the timing of the following processes: Rinse: When rinse operation completes Standby: Before the closed circuit leak check in the startup test Dialysis: End of UF Retrans: When Retrans switch is pressed			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty "PH5" and "PH6" or broken wire in the harness	A: Check to be sure on the Maintenance Mode 4 screen that the "PH5" and "PH6" respond when the dog plate shields light between the sensor slits. C: If the "PH5" and "PH6" do not respond, replace them.	BHSM**-109
Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101
Faulty drive PCB	C: Replace the drive PCB.	BHSM**-103

E66 BATTERY VOLTAGE TOO LOW

Detection condition	When the AC power is applied, a battery voltage below 22V used for power failure backup is detected. When a power failure occurs, a battery voltage below 22V used for power failure backup is detected for a period of consecutive two seconds.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Normal condition	- / - / -	○	○	-	-	-	○
Power failure condition	- / - / -	○ (60sec)	-	○	-	-	-
Reset method	Press the RESET switch.			Remark	During a power failure, BP: Stopped, Clamp: Closed, Air bubble detector: OFF(after 1min) To reset the alarm, set the "BP BACKUP BATTERY SETTING" function to "OFF" on the Maintenance Menu 2 screen.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The battery is not charged		
Operating the blood pump with the use of the battery for an extended period of time results in shortage of charge time	C: Set the Power Supply switch on the Device to ON to apply power. 48 hours are needed to fully charge the battery.	BHSM**-107
○ The battery has reached the end of its service life	C: Replace the battery.	BHSM**-105
○ Faulty battery charge circuit	C: Replace the power supply PCB.	CXSM**-101
○ Faulty I/O PCB	C: Replace the I/O PCB.	

E67 ARTERIAL PRESSURE MAX LIMIT



Detection condition	When the arterial pressure value exceeds the maximum set value for arterial pressure limiter						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	-	-	○
Rinse	- / - / -	○	○	-	-	-	○
Dia. Wait	- / - / -	○	○	○	-	-	○
Standby	- / - / -	○	○	○	-	-	○
Preparation	- / - / -	○	○	○	-	-	○
Dialysis	- / ○ / -	○	○	○	-	-	-
Retrans	- / - / -	○	○	○	-	-	○
Reset method	Press the RESET switch.			Remark	This alarm is not detected during the startup test in the standby mode.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
This error is accompanied with an error related to the fluid pressure	C: Remove the causes of the error related to the fluid pressure.	
This error is accompanied with an error related to the venous pressure	C: Remove the causes of the error related to the venous pressure.	
The tube in the blood circuit connecting between the arterial pressure air chamber and the dialyzer has bends, clogging, or blockage with Robert's clamp	C: Remove the causes of the blockage.	
The mesh of chamber on the arterial pressure side has blockage due to thrombus	C: Replace the blood circuit.	
Faulty PV/PA PCB	A: Disconnect the blood circuit from the arterial pressure transducer and open it to atmosphere. Then check whether the arterial pressure of 0 mmHg is displayed. C: If the arterial pressure of 0 mmHg is not displayed, perform the calibration of the PV/PA PCB. If the calibration cannot be performed, replace the PV/PA PCB.	BHAJ**-001 BHSM**-104
Faulty I/O PCB	C: If the arterial pressure does not return to its normal level even after the PA/PV PCB is replaced, replace the I/O PCB.	CXSM**-101
Faulty main CPU PCB	C: If the arterial pressure does not return to its normal level even after the I/O PCB is replaced, replace the main CPU PCB.	BHSM**-100

E68 ARTERIAL PRESSURE MIN LIMIT



Detection condition	When the arterial pressure value drops below the minimum set value for arterial pressure limiter						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	-	-	○
Rinse	- / - / -	○	○	-	-	-	○
Dia. Wait	- / - / -	○	○	○	-	-	○
Standby	- / - / -	○	○	○	-	-	○
Preparation	- / - / -	○	○	○	-	-	○
Dialysis	- / ○ / -	○	○	○	-	-	-
Retrans	- / - / -	○	○	○	-	-	○
Reset method	Press the RESET switch.			Remark	This alarm is not detected during the startup test in the standby mode.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
This error is accompanied with an error related to the fluid pressure	C: Remove the causes of the error related to the fluid pressure	
The tube in the blood circuit connecting between the arterial pressure air chamber and a patient has bends, clogging, or blockage with Robert's clamp	C: Remove the causes of the blockage	
The connector of the shunt part has blockage due to thrombus	C: Remove the thrombus	
Faulty PV/PA PCB	A: Disconnect the blood circuit from the arterial pressure transducer and open it to atmosphere. Then check whether the arterial pressure of 0 mmHg is displayed. C: If the arterial pressure close to 0 mmHg is not displayed, perform the calibration of the PV/PA PCB. If the calibration cannot be performed, replace the PV/PA PCB.	BHAJ**-001 BHSM**-104
Faulty I/O PCB	C: If the arterial pressure does not return to its normal level even after the PV/PA PCB is replaced, replace the I/O PCB.	CXSM**-101
Faulty main CPU PCB	C: If the arterial pressure does not return to its normal level even after the I/O PCB is replaced, replace the main CPU PCB.	BHSM**-100

E69 DIALYSATE FLOW SENSOR ERROR (for Type 1)



Detection condition	When dialysate flow “STOP” is not detected even after a lapse of 245 seconds since detect step is initiated $245 \text{ seconds} = 1,000 \text{ mL} \div 282 \text{ mL/min.} \times 60 \text{ sec.} \times 1.2 - (\text{Period of C1 or J1})$ (Chamber capacity: 500 mL $\times$ 2/Dialysate flow rate: 282 mL/min (A flow meter error 3% of full scale is taken into account)/Safety factor: 20%)						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
C2	- / - / -	○	○	-	-	-	-
J2	- / - / -	○	○	-	-	-	-
D2, D'2	- / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	Detect step · J2 in Standby · C2 in Hot Citric or Hot rinse · C2 in Cool Down · D2,D'2 in Disinfection or Acid rinse · C2 in Water Rinse/ Warm Rinse Only type 1 of fluid circuit		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Clogging of closed circuit	C: Overhaul or replace the clogged part(s).	
Faulty “P2”	C: Overhaul or replace the “P2”.	CXSM**-007
Failure of any valve out of the “V5a/b” to “V9a/b”	C: Overhaul or replace the faulty valve(s) out of the “V5a/b” to “V9a/b”.	CXSM**-001
Low dialysate flow rate	C: Make adjustment of dialysate flow rate in the range of 300 to 700 mL/min. (300 to 800 mL/min at Hot rinse option installation.)	CXAJ**-010
Clogging of closed circuit	C: Overhaul or replace clogged part(s).	
Faulty “P2”	C: Overhaul or replace the pump “P2”.	CXSM**-007
Broken wiring in the wiring to the “FS2” or loose connection of connector part	A: Lightly pull the lead wires of the CN213 on the I/O PCB to check them for disconnection or loose connection. Furthermore, check for the connector on the “FS2” side. C: If any disconnection or loose connection is observed, remedy it.	
Faulty “FS2”	A: Measure a power supply voltage of 24V and output voltage on the flow sensor “FS2”; • Between “+” and “-” of power supply voltage: Approx. 24V (Acceptance range: 24V $\pm$ 10%) • Between “OUT” and “-” of output voltage: Approx. 24V with the float located in its “up” position (Acceptance range: 24V $\pm$ 10%)/Approx. 0V with the float located in its “down” position (Acceptance range: Not more than 1V) C: Replace the “FS2” or the flow meter assembly.	

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty I/O PCB	A: If the "FS2" is normal, check for the operation display state and the display of the "FS2". C: If the display is abnormal, replace the I/O PCB.	CXSM**-101

E70 AIR BUBBLE SENSOR SETTING ERROR

Detection condition	When normal setting is not made to the air bubble CPU.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	○	-	-
Rinse	- / - / -	○	○	-	○	-	-
Dia. Wait	- / - / -	○	○	○ SW OFF	○	-	-
Standby	- / - / -	○	○	○ SW OFF	○	-	-
Preparation	- / - / -	○	○	○ SW OFF	○	-	-
Dialysis	- / ○ / -	○	○	○ SW OFF	○	-	-
Retrans	- / - / -	○	○	○ SW OFF	○	-	-
Reset method	Press the RESET switch.			Remark	The air bubble ON/OFF SW switch blinking		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Broken wire in the air bubble sensor harness or loose connection of connector part	A: Pull the lead wires of the CN210 on the I/O PCB to check them for loose connection, and further check whether the lead wires are all the way seated in the connector. C: Replace the air bubble sensor.	CXSM**-010
Malfunction due to external noises	C: Cut off a noise source or keep it at a distance.	
Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101
Faulty main CPU PCB	C: Replace the main CPU PCB.	BHSM**-100

E71 HP DRIVE ERROR (PH7)

Detection condition	When the response cycle of the PH7 exceeds $\pm 20\%$ of the set value after start up of the heparin pump.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dia. Wait	- / - / -	○	○	HP STOP	-	-	○
Standby	- / - / -	○	○	HP STOP	-	-	○
Preparation	- / - / -	○	○	HP STOP	-	-	○
Dialysis	- / - / -	○	○	HP STOP	-	-	-
Returns	- / - / -	○	○	HP STOP	-	-	○
Reset method	Reset by pressing the RESET switch.			Remark	This error is not detected while in pump F.F. mode		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The heparin pump motor normally rotates but stops rotating as soon as the alarm is issued	A: Check to be sure on the Operation display screen that the LED in the "PH7" turns ON when the dog plate shields light between the sensor slits.	
○ Defective detection of the dog "PH7"	A: Check the "PH7" for any stains. C: Wipe stains off it. C: Replace the HP sensor PCB.	BHSM**-003
○ Faulty sub I/O PCB	C: Replace the I/O PCB.	CXSM**-101
○ Dropping-off of dog plate	C: Remedy the dog plate.	BHSM**-003
The heparin pump motor rotates backward		
○ Faulty drive PCB	C: Replace the drive PCB.	BHSM**-103
○ Faulty motor	C: Replace the motor.	BHSM**-003

E72 HP DRIVE ERROR (PH8)

Detection condition	When the response cycle of the PH8 exceeds $\pm 20\%$ of the set value after the start up of the heparin pump.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dia. Wait	- / - / -	○	○	HP STOP	-	-	○
Standby	- / - / -	○	○	HP STOP	-	-	○
Preparation	- / - / -	○	○	HP STOP	-	-	○
Dialysis	- / - / -	○	○	HP STOP	-	-	-
Returns	- / - / -	○	○	HP STOP	-	-	○
Reset method	Reset by pressing the RESET switch.			Remark	This error is not detected while in pump F.F. mode		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The heparin pump motor normally rotates but stops rotating as soon as the alarm is issued	A: Check to be sure on the Operation Display screen that the LED in the "PH8" turns ON when the dog plate shields light between the sensor slits.	
○ Defective detection of the dog "PH8"	A: Check the "PH8" for any stains. C: Wipe stains off it, if any. C: Replace the HP sensor PCB.	BHSM**-003
○ Faulty sub I/O PCB	C: Replace the I/O PCB.	CXSM**-101
○ Dropping-off of dog plate	C: Remedy the dog plate.	BHSM**-003
The heparin pump motor rotates backward		
○ Faulty drive PCB	C: Replace the drive PCB.	CXSM**-101
○ Faulty motor	C: Replace the motor.	BHSM**-003

E73 MAIN CPU (HMI) ERROR

Detection condition	When any of the signals that are periodically exchanged with the control program stops responding						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	-	-	-
Rinse	○ / - / -	○	○	-	-	-	-
Dia. Wait	- / - / -	○	○	○	-	-	-
Standby	○ / - / -	○	○	○	-	-	-
Preparation	○ / - / -	○	○	○	-	-	-
Dialysis	○ / ○ / -	○	○	○	-	-	-
Retrans	○ / - / -	○	○	○	-	-	-
Reset method	Recovers by "I/O" switch OFF/ON. (Restart the Device.)			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty main CPU PCB	C: Replace the main CPU PCB.	BHSM**-100

E74 BLOOD PUMP ERROR



Detection condition	[When starting the BP] - If an error in the rotating direction of the blood pump is detected for one second - If an error in the revolutions of the blood pump is detected for two seconds (This error will be detected when the pump flow rate exceeds $\pm 20\%$ of the set value. If the actual revolution is not more than 2 rpm or when it exceeds $\pm 40\%$ of the set value in the case of any revolutions other than that aforementioned.) - The signal of one of U, V, W phases counted zero for one second. [When stopping the BP] - If normal or reverse rotation is detected for two seconds at rotation speed 2rpm or greater. - If normal or reverse rotation is detected for eight seconds at rotation speed less than 2rpm.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dia. Wait	- / - / -	○	○	○ SW OFF	-	-	-
Standby	- / - / -	○	○	○ SW OFF	-	-	-
Preparation	- / - / -	○	○	○ SW OFF	-	-	-
Dialysis	- / ○ / -	○	○	○ SW OFF	-	-	-
Retrans	- / - / -	○	○	○ SW OFF	-	-	-
Reset method	Press the RESET switch.			Remark	This error is not detected while the startup test is in progress. This error is not detected for 2seconds after BP cover is closed.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The motor shaft is locked		
○ The blood pump rotor part bites foreign matter	C: Set the Blood Pump Operation switch to OFF, and then remove the foreign matters.	
○ Damage to the gear head part of the motor	C: Replace the blood pump assembly.	BHSM**-013
○ Faulty power supply PCB	C: Replace the power supply PCB.	BHSM**-103
○ Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101
○ Faulty main CPU PCB	C: Replace the main CPU PCB.	BHSM**-100
The motor does not rotate		
○ Broken wire in the wirings of the blood pump and loose connection of connector	A: Pull the lead wires of the CN114 on the power supply PCB and of the connector on the blood pump side to check them for loose connection. C: Remedy or replace.	
○ Faulty blood pump motor	C: Replace the blood pump assembly.	BHSM**-013
○ Faulty power supply PCB	C: Replace the power supply PCB.	BHSM**-103

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
○ Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101
○ Faulty main CPU PCB	C: Replace the main CPU PCB.	BHSM**-100

E75 BLOOD PUMP FAULT LOAD ALARM



Detection condition	When a overload signal is detected for a certain time during the start up of the BP.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dia. Wait	- / - / -	○	○	○ SW OFF	-	-	-
Standby	- / - / -	○	○	○ SW OFF	-	-	-
Preparation	- / - / -	○	○	○ SW OFF	-	-	-
Dialysis	- / ○ / -	○	○	○ SW OFF	-	-	-
Retrans	- / - / -	○	○	○ SW OFF	-	-	-
Reset method	Press the RESET switch.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The motor shaft is locked		
○ The blood pump rotor part bites foreign matters	A: Check whether the blood pump rotor part bites any foreign matters. C: Set the Blood Pump Operation switch to OFF, and then remove the foreign matters.	
Disconnecting the connector from the blood pump harness and then rotating the rotor by hand causes a heavy feeling in it.	A: Disconnect the connector from the blood pump harness, and then rotate the rotor by hand to check whether it smoothly rotates.	
○ Damage to the gear head part of the motor	C: Replace the blood pump assembly.	BHSM**-013
Disconnecting the connector from the blood pump harness enables the rotor to smoothly rotate	A: Disconnect the connector from the blood pump harness, and then rotate the rotor by hand to check whether it smoothly rotates.	
○ Faulty power supply PCB	C: Replace the power supply PCB.	BHSM**-103
The motor does not rotate		
○ Broken wire in the wirings to the blood pump and loose connection of connector part	A: Pull the lead wires of the CN114 on the power supply PCB and those of the connector on the blood pump side to check them for loose connection. C: Remedy or replace.	
○ Faulty blood pump motor	C: Replace the blood pump assembly.	BHSM**-013
○ Faulty power supply PCB	C: Replace the power supply PCB.	BHSM**-103

E76 MAIN CPU (CONTROL) ERROR

Detection condition	The CPU2 detected the stop of the clock of the CPU1.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	-	-	-
Rinse	- / - / -	○	○	-	-	-	-
Dia. Wait	- / - / -	○	○	○	-	-	-
Standby	- / - / -	○	○	○	-	-	-
Preparation	- / - / -	○	○	○	-	-	-
Dialysis	- / ○ / -	○	○	○	-	-	-
Retrans	- / - / -	○	○	○	-	-	-
Reset method	Recovers by "I/O" switch OFF/ON. (Restart the Device.)			Remark	Turn V9b and V19 OFF. Stop P1 and P2.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty main CPU PCB	C: Replace the main CPU PCB.	BHSM**-100

E77 SUB CPU ALARM

Detection condition	When any of the signals that are periodically exchanged with the monitoring CPU stops responding						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	-	-	-
Rinse	○ / - / -	○	○	-	-	-	-
Dia. Wait	- / - / -	○	○	○ SW OFF	-	-	-
Standby	○ / - / -	○	○	○ SW OFF	-	-	-
Preparation	○ / - / -	○	○	○ SW OFF	-	-	-
Dialysis	○ / ○ / -	○	○	○ SW OFF	-	-	-
Retrans	○ / - / -	○	○	○ SW OFF	-	-	-
Reset method	Recovers by "I/O" switch OFF/ON. (Restart the Device.)			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101

E78 PANEL CPU ALARM

Detection condition	Error was detected in the panel CPU control						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	-	-	-
Rinse	○ / - / -	○	○	-	-	-	-
Dia. Wait	- / - / -	○	○	○ SW OFF	-	-	-
Standby	○ / - / -	○	○	○ SW OFF	-	-	-
Preparation	○ / - / -	○	○	○ SW OFF	-	-	-
Dialysis	○ / ○ / -	○	○	○ SW OFF	-	-	-
Retrans	○ / - / -	○	○	○ SW OFF	-	-	-
Reset method	Recovers by "I/O" switch OFF/ON. (Restart the Device.)			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty panel PCB	C: Replace the panel PCB.	CXSM**-106
Damage to the panel communication cable	A: Check the connector of the communication cable between the panel and the I/O PCB. Check the communication cable between the panel and the I/O PCB for any flaws. C: Replace the panel communication cable.	

E80 BPM RAM ERROR

Detection condition	When a RAM error of the BPM module is detected						
Detection process	Blood pressure measurement	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
System startup or Recovery from power failure	The measurement is not possible.	○	○	-	-	-	-
Reset method	Recovers by "I/O" switch OFF/ON. (Restart the Device.)			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty BPM module	C: Replace the BPM module.	BHSM**-108

E81 BPM ROM ERROR

Detection condition	When a RAM error of the ROM module is detected						
Detection process	Blood pressure measurement	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
System startup or Recovery from power failure	The measurement is not possible.	○	○	-	-	-	-
Reset method	Recovers by "I/O" switch OFF/ON. (Restart the Device.)			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty BPM module	C: Replace the BPM module.	BHSM**-108

E82 BPM ERROR

Detection condition	When the single failure state of the BPM is detected						
Detection process	Blood pressure measurement	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
While in blood pressure measurement	The measurement is not possible.	○	○	-	-	-	-
Reset method	Recovers by "I/O" switch OFF/ON. (Restart the Device.)			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty BPM module	C: Replace the BPM module.	BHSM**-108

E83 BPM COMMUNICATION ERROR



Detection condition	When a communication error is detected between the Device and the BPM module.						
Detection process	Blood pressure measurement	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
While in blood pressure measurement	The measurement is not possible.	○	○	-	-	-	-
Reset method	Press the RESET switch. Press the BPM MEASURE START switch to initiate re-measurement.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Normal communications between the dialyzer and the BPM module are disabled		
○ Disconnection of a communication cable or a power supply cable Communication cable/BPM interface harness Power supply cable/Power supply harness	C: Properly connect the connectors between the dialyzer and the BPM Check for the connectors listed below. • CN225 mounted on the I/O PCB • CN1, CN2, and CN3 mounted on the BPM interface PCB • Connector of the BPM module • CN108 mounted on the power supply PCB	CXSM**-101 BHSM**-108 BHSM**-105
○ Loose connection of the connector part of a communication cable or a power supply cable	C: Clean the connections with contact cleaner & rejuvenator or else.	
○ Broken wire in a communication cable or a power supply cable	A: Check for continuity using a tester.	
○ Faulty I/O PCB	C: If no continuity, solder again or replace the defective cable.	CXSM**-101
○ Faulty BPM interface PCB	C: Replace the I/O PCB.	BHSM**-108
○ Faulty BPM module	C: Replace the BPM interface PCB.	BHSM**-108
○ Influence of noises	C: Replace the BPM module. A: Identify noise sources. Examples of noise sources: Electric hotpacks, electrocardiograms, surgical laser, clean towers, etc. C: Keep the Device away from the noise sources. Take noise protective measures (e.g. install an isolating transformer).	

E84 BPM PRESSURIZATION ERROR

Detection condition	When the cuff pressure does not reach 10 mmHg even after a lapse of five seconds since pressurization is initiated.						
Detection process	Blood pressure measurement	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
While in blood pressure measurement	The measurement is not possible.	○	○	-	-	-	-
Reset method	Press the RESET switch. Press the BPM MEASURE START switch to initiate re-measurement.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The cuff pressure does not reach 10 mmHg even after a lapse of five seconds since pressurization is initiated	A: Check to be sure that the air hose is securely connected to the Device or the cuff. Furthermore, check to be sure that the cuff is tightly wound. C: Remedy loose connection or winding, if any.	

E85 HIGHEST-BLOOD-PRESSURE MAX ERROR

Detection condition	When the systolic blood pressure exceeds the maximum set value for the systolic blood pressure alarm						
Detection process	Blood pressure measurement	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
While in blood pressure measurement	The measurement is not possible.	○	○	-	-	-	-
Reset method	Press the RESET switch. Press the BPM MEASURE START switch to initiate re-measurement.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The systolic blood pressure exceeds the maximum set value for the systolic blood pressure alarm	C: Perform a given treatment under a physician's direction. According to circumstances, make resetting of the maximum set value for the systolic blood pressure.	

E86 HIGHEST-BLOOD-PRESSURE MIN ERROR

Detection condition	When the systolic blood pressure drops below the minimum set value for the systolic blood pressure alarm						
Detection process	Blood pressure measurement	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
While in blood pressure measurement	The measurement is not possible.	○	○	-	-	-	-
Reset method	Press the RESET switch. Press the BPM MEASURE START switch to initiate re-measurement.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The systolic blood pressure drops below the minimum set value for the systolic blood pressure alarm	C: Perform a given treatment under a physician's direction. According to circumstances, make resetting of the minimum set value for the systolic blood pressure.	

E87 LOWEST-BLOOD-PRESSURE MAX ERROR

Detection condition	When the diastolic blood pressure exceeds the maximum set value for the diastolic blood pressure alarm						
Detection process	Blood pressure measurement	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
While in blood pressure measurement	The measurement is not possible.	○	○	-	-	-	-
Reset method	Press the RESET switch. Press the BPM MEASURE START switch to initiate re-measurement.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The diastolic blood pressure exceeds the maximum set value for the diastolic blood pressure alarm	C: Perform a given treatment under a physician's direction. According to circumstances, make resetting of the maximum set value for the diastolic blood pressure.	

E88 LOWEST-BLOOD-PRESSURE MIN ERROR

Detection condition	When the diastolic blood pressure drops below the minimum set value for the diastolic blood pressure alarm						
Detection process	Blood pressure measurement	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
While in blood pressure measurement	The measurement is not possible.	○	○	-	-	-	-
Reset method	Press the RESET switch. Press the BPM MEASURE START switch to initiate re-measurement.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The diastolic blood pressure drops below the minimum set value for the diastolic blood pressure alarm	C: Perform a given treatment under a physician's direction. According to circumstances, make resetting of the minimum set value for the diastolic blood pressure.	

E89 PULSE-RATE MAX ERROR

Detection condition	When the pulse rate exceeds the maximum pulse rate set value						
Detection process	Blood pressure measurement	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
While in blood pressure measurement	The measurement is not possible.	○	○	-	-	-	-
Reset method	Press the RESET switch. Press the BPM MEASURE START switch to initiate re-measurement.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The pulse rate exceeds the maximum pulse rate set value	C: Perform a given treatment under a physician's direction. According to circumstances, make resetting of the maximum pulse rate set value.	

E90 PULSE-RATE MIN ERROR

Detection condition	When the pulse rate drops below the minimum pulse rate set value						
Detection process	Blood pressure measurement	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
While in blood pressure measurement	The measurement is not possible.	○	○	-	-	-	-
Reset method	Press the RESET switch. Press the BPM MEASURE START switch to initiate re-measurement.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The pulse rate drops below the minimum pulse rate set value	C: Perform a given treatment under a physician's direction. According to circumstances, make resetting of the minimum pulse rate set value	

E97 SHORT CHEMICAL SOLUTION ALARM (for Type 1)

Detection condition	- When the switch-OFF state of FS3 (no flow) is detected for 15msec or longer consecutively (but such the conditions do not amount to 1sec cumulatively) in 2 Sequence Steps in a row. - When the switch-OFF state of FS3 (no flow) is detected for 15msec or longer consecutively and such the conditions amount to 1sec or longer in single Sequence Step.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Disinfection rinse	- / - / -	○	○	-	-	-	-
Tepid disinfection	- / - / -	○	○	-	-	-	-
Acid rinse	- / - / -	○	○	-	-	-	-
Tepid Acid	- / - / -	○	○	-	-	-	-
Hot citric acid	○ / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	Only type 1 of fluid circuit.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The Device has run out of rinse solution	C: Refill the rinse solution.	
The Device has not run out of rinse solution		
○ Defective mounting of the FS3	A: Check the FS3 for the mounting conditions. C: If the FS3 disconnected from the tube, connect it to the tube.	
Faulty FS3	C: If the FS3 is not disconnected from the tube, check for the operation indicating LED. If the LED does not respond, replace the FS3.	
Faulty I/O PCB	C: If the LED responds, check the CN213 on the I/O PCB for the pin conditions, and then replace the I/O PCB.	CXSM**-101

E97 SHORT CHEMICAL SOLUTION ALARM (for Type 2)

Detection condition	1) FS3 OFF (no fluid) is detected in CS-8/9 for more than 200msec continuously. 2) FS3 OFF (no fluid) is detected during transfer to CS-10. 3) FS3 OFF (no fluid) is detected in CS-10 for more than 200msec continuously. Alarm is not issued or fluid is not stopped for the first detection but only resetting is done. If the same content is detected again after resetting, an alarm is issued.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Disinfection	○ / - / -	○	○	-	-	-	-
Tepid disinfection	○ / - / -	○	○	-	-	-	-
Acid rinse	○ / - / -	○	○	-	-	-	-
Tepid Acid	○ / - / -	○	○	-	-	-	-
Hot citric acid	○ / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	This error is monitored only in CS step. Operation after alarm reset: After priming of disinfectant port and deaeration of dialysate circuit, restart sucking the disinfectant. The rinsing time is not calculated at this time but the elapsed time is calculated after CIR1 is started. Only type 2 of fluid circuit.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The Device has run out of rinse solution	C: Refill the rinse solution.	
The Device has not run out of rinse solution		
○ Disconnect the port with the tank.	C: Connect the port with the tank.	
○ Defective mounting of the FS3	A: Check the FS3 for the mounting conditions. C: If the FS3 disconnected from the tube, connect it to the tube.	
○ Faulty FS3	C: If the FS3 is not disconnected from the tube, check for the operation indicating LED. If the LED does not respond, replace the FS3.	
○ Faulty I/O PCB	C: If the LED responds, check the CN213 on the I/O PCB for the pin conditions, and then replace the I/O PCB.	

E99 BATTERY ERROR

Detection condition	1. AC powered AND battery below 22V AND 10 minutes elapsed after "E66 BATTERY VOLTAGE TOO LOW" 2. AC powered AND battery over 30V						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	-	-	○
Rinse	- / - / -	○	○	-	-	-	○
Dia. Wait	- / - / -	○	○	-	-	-	○
Standby	- / - / -	○	○	-	-	-	○
Preparation	- / - / -	○	○	-	-	-	○
Dialysis	- / - / -	○	○	-	-	-	-
Retrans	- / - / -	○	○	-	-	-	○
Reset method	Press the RESET switch.			Remark	To reset the alarm, set the "BP BACKUP BATTERY SETTING" function to "OFF" on the Maintenance Menu 2 screen.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The battery is not charged		
Operating the blood pump with the use of the battery for an extended period of time results in shortage of charge time	C: Set the Power Supply switch on the Device to ON to apply power. 48 hours are needed to fully charge the battery.	
○ The battery has reached the end of its service life	C: Replace the battery.	BHSM**-107
○ Faulty battery charge circuit	C: Replace the power supply PCB.	BHSM**-105
○ Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101

E100 INTERNAL DATA ERROR

Detection condition	When there is a difference between data saved in the F-RAM and that saved in the shared memory.(detected by the HMI process in the CPU1). When there is a difference between data saved in the DP-RAM and bit-inverted data saved in the DP-RAM.(detected by the CPU2)						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	○	-	-
Rinse	○ / - / -	○	○	-	○	-	-
Dia. Wait	- / - / -	○	○	○	○	-	-
Standby	○ / - / -	○	○	○	○	-	-
Preparation	○ / - / -	○	○	○	○	-	-
Dialysis	○ / ○ / -	○	○	○	○	-	-
Retrans	○ / - / -	○	○	○	○	-	-
Reset method	Recovers by "I/O" switch OFF/ON. (Restart the Device.)			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101

E101 SOLUTION B CONDUCTIVITY ALARM (for Type 1)

Detection condition	•When a mean CD2 value (solution B conductivity value) in excess of the maximum set value for the solution B conductivity alarm or below the minimum set value for the solution B conductivity alarm is detected in two consecutive rounds. •When a mean CD2 value (solution B conductivity value) in excess of +10% for the solution B base conductivity or below -10% for the solution B base conductivity is detected.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Standby(CFP)	- / - / ○	○	○	-	-	-	-
Preparation	- / - / ○	○	○	-	-	-	-
Dialysis	- / ○ / ○	○	○	-	-	-	-
Retrans	- / - / ○	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	•While in the fluid stop state and after resetting the state, this alarm is not detected unless the two rounds are complete •While SQHD is in progress or after SQHD is reset, the alarm is not detected unless the two rounds are complete. Only type 1 of fluid circuit		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty "P4"	C: Replace the "P4".	CXSM**-021
Faulty "drive PCB"	C: Replace the "drive PCB".	BHSM**-103
Charge flow rate too high	A: Check to be sure that the charge flow rate is corresponding to the dialysate flow rate. C: If not, make adjustment of the charge flow rate.	CXAJ**-009
Bended, clogged, or crushed tube in the concentrate fluid line	C: Remedy the bended, clogged, or crushed tube.	
The conductivity is normal		
○ Stains (e.g. calcium carbonate) on the electrode of the conductivity sensor, or fluid leaks	C: Perform thorough acid rinse. If the stains cannot be removed, perform resolvent rinse with a specified chemical (e.g. CHARON or SHUMMA).	
○ Loose connection of the "CN233" conductivity sensor connector	A: Visually check the CN233 for any erroneous wire connections, corrosion of contact surface, or broken wires. C: Remedy erroneous wire connections, if any. If any corrosion or broken wires are observed, replace the connector.	
○ Broken wire in or short-circuit of the conductivity sensor	A: Disconnect the CD2 connector from the I/O PCB, and then make measurement of "CD2 - GND" resistance to check to be sure that the sensor has no short circuit, broken wires, or loose connection. C: Replace the harness of the CD2.	CXSM**-011
○ Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101

E101 SOLUTION B CONDUCTIVITY ALARM (for Type 2)

Detection condition	•When a mean CD2 value (solution B conductivity value) in excess of the maximum set value for the solution B conductivity alarm or below the minimum set value for the solution B conductivity alarm is detected in three consecutive rounds. •When a mean CD2 value (solution B conductivity value) in excess of +9.9% for the solution B base conductivity or below -9.9% for the solution B base conductivity is detected.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Preparation	- / - / ○	-	-	-	-	-	-
Dialysis	- / ○ / ○	○	○	-	-	-	-
Retrans	- / - / ○	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	In Prep. complete process, this alarm can be detected only in auto-bypass. No detection when P4 is stopped. Only type 2 of fluid circuit		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty "P4"	C: Replace the "P4".	CXSM**-021
Faulty "drive PCB"	C: Replace the "drive PCB".	BHSM**-103
Charge flow rate too high	A: Check to be sure that the charge flow rate is corresponding to the dialysate flow rate. C: If not, make adjustment of the charge flow rate.	CXAJ**-009
Bended, clogged, or crushed tube in the concentrate fluid line	C: Remedy the bended, clogged, or crushed tube.	
The conductivity is normal		
○ Stains (e.g. calcium carbonate) on the electrode of the conductivity sensor, or fluid leaks	C: Perform thorough acid rinse. If the stains cannot be removed, perform solvent rinse with a specified chemical (e.g. CHARON or SHUMMA).	
○ Loose connection of the "CN233" conductivity sensor connector	A: Visually check the CN233 for any erroneous wire connections, corrosion of contact surface, or broken wires. C: Remedy erroneous wire connections, if any. If any corrosion or broken wires are observed, replace the connector.	
○ Broken wire in or short-circuit of the conductivity sensor	A: Disconnect the CD2 connector from the I/O PCB, and then make measurement of "CD2 - GND" resistance to check to be sure that the sensor has no short circuit, broken wires, or loose connection. C: Replace the harness of the CD2.	CXSM**-011
○ Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101

E102 T5 TEMPERATURE SENSOR ERROR

Detection condition	When the T5 temperature sensor detects temperatures of not less than 99.8°C for a period of consecutive 5 seconds When the T5 temperature sensor detects temperatures of not more than 0.5°C for a period of consecutive 5 seconds						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	-	-	○
Rinse	- / - / -	○	○	-	-	-	○
Dia. Wait	- / - / -	○	○	-	-	-	○
Standby	○ / - / -	○	○	-	-	-	○
Preparation	○ / - / -	○	○	-	-	-	○
Dialysis	○ / ○ / -	○	○	-	-	-	-
Retrans	○ / - / -	○	○	-	-	-	○
Reset method	Press the RESET switch.			Remark	This alarm is not detected during the startup test in the standby mode.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The T5 and other analog displays are out of order	A: Check in the display of all the temperature sensors and other sensors (e.g. fluid pressure sensor) are working properly. If not, check 5.00V±0.01V at the test point on the I/O PCB. C: If the voltage does not come to 5.00V±0.01V, perform voltage calibration.	CXSM**-101 CXSM**-101
Loose connection of the T5 temperature sensor connector	C: Visually check the CN234 for any erroneous wire connections, corrosion of contact surface, or broken wires. C: Remedy erroneous wire connections, if any. If any corrosion or broken wires are observed, replace the harness.	BHSM**-008
Broken wire or short circuit in the "T5 temperature sensor"	A: Disconnect the T5 temperature sensor from the CN234 on the I/O PCB. Then measure the "Red - White" resistance, to check that the sensor has no short circuit, broken wires, or loose connection. C: If the sensor has any short circuit, broken wire, or loose connection, replace the "T5 temperature sensor".	BHSM**-008
Deviation from the display temperature calibration of the I/O PCB	A: Disconnect T5 temperature sensor from the PCB. Connect an another normal temperature (e.g. T4) to the connector of T5, to confirm condition of PCB. C: If the temperature display is abnormal, perform the calibration of the condensation sensor PCB. If the calibration cannot be performed, replace the condensation sensor PCB	BHAJ**-004 CXSM**-101

E103 I/O SWITCH ERROR

Detection condition	When an error occurs in the signal cable for the I/O switch (e.g. broken wire or disconnected connector)						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	-	-	-
Rinse	- / - / -	○	○	-	-	-	-
Dia. Wait	- / - / -	○	○	-	-	-	-
Standby	- / - / -	○	○	-	-	-	-
Preparation	- / - / -	○	○	-	-	-	-
Dialysis	- / - / -	○	○	-	-	-	-
Retrans	- / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch. If the I/O switch falls outside of the detection condition, the alarm will be automatically reset.		Remark				

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
CN102 disconnected from the power supply PCB or CN901 disconnected from the I/O switch PCB	C: Connect the connector again	BHSM**-105
Broken wire in the harness connecting between the connectors aforementioned	C: Replace the harness.	
Faulty power supply PCB	C: Replace the power supply PCB.	BHSM**-105
Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101

E104 BACK LIGHT ERROR

Detection condition	When one of the two backlights turn OFF						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	○ / - / -	○	○	-	-	-	○
Rinse	○ / - / -	○	○	-	-	-	○
Dia. Wait	○ / - / -	○	○	-	-	-	-
Standby	○ / - / -	○	○	-	-	-	-
Preparation	○ / - / -	○	○	-	-	-	-
Dialysis	○ / - / -	○	○	-	-	-	-
Retrans	○ / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.		Remark	· Not detect when “BACK LIGHT OFF TIME” effective. · Not detect the alarm after the detection for one hour.			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The backlight turns OFF		
○ CN306 disconnected from the panel CPU PCB or a connector disconnected from the inverter PCB	C: Connect the connector again.	
○ Broken wire in the harness connecting between the connectors aforementioned	C: Replace the harness.	
○ The battery has reached the end of its service life	C: Since the backlight cannot be independently replaced, replace the operation panel assembly. The backlights have a service life of approximately 30,000 hours. However, it varies with use conditions, environments, and luminance.	CXSM**-106
○ Faulty backlight inverter PCB	C: Replace the backlight inverter PCB.	CXSM**-106
○ Faulty panel CPU PCB	C: Replace the panel CPU PCB.	CXSM**-106
○ Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101
The backlight turns ON		
○ Faulty backlight inverter PCB	C: Replace the backlight inverter PCB.	CXSM**-106
○ Faulty panel CPU PCB	C: Replace the panel CPU PCB.	CXSM**-106
○ Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101

E105 CPU CHECK ERROR

Detection condition	When the “Memory” used for startup tests becomes “NG”						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Annunciator light (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm reset
Standby	○ / - / -	○	○	○	○	-	-
Reset method	Recovers by “I/O” switch OFF/ON. (Restart the Device.)		Remark				

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101

E106 FPGA2 POWER OF I/O ERROR

Detection condition	The VCCIO voltage (+3.3V) of the FPGA2 is out of the range between +3.0 and +3.6 volts.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	○	-	-
Rinse	- / - / -	○	○	-	○	-	-
Dia. Wait	- / - / -	○	○	○ (SW OFF)	○	-	-
Standby	○ / - / -	○	○	○ (SW OFF)	○	-	-
Preparation	○ / - / -	○	○	○ (SW OFF)	○	-	-
Dialysis	○ / ○ / -	○	○	○ (SW OFF)	○	-	-
Retrans	○ / - / -	○	○	○ (SW OFF)	○	-	-
Power failure	- / - / -	○	-	○ (SW OFF)	○	-	-
Reset method	Press the RESET switch.			Remark	The alarm is detected while Battery operation.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101

E107 FPGA2 POWER OF INT ERROR

Detection condition	The INT power supply voltage (+2.5V) of the FPGA2 is out of the range between +2.375 and +2.635 volts.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	○	-	-
Rinse	- / - / -	○	○	-	○	-	-
Dia. Wait	- / - / -	○	○	○ (SW OFF)	○	-	-
Standby	○ / - / -	○	○	○ (SW OFF)	○	-	-
Preparation	○ / - / -	○	○	○ (SW OFF)	○	-	-
Dialysis	○ / ○ / -	○	○	○ (SW OFF)	○	-	-
Retrans	○ / - / -	○	○	○ (SW OFF)	○	-	-
Power failure	- / - / -	○	-	○ (SW OFF)	○	-	-
Reset method	Press the RESET switch.			Remark	The alarm is detected while Battery operation.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101

E108 UFP STOP ERROR

Detection condition	A change in the rotation signal of the UF pump was detected while the UF pump (UFP) was being stopped.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Preparation	○ / - / -	○	○	-	-	-	-
Dialysis	○ / - / -	○	○	-	-	-	-
Retrans	○ / - / -	○	○	-	-	-	-
Power failure	- / - / -	○	-	-	-	-	-
Reset method	Press the RESET switch.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty "PH5"/"PH6", or broken wire in the harness	A: Check to be sure on the Maintenance Mode 4 screen that the "PH5"/"PH6" responds when the dog plate shields light between the sensor slits. C: If the "PH5"/"PH6" does not respond, replace it.	BHSM**-109
Faulty I/O PCB	C: Replace the I/O PCB	CXSM**-101
Faulty main CPU PCB	C: Replace the main CPU PCB.	BHSM**-103

E110 CONC CONNECTOR TO RINSING PORT (for Type 1)

Detection condition	· The AS2 detected air for three seconds within two minutes since the rinse port connection check process (RPC2) started · Dialysate conductivity or B-conductivity exceeded 3.0mS/cm two minutes after the rinse port connection check process (RPC2) started						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Water rinse	○ / - / -	○	○	-	-	○	-
Tepid water rinse	○ / - / -	○	○	-	-	○	-
Hot rinse	○ / - / -	○	○	-	-	○	-
Hot citric acid	○ / - / -	○	○	-	-	○	-
Reset method	Press the RESET switch.			Remark	Supply of dialysate stops when the air emission sequence ends. Only type 1 of fluid circuit.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The Connector is disconnected from the Rinse Port.	C: Connect the Connector to the Rinse Port.	
The Connector is connected to the Stock Solution Tank.	C: Connect the Connector to the Rinse Port.	

E110 CONC CONNECTOR TO RINSING PORT (for Type 2)

Detection condition	1) Air was detected at the AS2 for 3 seconds continuously during the Rinse Port Check process. 2) The conductivity of dialysate or B-solution was 3.0mS/cm or more during the Rinse Port Check process.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Water rinse	○ / - / -	○	○	-	-	○	-
Tepid water rinse	○ / - / -	○	○	-	-	○	-
Hot rinse	○ / - / -	○	○	-	-	○	-
Hot citric acid	○ / - / -	○	○	-	-	○	-
Reset method	Press the RESET switch.		Remark	Supply of dialysate stops when the air emission sequence ends. Only type 2 of fluid circuit.			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The Connector is disconnected from the Rinse Port.	C: Connect the Connector to the Rinse Port.	
The Connector is connected to the Stock Solution Tank.	C: Connect the Connector to the Rinse Port.	

E111 T6 TEMPERATURE SENSOR ERROR

Detection condition	When the T6 temperature sensor detects temperatures of not less than 99.8°C for a period of consecutive 5 seconds. When the T6 temperature sensor detects temperatures of not more than 0.5°C for a period of consecutive 5 seconds.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	-	-	-
Rinse	- / - / -	○	○	-	-	-	-
Dia. Wait	- / - / -	○	○	-	-	-	-
Standby	○ / - / -	○	○	-	-	-	-
Preparation	○ / - / -	○	○	-	-	-	-
Dialysis	○ / ○ / -	○	○	-	-	-	-
Retrans	○ / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	Available only with the B-cartridge option. This alarm is not detected during the startup test in the standby mode.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The T6 and other analog displays are out of order	A: Check in the display of all the temperature sensors and other sensors (e.g. fluid pressure sensor) are working properly. If not, check 5.00V±0.01V at the test point on the I/O PCB. C: If the voltage does not come to 5.00V±0.01V, perform voltage calibration.	CXSM**-101
Loose connection of the T6 temperature sensor connector	A: Check visual the CN239 for any erroneous wire connections, corrosion of contact surface, or broken wires. C: Remedy erroneous wire connections, if any. If any corrosion or broken wires are observed, replace the connector.	BHSM**-008
Broken wire or short circuit in the "T6 temperature sensor"	A: Disconnect the T6 temperature sensor from the CN239 on the I/O PCB. Then measure the "Red - White" resistance, to check that the sensor has no short circuit, broken wires, or loose connection. C: If the sensor has any short circuit, broken wire, or loose connection, replace the "T6 temperature sensor".	BHSM**-008
Deviation from the display temperature calibration of the I/O PCB	A: Disconnect T6 temperature sensor from the PCB. Connect an another normal temperature (e.g.T5) to the connector of T6, to confirm condition of PCB. C: If the temperature display is abnormal, perform the calibration of the temperature sensor module PCB (CD3). If the calibration cannot be performed, replace the I/O PCB.	BHAJ**-005 CXSM**-101

E112 T7 TEMPERATURE SENSOR ERROR

Detection condition	When the T7 temperature sensor detects temperatures of not less than 99.8°C for a period of consecutive 5 seconds. When the T7 temperature sensor detects temperatures of not more than 0.5°C for a period of consecutive 5 seconds.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	-	-	-
Rinse	- / - / -	○	○	-	-	-	-
Dia. Wait	- / - / -	○	○	-	-	-	-
Standby	○ / - / -	○	○	-	-	-	-
Preparation	○ / - / -	○	○	-	-	-	-
Dialysis	○ / ○ / -	○	○	-	-	-	-
Retrans	○ / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	This alarm is not detected during the startup test in the standby mode. Heater OFF when the heater is on.		

Symptom/Cause	A:Analysis / C:Cuntermesure	Work No.
The T7 display and other analog displays are out of order	A: Check in the display of all the temperature sensors and other sensors (e.g. fluid pressure sensor) are working properly. If not, check 5.00V±0.01V at the test point on the I/O PCB. C: If the voltage does not come to 5.00V±0.01V, perform voltage calibration.	CXSM**-101
Loose connection of the T7 temperature sensor connector	A: Check visual the CN246 for any erroneous wire connections, corrosion of contact surface, or broken wires. C: Remedy erroneous wire connections, if any. If any corrosion or broken wires are observed, replace the connector.	BHSM**-008
Broken wire or short circuit in the "T7 temperature sensor"	A: Disconnect the T7 temperature sensor from the CN246 on the I/O PCB. Measure the "Red - White" resistance, to check that the sensor has no short circuit, broken wires, or loose connection. C: If the sensor has any short circuit, broken wire, or loose connection, replace the "T7 temperature sensor".	BHSM**-008
Deviation from the display temperature calibration of the I/O PCB	A: Disconnect T7 temperature sensor from the PCB. Connect an another normal temperature (e.g. T6) to the connector of T7, to confirm condition of PCB. C: If the temperature display is abnormal, perform the calibration of the temperature sensor module PCB.	BHAI**-003 CXSM**-101

E113 T5&T6 DIFFERENCE ALARM (for Type 1)

Detection condition	When a difference between the T5 and T6 temperatures exceeds 1.2°C for a period of consecutive five seconds						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dialysis	- / ○ / ○	○	○	-	-	-	-
Reset method	Press the RESET switch.		Remark	While SQHD is in progress or after SQHD is reset, the alarm is not detected unless the bypass operation is reset. While Fluid Stop is in progress, the alarm is not detected. The alarm is detected with the B-powder option. Only type 1 of fluid circuit			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
There is an actual difference in temperatures		
○ Factors causing temperature changes present in the vicinity of the temperature sensor	A: Check for piping conditions. C: Remedy piping so as to be immune to disturbance.	
There is no actual difference in temperatures		
○ Deviation from the display temperature calibration of the I/O PCB	C: For detail, refer to information in the “E102 T5 TEMPERATURE SENSOR ERROR” C: For detail, refer to information in the “E111 T6 TEMPERATURE SENSOR ERROR”	CXTS**-E102 CXTS**-E106
○ Faulty T5 or T6 temperature sensor	C: For detail, refer to information in the “E102 T5 TEMPERATURE SENSOR ERROR” C: For detail, refer to information in the “E111 T6 TEMPERATURE SENSOR ERROR”	CXTS**-E102 CXTS**-E106

E114 TEMPERATURE CANNOT REACH (for Type 1)

Detection condition	T7 failed to rise to (setting – 3) °C in 50 min from the start of the circulation sequence.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Hot Citric Acid Hot Rinse	- / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.		Remark	The system shifts to Cool-Down Only type 1 of fluid circuit			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
There is an actual difference in temperatures		
○ Factors causing temperature changes present in the vicinity of the temperature sensor	A: Check for tubing conditions. C: Remedy tubing so as to be immune to disturbance.	
There is no actual difference in temperatures		
○ Deviation from the display temperature calibration of the I/O PCB	C: For detail, refer to information in the “E112 T7 TEMPERATURE SENSOR ERROR”	CXTS**-E112
○ Faulty T7 temperature sensor	C: For detail, refer to information in the “E112 T7 TEMPERATURE SENSOR ERROR”	CXTS**-E112

E114 TARGET TEMPERATURE NOT REACHED (for Type 2)

Detection condition	Cleaning time is extended over 20 min which is the setting time of Heat up/Circulation due to "M66 Hot disinfection time is extended."						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Hot Citric Acid Hot Rinse	- / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	The system shifts to Cool-Down Only type 2 of fluid circuit		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
There is an actual difference in temperatures		
○ Factors causing temperature changes present in the vicinity of the temperature sensor	A: Check for tubing conditions. C: Remedy tubing so as to be immune to disturbance.	
There is no actual difference in temperatures		
○ Deviation from the display temperature calibration of the I/O PCB	C: For detail, refer to information in the "E112 T7 TEMPERATURE SENSOR ERROR"	CXTS**-E112
○ Faulty T7 temperature sensor	C: For detail, refer to information in the "E112 T7 TEMPERATURE SENSOR ERROR"	CXTS**-E112

E115 B CARTRIDGE HOLDER OPEN

Detection condition	· When push "Prep" SW, LS7a or LS7b is off, or LS7a and LS7b is off. · When push "Rinse" SW, LS7a or LS7b is off, or LS7a and LS7b is off. · While Rinse operation, LS7a or LS7b is off, or LS7a and LS7b is off.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
· Rinse Wait (When push "Rinse" SW)	- / - / -	○ (3 times)	○	-	-	-	-
· Rinse	○ / - / -	○	○	-	-	-	-
· Dia. Wait (When push "Prep" SW)	- / - / -	○ (3 times)	○	-	-	-	-
Reset method	Press the RESET switch Auto reset after release from alarm condition.			Remark	Available only with the B-powder option. PREP. switch operation is refused. RINSE switch operation is refused.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The B-cartridge Holder open. ○ Closing the cartridge holder forgotten.	C: Close the B-cartridge Holder.	
The B-cartridge Holder close. ○ Disconnection of the LS7a or LS7b harness connector.	A: Check the connection of the LS7a or LS7b harness connector. C: Connect the connector.	
○ Wire breakage of the LS7a or LS7b harness.	A: Check the damage of the LS7a or LS7b harness. C: Connect the connector.	
○ Failure of the LS7a or LS7b.	C: Replace the LS7a or LS7b.	BHSM**-024
○ Failure of the I/O board.	A: If the problem was not solved by the replacement of the B-cartridge Holder Assy, C: Replace the I/O board.	CXSM**-101

E117 B CONDUCTIVITY CANNOT REACH (for Type 1)

Detection condition	CD3 < setting B-conductivity, AND priming up to the outlet of P4 is in operation, AND 2 minutes elapsed.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Standby	○ / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.		Remark	Available only with the B-powder option and type 1 of fluid circuit.			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Water empty in the B-powder cartridge.		
○ Failure of V11, V27b or V27a.	A: Operate the valves manually on the MAINT4 screen and check their working conditions. C: Disassemble and repair the valves, or replace them.	CXSM**-001
Water remaining in the B-powder cartridge.		
○ Failure of V27a or V28b.	A: Operate the valves manually on the MAINT4 screen and check their working conditions. C: Disassemble and repair the valves, or replace them.	CXSM**-001
○ F8 clogged.	A: Check if the F8 is clogged. C: If the F8 is clogged, replace it.	
○ Failure of P4.	A: Operate the P4 manually on the MAINT4 screen and check its working condition. C: If the pump does not work, replace it.	CXSM**-021
○ Contamination of CD3 electrode (ex. by calcium carbonate)	C: Remove the contamination by performing acid rinse operation.	
○ Imperfect contact of the CN238 connector of the conductivity sensor.	A: Disconnect the connector and check by sight the corrosion on the contact faces of the connector terminals. C: If the connector is faulty, replace it.	CXSM**-011
○ Failure of the I/O board.	A: Check if the display value of conductivity or temperature changes by changing the ambient temperature of the I/O board. C: If the I/O board is faulty, replace it.	CXSM**-101

E118 T2 OVERHEAT (for Type 1)

Detection condition	T2 exceeded 99.0°C for a period of consecutive 1 second while the heater was ON.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Hot rinse Hot citric acid	- / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	Heater OFF Only type 1 of fluid circuit.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
There is an actual difference in temperatures		
○ Factors causing temperature changes present in the vicinity of the temperature sensor	A: Check for tubing conditions. C: Clean/replace the tubing to avoid disturbance.	
There is no actual difference in temperatures		
○ Deviation from the display temperature calibration of the I/O PCB	C: For detail, refer to information in the “E35 T2 TEMPERATURE SENSOR ERROR”	CXTS**-E35
○ Faulty T2 temperature sensor	C: For detail, refer to information in the “E35 T2 TEMPERATURE SENSOR ERROR”	CXTS**-E35

E118 T2 OVERHEATED (for Type 2)

Detection condition	T2 exceeded 100.0°C for a period of consecutive 1 second while the heater was ON.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Hot rinse Hot citric acid	- / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	Heater OFF Only type 2 of fluid circuit.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
There is an actual difference in temperatures		
○ Factors causing temperature changes present in the vicinity of the temperature sensor	A: Check for tubing conditions. C: Clean/replace the tubing to avoid disturbance.	
There is no actual difference in temperatures		
○ Deviation from the display temperature calibration of the I/O PCB	C: For detail, refer to information in the “E35 T2 TEMPERATURE SENSOR ERROR”	CXTS**-E35
○ Faulty T2 temperature sensor	C: For detail, refer to information in the “E35 T2 TEMPERATURE SENSOR ERROR”	CXTS**-E35

E119 T7 TEMP LOW (for Type 1)

Detection condition	T7 is lower than (setting temperature – 4) °C for 5 min during the KEEP CYCLE.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse (Hot Citric/Hot Rinse)	- / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	Only type 1 of fluid circuit.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
There is an actual difference in temperatures		
○ Factors causing temperature changes present in the vicinity of the temperature sensor	A: Check for tubing conditions. C: Clean/replace the tubing to avoid disturbance.	
There is no actual difference in temperatures		
○ Deviation from the display temperature calibration of the I/O PCB	C: For detail, refer to information in the “E112 T7 TEMPERATURE SENSOR ERROR”	CXTS**-E112
○ Faulty T7 temperature sensor	C: For detail, refer to information in the “E112 T7 TEMPERATURE SENSOR ERROR”	CXTS**-E112

E119 TEMPERATURE IS LOW (for Type 2)

Detection condition	T7 is lower than (setting temperature – 4) °C for 5 min during the KEEP CYCLE.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Hot Rinse	- / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.		Remark	Masking when T1 is below 80 °C. Only type 2 of fluid circuit.			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
There is an actual difference in temperatures		
○ Factors causing temperature changes present in the vicinity of the temperature sensor	A: Check for tubing conditions. C: Clean/replace the tubing to avoid disturbance.	
There is no actual difference in temperatures		
○ Deviation from the display temperature calibration of the I/O PCB	C: For detail, refer to information in the “E112 T7 TEMPERATURE SENSOR ERROR”	CXTS**-E112
○ Faulty T7 temperature sensor	C: For detail, refer to information in the “E112 T7 TEMPERATURE SENSOR ERROR”	CXTS**-E112

E120 CIRCURATION FLOW STOP (for Type 1)



Detection condition	“FS1” and “FS2” detected no flow for 5 seconds during the circulation sequence. (Normal spec.) “FS1” detected no flow or FS5 voltage < 1.0V for 5 seconds during the circulation sequence. (Hot rinse spec.)						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
[Normal spec]							
· Disinfection	- / - / -	○	○	-	-	-	-
· Acid Rinse	- / - / -	○	○	-	-	-	-
[Hot rinse spec]							
· Disinfection	- / - / -	○	○	-	-	-	-
· Acid Rinse	- / - / -	○	○	-	-	-	-
· Hot rinse	- / - / - (Heater off)	○	○	-	-	-	-
· Hot citric	- / - / - (Heater off)	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	Only type 1 of fluid circuit.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Dialysate flows		
○ Broken wiring in the wiring to the “FS1” or loose connection of connector part	A: Lightly pull the lead wires of the CN213 on the I/O PCB to check them for disconnection or loose connection. Furthermore, check for the connector on the “FS1” side. C: If any disconnection or loose connection is observed, remedy it.	CXSM**-012
○ Broken wiring in the wiring to the “FS2” or loose connection of connector part	A: Lightly pull the lead wires of the CN213 on the I/O PCB to check them for disconnection or loose connection. Furthermore, check for the connector on the “FS2” side. C: If any disconnection or loose connection is observed, remedy it.	CXSM**-012
○ Broken wiring in the wiring to the “FS5” or loose connection of connector part	A: Lightly pull the lead wires of the CN241 on the I/O PCB to check them for disconnection or loose connection. Furthermore, check for the connector on the “FS5” side. C: If any disconnection or loose connection is observed, remedy it.	
○ Faulty “FS1”	A: Measure a power supply voltage of 24V and output voltage on “FS1”. - Between “+” and “-” of power supply voltage: Approx. 24V (Acceptance range: 24V±10%) - Between “OUT” and “-” of output voltage: Approx. 24V with the float located in “up” position (Acceptance range: 24V±10%)/Approx. 0V with the float located in “down” position (Acceptance range: Not more than 1V) C: Replace the “FS1” or the flow meter assembly.	CXSM**-012

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
○ Faulty "FS2"	A: Measure a power supply voltage of 24V and output voltage on "FS2". - Between "+" and "-" of power supply voltage: Approx. 24V (Acceptance range: $24V \pm 10\%$) - Between "OUT" and "-" of output voltage: Approx. 24V with the float located in "up" position (Acceptance range: $24V \pm 10\%$)/Approx. 0V with the float located in "down" position (Acceptance range: Not more than 1V) C: Replace the "FS2" or the flow meter assembly.	CXSM**-012
○ Faulty "FS5"	C: Replace the FS5.	
○ Faulty I/O PCB "	A: If the "FS2" is normal, check for the operation display state and the display of the "FS2". C: If the display is abnormal, replace the I/O PCB.	CXSM**-101
Dialysate does not flow		
○ Entry of large amount of air in the blood leak sensor	A: Visually check for the entry of air. C: Repair part(s) in which air enters.	
○ Leaks in the closed circuit	C: Repair leak part(s).	
○ Faulty "P1"	C: Overhaul or replace the P1.	CXSM**-007
○ Faulty P2	C: Overhaul or replace the P2.	CXSM**-007

E120 CIRCURATION FLOW STOP (for Type 2)



Detection condition	FS1 and FS2 detects no flow for 2 seconds continuously.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
· Disinfection	- / - / -	○	○	-	-	-	-
· Acid Rinse	- / - / -	○	○	-	-	-	-
· Hot rinse	- / - / - (Heater off)	○	○	-	-	-	-
· Hot citric	- / - / - (Heater off)	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	Only type 2 of fluid circuit.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Dialysate flows		
○ Broken wiring in the wiring to the "FS1" or loose connection of connector part	A: Lightly pull the lead wires of the CN213 on the I/O PCB to check them for disconnection or loose connection. Furthermore, check for the connector on the "FS1" side. C: If any disconnection or loose connection is observed, remedy it.	CXSM**-012
○ Broken wiring in the wiring to the "FS2" or loose connection of connector part	A: Lightly pull the lead wires of the CN213 on the I/O PCB to check them for disconnection or loose connection. Furthermore, check for the connector on the "FS2" side. C: If any disconnection or loose connection is observed, remedy it.	CXSM**-012
○ Faulty "FS1"	A: Measure a power supply voltage of 24V and output voltage on "FS1". - Between "+" and "-" of power supply voltage: Approx. 24V (Acceptance range: 24V±10%) - Between "OUT" and "-" of output voltage: Approx. 24V with the float located in "up" position (Acceptance range: 24V±10%)/Approx. 0V with the float located in "down" position (Acceptance range: Not more than 1V) C: Replace the "FS1" or the flow meter assembly.	
○ Faulty "FS2"	A: Measure a power supply voltage of 24V and output voltage on "FS2". - Between "+" and "-" of power supply voltage: Approx. 24V (Acceptance range: 24V±10%) - Between "OUT" and "-" of output voltage: Approx. 24V with the float located in "up" position (Acceptance range: 24V±10%)/Approx. 0V with the float located in "down" position (Acceptance range: Not more than 1V) C: Replace the "FS2" or the flow meter assembly.	CXSM**-012
○ Faulty I/O PCB	A: If the "FS2" is normal, check for the operation display state and the display of the "FS2". C: If the display is abnormal, replace the I/O PCB.	CXSM**-012

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Dialysate does not flow		
○ Entry of large amount of air in the blood leak sensor	A: Visually check for the entry of air. C: Repair part(s) in which air enters.	CXSM**-101
○ Leaks in the closed circuit	C: Repair leak part(s).	
○ Faulty "P1"	C: Overhaul or replace the P1.	CXSM**-007
○ Faulty P2	C: Overhaul or replace the P2.	CXSM**-007

E125 DIALYSYS FLOW FAILURE (for Type 1)



Detection condition	Actual "FS5" flow deviated from the value calculated from the closing time by 20% or more.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Preparation	○ / - / -	○	○	-	-	-	-
Dialysis	○ / - / -	○	○	-	-	-	-
Retrans	○ / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.		Remark		· Available only with the Hot Citric Acid option · Calculate when the chamber switches. · Available with "Feed back: ON" in adjust5. · The alarm is not detected in the round of feedback if the dialysate flow setting was changed. Only type 1 of fluid circuit.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Dialysate flows		
○ Broken wiring in the wiring to the "FS2" or loose connection of connector part	A: Lightly pull the lead wires of the CN213 on the I/O PCB to check them for disconnection or loose connection. Furthermore, check for the connector on the "FS2" side. C: If any disconnection or loose connection is observed, remedy it.	CXSM**-012
○ Broken wiring in the wiring to the "FS5" or loose connection of connector part	A: Lightly pull the lead wires of the CN241 on the I/O PCB to check them for disconnection or loose connection. Furthermore, check for the connector on the "FS5" side. C: If any disconnection or loose connection is observed, remedy it.	
○ Faulty "FS2"	A: Measure a power supply voltage of 24V and output voltage on "FS2". - Between "+" and "-" of power supply voltage: Approx. 24V (Acceptance range: 24V±10%) - Between "OUT" and "-" of output voltage: Approx. 24V with the float located in "up" position (Acceptance range: 24V±10%)/Approx. 0V with the float located in "down" position (Acceptance range: Not more than 1V) C: Replace the "FS2" or the flow meter assembly.	CXSM**-012
○ Faulty "FS5"	C: Replace the FS5.	
○ Faulty I/O PCB	A: If the "FS2 and FS5" are normal, check for the operation display state and the display of the "FS2 and FS5". C: If the display is abnormal, replace the I/O PCB.	CXSM**-101
Dialysate does not flow		
○ Entry of large amount of air in the blood leak sensor	A: Visually check for the entry of air. C: Repair part(s) in which air enters.	
○ Leaks in the closed circuit	C: Repair leak part(s).	
○ Faulty "P1"	C: Overhaul or replace the P1.	CXSM**-007

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
○ Faulty P2	C: Overhaul or replace the P2.	CXSM**-007

E125 DIALYSYS FLOW FAILURE (for Type 2)



Detection condition	(1) The average of the past 26 rounds of F_OUT is lowered than the dialysate setting value. (2) The average of the past 26 rounds of F_OUT exceeds the dialysate setting value by more than 10%.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Preparation	○ / - / -	○	○	-	-	-	-
Dialysis	○ / - / -	○	○	-	-	-	-
Retrans	○ / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.		Remark	This error is masked at the following rounds. 1) A round when dialysate flow rate setting is changed. 2) A round when feedback of flow is changed. 3) A round when closed circuit leak check is executed during Dialysis process. 4) A round after SQUF, HF is executed and returned. 5) A round when bypass is done. Only type 2 of fluid circuit.			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Dialysate flows		
○ Broken wiring in the wiring to the "FS2" or loose connection of connector part	A: Lightly pull the lead wires of the CN213 on the I/O PCB to check them for disconnection or loose connection. Furthermore, check for the connector on the "FS2" side. C: If any disconnection or loose connection is observed, remedy it.	CXSM**-012
○ ○ Faulty "FS2"	A: Measure a power supply voltage of 24V and output voltage on "FS2". - Between "+" and "-" of power supply voltage: Approx. 24V (Acceptance range: 24V±10%) - Between "OUT" and "-" of output voltage: Approx. 24V with the float located in "up" position (Acceptance range: 24V±10%)/Approx. 0V with the float located in "down" position (Acceptance range: Not more than 1V) C: Replace the "FS2" or the flow meter assembly.	CXSM**-012
○ ○ Faulty I/O PCB	A: If the "FS2 and FS5" are normal, check for the operation display state and the display of the "FS2 and FS5". C: If the display is abnormal, replace the I/O PCB.	CXSM**-101
Dialysate does not flow		
○ Entry of large amount of air in the blood leak sensor	A: Visually check for the entry of air. C: Repair part(s) in which air enters.	
○ Leaks in the closed circuit	C: Repair leak part(s).	
○ Faulty P2	C: Overhaul or replace the P2.	CXSM**-007
○ Defective adjustment of closed circuit.	C: Adjust the closed circuit.	CXSM**-007

E126 CD2&CD3 DIFFERENCE ALARM (for Type 1)



Detection condition	CD2 - CD3 ≥ 0.2mS/cm, AND 10 seconds elapsed when “FS1” detected flow.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Preparation	○ / - / -	○	○	-	-	-	-
Dialysis	○ / - / -	○	○	-	-	-	-
Retrans	○ / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch..			Remark	Available only with the B-powder option and type 1 of fluid circuit. Not detect while Fluid Stop is in progress. Not detect while SQHD is sin progress.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Water empty in the B-powder cartridge.		
○ Failure of V11, V27b or V27a.	A: Operate the valves manually on the MAINT4 screen and check their working conditions. C: Disassemble and repair the valves, or replace them.	CXSM**-001
Water remaining in the B-powder cartridge.		
○ Failure of V27a or V28b.	A: Operate the valves manually on the MAINT4 screen and check their working conditions. C: Disassemble and repair the valves, or replace them.	CXSM**-001
○ F8 clogged.	A: Check if the F8 is clogged. C: If the F8 is clogged, replace it.	
○ Failure of P4.	A: Operate the P4 manually on the MAINT4 screen and check its working condition. C: If the pump does not work, replace it.	CXSM**-021
○ Contamination of CD3 electrode (ex. by calcium carbonate)	C: Remove the contamination by performing acid rinse operation.	
○ Imperfect contact of the CN238 connector of the conductivity sensor.	A: Disconnect the connector and check by sight the corrosion on the contact faces of the connector terminals. C: If the connector is faulty, replace it.	CXSM**-011
○ Failure of the I/O board.	A: Check if the display value of conductivity or temperature changes by changing the ambient temperature of the I/O board. C: If the I/O board is faulty, replace it.	CXSM**-101

E127 DIALYSIS FLOW CONTROL FAILURE (for Type 1)

Detection condition	Feedback rate of flow rate was $\geq 50\%$.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Preparation	○ / - / -	○	○	-	-	-	-
Dialysis	○ / - / -	○	○	-	-	-	-
Retrans	○ / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.		Remark	Available only with the Hot Citric Acid option Available with "Feed back: SET" in adjust5. Only type 1 of fluid circuit.			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty "FS5"	C: Replace the FS5.	

E127 DIALYSIS FLOW CONTROL ERROR (for Type 2)

Detection condition	1) F_OUT does not reach the setting flow even at P2 control voltage of 5V 2) F_OUT exceeds the setting flow by more than +10% even at P2 control voltage of 1V.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Preparation	○ / - / -	○	○	-	-	-	-
Dialysis	○ / - / -	○	○	-	-	-	-
Retrans	○ / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	Masking in the following cases 1) Before connection of dialyzer 2) Gas purged Only type 2 of fluid circuit.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
○ Broken wire in the FS2 or loose harness of connector.	A: Lightly pull the lead wires of the CN213 on the I/O PCB to check them for disconnection or loose connection. Furthermore, check for the connector on the "FS2" side. C: If any disconnection or loose connection is observed, remedy it.	
○ Faulty FS2	C: Overhaul or repair the FS2.	
○ The closed circuit has leaks.	For detail, refer to information in "E60 closed circuit leak error" section.	
○ Faulty P2	C: Overhaul or repair the P2.	
○ Defective adjustment of closed circuit.	C: Adjust the closed circuit.	

E128 T2&T5 DIFFERENCE ALARM (for Type 1)

Detection condition	T5 exceeded T2 by 2°C or more for 5 sec after T7 had risen to setting temperature or more and T2 or T5 is higher than 70°C						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Hot Citric Hot Rinse	- / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.		Remark	The system shifts to Cool-Down Turn the Heater OFF Only type 1 of fluid circuit.			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
There is an actual difference in temperatures		
○ Factors causing temperature changes present in the vicinity of the temperature sensor	A: Check for tubing conditions. C: Remedy tubing so as to be immune to disturbance.	
There is no actual difference in temperatures		
○ Deviation from the display temperature calibration of the I/O PCB	C: For detail, refer to information in the "E35 T2 TEMPERATURE SENSOR ERROR" C: For detail, refer to information in the "E102 T5 TEMPERATURE SENSOR ERROR"	CXTS**-E35 CXTS**-E102
○ Faulty T2 or T5 temperature sensor	C: For detail, refer to information in the "E35 T2 TEMPERATURE SENSOR ERROR" C: For detail, refer to information in the "E102 T5 TEMPERATURE SENSOR ERROR"	CXTS**-E35 CXTS**-E102

E128 T2&T5 DIFFERENCE ALARM (for Type 2)

Detection condition	T5 exceeded T2 by 2°C or more for 5 sec after T7 had risen to setting temperature or more and T2 and T5 is higher than 50°C						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Hot Citric Hot Rinse	- / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.		Remark	The system shifts to Cool-Down Turn the Heater OFF Only type 2 of fluid circuit.			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
There is an actual difference in temperatures		
○ Factors causing temperature changes present in the vicinity of the temperature sensor	A: Check for tubing conditions. C: Remedy tubing so as to be immune to disturbance.	
There is no actual difference in temperatures		
○ Deviation from the display temperature calibration of the I/O PCB	C: For detail, refer to information in the “E35 T2 TEMPERATURE SENSOR ERROR” C: For detail, refer to information in the “E102 T5 TEMPERATURE SENSOR ERROR”	CXTS**-E35 CXTS**-E102
○ Faulty T2 or T5 temperature sensor	C: For detail, refer to information in the “E35 T2 TEMPERATURE SENSOR ERROR” C: For detail, refer to information in the “E102 T5 TEMPERATURE SENSOR ERROR”	CXTS**-E35 CXTS**-E102

E129 T4&T7 DIFFERENCE ALARM

Detection condition	The system detected "T7-T4 $\geq$ 2 (°C)" for 5 seconds continuously after 5 minutes from the start of circulation sequence when the Heater is operated.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
· Rinse (Hot Citric/Hot Rinse)	- / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	The system shifts to Cool-Down Turn the Heater OFF		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
There is an actual difference in temperatures		
○ Factors causing temperature changes present in the vicinity of the temperature sensor	A: Check for tubing conditions. C: Remedy tubing so as to be immune to disturbance.	
There is no actual difference in temperatures		
○ Deviation from the display temperature calibration of the I/O PCB	C: For detail, refer to information in the "E37 T4 TEMPERATURE SENSOR ERROR" C: For detail, refer to information in the "E112 T7 TEMPERATURE SENSOR ERROR"	CXTS**-E37 CXTS**-E112
○ Faulty T4 or T7 temperature sensor	C: For detail, refer to information in the "E37 T4 TEMPERATURE SENSOR ERROR" C: For detail, refer to information in the "E112 T7 TEMPERATURE SENSOR ERROR"	CXTS**-E37 CXTS**-E112

E137 FAN3 STOP ALARM

Detection condition	Error stop signal was sent by the fan3a. 1 The FAN3 stop was detected within one second since the Hot Rinse or Hot Citric process started. 2 The FAN3 stop was detected while the FAN3 was operating during the Hot Rinse or Hot Citric process.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
· Rinse (Hot Citric/Hot Rinse)	○ / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Clogged fan3a	A: Check to be sure that the fan3a stops operation due to foreign matters caught in the clamp. C: Remove the foreign matters.	
Faulty fan3a	A: Disconnect the CN248 from the I/O PCB, and then make measurement of a voltage between WJ1 and Pin 1 of the CN248. C: If a 24V voltage is output, replace the fan3a. (Acceptance range: 24V±10%)	

E138 DIALYSYS FLOW SENSOR ALARM (for Type 1)

Detection condition	FS5 output was ≥ 680 mL/min for 5 consecutive sec during JB step after J2 step or C3 step (Water rinse).						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
C3 step (Water Rinse)	- / - / -	○	○	-	-	-	-
JB step (Standby, Preparation, Retrans)	○ / - / -	○	○	-	-	-	-
JB step (Dialysis)	○ / ○ / -	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	Available only with the Hot Citric Acid option and type 1 of fluid circuit.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Faulty "FS5"	C: Replace the FS5.	

E139 VALVE CHECK ERROR (for Type 1)

Detection condition	Fluid pressure does not reach 250mmHg when the pressure rises during valve check. Fluid pressure changes by 15mmHg or more within 30 seconds during valve check.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Waiting	- / - / -	○	○	-	-	-	-
Standby	- / - / -	○	○	-	-	-	-
Preparation	- / - / -	○	○	-	-	-	-
Dialysis	- / - / -	○	○	-	-	-	-
Retrans	- / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	The valve check is performed right before the drain operation of B-powder cartridge. Only type 1 of fluid circuit		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Fluid pressure does not reach 250mmHg when the pressure rises during valve check.		
○ Leaks in the filling circuit	A: Check visual or with the fingertips for fluid leaks at the filling circuit. A: Replace the part(s) from which the fluid leaks.	
○ Entry of large amount of air in the circuit	A: Check visual for the entry of air. C: Repair part(s) in which air enters.	
○ Deviation from the adjustment of or failure in "PR"	A: Check whether the fluid pressure value reaches not less than 250 mmHg when the "V1" and "V10" turn ON while in the standby-to-dialysis process. C: Make adjustment of the "PR". If this adjustment cannot be made, replace the "PR".	
○ Faulty fluid pressure sensor assembly	A: Disconnect the tube from the fluid pressure sensor assembly .Open it to atmosphere and check if the fluid pressure of 0 mmHg is displayed. Then, use a syringe to pressurize it until the reference pressure gauge reads 0 mmHg and/or 300 mmHg. C: If the fluid pressure of 300 mmHg is not displayed, perform the calibration of the fluid pressure sensor PCB. If the calibration cannot be performed, replace the fluid pressure sensor assembly.	
Fluid pressure changes by 15mmHg or more within 30 seconds during valve check.		
○ Leaks from the valve	C: Repair or replace the valves (V12, V17, V21, V27a, V27b, V29, V31, V33, V34 or V35) from which the fluid leaks.	
○ Leaks in the filling circuit	A: Check visual or by fingertips for fluid leaks at the filling circuit. C: Replace the part(s) from which the fluid leaks.	

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
○ Entry of large amount of air in the circuit	A: Visually check for the entry of air. C: Repair part(s) in which air enters.	

E140 AVE. STROKE VOLUME MIN ALARM

Detection condition	During S/N operation, AVERAGE STROKE VOLUME is equal to or lower than S/N AVERAGE STROKE VOLUME MIN in "SINGLE/N" window.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dialysis (SN)	- / ○ / -	○	○	○	○	-	-
Reset method	Press the RESET switch.		Remarks	The V4 is turned ON for two seconds if the fluid pressure reaches 300mmHg or greater in five seconds after the switch-on of the V19 at the event of the alarm.			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
SN AVG STROKE VOLUME MIN value is not appropriate.	C: Change SN AVG STROKE VOLUME MIN value to appropriate value.	
Used blood line is not complied with this machine.	C: Change to appropriate blood line for this machine.	
Faulty power supply PCB	C: Replace the power supply PCB.	
Faulty I/O PCB	C: Replace the I/O PCB.	
Faulty main CPU PCB	C: Replace the main CPU PCB.	

E141 CLOSED CIRCUIT LEAK ERROR (for Type 1)

Detection condition	When either of low or fluctuating positive pressure of the chamber A or low or fluctuating negative pressure of the chamber B is detected * Refer to information in the “Detection condition for leaks from closed circuit using the Maintenance Menu 7 screen” section.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dialysis	○ / - / -	○	○	-	-	-	-
Reset method	The check is resumed by the Reset SW.			Remark	If "E141" is detected twice continuously, treatment cannot be continued. Only type 1 of fluid circuit		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The pressure rising check results in NG		
○ Leaks in the closed circuit	A: Check visual or with the fingertips for fluid leaks out of the closed circuit. C: Replace the part(s) from which the fluid leaks.	
○ Faulty fluid pressure sensor assembly	A: Disconnect the hose from the fluid pressure sensor assembly to open it to the atmosphere, and then check whether the fluid pressure of 0 mmHg is displayed. Then, use a syringe to pressurize it until the reference pressure gauge reads 300 mmHg. C: If the fluid pressure of 300 mmHg is not displayed, perform the calibration of the fluid pressure sensor PCB. If the calibration cannot be performed, replace the fluid pressure sensor assembly.	
○ Faulty “UFP”	C: Refer to information in the “E49 UFP ERROR” section.	
○ Malfunction of valve	A: Check each valve for operation on the Maintenance 4 screen. C: If any valve does not operate, repair or replace the valve concerned.	
The leak check results in NG		
○ Leaks from the coupler	C: Replace the O-ring of the coupler or the coupler by the set.	
○ Leaks from the valve	C: Repair or replace the valve(s) from which the fluid leaks.	

E143 BLOOD DETECTED



Detection condition	[Preparation/Priming] Blood is detected while blood is absent. BS Input Voltage<Detection Voltage and Detection Time Continuity [Preparation comp.] Blood is detected while blood is absent. BS Input Voltage<Detection Voltage and Detection Time Continuity and during BP Stop and Before M24 Detection						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Annunciator light (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm reset
Preparation/Priming	- / - / -	○	○	○	○	-	○
Preparation comp.	- / - / -	○	○	○	○	-	○
Reset method	Press the RESET switch.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Blood is detected while it is absent.		
○ The detected voltage is not appropriate.	C: Input the detected voltage appropriately.	
○ Calibration of the blood sensor is deviated.	C: Calibrate the blood sensor.	CXAJ**-023
○ Malfunction is caused by ambient light.	C: Check that the cover of the blood sensor housing is fixed securely.	
○ Wiring to the sensor is disconnected and the connector is not connected properly.	A: Remove the sensor harness from the I/O board connector and measure the resistance to check that there is no disconnection or fluctuation. C: Replace the blood sensor in case of disconnection or fluctuation.	CXSM**-010
○ The blood sensor is faulty.	C: Replace the blood sensor.	CXSM**-010
○ The I/O board is faulty.	C: Replace the I/O board if adjustment is impossible.	CXSM**-101



E144 BLOOD NOT DETECTED

Detection condition	[Dialysis/Retrans] Blood sensor detected that there is no blood. BS Input Voltage $\geq$ Detection Voltage (DIALYSIS VOLTAGE) and Detection Time (DIALYSIS TIME) Continuity *1 [Retrans Screen] Blood sensor detected that there is no blood. BS Input Voltage $\geq$ Detection Voltage (RETRANS VOLTAGE) and Detection Time (RETRANS TIME) Continuity						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Annunciator light (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm reset
Dialysis/Retrans	- / - / -	○	○	-	-	-	○
Retrans Screen	- / - / -	○	○	○	○	-	○
Reset method	Press the RESET switch.			備考	*1 Alarm is masked until the settled time "NO BLOOD MASK TIME (DIALYSIS)" is passed after RESET Switch is pressed.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
There is no blood.		
○ Arterial puncture is not correctly done.	C: Conduct arterial puncture correctly.	
There is blood.		
○ The detected voltage is not appropriate	C: Input the detected voltage appropriately.	
○ Calibration of the blood sensor is deviated.	C: Calibrate the blood sensor.	CXAJ**-023
○ The blood sensor is faulty.	C: Replace the blood sensor.	CXSM**-010
○ The I/O board is faulty.	C: Replace the I/O board if adjustment is impossible.	CXSM**-101

E145 CENTRAL SYSTEM VALVE LEAK ERROR (for Type 2)

Detection condition	PD value is not under -250mmHg in 10 seconds after starting C4-1 or C4-3.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Annunciator light (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm reset
Rinse	○ / - / -	○	○	-	-	-	○
Reset method	Press the RESET switch.			Remark	Only type 2 of fluid circuit		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Leakage from V45	A: Check that the trouble is improved by closing the V45 line with pean. C: Disassemble/clean V45 or replace it.	CXSM**-002
Leakage from V46	A: Check that the trouble is improved by closing the V46 line with pean. C: Disassemble/clean V46 or replace it	CXSM**-001
P2 is faulty (does not function).	C: Disassemble/repair P2 or replace it.	CXSM**-007
The I/O board is faulty.	C: Replace the I/O board if the output to P2 is faulty.	CXSM**-101

E147 HP BOLUS OPERATION ERROR

Detection condition	HP bolus operation is continuously done even after the target value was reached.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Annunciator light (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm reset
Dialysis, Preparation, Preparation comp, Retrans	- / - / -	○	○	HP stop	-	-	-
Reset method	Press the RESET switch.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The I/O board is faulty.	C: Replace the I/O board if the output to HP is faulty.	CXSM**-101

E148 VALVE ERROR (DISINFECTION VALVE) (for Type 2)

Detection condition	(1) V33 operation check (when B powder option is ON) FS1 does not detect 1 consecutive sec ON for 120sec in R3-2. (2) V33 operation check (when B powder option is OFF) FS-1 does not detect 1 consecutive sec ON for 120 sec in R3-1. (3) V31 operation check (only when B powder option is ON) PD1 variable value does not exceed +20mmHg or more within 10 sec after starting R3-5. (4) V21 operation check PD1 variable value does not exceed +20mmHg or more within 10 sec after starting R4-3.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Annunciator light (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm reset
Disinfection, Acid Rinse, Hot Rinse, Hot Citric Rinse	○ / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	Only type 2 of fluid circuit		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
A valve to operate does not start normally in each disinfection process.	A: Operate each process independently on the manual operation screen and locate a valve that opens/closes abnormally. C: Replace the faulty valve.	CXSM**-002
The I/O board is faulty.	C: Replace the I/O board if the output to the valve is faulty.	CXSM**-101



E149 FS1 SENSOR ALARM (for Type 2)

Detection condition	1) C1 step 1 does not turn OFF even after 20 sec. (water rinse) 2) C" step 3 does not turn OFF even after 120 sec. 3) CA/CB step 7 does not turn OFF even after 120 sec. 4) CS step 3 does not turn OFF even after 120 sec. 5) J" does not turn OFF even after 120 sec. 6) C1 step 2-1 does not turn ON even after 2 sec. 7) C5 step 1 does not detect turn ON 1 sec continuity after 2 sec. 8) CS step 5 does not turn OFF even after 120sec.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Annunciator light (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm reset
Rinse, Preparation	○(heater off) / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	Only type 2 of fluid circuit		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
FS1 is faulty.	A: Measure the power voltage 24V and output voltage of FS1. Measure the terminal voltage of the flow sensor. * Between + and - power voltage sides: About 24V (OK range: 24V±10%) * Between OUT and - output voltage sides: About 24V with the float up (OK range: 24V±10%) About 0V with the float down (OK range: Below 1V) C: Replace F1.	CXSM**-012
The valve does not operate normally.	C: Check the related valve operation and replace the faulty valve.	CXSM**-001
The I/O board is faulty.	A: When FS1 is normal, check the operation display status and check the display of FS1. C: Replace the I/O board if the display is abnormal.	CXSM**-101



E150 FS2 SENSOR ALARM (for Type 2)

Detection condition	1) C1 step 1 does not turn OFF even after 20 sec. (water rinse) 2) C6 step 3 does not turn OFF even after 120 sec. 3) C' step 3 does not turn OFF even after 120 sec. 4) CA/CB step 7 does not turn OFF even after 120 sec. 5) R3 step 4 does not turn OFF even after 120 sec. 6) CS step 1 does not turn OFF even after 120 sec. 7) J" does not turn OFF even after 120 sec. 8) J'" does not turn OFF even after 120 sec. 9) C1 step 3 does not turn ON even after 2 sec.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Annunciator light (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm reset
Rinse, Preparation	○(heater off) / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	Only type 2 of fluid circuit		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
FS2 is faulty.	A: Measure the voltage 24V and output voltage of FS2. Measure the terminal voltage of the flow sensor. * Between + and - power voltage sides: About 24V (OK range: 24V±10%) * Between OUT and - output voltage sides: About 24V with the float up (OK range: 24V±10%) About 0V with the float down (OK range: Below 1V) C: Replace FS2.	CXSM**-012
The valve does not operate normally.	C: Check the related valve operation and replace the faulty valve.	CXSM**-001
The I/O board is faulty.	A: When FS2 is normal, check the operation display status and check the display of FS2. C: Replace the I/O board if the display is abnormal.	CXSM**-101

E151 UFP SPEED ERROR (1 CYCLE) (for Type 2)



Detection condition	A deviation of one rotational period is monitored compared to the setting UF speed (including UF compensation). (Time required for one rotational period exceeding by ± 0.1 L/H is monitored compared to each UF speed.)						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Annunciator light (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm reset
Dialysis	○ / ○ / ○	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	Machine detects with ± 0.08 L/H considering 20% of margin Only type 2 of fluid circuit		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
UFP is faulty.	A: Hold the spindle lightly with a hand, while the motor is rotating or tries to rotate, and check if it stops. C: Tighten the headless set bolt for fixing securely or replace the UF pump.	CXSM**-021
PH5 or PH6 are faulty.	A: Check that PH5 and PH6 are reactive on the maintenance 4 screen while the sensor slit is shielded by dog plate. C: Replace PH5 or PH6 if there is no response.	BHSM**-109
The I/O board is faulty.	C: Replace the I/O board.	CXSM**-101
The drive board is faulty	C: Replace the drive board.	BHSM**-103

E152 DISINFECTANT TEMPERATURE ALARM (for Type 2)



Detection condition	T4 value is out of preset temperature value $\pm 2^{\circ}\text{C}$.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Annunciator light (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm reset
Hot Disinfection, hot acid	-(heater off) / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	1) Mask (hide) for 8 min after circulation starts. 2) Mask (hide) when temp setting is OFF. 3) Mask (hide) for 8 min after resetting alarm. Only type 2 of fluid circuit		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The heater is faulty.	A: Turn OFF the main power and measure the resistance on both ends of the heater. C: Replace the heater if the resistance becomes out of the following range. <Standard type> 230V spec. : 92.0 to 105.8 Ω (500W $\times$ 2) 110V spec. : 10.5 to 12.1 Ω (1000W) <Optional hot rinse type> 230V spec. : 92.0 to 105.8 Ω and 46.0 to 52.9 Ω (500W + 1000W) 110V spec. : 7.0 to 8.1 Ω (1500W)	BHSM**-004
T1 is faulty.	A: Shake the harness of the T1 temperature sensor to check if the T1 temperature display fluctuates. Check if the T1 temperature display is deviated when the temperature calibrator is connected. C: Replace the T1 temperature sensor if the analysis result is faulty.	BHSM**-008
T4 is faulty.	A: Shake the harness of the T4 temperature sensor to check if the T4 temperature display fluctuates. Check if the T4 temperature display is deviated when the temperature calibrator is connected. C: Replace the T4 temperature sensor if the analysis result is faulty.	BHSM**-008
Flow of the dialysate circuit is unstable.	C: Check if any valve or P2 pump is faulty. If any, repair or replace it.	
The I/O board is faulty.	A: Check if the T4 temperature display is deviated when the temperature calibrator is connected. C: Calibrate the T4 circuit of the I/O board, using a temperature calibrator. Replace the I/O board if calibration is impossible.	CXSM**-101

E153 VALVE ERROR (CHAMBER PERIPHERY VALVE) (for Type 2)



Detection condition	A faulty valve status (ON-OFF timing) is detected continuously for 500msec. by CPU-S. Monitoring Valves are V5a, V5b, V6a, V6b, V7a, V7b, V8a, V8b.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Annunciator light (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm reset
Preparation comp, Dialysis, Retrans	○ / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch.			Remark	Only type 2 of fluid circuit		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The I/O board is faulty.	C: Replace the I/O board.	CXSM**-101

E154 DIALYSATE PRESSURE GAUGE CHECK ERROR (for Type 2)



Detection condition	The PD value does not increase by 400mmHg from the value at the time of changing to C2-4 before eight rotations of the UFP in C2-4. Or the PD value does not reach 500mmHg within 10 seconds after the start of C2-5.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Annunciator light (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm reset
Rinse	○ / - / -	○	○	-	-	-	○
Reset method	Press the RESET switch.			Remark	Only type 2 of fluid circuit		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Adjustment of dialysate pressure sensor gain is deviated.	A: Check that the dialysate pressure display is 0 ± 2mmHg when the dialysate pressure display offset on the maintenance 6 screen to "0" and the atmosphere is released by removing the tube from the dialysate pressure sensor discharge side. Check also that the dialysate pressure display is $+300 \pm 5$mmHg when $+300$mmHg is externally added. C: Calibrate the dialysate pressure sensor when the adjustment value is deviated.	BHAJ**-002
The UFP is faulty.	A: Hold the spindle lightly with a hand, while the motor is rotating or tries to rotate, and check if it stops. C: Tighten the headless set bolt for fixing securely or replace the UF pump.	BHSM**-021
Leakage from the coupler port	C: Replace the coupler O ring or the entire coupler.	
Leakage from the CF coupler port	C: Replace the coupler O ring or the entire coupler.	
The I/O board is faulty.	C: Replace the I/O board.	CXSM**-101

E155 CLEANING VALVE ERROR (for Type 2)



Detection condition	The PD value decreases by 12mmHg or more from the value at the time of changing to C2-7 within 10 seconds after C2-7 is started.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Annunciator light (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm reset
Preparation comp, Dialysis, Retrans	○ / - / -	○	○	-	-	-	○
Reset method	Press the RESET switch.			Remark	Only type 2 of fluid circuit		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Leakage from V12	A: Check that the trouble is improved by closing the V12 line with pean. C: Disassemble/clean V12 or replace it.	CXSM**-002
Leakage from V56	A: Check that the trouble is improved by closing the V56 line with pean. C: Disassemble/clean V56 or replace it.	CXSM**-001
Leakage from V17	A: Check that the trouble is improved by closing the V17 line with pean. C: Disassemble/clean V17 or replace it.	CXSM**-002
Leakage from V57	A: Check that the trouble is improved by closing the V57 line with pean. C: Disassemble/clean V57 or replace it.	CXSM**-001
The I/O board is faulty.	C: Replace the I/O board.	CXSM**-101

M01 COUPLER SW IS ON

Detection condition	· The Dialyse switch was pressed while the coupler was set (LS2 is ON) in the Preparation or Retrans process. · The coupler was set (LS2 is ON) in the Dialysis process.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Preparation	- / - / -	○	○	-	-	-	-
Dialysis	- / ○ / -	○	○	-	-	-	-
Retrans	- / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch. Auto reset after release from alarm condition. (Preparation ,Retrans)			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The “LS2” coupler switch is activated		
○ DIALYSIS switch pressed with the coupler mounted to the coupler holder	A: Failure to operate the switch. C: Mount the coupler to the dialyser	
○ “LS2” coupler switch set to ON while in the dialysis process	A: Failure to operate the switch C: Mount the coupler to the dialyser	
The “LS2” coupler switch is not activated	A: Monitoring the Operation Display screen, manually set the “LS2” to ON or OFF to check it for the operating state.	
○ Faulty “LS2”	A: Disconnect the connector from the “LS2”, and then check for continuity between the Pins 1 and 2. • Short-circuited when the “LS2” turns ON. • Open when the “LS2” turns OFF. C: If the “LS2” malfunctions, replace it	
○ Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101

M02 COUPLER SW IS OFF

Detection condition	The LS2 was turned OFF in each the operation.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	-	-	-
Rinse	○ / - / -	○	○	-	-	-	-
Preparation (Start up test)	○ / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch Auto reset after release from alarm condition.		Remark	All relevant operations are the followings: · Rinse Waiting process · Rinse in progress · Closed leak check in the startup test			

Symptom/Cause	A:Analysis/C:Countermeasure	Work No.
No coupler is mounted to the coupler holder		
○ The Device is put in the standby-to-rinse, rinse, or preparation process without mounting the coupler to the coupler holder	A: Failure to operate the switch. C: Mount the coupler to the dialyser.	
○ The coupler is disconnected while the standby-to-rinse, rinse, or preparation process is in progress	C: Properly mount the coupler to the coupler holder.	
The coupler is mounted to the coupler holder	A: Monitoring the Operation Display screen, manually set the "LS2" to ON or OFF to check it for the operating state.	
○ Damage to the button of the "LS2"	C: Replace the "LS2".	
○ Unfastening or displacement of the screw that fixed the "LS2"	C: Remedy.	
○ Faulty "LS2"	A: Disconnect the connector from the "LS2", and then check for continuity between the Pins 1 and 2. • Short-circuited when the "LS2" turns ON. • Open when the "LS2" turns OFF. C: If the "LS2" malfunctions, replace it.	
○ Broken wire in wirings to the "LS2" and loose connection of connector part	A: Lightly pull the lead wires of the connector on the "LS2" side to check them for disconnection or loose connection. Furthermore, check the CN213 on the I/O PCB for the same. C: Remedy or replace the harness.	
○ Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101

M03 UF SPEED SETTING EXCEEDED MAX LIMIT



Detection condition	Calculation result of the BP flow rate and the UF max limit alarm setting was less than the UF rate. $\frac{\text{BP Flow Rate(mL/min)}}{1000} \times 60 \times \frac{\text{The maximum UF setting alarm value (\%)}}{100} < \text{UF RATE (L/h)}$						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dialysis	- / ○ / -	○	○ (Green)	-	-	-	-
Reset method	Press the RESET switch Auto reset after release from alarm condition.			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The detection condition is satisfied		
○ Blood pump flow rate reduced	C: Make setting according to the patient's conditions. (Reduce the UF rate or increase the blood pump flow rate.)	
○ UF rate increased	C: Make setting according to the patient's conditions. (Increase the blood pump flow rate or reduce the UF rate.)	
○ Improper setting of "UF RATE MAX ALARM SET V."	C: Enter a proper value to "UF RATE MAX ALARM SET V."	

Detection condition	The following condition became right while heparin was being infused. Remaining dialysis time <=Heparin Stop Time						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dialysis	- / - / -	○ (3 times)	○ (YELLOW)	HP Stop (SW blink)	-	-	○ (SETTING1)
Reset method	Press the RESET switch Auto reset after release from alarm condition.			Remark	- Turn OFF the function of “Syringe Stop” to disable the alarm. - After the reset, the yellow light of the signal tower turns on until the detection conditions become wrong.		

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
The syringe infusion reaches the programmed period of time. ○ The remaining dialysis time reaches the syringe pre-stop time ○ The syringe pre-stop time setting is changed longer than the remaining dialysis time ○ The remaining UF time and remaining dialysis time settings are changed, and then the remaining dialysis time is set longer than the syringe pre-stop time ○ The UF rate and goal UF volume settings are changed, and then the remaining dialysis time is set longer than the syringe pre-stop time.	A: Normal C: To infuse again, set the Syringe Pump Operation switch to ON.	

M05 PERFORMED TARGET UF

Detection condition	· Current UF volume $\geq$ Target UF volume · Remaining UF time became 0:00						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dialysis	- / ○ / -	○ (Melody)	○ (YELLOW)	-	-	-	-
Reset method	Press the RESET switch Auto reset after release from alarm condition.			Remark	· The alarm is not detected if the target UF volume is 0.00L · After the buzzer stops to sound, the yellow light turns on until the detection conditions become wrong.		

Symptom/Cause	A:Analysis／C:Countermeasure	Work No.
The UF removed volume reaches the goal UF volume or the goal UF volume is reduced to the value of the UF removed volume	A: Normal A: To perform the UF process again, increase the goal UF volume setting	

M06 PRIMING IS COMPLETE

Detection condition	Target volume of priming was met.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Priming	- / - / -	○ (3 times)	○ (BLUE)	○ (SW OFF)	○	-	-
Reset method	Press the RESET switch Auto reset after release from alarm condition.			Remark			

Symptom/Cause	A:Analysis／C:Countermeasure	Work No.
The current priming volume reaches the goal priming volume or the goal priming volume is reduced to the value of the current priming volume.	A: Normal C: To perform the priming operation again, increase the goal priming volume setting.	

M07 UFR IS SET TO ZERO

Detection condition	The process transferred to the Dialysis while the UF speed was set to 0.00L/h. X minutes elapsed while the UF speed was set to 0.00L/h. (X is a setting time of the UF Speed Zero Setup Alarm)						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dialysis	- / - / -	○ (3 times)	○ (GREEN)	-	-	-	○ (MAINT2)
Reset method	Press the RESET switch Auto reset after release from alarm condition.			Remark	· Turn OFF the setting time of the “UF Speed Setup Alarm” to disable the alarm. · The alarm is activated only if Target UF volume > Current UF volume.		

Symptom/Cause	A:Analysis/C:Countermeasure	Work No.
The Device shifts to the dialysis process with the UF rate set to 0.00 L/h. (Failure to make UF rate setting)	C: Make the UF rate setting.	
While the dialysis process is in progress, the UF rate is set to 0.00 L/h	C: If an erroneous UF rate setting is made, make resetting of the UF rate. If the UF operation is intentionally stopped, provide a follow-up to that.	

M08 HP SPEED IS SET TO ZERO

Detection condition	The heparin flow setting was set to 0.0mL/h during the dialysis process.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dialysis	- / - / -	○ (3 times)	○ (GREEN)	-	-	-	○ (MAINT2)
Reset method	Press the RESET switch Auto reset after release from alarm condition.			Remark	· Turn OFF the “HP Speed Zero Setup Alarm” to disable the alarm. · The alarm is not detected if the heparin pump switch is turned OFF.		

Symptom/Cause	A:Analysis／C:Countermeasure	Work No.
With the Heparin Pump Operation switch set to ON and the heparin pump flow rate set to 0.0 mL/h, the Device shifts to the dialysis process. (Failure to make setting of heparin pump flow rate)	C: If the heparin pump is not used, set the Heparin Pump Operation switch to OFF. In any case other than that, make setting of heparin pump flow rate.	

M09 UF GOAL IS SET TO ZERO

Detection condition	The process transferred to the Dialysis while the target UF volume was set to 0.00L.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dialysis	- / - / -	○ (3 times)	○ (GREEN)	-	-	-	-
Reset method	Press the RESET switch Auto reset after release from alarm condition.			Remark	· The alarm is not detected if the UF speed is set to 0.00L/h.		

Symptom/Cause	A:Analysis／C:Countermeasure	Work No.
While in the dialysis process, the goal UF volume is set to 0.00 L with the UF rate set to not less than 0.10 L/h	C: To disable the UF operation, set the UF rate to 0.00 L/h..	
While in the dialysis process, the UF rate is set to not less than 0.10 L with the goal UF volume set to 0.00 L/h	C: To perform the UF operation, make setting of goal UF volume.	

M10 TIMER COMPLETION

Detection condition	Timer expired.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○ (Melody)	○ (BLUE)	-	-	-	-
Rinse	- / - / -	○ (Melody)	○ (BLUE)	-	-	-	-
Dia. Wait	- / - / -	○ (Melody)	○ (BLUE)	-	-	-	-
Standby	- / - / -	○ (Melody)	○ (BLUE)	-	-	-	-
Preparation	- / - / -	○ (Melody)	○ (BLUE)	-	-	-	-
Dialysis	- / - / -	○ (Melody)	○ (BLUE)	-	-	-	-
Retrans	- / - / -	○ (Melody)	○ (BLUE)	-	-	-	-
Reset method	Press the RESET switch			Remark			

Symptom/Cause	A:Analysis/C:Countermeasure	Work No.
TIMER COMPLETION	C: Press [I/O] SW.	

M11 CONSUMPTION EXCHANGE TIME. **REFER TO MAINTENANCE 8**



Detection condition	Used hours of the consumable parts reached recommended replacement hours. (On the usage indicator, the following condition became right. "Used hours" >= "Recommended replacement hours")						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
· Rinse Wait	- / - / -	○	○	-	-	-	○ (MAINTENANCE 8)
· Rinse	- / - / -	○	○	-	-	-	○ (MAINTENANCE 8)
Reset method	Press the RESET switch			Remark	· Set the "recommended replacement hours" to zero to disable the alarm. · Display color of the "Used hours" on the MAINTENANCE 8 screen was changed.		

Symptom/Cause	A:Analysis/C:Countermeasure	Work No.
The consumable part reaches operating hours in excess of the recommended replacement hours.	A: Check on the "Use Frequency Display" screen for consumable parts that exceeds their recommended replacement hours. C: Replace the consumable part. After that, put the cursor on the consumer part replaced on the "Use Frequency Display" screen, and then press the CLR (Clear) switch. Consequently, the operating hours of the consumable part concerned will be cleared to count the replacement frequency by "1". In order not to replace the consumable part, press the BUZZER STOP / RESET switch to directly use it or make resetting of the recommended replacement hours. Memo ● If any consumable part reaches operating hours in excess of its recommended replacement hours, that does not mean it immediately gets faulty or becomes unusable. In this connection, make judgment with your responsibility on whether or not it should be replaced. However, it is recommended to replace it to maintain the performance of the Device.	
Due to a change in the setting of operating hours or recommended replacement hours, the operating hours exceeds the recommended replacement hours.	C: Make resetting of the operating hours or the recommended replacement hours.	

M12 AFTER BUBBLE REMOVAL. BP IS OPERATED

Detection condition	The BUBBLE DETECTOR switch was turned on after the air bubble alarm went off						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
· After detection of E18	- / ○ / -	○	○	○ (SW OFF)	-	-	-
Reset method	Reset by the BP switch.			Remark	Detected only during SYSTEM1 "AIR BUBBLE ALARM RESET TYPE: SPECIAL".		

Symptom/Cause	A:Analysis／C:Countermeasure	Work No.
After air bubbles are detected to give the alarm, the Air Bubble Sensor Operation switch is pressed to unclamp	C: Be sure to remove air bubbles, and then set the Blood Pump Operation switch to ON.	

M13 AFTER SETTING TUBE. BLOOD PUMP IS OPERATED



Detection condition	The clamp was released by turning OFF the bubble sensor switch after the tube-disconnection alarm was detected.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
· After detection of E20	- / ○ / -	○	○	○ (SW OFF)	-	-	-
Reset method	Reset by the BP switch.		Remark	Detected only during SYSTEM1 "AIR BUBBLE ALARM RESET TYPE: SPECIAL".			

Symptom/Cause	A:Analysis/C:Countermeasure	Work No.
After the "tube not mounted" alarm is detected to give the alarm, the Air Bubble Sensor Operation switch is pressed to unclamp	C: Mount the tube, and then set the Blood Pump Operation switch to ON	

M14 PERFORMED TARGET DIALYSES

Detection condition	Remaining dialysis time (expected dialysis time) became 0:00.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dialysis	- / - / -	○ (Melody)	○ (YELLOW)	-	-	-	-
Reset method	Press the RESET switch Auto reset after release from alarm condition.			Remark			

Symptom/Cause	A:Analysis／C:Countermeasure	Work No.
The remaining dialysis time comes to 0:00	A: Normal. C: To perform dialysis again, make the remaining dialysis time setting longer.	

M15 WATER RINSE NOT SUFFICIENT

Detection condition	(1) After the Disinfection (including Tepid Disinfection) or the Acid Rinse (including Tepid Acid Rinse), the process transferred to the Dialysis Waiting without performing the Water Rinse, Tepid Rinse for 20 minutes.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse Wait	- / - / -	○	○	-	-	-	-
Reset method	Perform the Water Rinse, Tepid Rinse for required minimum period.		Remark	DIA.MODE switch operation is refused For the minimum operation time of water rinsing, refer to the time written in the section 7.5 Check after the rinse operation. Only type 1 of fluid circuit.			

Symptom/Cause	A:Analysis/C:Countermeasure	Work No.
The system entered the Dialysis Waiting mode with completing water rinse insufficiently.	C: Perform Water Rinse, Tepid-Water Rinse or Hot Water Rinse for required minimum time.	

M16 RETRANS COMPLETE

Detection condition	Current Retrans volume reached the target Retrans volume.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Retrans	- / - / -	○ (3 times)	○	○ (SW OFF)	○	-	-
Reset method	Press the RESET switch Auto reset after release from alarm condition.			Remark	Only applicable to the Retrans operation in the Retrans window.		

Symptom/Cause	A:Analysis／C:Countermeasure	Work No.
The current retransfusion volume reaches the goal retransfusion volume or the goal retransfusion volume is reduced to the current retransfusion volume	A: Normal. C: To continue retransfusion, increase the goal retransfusion volume.	

M20 STEP TIME: LESS THAN 5 MINUTES

Detection condition	- The process (Preparation or Retrans) transferred to the Dialysis while each the step time was set to less than five minutes with enabling the UF-PRO function. - Each the step time (except the last step time) became less than five minutes in the Dialysis process while the UF-PRO function was enabled.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dialysis	- / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch Auto reset after release from alarm condition.			Remark	In the Preparation or Retrans process, the start switch will not be accepted with displaying the message "STEP TIME: LESS THAN 5 MINUTES."		

Symptom/Cause	A:Analysis/C:Countermeasure	Work No.
The detection condition is satisfied	C: Make resetting of the UF rate (mean UFR) and the goal UF volume so that each step time will come to not less than five minutes.	

M21 F-RAM DATA WAS OVERWRITTEN

Detection condition	Data was overwritten due to data error in the F-RAM.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Upon startup	- / - / -	○	○	-	-	-	-
Reset method	Press the RESET switch			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
Data transfer to the F-RAM was executed.	A: Normal C: Reset by the Mute/Reset.	
Data transfer to the F-RAM was not executed. ○Data check of the S-RAM and F-RAM during startup failed. ○Failure of the "I/O board."	C: Turn off and on the I/O switch. C: Replace the "I/O board."	CXSM**-101

M23 SOLUTION PREP MODE IS INCORRECT. CHECK THE MAINTENANCE 5 SCREEN



Detection condition	When push the "PREP." SW, Selected dialysate type mode does not match with the dialysate preparation mode.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Dia. Waiting	- / - / -	○ (3 times)	○	-	-	-	-
Reset method	Press the RESET switch			Remark	A standby process is not started.		

Symptom/Cause	A:Analysis/C:Countermeasure	Work No.
Selected dialysate type and dialysate preparation mode do not match	A: CHECK MAINTENANCE 5 SCRIN Unmatched conditions: - While selected dialysate type code is one of those from BC1 to BC4, the dialysate preparation mode is "water" or "water + A." - While selected dialysate type code is AC1 or AC2, the dialysate preparation mode is "water."	

M24 BLOOD DETECTED, please ensure venous line is connected to patient and press the DIALYSE Switch to start treatment



Detection condition	The blood sensor detected blood while the blood pump was operating.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Preparation	- / - / -	○	○	○	○	-	○
Reset method	Press the RESET switch Press the DIALYSE switch			Remark			

Symptom/Cause	A:Analysis/C:Countermeasure	Work No.
Blood is detected while it is absent.		
○ The detected voltage is not appropriate.	C : Input the detected voltage appropriately.	
○ Calibration of the blood sensor is deviated.	C : Calibrate the blood sensor.	CXAJ**-023
○ Malfunction is caused by ambient light.	C : Check that the cover of the blood sensor housing is fixed securely.	
○ Wiring to the sensor is disconnected and the connector is not connected properly.	A : Remove the sensor harness from the I/O board connector and measure the resistance to check that there is no disconnection or fluctuation. C : Replace the blood sensor in case of disconnection or fluctuation.	CXSM**-010
○ The blood sensor is faulty.	C : Replace the blood sensor.	CXSM**-010
○ The I/O board is faulty.	C : Replace the I/O board if adjustment is impossible.	CXSM**-101

M50 CHECK THE CUFF&HOSE

Detection condition	The cuff pressure did not reach a preset pressure level within a preset time from the beginning of pressurization.						
Detection process	measurements	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
BP measurement	The measurement is not possible.	○	○	-	-	-	-
Reset method	Press the BP measurement start switch to restart.		Remark	No retry is performed before alarm detection.			

Symptom/Cause	A:Analysis／C:Countermeasure	Work No.
The cuff pressure does not reach the set value within a period of 30 seconds since pressurization is initiated	A: Check to be sure that the air hose is securely connected to the Device or the cuff. Furthermore, check to be sure that the cuff is tightly wound. C: Remedy loose connection or winding, if any.	

M51 CHECK THE CONNECTION OF CUFF



Detection condition	The cuff pressure decreased to 10 mmHg before the measurement was completed.						
Detection process	measurements	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
BP measurement	The measurement is not possible.	○	○	-	-	-	-
Reset method	Press the BP measurement start switch to restart.		Remark	Two retries are performed before alarm detection.			

Symptom/Cause	A:Analysis／C:Countermeasure	Work No.
The blood pressure measurement is not complete but the cuff pressure falls to 10 mmHg	A: Check for patient's conditions and the cuff mounting state. C: Give a treatment under a physician's direction.	

M52 IT CANNOT MEASURE BECAUSE OF PATIENT MOVEMENT



Detection condition	Deflation was interrupted for more than 15 seconds due to patient's movements.						
Detection process	measurements	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
BP measurement	The measurement is not possible.	○	○	-	-	-	-
Reset method	Press the BP measurement start switch to restart.		Remark	Two retries are performed before alarm detection.			

Symptom/Cause	A:Analysis／C:Countermeasure	Work No.
Air is not exhausted for a period of 15 or more seconds due to patient's body motion	A: Check the patient for any shakes or arrhythmia. Note that shakes or arrhythmia may result from patient's high blood pressure or body motions. C: Give a treatment under a physician's direction.	

M53 CUFF PRESSURE IS INSUFFICIENT, IT CANNOT MEASURE



Detection condition	Pressurization was insufficient.						
Detection process	measurements	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
BP measurement	The measurement is not possible.	○	○	-	-	-	-
Reset method	Press the BP measurement start switch to restart.		Remark	Three retries are performed before alarm detection.			

Symptom/Cause	A:Analysis／C:Countermeasure	Work No.
Insufficient pressurization	A: Check the patient for any shakes or arrhythmia. Note that shakes or arrhythmia may result from patient's high blood pressure or body motions. C: Give a treatment under a physician's direction.	

M54 IT CANNOT MEASURE BECAUSE OF MOVEMENT OR ARRHYTHMIC



Detection condition	Signals could not be collected properly.						
Detection process	measurements	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
BP measurement	The measurement is not possible.	○	○	-	-	-	-
Reset method	Press the BP measurement start switch to restart.		Remark	Two retries are performed before alarm detection.			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
No right signals are taken	A: Check the patient for conditions such as any shakes or arrhythmia. C: Give a treatment under a physician's direction.	

M55 IT CANNOT MEASURE BECAUSE OF MOVEMENT OR ARRHYTHMIC



Detection condition	The measurement was impossible due to arrhythmia or patient's movements.						
Detection process	measurements	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
BP measurement	The measurement is not possible.	○	○	-	-	-	-
Reset method	Press the BP measurement start switch to restart.		Remark	Two retries are performed before alarm detection.			

Symptom/Cause	A:Analysis/C:Countermeasure	Work No.
No blood pressure measurement can be made due to patient's arrhythmia and body motions	A: Check the patient for conditions such as any shakes or arrhythmia. C: Give a treatment under a physician's direction.	

M56 CHECK THE CUFF AND THE BODY MOTION OF PATIENT



Detection condition	The time including the duration of measurement retries exceeded 160 seconds from the beginning of the measurement.						
Detection process	measurements	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
BP measurement	The measurement is not possible.	○	○	-	-	-	-
Reset method	Press the BP measurement start switch to restart.		Remark	No retry is performed before alarm detection.			

Symptom/Cause	A:Analysis/C:Countermeasure	Work No.
A period of 180 or more seconds including that of remeasurement elapses after the blood pressure measurement is initiated	A: Check for any reasons of not exhausting air, such as patient's body motions or bended air hose. C: Give a treatment under a physician's direction. Remedy the bended air hose or else.	

M57 PLEASE CHECK CUFF&MOVEMENT

Detection condition	Pulse count exceeded 160 during the measurement.						
Detection process	measurements	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
BP measurement	The measurement is not possible.	○	○	-	-	-	-
Reset method	Press the BP measurement start switch to restart.		Remark	One retry is performed before alarm detection.			

Symptom/Cause	A:Analysis/C:Countermeasure	Work No.
160 or more pulses are detected while blood pressure measurement is in progress	A: Check for any reasons of not exhausting air, such as patient's body motions or bended air hose. C: Give a treatment under a physician's direction. Remedy the bended air hose or else.	

M58 PLEASE CHECK CUFF&MOVEMENT

Detection condition	The cuff pressure exceeded 300 mmHg.						
Detection process	measurements	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
BP measurement	The measurement is not possible.	○	○	-	-	-	-
Reset method	Press the BP measurement start switch to restart.		Remark	Two retries are performed before alarm detection.			

Symptom/Cause	A:Analysi/C:Countermeasure	Work No.
The cuff pressure exceeds 320 mmHg	A: Check whether a patient bends its arm or any hose is bended. C: If so, remedy it.	

M59 PULSE IS SMALL,IT CANNOT MEASURE

Detection condition	Pulse signals were too small to measure						
Detection process	measurements	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
BP measurement	The measurement is not possible.	○	○	-	-	-	-
Reset method	Press the BP measurement start switch to restart.		Remark	No retry is performed before alarm detection.			

Symptom/Cause	A:Analysis/C:Countermeasure	Work No.
Pulse signals are too weak to make blood pressure measurement	A: Check a patient for conditions and the cuff for the mounting conditions. C: Give a treatment under a physician's direction.	

M60 PLEASE CHECK CUFF&HOSE

Detection condition	An infant cuff was used.						
Detection process	measurements	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
BP measurement	The measurement is not possible.	○	○	-	-	-	-
Reset method	Press the BP measurement start switch to restart.		Remark	Two retries are performed before alarm detection.			

Symptom/Cause	A:Analysi/C:Countermeasure	Work No.
A cuff used for newborns is connected	C: Replace with a cuff used for adults.	

M61 BLOOD PRESSURE IS CHANGED

Detection condition	At the end of the Smart Measurement, the fluctuation of BP was large compared with previous measurement.						
Detection process	measurements	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
BP measurement	The measurement is not possible.	○ (3 time)	○	-	-	-	-
Reset method	Press the RESET switch		Remark		· Turn OFF the “SMART MODE” to disable the alarm. · Turn ON the “SMART MODE” to perform measurement while the fluctuation is large.		

Symptom/Cause	A:Analysis/C:Countermeasure	Work No.
Blood pressure shows no fluctuations. (No abnormal blood pressure is observed.)	A: Check a patient for any body motions, a change of the cuff, remounting of the cuff, or the position. C: Give a treatment under a physician's direction. According to circumstances, make remeasurement of blood pressure.	
Blood pressure actually shows fluctuations	C: Give a treatment under a physician's direction. According to circumstances, make remeasurement of blood pressure.	

M65 PLEASE CONFIRM THE SUPPLY CONDITION OF CONCENTRATION



Detection condition	1) CD3 does not detect 1.0mS/cm or more even when 90 seconds (or 60 seconds in case of optional B powder or central) has elapsed after the execution of J' step3. 2) CD1 does not detect 5.0mS/cm or more even when 90 seconds (or 60 seconds in case of optional B powder or central) has elapsed after the execution of J' step3. 3) CD1 becomes out of the dialysate conductivity alarm setting value continuously in 20 rounds in JA, JB, JKA, JKB, TA, TB, KA and KB.						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Preparation, Preparation comp, Dialysis, Retrans	○ / - / -	○	-	-	-	-	-
Reset method	Press the RESET switch			Remark			

Symptom/Cause	A:Analysis / C:Countermeasure	Work No.
○ The concentrate connector is not disconnected from the rinse port.	C : Disconnect the concentrate connector from the rinse port and connect it to the concentrate tank.	
○ Leakage from the rinse port	A : Visually check that there is no leakage from the rinse port. C : Replace the rinse port.	

LCD screen is not displayed



Detection condition	After power loss, the power returns again while the device is in operation						
Detection process	Fluid stop/UF stop/Bypass	Audible alarm	Signal (Blink)	BP/HP stop	Clamp	V4 & V10: ON (Depressurization)	Alarm Forced off
Rinse	Continues with the process immediately before failure.	○ (3 times)	○	-	-	-	-
Standby		○ (3 times)	○	-	-	-	-
Preparation		○ (3 times)	○	-	-	-	-
Dialysis		○ (3 times)	○	-	-	-	-
Retrans		○ (3 times)	○	-	-	-	-
Reset method	Press the RESET switch.			Remark			

Symptom/Cause	A:Analysis/C:Countermeasure	Work No.
The LCD screen turns OFF while in the standby-to-rinse, rinse, or standby-to-priming process.	A: Normal If the touch panel switch is not operated during a period of time longer than that set with the "BACKLIGHT OFF TIME" function on the System 1 screen, the backlight will turn OFF	
○ The backlight OFF function is activated.	C: Operate the touch panel switch to turn ON the backlight.	
The LCD screen remains OFF.		
○ Replacement of the operation panel assembly	C: Replace the operation panel PCB.	CXSM**-106
○ Broken wire in the LCD harness and loose connection of connector part	A: Lightly pull the harness part of the CN306 connector on the panel CPU board to check the connector for disconnection or loose connection. Furthermore, check whether or not the connector is unlocked. C: Remedy or replace the harness and/or the connector.	CXSM**-106

Touch panel switch cannot operate



Symptom/Cause	A:Analysis／C:Countermeasure	Work No.
The switch is not accepted or the switch pressed is not corresponding to that accepted.		
○ Faulty operation panel assembly	C: Replace the operation panel assembly.	CXSM**-106
○ Broken wire in any cable coming from the touch panel and the loose connection of connector part	A: Lightly pull the cables coming from the touch panel to check it for disconnection or loose connection. Furthermore, check to be sure that the cables are securely put in the CN307, CN308, and CN309 connectors on the panel CPU PCB. C: Remedy the cable and/or the connector.	CXSM**-106

Heparin pump does not operate



Symptom/Cause	A:Analysis／C:Countermeasure	Work No.
The motor does not rotate.		
○ Broken wire in any cable coming from the Heparin pump harness and the loose connection of connector part.	A: Lightly pull the lead wires of the CN1002 connector on the drive PCB to check the connector for disconnection or loose connection. Furthermore, check the connector on the “HP” (high pressure) side for the same. C: Remedy or replace the harness and/or the connector.	
○ Faulty drive PCB	C: Replace the drive PCB.	BHSM**-103
○ Faulty power supply PCB	C: Replace the power supply PCB.	BHSM**-105
○ Faulty CPU PCB	C: Replace the CPU PCB.	BHSM**-100
○ Faulty motor	C: Replace the motor.	BHSM**-003
The motor rotates.		
○ Slip due to worn-out half nut	C: Replace the half nut.	BHSM**-003
○ Loose pulley	A: Check the fixed part for loose connection. C: Remedy the fixed part if it is loosened.	BHSM**-003
○ Broken belt	C: Replace the half nut.	BHSM**-003
○ Slack belt	C: Remedy the belt.	BHSM**-003

Blood pump error



Symptom/Cause	A:Analysis/C:Countermeasure	Work No.
The pump shows uneven rotation		
○ Improper rotor spacing	C: Remedy or replace the rotor assembly.	BHSM**-013
The pump revolutions level off	A: For detail, refer to information in the “E74 BLOOD PUMP ERROR” section.	CXTS**-E74
○ Faulty power supply PCB	C: Replace the power supply PCB.	BHSM**-105
○ Faulty CPU PCB	C: Replace the CPU PCB.	BHSM**-100
○ Faulty motor	C: Replace the blood pump assembly.	BHSM**-013
The pump rotates at abnormally high revolutions	A: For detail, refer to information in the “E74 BLOOD PUMP ERROR” section.	CXTS**-E74
○ Defective adjustment of blood pump flow rate	C: Make adjustment of the blood pump flow rate	CXAJ**-015
○ Faulty power supply PCB	C: Replace the power supply PCB.	BHSM**-105
○ Faulty CPU PCB	C: Replace the CPU PCB.	BHSM**-100
○ Faulty motor	C: Replace the blood pump assembly.	BHSM**-013
Rotating the pump by hand causes heavy feeling in it		
○ Faulty power supply PCB	C: Replace the power supply PCB.	BHSM**-105
○ Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101
○ Faulty motor	C: Replace the blood pump assembly.	BHAJ**-103
The pump makes abnormal noises		
○ Worn-out gear head	C: Replace the blood pump assembly.	BHSM**-013

Fluid still flows out from coupler after bypassed



Symptom/Cause	A:Analysis／C:Countermeasure	Work No.
○Faulty V9b	A: Manually operate the V9b on the Operation Display screen to check to be sure that it is in operation. A: If the V9b is not in operation, disconnect the harness from the V9b, and then connect it to a different valve to check this valve for operation in the same manner as that described above. C: If this different valve operates, replace the V9b. C: If this different valve does not operate, replace the harness connected to the V9b or the I/O PCB.	CXSM**-001 CXSM**-101
○ Faulty V19	A: Check the V19 in the same procedure as that for the V9b.	

Venous chamber shows fluctuations in the level



Symptom/Cause	A:Analysis/C:Countermeasure	Work No.
The chamber level rises		
○ Leaks between the venous chamber and the pressure receiving port (Particularly, leaks from the filter joint part)	A: Clamp the line with forceps to check for any leaks. C: If the line has leaks, remedy the leak point.	
○ Leaks between the pressure receiving port and the pressure sensor	A: Clamp the line with forceps to check for any leaks. C: If the line has leaks, remedy the leak point.	
○ Leaks due to damage to the pressure sensor	C: Replace the PV/PA PCB.	BHSM**-104
○ Bend or clogging in lines provided behind the venous chamber	C: Remedy the circuit.	
The chamber level drops		
○ Mixing of air due to inadequate blood removal	C: Take countermeasures.	
○ Mixing of air from the arterial line	C: Take countermeasures.	
○ Transfer of gas from the dialyser	C: Remove the causes of aeration on the Device side.	

Air gets mixed in dialysate



Symptom/Cause	A:Analysis／C:Countermeasure	Work No.
Air flows to the dialyser		
○ Inadequate deaeration from the water treatment system or the feeder	A: Make measurement of PO ₂ value. C: Remedy.	
○ Air not removed through the “AS1”	C: Overhaul or replace the AS1.	CXSM**-016
○ Air sucked from pipe joint	C: Remedy the pipe joints.	
○ Worn-out O-ring of the coupler	C: Replace the O-ring.	
○ UF of a high rate performed with a dialyser having a low UFRP	C: Make a UF setting change or replace the dialyser with that having a larger UFRP.	
○ Drop of the deaeration pump capacity	A: Make measurement of the deaeration pressure, and then adjust it to any value. C: Overhaul or replace the “P1” deaeration pump.	CXSM**-007

Dialysate flow rate shows instability (for Type 2)



Symptom/Cause	A:Analysis／C:Countermeasure	Work No.
The flow rate shows fluctuations with the shift of chambers		
○ Failures in valves “V5a” to “V8b”	C: Remedy the faulty valves.	CXSM**-001
The flow rate constantly shows delicate fluctuations		
○ Leaks from the “V4” valve	C: Remedy the faulty valves.	CXSM**-001
○ Leaks from the “V10” valve	C: Remedy the faulty valves.	CXSM**-001
○ Degraded performance of the “P2” fluid pressure pump	C: Overhaul or replace the “P2”.	CXSM**-007
The flow rate does not reach 500 mL/min		
○ Broken piping tube	C: Remedy the faulty valves.	
○ Degraded performance of the “P2” fluid pressure pump	C: Overhaul or replace the “P2”.	CXSM**-007
○ Clogging of CF(Cut Filter)	C: Back-rinse or replace the CF.	

Normal fluid drain is not achieved even by pressing the DRAIN switch (for Type 2)



Symptom/Cause	A:Analysis／C:Countermeasure	Work No.
The fluid drain rate is low		
○ Deformed diaphragm of the “V4” or plunger caught in the “V4”	C: Overhaul or replace the “V4”.	CXSM**-001
○ Nearly closed “NV4”	C: Make adjustment of the gas purge pressure.	CXAJ**-020
○ Degraded performance of the “P2”	C: Overhaul or replace the “P2”.	CXSM**-007
○ Broken or bended tubes provided behind the “AS2”	C: Remedy the tubes.	
The fluid is not drained		
○ Faulty “V4”	C: Overhaul or replace the “V4”.	CXSM**-001
○ Broken wire in the valve harness and loose connection of connector part	A: Lightly pull the lead wires of the CN212 connector on the I/O PCB to check the connector for disconnection or loose connection. Furthermore, check the connector on the “V4” side for the same. C: Remedy or replace the harness and/or the connector.	
○ Faulty I/O PCB	A: Energize the “V4” manually on the Operating State screen, and then check to be sure that a 24V voltage is output to the connector of the “V4”. (Acceptance range: 24V±10%) C: If a 24V voltage is not output, replace the I/O PCB.	CXSM**-101

Error in the UF volume exceeds the specified limit



Symptom/Cause	A:Analysis/C:Countermeasure	Work No.
The closed circuit has leaks	A: For detail, refer to information in the “E60 CLOSED CIRCUIT LEAK ERROR” section. C: For detail, refer to information in the “E60 CLOSED CIRCUIT LEAK ERROR” section.	CXTS**-E60 CXTS**-E60 BHTS**-E60 CXTS**-E60
The UF pump causes faulty rotation	A: For detail, refer to information in the “E61 UFP ROTATION ERROR” and “E62 UFP SPEED ERROR” sections. C: For detail, refer to information in the “E61 UFP ROTATION ERROR” and “E62 UFP SPEED ERROR” sections.	CXTS**-E61 BHTS**-E62 CXTS**-E62 CXTS**-E61 BHTS**-E62 CXTS**-E62
○ Faulty motor	C: Replace the UF pump.	CXSM**-021
○ Faulty I/O PCB	C: Replace the I/O PCB.	CXSM**-101
The minimum UF volume to allow adjustment does not fall in the specified range	A: For detail, refer to information in the “Checking for minimum UF volume to allow adjustment section”. C: For detail, refer to information in the “Checking for minimum UF volume to allow adjustment section”.	CXAJ**-016 CXAJ**-016
The UF pump flow rate comes to outside of the set value	A: For detail, refer to information in the “UFP flow rate measurement” section. C: For detail, refer to information in the “UFP flow rate measurement” section.	BHAJ**-017 BHAJ**-017
UF correction coefficient comes to outside of the set value	A: Check the coefficient with the set value label affixed on the front face of the electric component box, and then make a comparison with a value entered to the Adjustment 2 parameter. C: If any coefficient different from that indicated on the set value label is entered, reenter a right coefficient.	
The pressure adjustment of the fluid circuit section has a deviation		
○ Adjustment deviation of “PR”	C: Make adjustment of shutoff pressure.	
○ Adjustment deviation of “RV2”	C: Make adjustment of the closed circuit pressure.	CXAJ**-010

Pump does not operate



Symptom/Cause	A:Analysis／C:Countermeasure	Work No.
The deaeration pump does not operate		
○ Damage to the magnet can due to worn-out bearing	A: Disassemble the pump head. C: Replace necessary parts.	CXSM**-007 CXSM**-007
The fluid pressure pump does not operate		
○ Damage to the magnet can due to worn-out bearing	A: Disassemble the pump head. C: Replace necessary parts.	CXSM**-007 CXSM**-007

2. Service manual (Machine parts)

Service Manual List (Machine Parts)

No.	Identification Number	Document Name	Option
1	CXSM01-001	Two-way Solenoid Valve	
2	CXSM00-002	Two-way Solenoid Valve (Three way)	
3	BHSM01-003	Heparin Pump	
4	BHSM01-004	Heater	
5	CXSM00-005	Relief Valve	
6	CXSM02-006	Chamber	
7	CXSM00-007	Magnet Gear Pump	
8	BHSM00-008	Temperature Sensor	
9	BHSM00-009	Clamp	
10	CXSM00-010	Air Bubble Sensor	
11	CXSM00-011	Conductivity Sensor	
12	CXSM01-012	Flow Sensor	
13	BHSM01-013	Blood Pump	
14	CXSM00-014	Reducing Valve	
15	CXSM00-015	Air Elimination Tank	
16	CXSM00-016	Degassing Tank	
18	BHSM00-018	Blood Leakage Sensor	
19	BHSM00-019	Dialysate Pressure Sensor	
20	CXSM00-020	Needle Valve	
21	CXSM00-021	Piston Pump	
22	BHSM00-022	Relief Solenoid Valve	○
23	CXSM00-023	AS3	○
24	BHSM00-024	B-powder Cartridge Holder	○

⚠ Caution

Each part and part of some parts can be modified for improvement without preceding notification. Therewith, your understanding will be requested in advance if the parts of the product you are using are different from those listed above.

Tightening Torque List

Tightening Torque

- Torque specified in the main body of the service manual is as follows:

ID Number	Usage	Requirement
CXSM**-001	Tighten screws 10and 11.	0. 69N·m(7kgf·cm)
CXSM**-002	Tighten screw 11.	0. 69N·m(7kgf·cm)
CXSM**-006	Tighten screw 11.	5N·m(50kgf·cm)
	Tighten hose opening 3.	0. 8N·m(8kgf·cm)
CXSM**-007	Tighten screw 36.	0. 6N·m(6kgf·cm)
	Tighten screw 32.	0. 8N·m(8kgf·cm)
CXSM**-011	Tighten screws 4 and 10.	0. 39N·m(4kgf·cm)
CXSM**-014	Tighten hose opening 2.	2N·m(20kgf·cm)
CXSM**-015	Tighten nut 11.	After tightening manually, give 1/4 revolution by tool.
CXSM**-020	Attach to needle valve device.	4N·m(40kgf·cm)
BHSM**-022	Tighten screw 10.	0. 69N·m(7kgf·cm)

- For tightening torque unspecified in the service manual, refer to the followings:

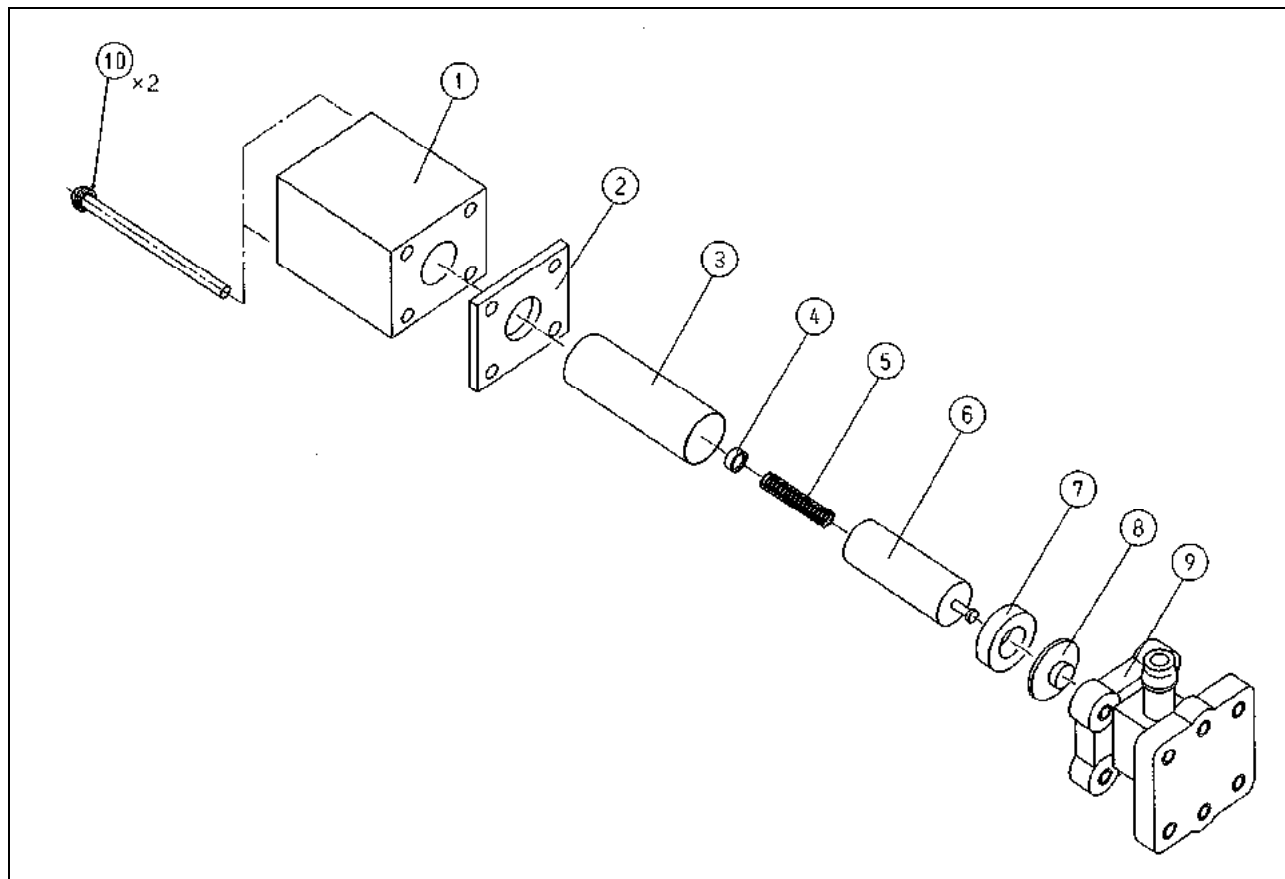
Object	Application	Requirement
Metal	Tighten M2 screw and nut.	0. 2N·m(2kgf·cm)
Metal	Tighten M2.3 screw and nut.	0. 2N·m(2kgf·cm)
Metal	Tighten M2.5 screw and nut.	0. 3N·m(3kgf·cm)
Metal	Tighten M3 screw and nut.	0. 8N·m(8kgf·cm)
Metal	Tighten M4 screw and nut.	1. 2N·m(12kgf·cm)
Metal	Tighten M5 screw and nut.	2. 9N·m(29kgf·cm)
Metal	Tighten M6 screw and nut.	4. 9N·m(49kgf·cm)
Resin	Tighten M4 screw and nut.	0. 8N·m(8kgf·cm)

- Ensure the above requirement so as not to tighten them too much or tighten them less than needed.

1. Two-way valve



1.1. Overall configuration (Type SV2-322)



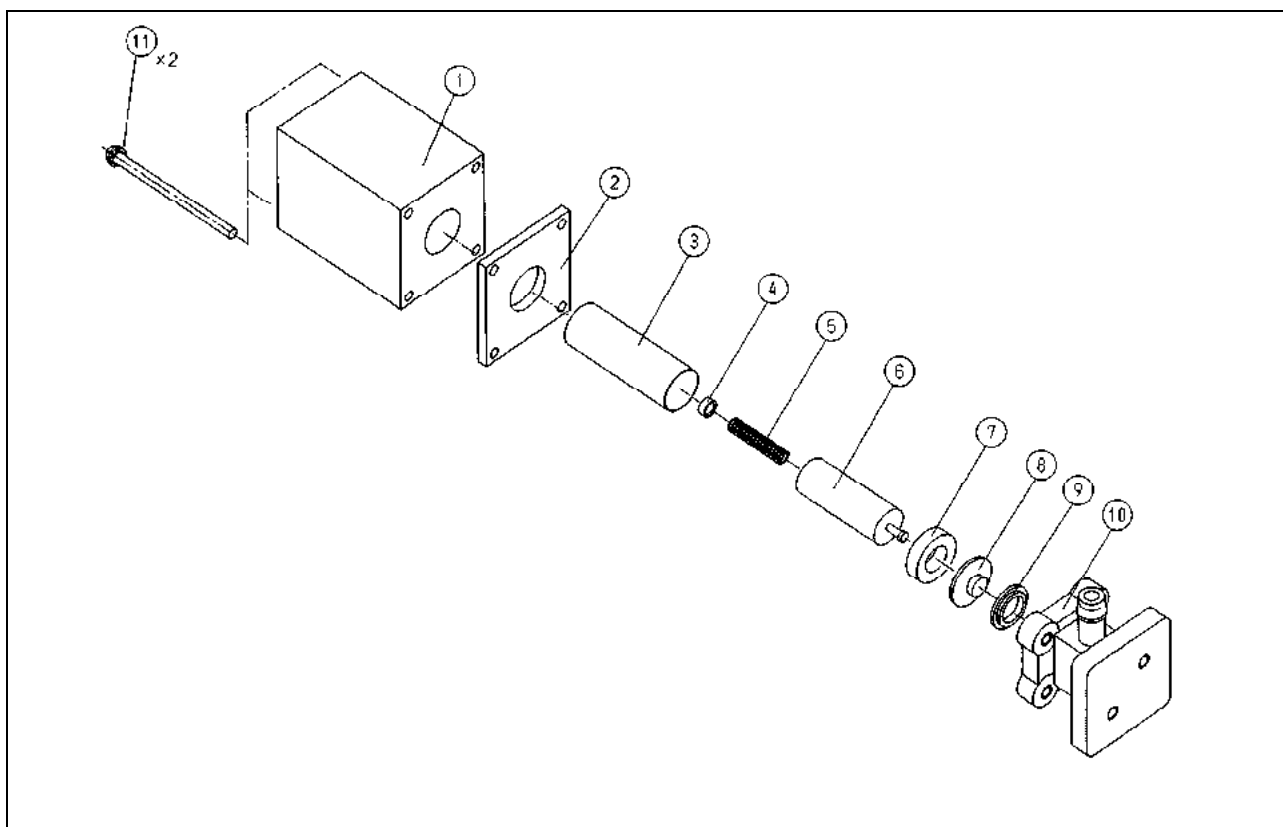
1.2. List of components (SV2-322-S)

No.	Name	Model No.	Code No.	Note
1	Actuator Assy	D3Y570		
2	Core	D3E370		
3	Pacless pipe	D3I2600		
4	Sound damp sheet	D3L490		
5	Spring	D3K680		
6	Plunger	D3I2430		
7	Diaphragm support	D3H370		
8	Diaphragm	D3A540	0-501-117	
9	Valve body	D3B1811	0-501-118	
10	Screw	M3x45P2	0-501-119	0.69N・m
11	Two-way solenoid valve	SV2-322-S	2-006-008	SSM-05050 (V46)

1.3. List of components (SV2-322-SHW)

No.	Name	Model No.	Code No.	Note
1	Actuator Assy	D3Y620		
2	Core	D3E370		
3	Pacless pipe	D3I2600		
4	Sound damp sheet	D3L490		
5	Spring	D3K680		
6	Plunger	D3I2430		
7	Diaphragm support	D3H370		
8	Diaphragm	D3A540	0-501-117	
9	Valve body	D3B1811	0-501-118	
10	Screw	M3x45P2	0-501-119	0.69N·m
11	Two-way solenoid valve	SV2-322-SHW	0-501-126	SSM-08022 V4,V10,V11,V21,(V27), (V28),(V29),(V30),(V31), (V33),(V43),V56,(V57)

1.4. Overall configuration (Type SV2-422)



1.5. List of components (SV2-422-S)

No.	Name	Model No.	Code No.	Note
1	Actuator Assy	D4Y190		
2	Core	D4E230		
3	Pack-less pipe	D4I1320		
4	Sound damp sheet	D3L490		
5	Spring	D4K250		
6	Plunger	D4I1330		
7	Diaphragm support	D4H230		
8	Diaphragm	D4A350	0-501-121	
9	Ring	D4H240		
10	Valve body	D4B1000	0-501-122	
11	Screw	M3x53P2		0.69N・m
12	Two-way solenoid valve	SV2-422-S	0-501-120	SSM-08024 V1,V9a,V9b,V19,V34,V35, V55

1.6. List of components (SV2-422-SH)

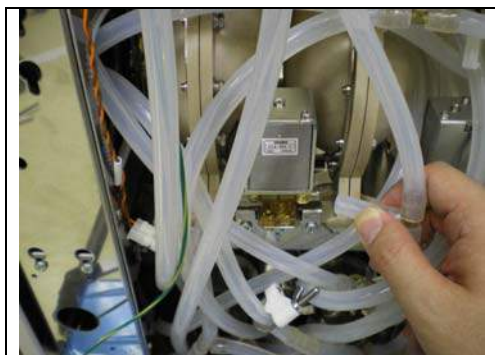
No.	Name	Model No.	Code No.	Note
1	Actuator Assy	D4Y190		
2	Core	D4E230		
3	Pack-less pipe	D4I1320		
4	Sound damp sheet	D3L490		
5	Spring	D4K250		
6	Plunger	D4I1330		
7	Diaphragm support	D4H230		
8	Diaphragm	D4A350	0-501-121	
9	Ring	D4H240		
10	Valve body	D4B1021	0-501-124	
11	Screw	M3x53P2		0.69N・m
12	Two-way solenoid valve	SV2-422-SH	0-501-123	SSM-08025

1.7. List of components (SV2-422-SHP)

No.	Name	Model No.	Code No.	Note
1	Actuator Assy	D4Y190		
2	Core	D4E230		
3	Pack-less pipe	D4I1320		
4	Sound damp sheet	D3L490		
5	Spring	D4K250		
6	Plunger	D4I1330		
7	Diaphragm support	D4H230		
8	Diaphragm	D4A350	0-501-121	
9	Ring	D4H240		
10	Valve body	D4B1021	0-501-124	
11	Screw	M3x53P2		0.69N・m
12	Two-way solenoid valve	SV2-422-SHP	0-501-125	SSM-08026 V5a,V5b,V6a,V6b,V7a,V7b, V8a,V8b

1.8. Removal & installation

<Illustration>



<Removal procedure>

⚠ Caution:

-
- Be sure to turn OFF the breaker.
-

Remove the front cover.

Remove the right and left side face covers at the lower parts.

Remove the connector to the Two-way valve that is to be removed.

Shut off the silicon tubes on both edges by a pean pincer, and remove the silicon tubes.

⚠ Caution:

-
- Put waste cloths so that liquids are not spilled in the device
-

Loosen the screws fixing the valve bracket and remove the Two-way valve together with it.

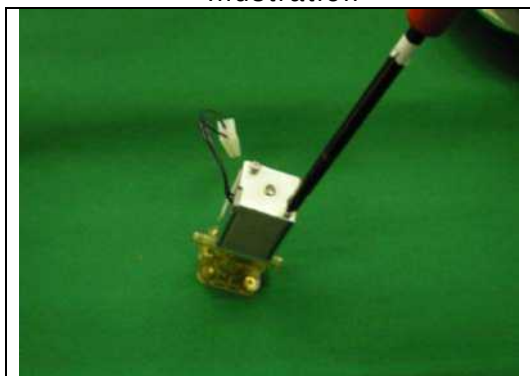
Remove the Two-way valve off the valve bracket.

<Installation>

Install the valve in the reverse order of the removal procedure.

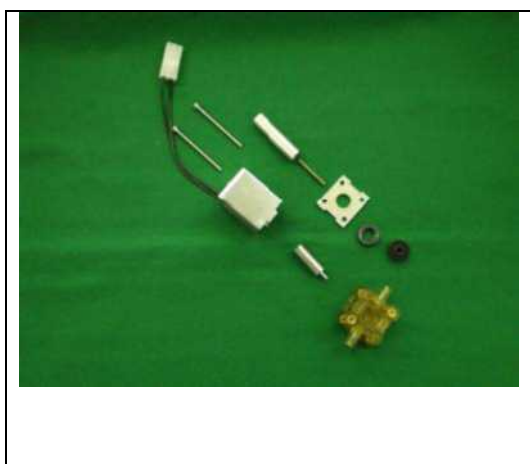
1.9. Disassembly and assembly

<Illustration>



<Removal procedure>

Remove the screws.



To be disassembled into the followings:

Actuator Assy

Core

Pack-less pipe

Spring

Plunger

Diaphragm presser

Diaphragm

Valve body

Ring (in the cases of SV2-422-***-types)

Remove the diaphragm off the plunger.

<Assembly>

Assemble the parts in the reverse order of the disassembly procedure.

⚠ Caution:

- When assembling the diaphragm, be sure to secure it tightly to the plunger by rotating two or three times. Make sure that the screw tightening torque is $0.69\text{N}\cdot\text{m}$ ($7\text{kgf}\cdot\text{cm}$) .

1.10. Inspection and replacement

- Replace parts whose regular replacement period has come.
- Make sure that no foreign object is inside.
- Replace parts that have any modification, damage, or malfunction.

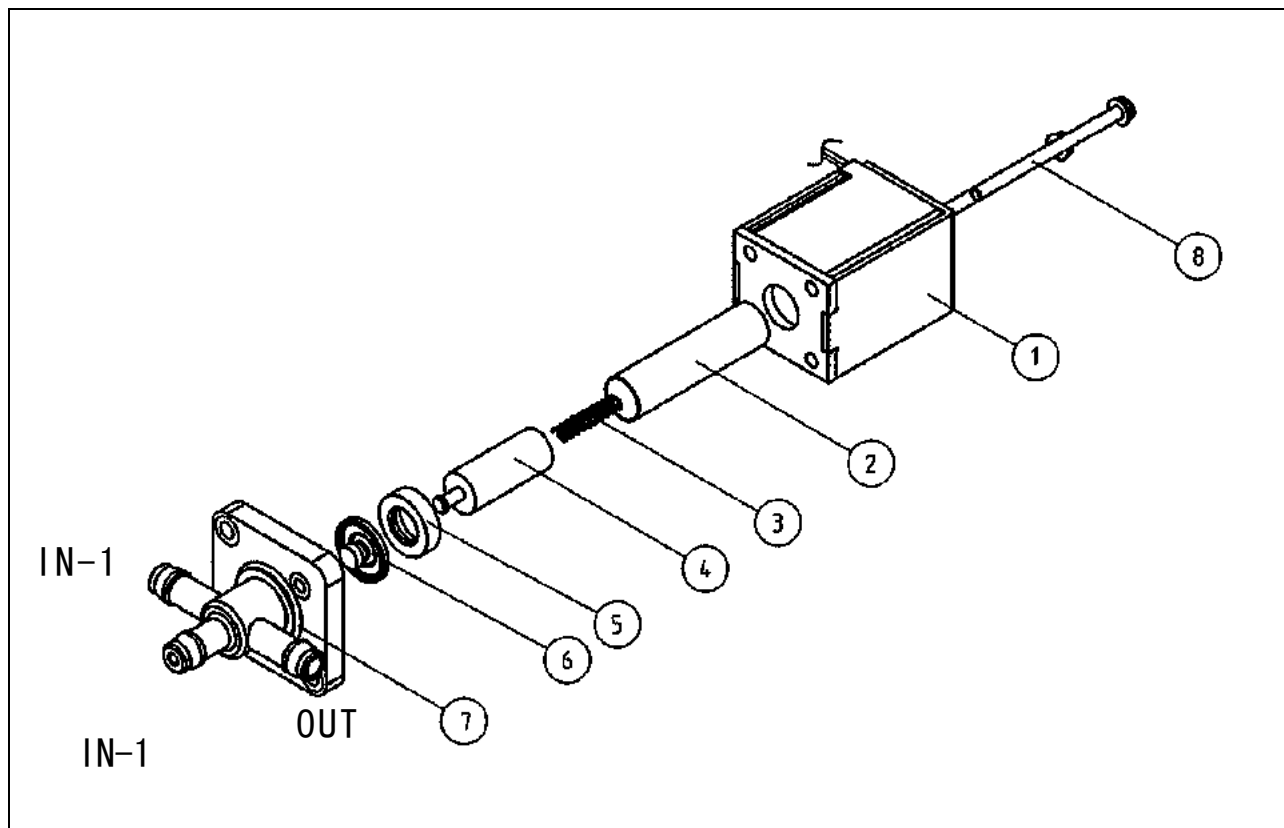
1.11. Confirmation procedure

- In Maintenance 4 "Operation Status," manually turn ON/OFF the Two-wayvalve and confirm the operation sound.
- Attach pressure gauge on the IN side of the Two-way valve, and maintain it by putting pressure of 0.3MPa (3.0kgf/cm²) on the IN side of the solenoid valve by using syringe, with no current applied, confirming there is no leakage.
- Confirm there is no fluid leakage in operation.

2. Mixing Solenoid Valve (Three way)



2.1. Overall configuration



2.2. List of components (SV2-322MX5-S)

No.	Name	Model No.	Code No.	Note
1	Actuator Assy			
2	Pack-less pipe			
3	Spring			
4	Plunger			
5	Diaphragm support			
6	Diaphragm	D3A540	0-501-117	
7	Valve body	D3B1950		
8	Screw	M3x45P2		0.69N·m
9	Two-way solenoid valve	SV2-322MX5-S	-	SSM-10038 V12,V17,V45, (V24)

2.3. Removal & installation

<Illustration>



<Removal procedure>

1. Open the target side cover.
2. Remove the fixing screws from the bracket that hold several solenoid valves.



3. Take out the bracket, disconnect the connector and tube of the target solenoid valve, and remove the target solenoid valve.

⚠ CAUTION

- Put waste cloths so that liquids are not spilled in the machine.
-

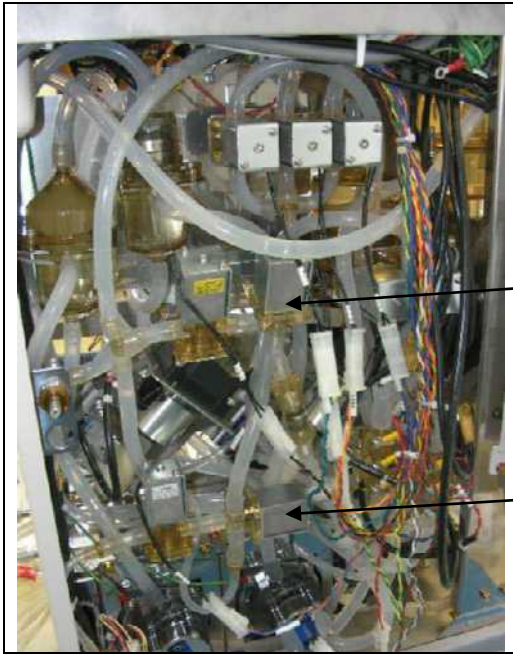


[Locations of the solenoid valves]

-Right side of the machine-

V17

V12



[Locations of the solenoid valves]
- Left side of the machine-

V24

V45

<Installation>

1. Install the valves in the reverse order of the removal procedure.

2.4. Disassembly and assembly

<Illustration>



<Disassembly>

1. Remove two fixing screws and disassemble the assembly into the followings:

Actuator Assy
 Core
 Pack-less pipe
 Spring
 Plunger
 Diaphragm presser
 Diaphragm
 Valve body

2. Replace the parts such as diaphragm.

<Assembly>

1. Assemble the parts in the reverse order of the disassembly procedure.

2.5. Inspection and replacement

- Replace parts at their time for replacement.
- Make sure that no foreign object is inside.
- Replace the deformed or damaged parts.

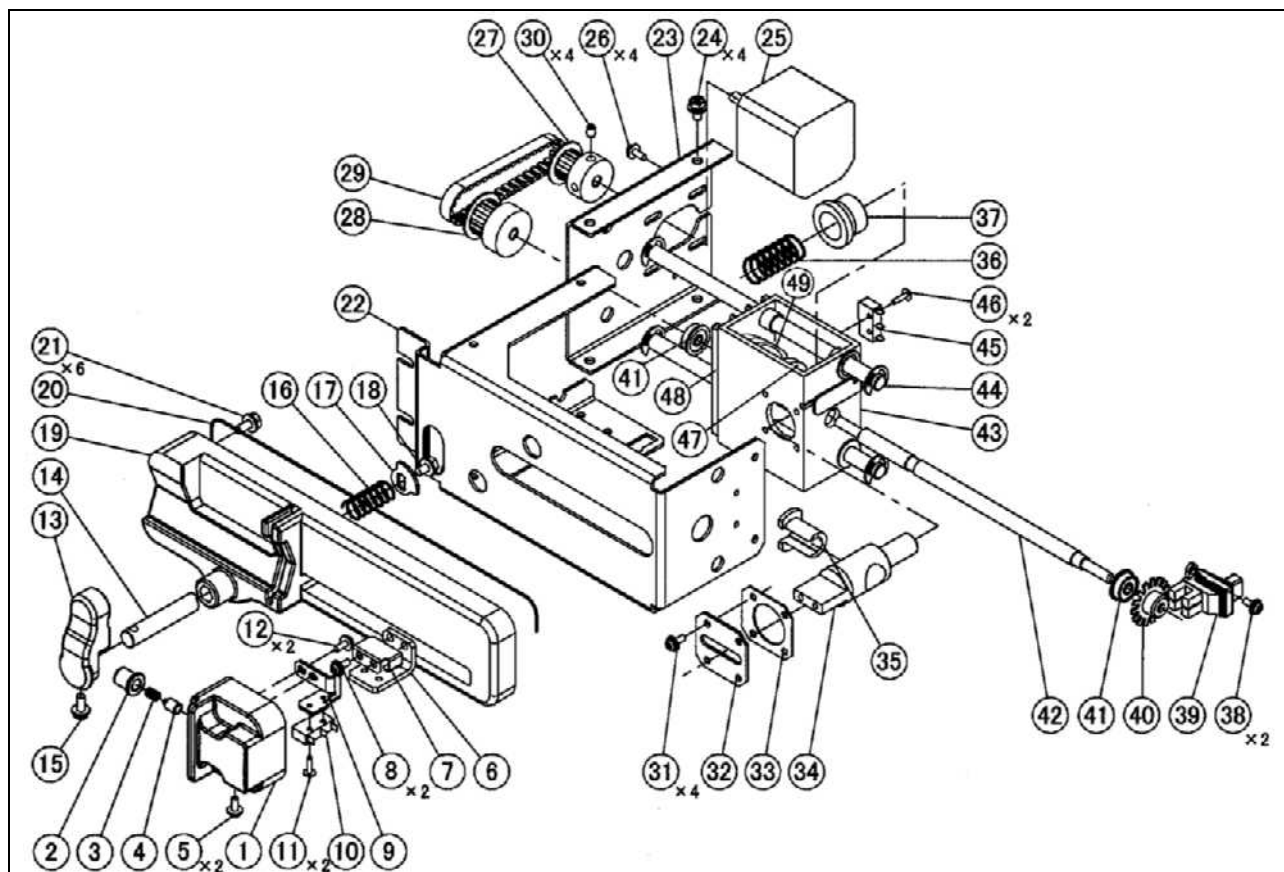
2.6. Confirmation procedure

- On the Flowchart screen, manually turn ON/OFF the Two-way solenoid valve and check the operation sound.
- Install a pressure gauge on the IN-2 side of the Two-way solenoid valve, without applying current and by using syringe, apply the pressure of 0.13MPa (1.3kgf/cm²) on the IN-2 side of the solenoid valve and keep the pressure. Then confirm that is no leakage.
- Confirm that there is no liquid leakage in the machine operation.

3. Hepar in Pump



3.1. Overall configuration



3.2. List of components

No	Name	Model Number	Torque	Note
1	Pusher E	93896511100		
2	Button	93896538100		
3	Coil spring	No. 1043		
4	Pin	93896537001		
5	Screw	M3x8	0.8N·m	With a flat washer and a spring washer
6	Metal fitting	93896533100		
7	Harness presser B	93896530003		
8	Screw	M3x8	0.8N·m	With a flat washer and a spring washer
9	Sensor bracket	93886536003		
10	Limit switch	SS-01		
11	Screw	M2.3x12	0.2N·m	With a flat washer and a spring washer

No	Name	Model Number	Torque	Note
12	Screw	M3x8	0.8N·m	With a flat washer and a spring washer
13	Syringe presser	93886513100		
14	Shaft A	93896524001		
15	Screw	M4x8	1.2N·m	
16	Coil spring	No.1145		
17	Stopper	93886535003		
18	Screw	M3x8	0.8N·m	With a flat washer and a spring washer
19	Syringe base	93896532100		
20	Packing	93146537197		
21	Screw	M4x16	1.2N·m	
22	Chassis	93886514003		
23	Motor plate	93886515100		
24	Screw	M4x8	1.2N·m	With a flat washer and a spring washer
25	Stepping motor	KT42JM4-004		
26	Screw	M3x8	0.8N·m	With a flat washer and a spring washer
27	Timing pulley	10XL037		Small in diameter
28	Timing pulley	12XL037		Large in diameter
29	Timing belt	70-XL-025U		
30	Setscrew	M4x4	1.2N·m	
31	Screw	M3x8	0.8N·m	
32	Plate	93886520003		
33	Plate 2	93886529003		
34	Shaft	93896519001		
35	Half nut	91936519010		
36	Clutch spring	91936523010		
37	Spring case	93146521251		
38	Screw	M3x10	0.8N·m	With a flat washer and a spring washer
39	Rotation detection sensor	—		
40	Dog	93896534100		
41	Bearing	RF-1450ZZ		
42	Feed screw	93896517001		
43	Housing	93896516001		
44	Guide shaft	93896518001		
45	Limit switch	SS-01		
46	Screw	M2.3x12	0.2N·m	With a flat washer and a spring washer
47	Nut	93886523001		
48	Lever	91936927010		
49	O.L spring	01930600121		
50	Heparin pump Assy	93896500		HP

3.3. Removal and installation

<Illustration>

<Removal procedure>

⚠ Caution:

-
- Be sure to turn OFF the breaker.
-

Remove the right and left side face covers at the upper parts.



Remove the screw ⑤ and then pusher ①.



Remove the screw ⑫ and then sensor bracket ⑨.



Remove the screw ⑪ and then limit switch ⑩ off from sensor bracket ⑨.



Move the metal fitting ⑥ to the full right.

Remove the followings:

UFP sensor board connector (CN701)

Motor drive connector (HP)

Limit switch connector (LS3, LS6)

Grounding cable



Remove all the four screws that fix the heparin pump.

Draw out the heparin pump from the left side of the device.

<Installation>

Install the pump in the reverse order of the removal procedure.

3.4. Disassembly and assembly

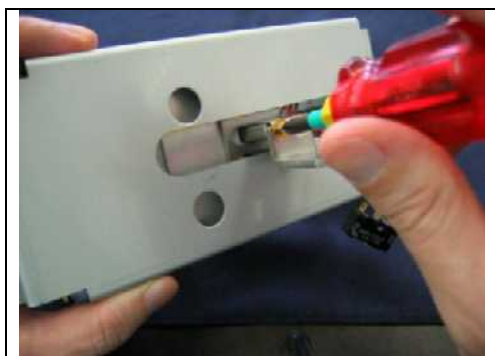
<Illustration>



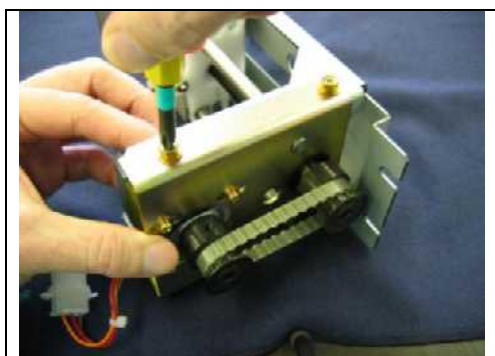
[Replacement of half nuts]

<Disassembly>

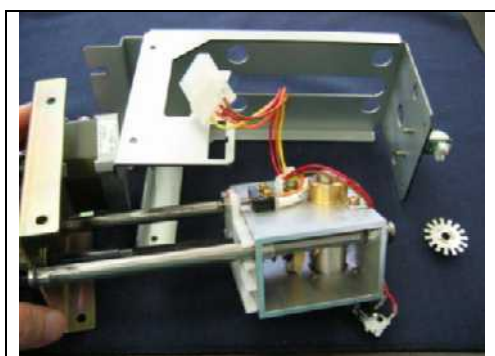
Loosen the screws that fix dog ④⑩.



Remove the screw ⑧ and then metal fitting ⑥ and harness presser B ⑦.

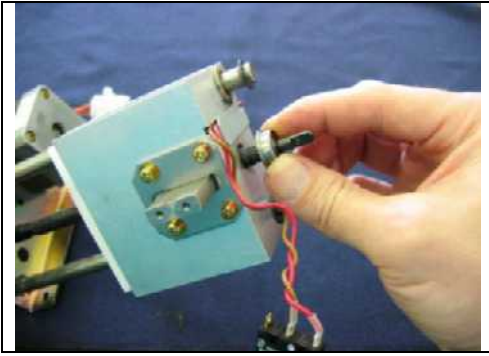


Remove the four screws of ②④.

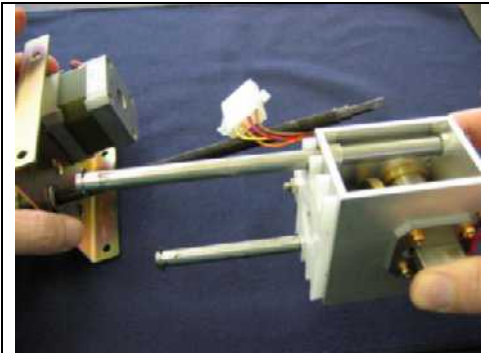


Disassemble the chassis ②② and motor plate ②③.

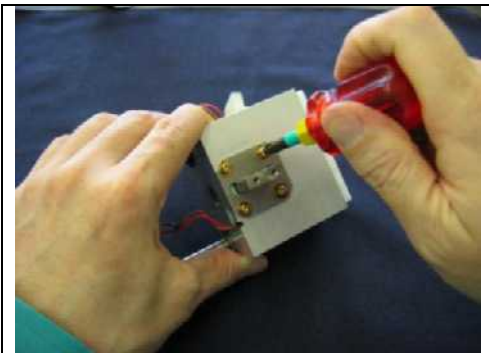
Then remove the dog ④⑩.



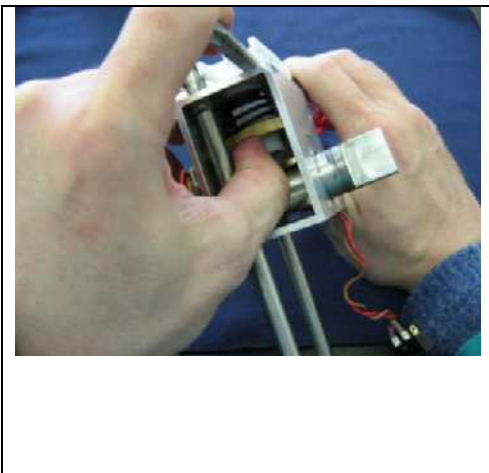
Pull out the bearing ④①.



Push in the shaft ③④ to make it free, and then remove the whole set of housing parts off from the feed screw ④②.

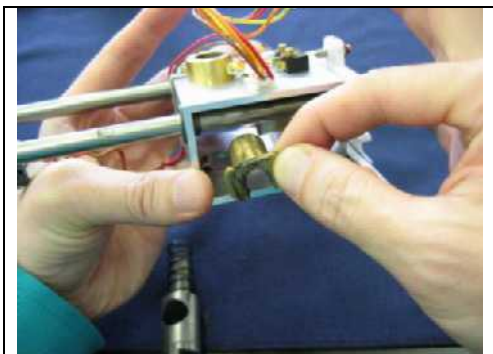


Remove the screw ③① and then plate ③② and plate ③③



While holding the housing ④③, push the nut ④⑦ in the housing toward the direction that O.L. spring ④⑨ shrinks, and then insert a tool like screwdriver between the housing ④③ and lever ④⑩ to keep the O.L. spring ④⑨ in the shrinking state.

Unless there is a space between the lever ④⑩ and half nut ③⑤, the half nut ③⑤ never turns around and it is impossible to remove it, and so it is necessary to maintain this status.



Rotate the shaft ③④ for 90 degrees, and then pull out the half nut ③⑤ in the shaft.

<Assembly>

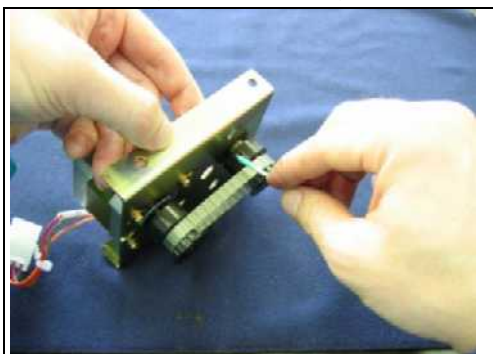
Assemble the parts in the reverse order of the disassembly procedure.

[Motor Part]

<Disassembly>

Loosen the screw ③⑩, and then remove the timing pulleys ②⑦ and ②⑧.

<Illustration>



Remove the screw ②⑥ and remove motor ②⑤ from the motor plate ②③.

<Assembly>

Install the parts in the reversal of the disassembly procedure.

Make sure that the tension of timing belt ②⑨ is 10 to 15mm when pinched in the middle by fingers.

memo

- The timing pulleys are small in diameter (②⑦) on the motor side; large (②⑧) on the driven side.

3.5. Inspection and replacement

- Make sure that disassembled parts are fully cleaned up.

- Replace parts whose regular replacement period has come.

Half nut:

- Extremely worn half nut is to be replaced.

Timing belt:

- Replace them when damaged or cracked.
- Replace them if their bent is extremely large and it is impossible to adjust it.

Limit switch:

- Check conductivity by using tester; replace those with malfunction by using solder and soldering iron.

Feed screw:

- Grease them up every six months.

Grease to use: Mori coat grease (BR2-PLUS)

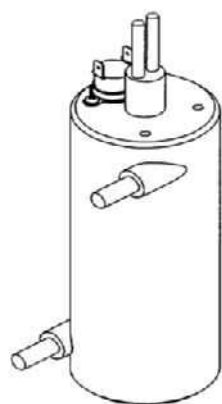
3.6. Adjustment

(1) Syringe test BHAJ**-013

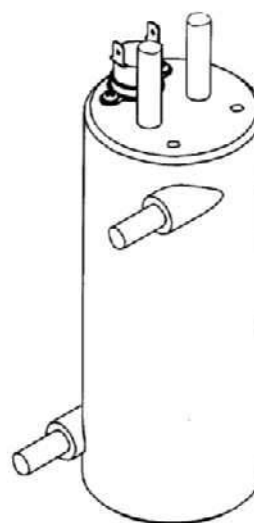
4. Heater



4.1. Overall configuration



(1) Aluminum cast heater (for 230V)



(2) Aluminum cast heater (for 110V)

4.2. List of components

<Normal>

No.	Name	Model Number	Note
1	Aluminum cast heater	N96237601	For 230V, 500W×2
	Aluminum cast heater	N96237503	For 110V, 1000W

<Hot disinfection with citric acid ASSY, Dialysate flow ASSY and high flow ASSY>

No.	Name	Model Number	Note
1	Aluminum cast heater	N96237601	For 230V, 500W, H1
	Aluminum cast heater	N96237503	For 230V, 1000W, H2
2	Aluminum cast heater	N96238201	For 110V, 1500W, H1

4.3. Removal and installation

<Illustration>

<Removal procedure>

Prepare a container to receive fluid in the heater.

⚠ Caution:

- Be sure to turn OFF the breaker.

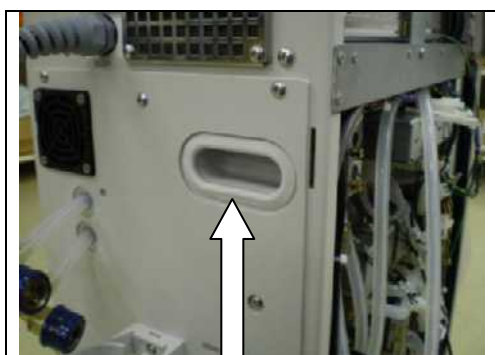
Remove the right and left side face covers at the upper parts.

Remove the right and left side face covers at the lower parts

Remove the chamber. Refer to BHSM**-006.
(For securing working space)



Remove the fixing screws of AS1 and AS2.
(For securing working space)



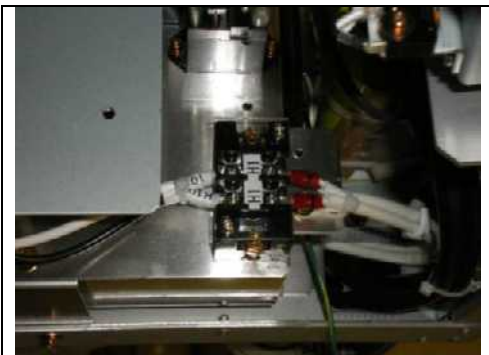
Embedded vinyl handle

Remove the embedded vinyl handle.
(For securing working space)

<Illustration>



<230V terminal block>



<110V terminal block>

<Removal>

Remove the harness on the side of the heater off from the terminal block on the left side of the electric component part.

Pull the heater harness to the heater part.

⚠ Caution:

- Handle them very carefully not to damage parts including the harness.

Shut off the silicon tubes on both sides of the heater by a pean pincer, and pull out the tubes.

Receive the fluid in the heater by container.

⚠ Caution:

- Put waste cloths so that there is no fluid spilled in the device.

Remove the terminals connected to the limit switch of the heater.

Remove the heater together with the heater bracket.

Remove the grounding cable, and remove the heater from the main body of the device.

Remove thermal insulation material.

<Installation>

Attach the device in the reverse of the removal procedure.

<In the case of 230V :hot disinfection with citric acid ASSY, Dialysate flow ASSY and high flow ASSY>

H1 (500W) is on the lower side.

H2 (1000W) is on the upper side.

4.4. Inspection and replacement

- Cleanse the inside of the heater.
- Remove calcium carbonate adhesion by acid cleansing.

Heater:

Measure insulation resistance and if it is under $50M\Omega$, replace it.

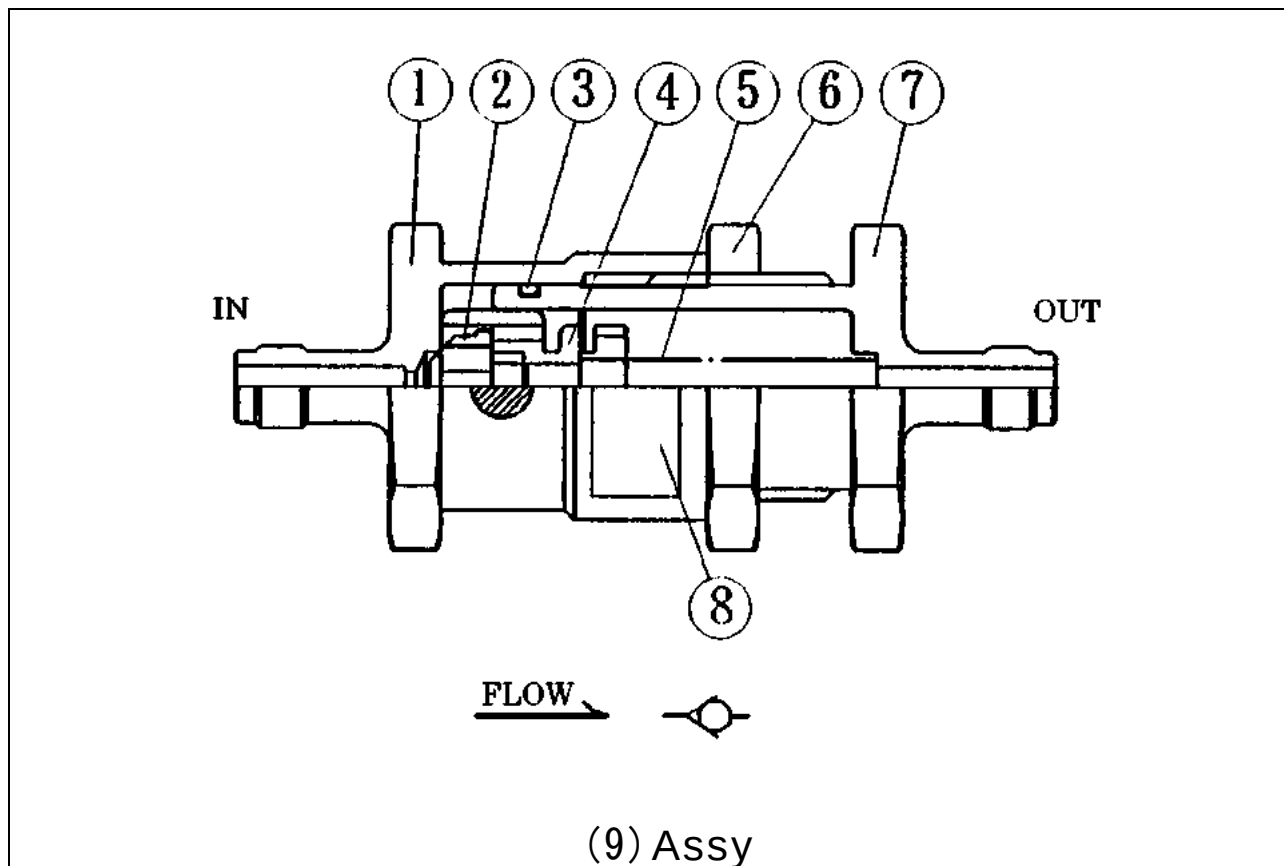
4.5. Confirmation procedure

- Confirm that there is no fluid leakage in operation.
- Confirm that temperature rises during the process of preparation.
- Confirm that fluid temperature is correct during the process of preparation.

5. Relief Valve



5.1. Overall configuration



5.2. List of components

No.	Name	Model No.	Code No.	Note
1	Valve body	ACB220	0-101-105	
2	Check seat	-		
3	Packing	ACL040	0-101-107	
4	Seat holder	-		
5	Spring	ACK340	0-101-109	
6	Lock nut	-		
7	Sub body	-		
8	Number Plate	-		
9	Relief valve Assy	AC-3200-63H-PA		SSM-5035 RV1,RV2

5.3. Removal & installation

<Illustration>



↑ RV2

↑ RV1

<Removal procedure>

Remove the lower right side face cover.

[Position of relief valve]

RV1

The valve contacts the OUT-side line of AS1.

RV2

The valve contacts the OUT-side line of AS2.

Shut off the silicon tubes on both sides of the relief valve by a pean pincer, and pull out the tubes.

⚠Caution:

- Especially when removing RV2, make sure to shut off tubes tightly by tools like a pean pincer so that there is no fluid spilled in the device from AS2 and dialysate pressure pump.
-

<Installation>

Install the parts in the reverse order of the removal procedure.

5.4. Disassembly and assembly

<Illustration>



<Disassembly>

Loosen the lock nut and turn the valve body counterclockwise.

Then disassemble the parts as if taking out parts from the case.

<Assembly>

Assemble the parts in the reversal order of the disassembly procedure.

5.5. Inspection and replacement

Wash the disassembled parts well.

Replace the whole assembly if any of parts are damaged.

5.6. Procedure of adjustment

Implement adjustment in the following order:

< RV1 >

- | | |
|---|------------|
| (1) Adjustment of Supplying Flow and Fill-up Pressure | CXAJ**-009 |
| (2) Balance test | CXAJ**-016 |
| (3) UF precision test 1 | BHAJ**-018 |
| (4) UF precision test 2 | BHAJ**-019 |

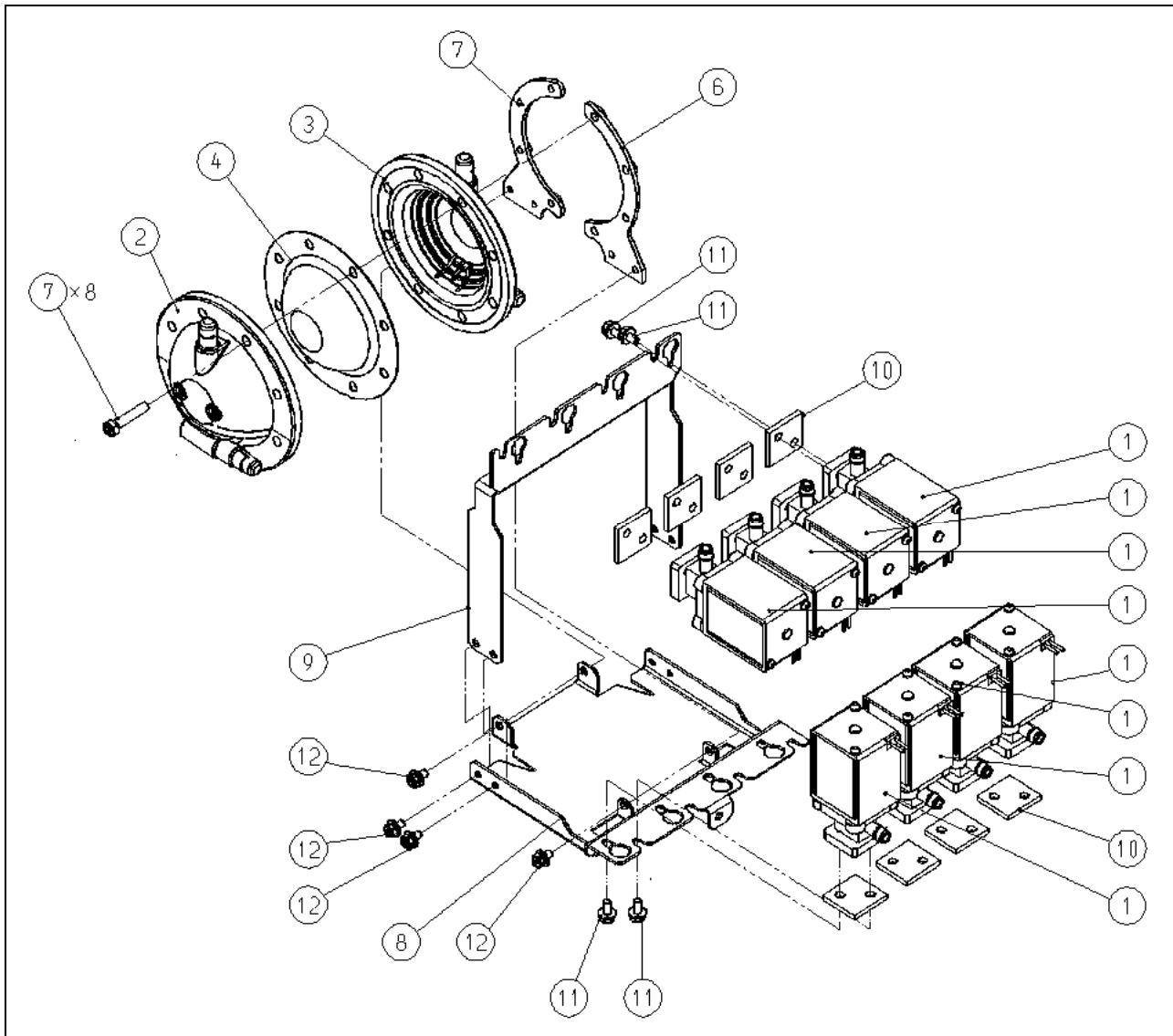
< RV2 >

- | | |
|---|------------|
| (1) Adjustment of Closed Circuit Pressure | CXAJ**-010 |
| (2) Balance test | CXAJ**-016 |
| (3) UF precision test 1 | BHAJ**-018 |
| (4) UF precision test 2 | BHAJ**-019 |

6. 100mL Chamber



6.1. Overall Configuration



6.2. Purchased parts list

No.	ITEM NAME	MODEL #	CODE #	REMARKS
1	2-port solenoid valve	SV2-422-SHP	0-501-125	
2	Case R	90351011100	0-703-002	
3	Case L	90351012100	0-703-004	
4	Film	90351014100	0-703-003	
5	nut plate A	90351015100	0-703-005	
6	nut plate B	90351016100	0-703-006	
7	Screw	M5*25	0-703-012	4.91N·m(50kgf·cm)
8	Chamber frame A	90351022100	0-501-286	
9	Chamber frame B	90351023100	0-501-287	
10	Rubber sheet	01931000096	0-703-011	
11	Screw	M4*10	0-201-128	1.18N·m(12kgf·cm)
12	Screw	M4*8	0-201-071	1.18N·m(12kgf·cm)
13	Chamber assy	90351000	0-501-285	Ca,Cb

6.3. Removal • Installation

<Illustration>

<Removal>

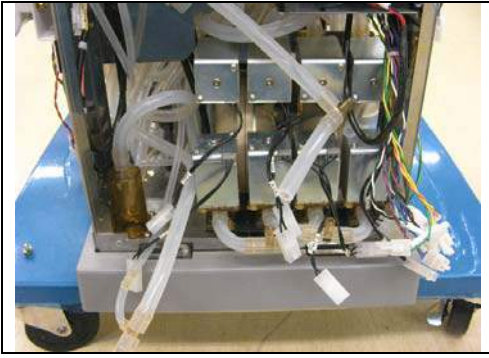
Prepare a container to receive the solution in the chamber.

⚠ Caution

- Turn OFF the power breaker.

Remove the lower right side cover and the lower left side cover.

Remove connectors of 2-port solenoid valve V5(a, b), V6(a, b), V7(a, b), V8(a, b).

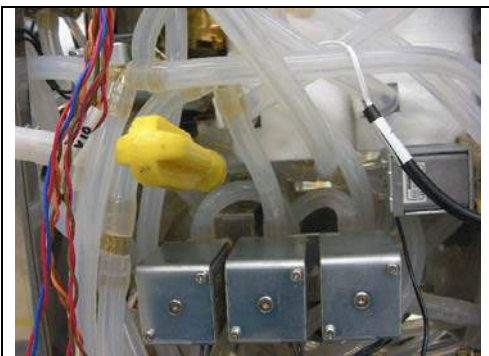


Pair silicone tubes connected to 2-port solenoid valves and block them, and remove the following tubes.

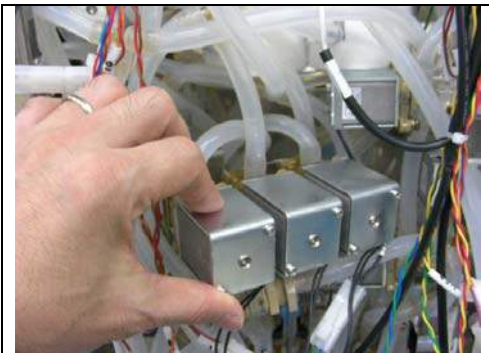
IN side of V5a, V6a, V7a, V8a



Remove fixing screws of the chamber assembly.



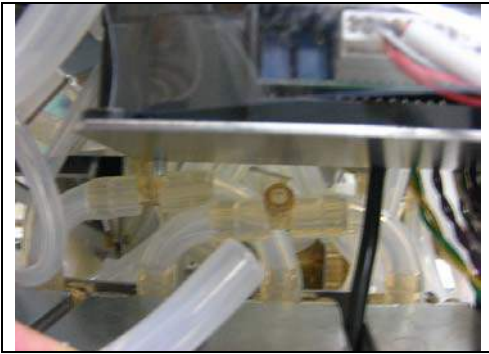
Remove the bracket fixing V55, V34, and V1.
(Loosen 2 screws, not remove)



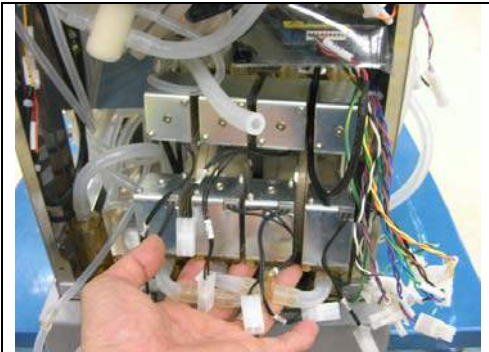
Raise the valve assembly to make room for work.



Remove a silicone tube at V6b, V8b OUT side.



Remove a silicone tube at V5b, V7b OUT side.



Remove the silicone tube connected to UFP and RV2.
Pull the chamber assembly gradually, lifting it from the bottom.



Remove the chamber assembly from the machine.

<Illustration>

<Removal>

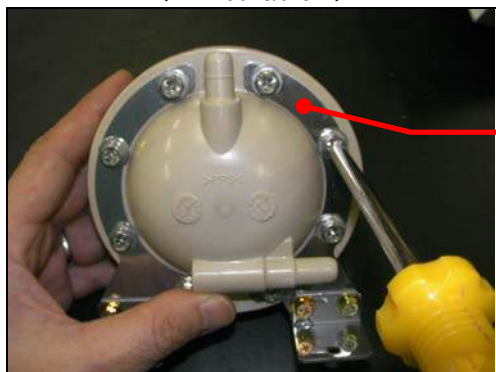
Remove the silicone tubes from the chamber and discharge the chamber solution to the container.

<Installation>

Reverse the removal procedures.

6.4. Disassembly • Assembly

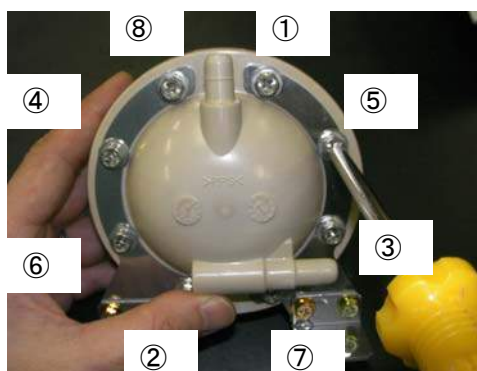
<Illustration>



<Disassembly>

Remove screws fixing the casing.

※Not to use this part



<Assembly>

Reverse the disassemble procedures.

Tighten screws diagonally.

● Tightening torque is 4.91N·m(50kgf·cm).

<Tightening order>

6.5. Inspection • Replacement

Remove the diaphragm according to the disassemble procedure.

☐ Diaphragm

Replace if scratched or deformed

☐ Casing

Replace if cracked.

- Clean disassembled parts thoroughly.
- Replace ones which reached to periodic replacement time.

6.6. Confirmation

- Visually confirm fixing screws are tightened.
- Visually confirm piping connection is firmly connected.
- Confirm no leak during operation.

6.7. Adjustment

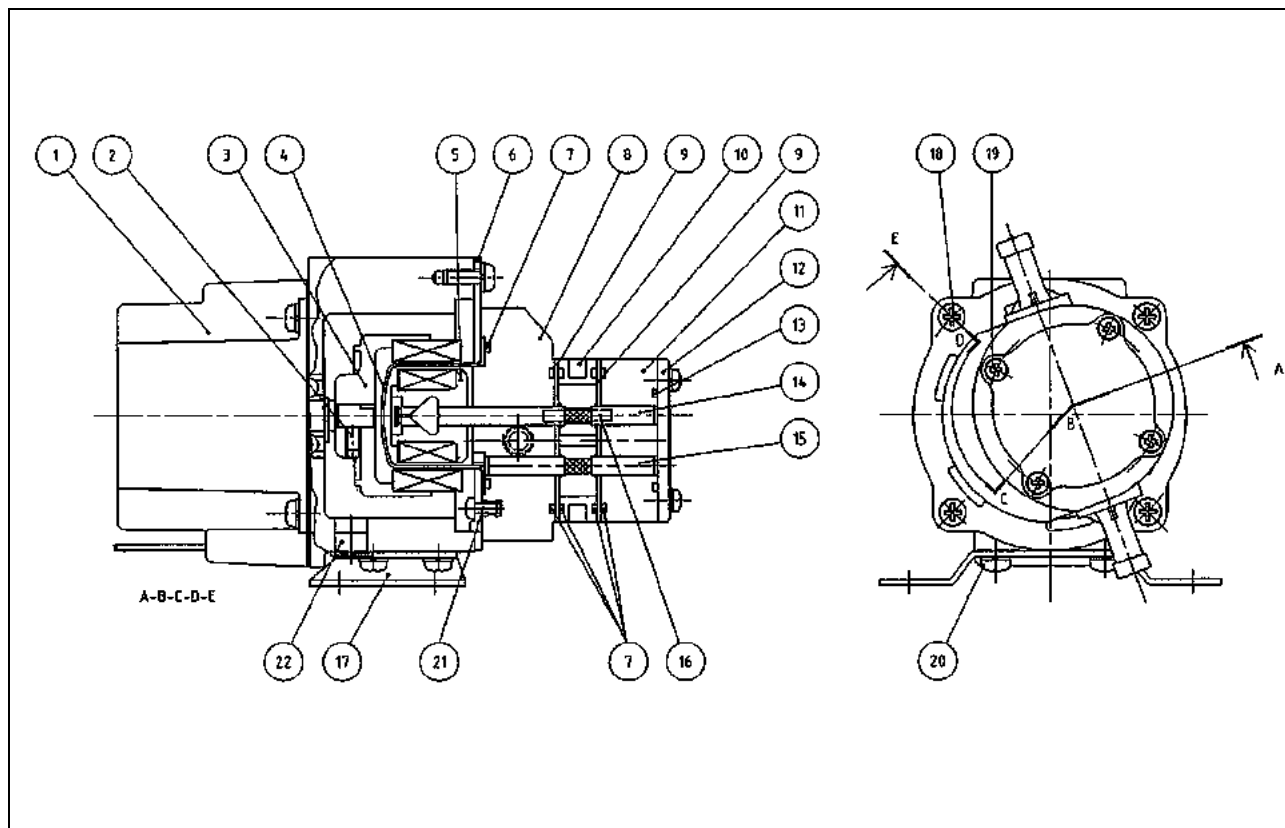
Implement as following order:

- | | |
|--|------------------------------|
| (1) Reducing valve | CXAJ**-008 |
| (2) Filling flow • filling complete pressure | CXAJ**-009 |
| (3) Closed circuit pressure | CXAJ**-010 |
| (4) Confirmation of adjust low limit UF amount | CXAJ**-016 |
| (5) UF accuracy test 1 | CXAJ**-018 |
| (6) UF accuracy test 2 | CXAJ**-019 |
| (7) Concentration | Refer to instruction manuals |

7. Magnet gear pump



7.1. Overall Configuration



7.2. Purchased parts list

No.	ITEM NAME	MODEL #	CODE #	REMARKS
1	Motor	—		
2	Set screw	M4×6		1.37N·m(14kgf·cm)
3	Drive magnet unit	—		
4	Rear casing	—		
5	Magnet can	—		
6	Mounting plate	—		
7	O ring	AS568-028		
8	Mounting disk	—		
9	Plate	—		
10	Gear case	—		
11	End disk	—		
12	Cover	—		
13	O ring	AS568-021		
14	Drive gear	—		
15	Driven gear	—		
16	Parallel pin	—		
17	Base	—		
18	Screw	M4×8		1.37N·m(14kgf·cm)
19	Screw	M3×35		0.78N·m(8kgf·cm)
20	Screw	M4×8		1.37N·m(14kgf·cm)
21	Screw	M3×6		0.78N·m(8kgf·cm)
22	Cap	—		
23	Spare kit	—		⑦ × 7, ⑧, ⑪, ⑬, ⑭, ⑮
24	Magnet gear pump	MDG-R3SD24V-01		P1,P2

7.3. Removal • Installation

< Illustration >



Refer to the left for P1 and P2 locations
P2
P1

< Removal >

⚠ Caution

- Turn off the power breaker.

1. Remove the machine covers (back side, right side, left side)



【Replacement of whole pump】

1. Remove IN, OUT tubes.
2. Remove 2 fixing screws tightening the bracket holding the pump.



3. Remove the bracket tightening V45 and V46. (just loosen 2 screws)



4. Loosen the valve assembly.



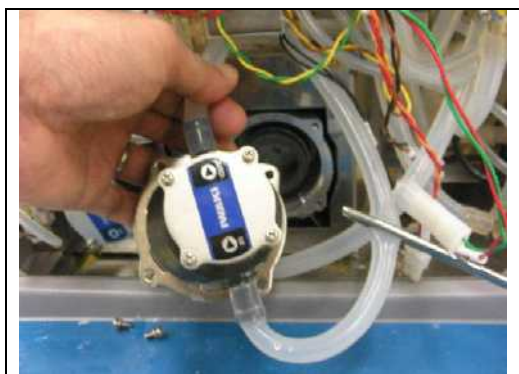
5. Remove the connector of the harness and pull the pump unit out.

⚠ Caution

- Put a cloth not to drip the fluid inside the machine.
-

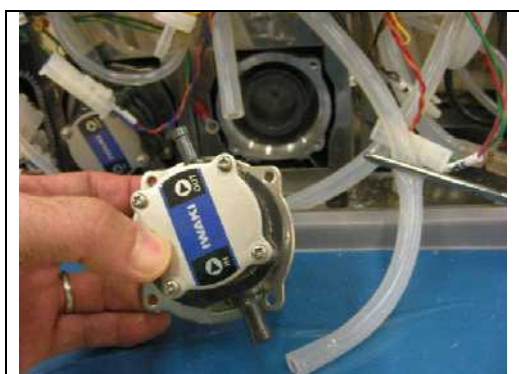
< Installation >

1. Reverse the removal procedure.



【 Replacement of the pump unit 】

1. Clamp IN and OUT tubes.
2. Remove 4 fixing screws ⑩ tightening the pump unit.



3. Remove the tubes and pump unit.

< Installation >

1. Reverse the removal procedures.

⚠ Caution

- Install the pump unit after confirming no foreign objects in the drive magnet.
-

7.4. Disassembly · Assembly

< Illustration >



< Disassembly >

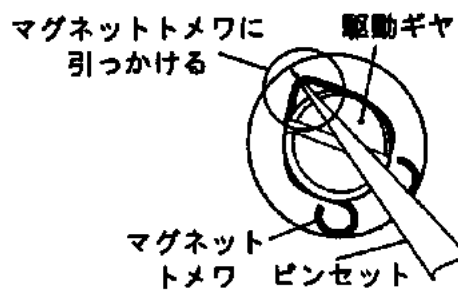
1. Unscrew ⑮ to remove the pump body unit.



2. Unscrew 6 screws ⑰ to remove mounting plate⑥.



4. Remove the magnet retaining ring with a pair of tweezers and remove the magnet can ⑤.



⚠ Caution

Carefully remove the magnet retaining ring. Pulling forcibly deforms the ring.



5. Remove the magnet can ⑤ from the drive gear ⑭.



6. Reverse it, remove the screw ⑬, and remove the cover ⑫.

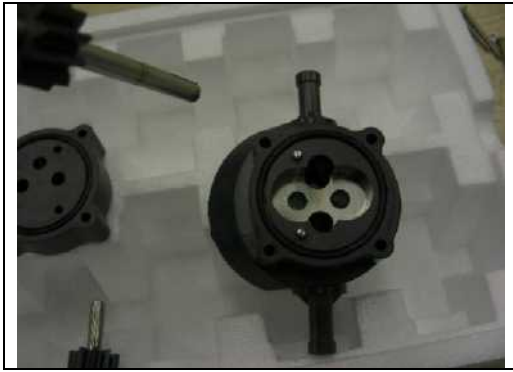
7. Remove the end disk ⑪.



8. Remove the plate ⑨.



9. Remove the driven gear ⑮.



10. Remove the drive gear ⑭.



11. Remove the gear case ⑩.



12. Remove the plate and parallel pin.



13. Disassembly complete.

Store removed magnet can ⑤ to keep it from iron powder.

< Assembly >



1. Insert the parallel pin ⑯ to the mounting disk⑧.

2. Insert O ring ⑦.

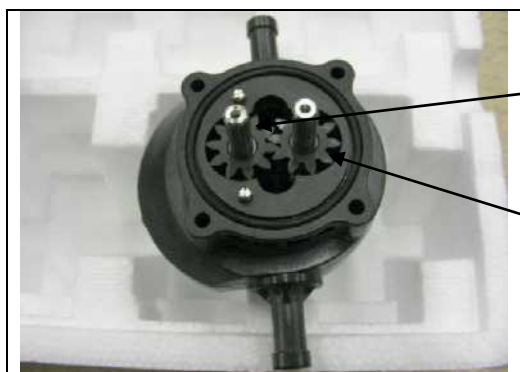


3. Place the Plate⑨.



4. Insert O ring ⑦ to the both end of the gear case ⑩.

5. Place the gear case⑩.



6. Insert drive gear⑭ and driven gear⑮.

Drive gear⑭

Driven gear⑮



7. Place the plate⑨.



8. Insert O ring⑦ and ⑬ to the both end of the end disk⑪.

9. Place the end disk⑪.



10. Place the cover ⑫ and tighten the screw ⑬.

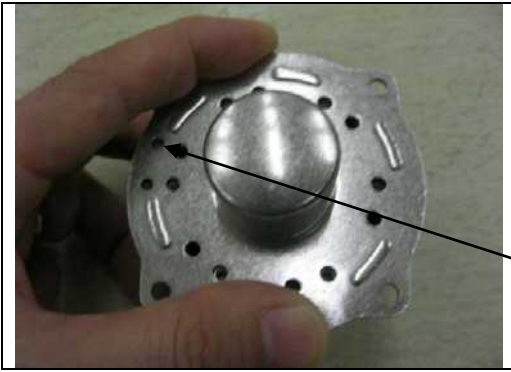
Tightening torque is $0.78\text{N}\cdot\text{m}$ ($8\text{kgf}\cdot\text{cm}$).



11. Reverse, insert the magnet can⑤ to the slit of the drive gear ⑭, and insert the magnet retaining ring to the groove of the drive gear.

※ The retaining ring is a little extended once it is removed. Squeeze when inserting into the grove.

※ Remove foreign particles attached to the magnet can with adhesive tape.



12. Place the rear casing ④ and the mounting plate⑥.

13. Fix with the 6 screws ②①.

Beware of the tightening angle.

Tightening torque is $0.78\text{N}\cdot\text{m}$ ($8\text{kgf}\cdot\text{cm}$).

Align the positioning pin and the insert hole.



14. Fix the pump unit to the motor with screws⑩.

Tightening torque is $0.78\text{N}\cdot\text{m}$ ($8\text{kgf}\cdot\text{cm}$).

7.5. Inspection • Replacement

- Clean the disassemble parts thoroughly

- ☐ For drive gear, driven gear, mounting disk, end disk, replace after 5000 hours or if broken or damaged.

- ☐ O ring

Replace with new one once disassembled.
Disassembly weakens the sealing.

- ☐ Others

Repair or replace if deformed, damaged or with defects.

7.6. Adjustment

Implement as following order:

Deaeration pump

(1) Filling flow • Complete pressure CXAJ**-009

(2) Confirmation of adjust low limit UF amount
CXAJ**-016

(3) UF accuracy test CXAJ**-018

Dialysate pressure pump

(1) Closed circuit CXAJ**-010

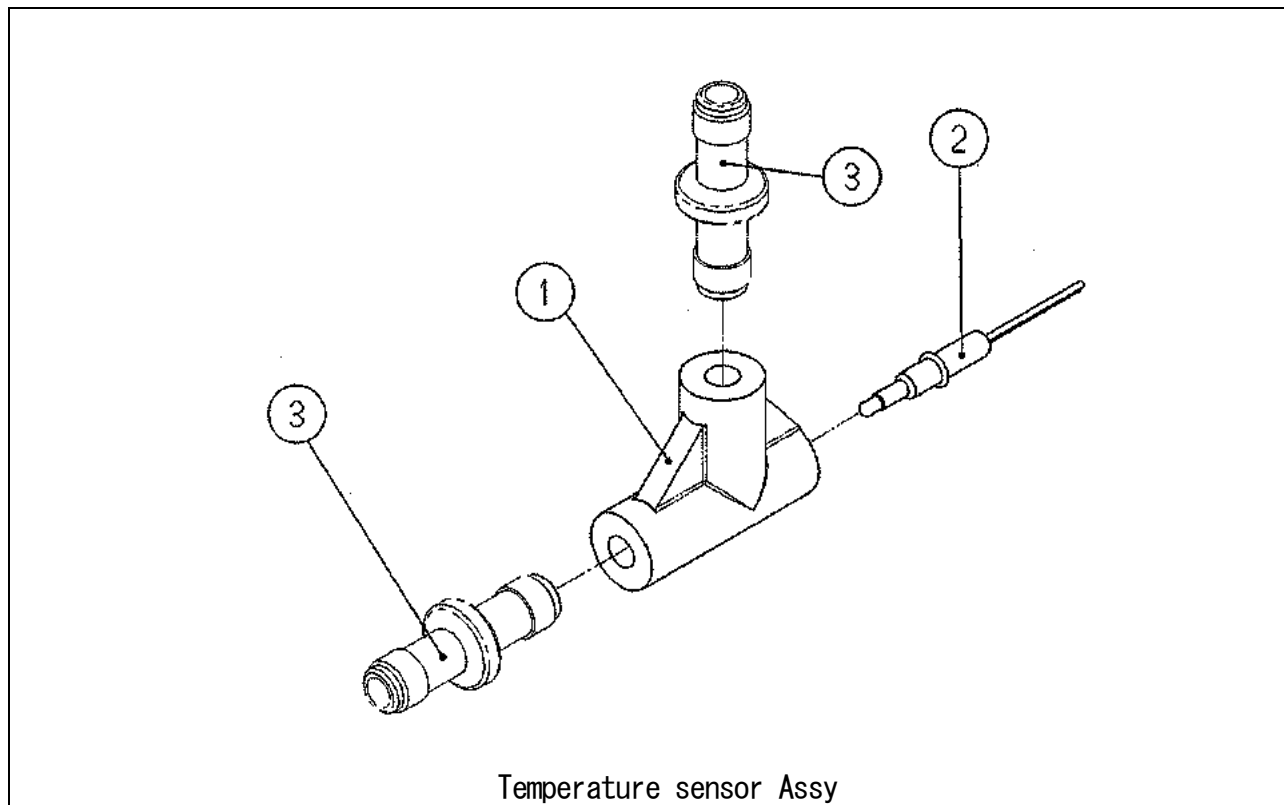
(2) Confirmation of adjust low limit UF amount
CXAJ**-016

(3) UF accuracy test CXAJ**-018

8. Temperature Sensor



8.1. Overall configuration

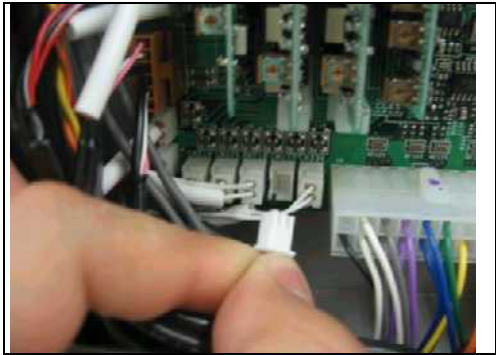


8.2. List of components

No	Name	Model No.	Torque
1	Sensor holder	93892511001	
2	Thermistor	E3M-42D-S36	
3	Hose opening	01930600070	Use arbitrarily.
4	Temperature sensor Assy	—	

8.3. Removal and installation

<Illustration>



<Removal procedure>

⚠ Caution:

- Be sure to turn off the breaker.

Remove the front cover.

Remove the right and left side face covers at the upper part.

Remove the right and left side face covers at the lower part.

Remove the right side face electric cover.

Remove the connectors for the temperature sensors to be worked on, which is connected onto the I/O board.

Temperature sensor		Connector
T1	Incoming water temperature	CN214
T2	Heater outlet temperature	CN215
T3	Dialysis fluid temperature	CN216
T4	CD1 temperature compensation	CN217
T5	CD2 temperature compensation	CN234
T6	CD3 temperature compensation	CN238
T7	Coupler temperature	CN246

⚠ Caution

- Be careful in handling harnesses and others so as not to damage them.



Cut the sensor holder of the sensor to be worked on, and take out the temperature sensor.

⚠ Caution

- Do not pull the harness. Lines in temperature sensors may be cut.
-

<Installation>

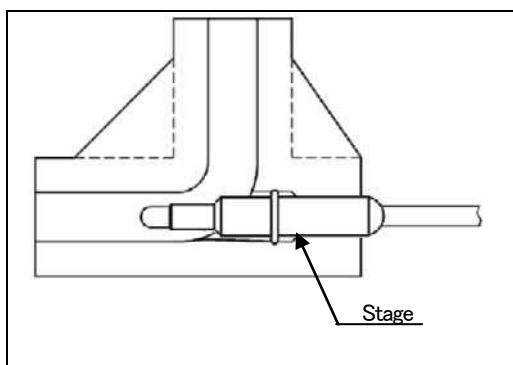
Install the parts in the reverse order of the removal procedure.

memo

- Sensor holders are to be replaced.
-

8.4. Disassembly and assembly

<Illustration>



<Assembly>

Insert the thermistor ② into the sensor holder ①.

At this time, make sure that the flange part of the thermistor ② is stopped at the position of the sensor holder ① shown in the left picture (about in the middle of the stage).

Insert the hose opening ③.

8.5. Inspection and replacement

<Illustration>

- Cleanse disassembled parts very well.
- Confirm that the difference between the thermometer display and the temperature sensor display is within ± 0.5 °C by preparing 37°C water according to Maintenance 1 "Dialysis data display."

Temperature Sensor	Maintenance Mode Display
T1	Incoming water temperature
T2	Heater outlet temperature
T3	Dialysis fluid temperature
T4	CD1 temperature compensation
T5	CD2 temperature compensation
T6	CD3 temperature compensation
T7	Coupler temperature

Thermistor:

- If there is any pinhole, replace thermistor with them.
- Cleanse them and replace those with rust that cannot be removed.
- Replace those heavily corroded.

Connector:

- Replace those with modification, rust, corrosion, etc.

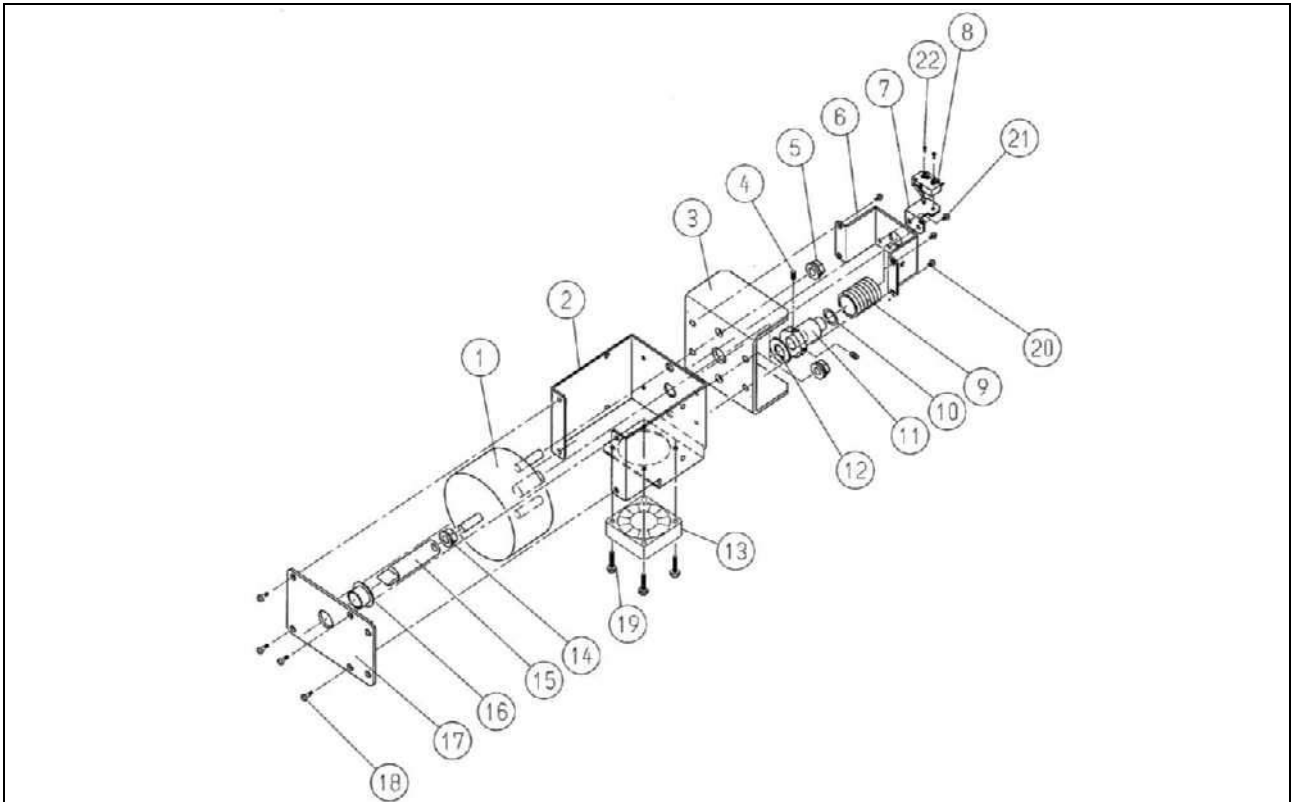
8.6. Adjustment procedure

- (1) Temperature sensor calibration ① BHAJ**-003
- (2) Temperature sensor calibration ② BHAJ**-004
- (3) Temperature sensor calibration ③ BHAJ**-022

9. Clamp



9.1. Overall configuration



9.2. List of components

No.	Name	Model No.	Torque	Note
1	Solenoid	7SC AWG. 25x3x4 long-life		
2	Solenoid bracket	93896211101		
3	Radiator board	93896214001		
4	Setscrew	M3x4	0. 8N·m	
5	Nut	For M6	4. 9N·m	Flat washer, spring washer
6	Spring bracket	93896213101		
7	Sensor bracket	93896218101		
8	Subminiature base	SS-01GL13		
9	Spring	No. 4068		
10	O-ring	P10		
11	Rod	93896217001		
12	Sheet	93896220001		
13	Fan	EUDC24B8S		
14	Nut	For M6	4. 9N·m	Spring washer
15	Head	93896215100		
16	Flange bush	80F-1212		
17	Solenoid base	93896212101		
18	Screw (S tight)	M4x8	Not removable	
19	Screw	M3x20	0. 8N·m	
20	Screw	M3x6	0. 8N·m	
21	Screw (S tight)	M3x8	Not removable	
22	Screw	M2. 3x12	0. 2N·m	
23	Clamp Assy	93896200		

9.3. Removal and installation

<Illustration>



<Removal>

⚠ Caution:

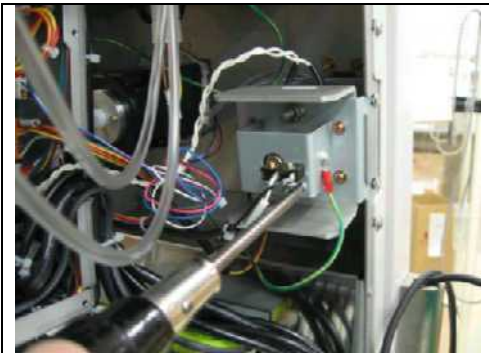
-
- Be sure to turn OFF the breaker.
-

Remove the upper left side face cover.

Remove the left side face electric cover.

Remove CN117 and CN118 on the power board.

Remove all the bands that bind clamp harness and L5 harness.



Remove the screw ②① and limit switch (L35) together with the bracket.



Remove the four screws that fix clamp Assy, and then take out clamp Assy.



Temporarily hold the clamp Assy so that its teeth do not hit the upper, lower, right, and left frame edges of the bubble clamp case when the clamp is shut off.

Manually move solenoid ① back and forth to confirm it smoothly slides and fix clamp Assy.

memo

- Adjust the fixation position of the clamp Assy if it does not slide in a smooth manner.
-

⚠ Caution

- If the head somehow does not stop hitting the framework of the bubble clamp case, adjust the set position of bubble clamp parts.
-

【Clamp interval adjustment】

Adjustment items: implement clamp leak test (BHAJ**-012), and if there is leakage, adjust clamp intervals.



Install adjustment jig (t=1.5) onto the inside of the cover and close the cover.

Turn the solenoid ① around and stop it when the head ⑮ hits the jig.



Hold solenoid ① so that it may not turn around and fasten the nut ⑭ to fix the teeth.

Repeat pushing and leaving the solenoid ①, and confirm again that head ⑮ does not hit the bubble clamp case window and it slides in a smooth manner.

Make sure that there is no clamp position irregularity shown immediately after startup.

When clamp position irregularity is detected, adjust the clamp attachment position again.

9.4. Adjustment item

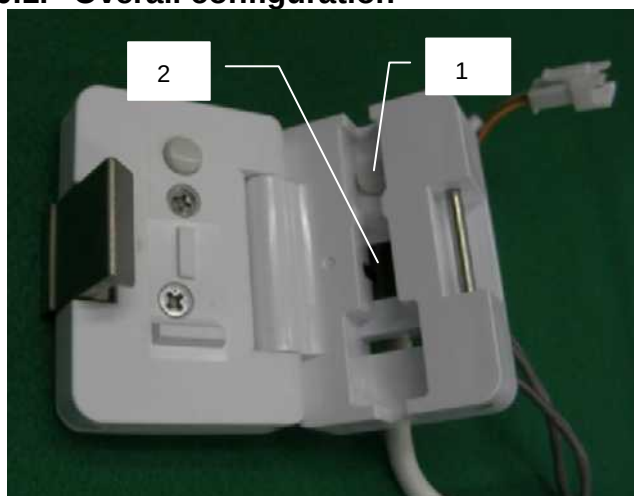
(1) Clamp leakage test

BHAJ**-012

10. Bubble Detector and Blood Sensor



10.1. Overall configuration



10.2. List of components

No.	Name	Model No.	Code No.	Note
1	Bubble Detector	-		BD
2	Blood Sensor	-		BS

10.3. Removal and installation

<Illustration>



<Removal>

⚠ Caution:

- Be sure to turn OFF the breaker

Remove the right and left side face cover at the upper part.

Remove the right side face electric cover.



Remove the screws that fix clamp Assy and place the clamp Assy in the device.

⚠ Caution:

- Be careful not to cause damage on the clamp harness and L5 harness



Remove CN210 on the I/O board.

Remove all the bands that bind bubble sensor harness.



Remove the four screws that fix bubble sensor.



Remove the bubble sensor from the device body.

<Installation>

Install the parts in the reversal order of the removal procedure.

10.4. Disassembly and assembly

<Illustration>



<Assembly>

CAUTION

- Do not touch the solenoid with your bare hands just after the operation since it is extremely hot.

1. Remove two fixing screws and take out the blood sensor.

2. Remove the plate that fix the bubble detector, and take out the ceramic oscillator.

3. Remove the inner cover at the top cover side.

4. Take out the ceramic oscillator.

<Assembly>

1. Install the parts in the reversal order of the removal procedure.

10.5. Inspection and replacement

Clean the oscillation device well.

I/O board parts:

If the voltage between TP1 on the bubble detector module on the I/O board and WJ1 on I/O board is 0V after inserting a water tube in the bubble detector,
replace the bubble detector sensor unit.

If the TP1-WJ1 voltage is still 0V at the re-test even after replacing the sensor unit,
replace the board since the board may be defective

Others:

Modify or replace the deformed, damaged or defective parts.

10.6. Adjustment procedure

Test to be performed:

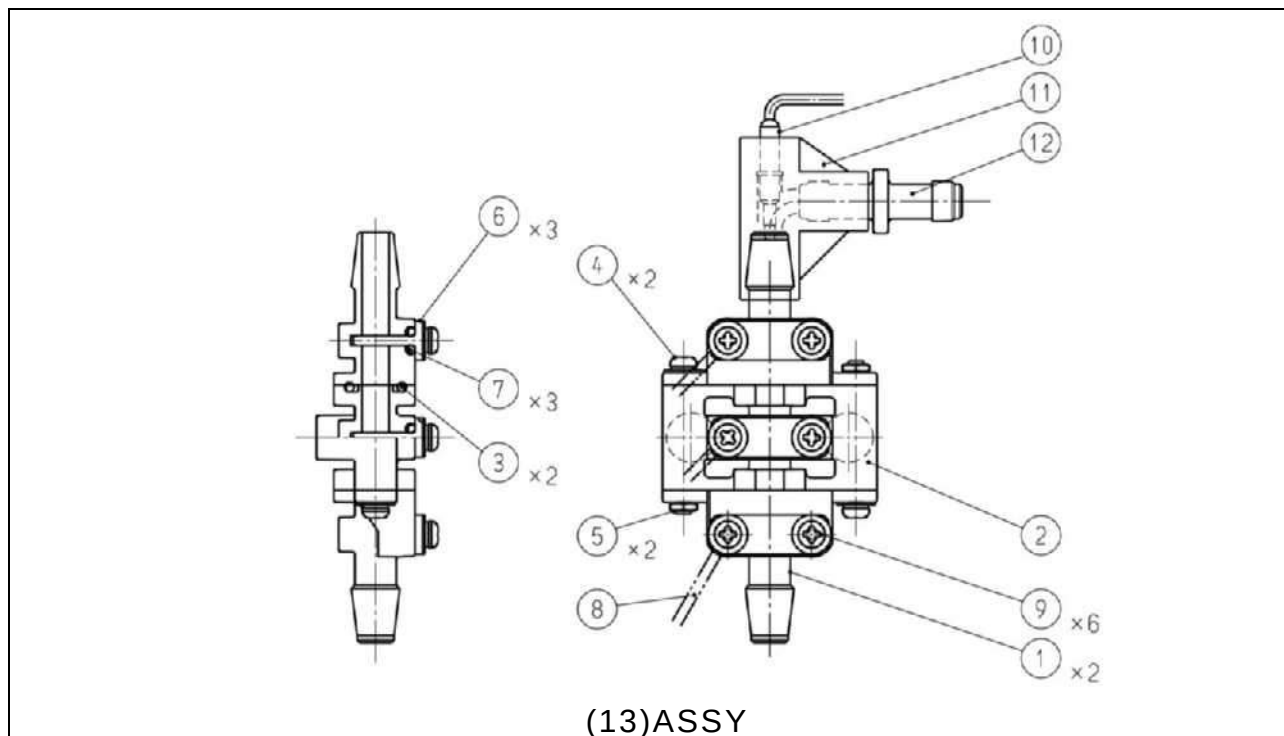
- (1)Bubble Detector calibration
- (2)Blood Sensor calibration

CXAJ**-006
CXAJ**-023

11. Conductivity Sensor



11.1. Overall configuration

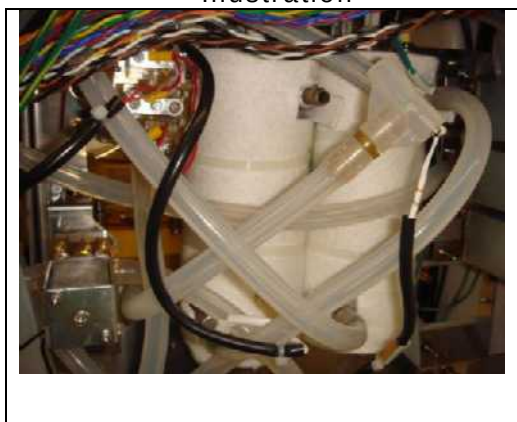


11.2. List of components

No.	Name	Model No.	Code No.	Note
1	Sensor holder A	01931100191	-	
2	Sensor holder B	01931100192	-	
3	O-ring	P9	-	Silicon rubber
4	Screw	M3x35	-	With a flat washer and a spring washer, SUS304 0.39N·m(4kgf·cm)
5	Nut	For M3	0-201-121	SUS304
6	Pin	01931000386	-	
7	O-ring	P2	-	Silicon rubber
8	harness	-	-	
9	Screw	M3x10	-	With a flat washer and a spring washer, SUS304 0.39N·m(4kgf·cm)
10	Thermistor	E3M-42D-S36		
11	Sensor holder	93892511001	-	Silicon rubber
12	Hose connector	01930600070	-	-
13	Conductivity sensor Assy	90573800	-	CD1,CD2,CD3

11.3. Removal and installation

<Illustration>



<CD1>



<CD2><CD3>

<Removal>

Open the right side cover.

Block the tubes on the both side of the conductivity sensor with forceps.

Then remove the tubes.

Loosen two screws that fix the assembly, and take out the sensor assy.

Remove the screws that fix the electrode on the harness side. Then remove the harness.

Remove two screws that fix the temperature sensor. Then remove the harness.

Remove CD2 and CD3 as the same step as CD1.

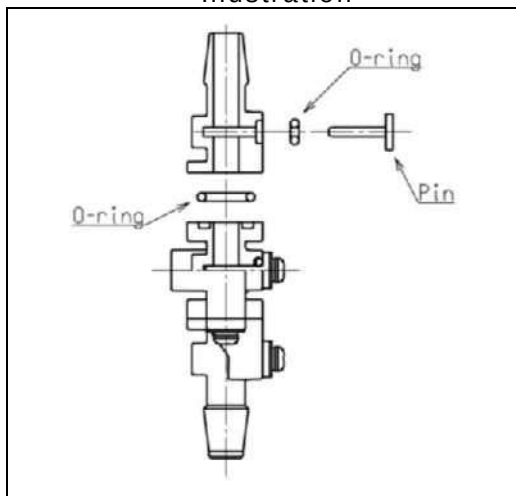
You may open the left side cover to remove the CD2 and CD3.

<Installation>

Install the parts in the reverse order of the removal procedure.

11.4. Disassembly and assembly

<Illustration>



<Disassembly>

Remove the six screws (9).

Draw out the pin (6).

Remove the O-ring(7) from the pin(6), and replace it.

Remove the two screws (4).

Remove the O-ring(3) from the sensor holder B(6), and replace it.

<Illustration>

<Assembly>

Assemble the parts in the reverse order of the disassembly procedure.

Tightening torque for the screw(4) and screw(9) shall be $0.39\text{N}\cdot\text{m}$ ($4\text{kgf}\cdot\text{cm}$) at assembling them.

 CAUTION

- When installing the O-ring(7) to the pin(6), make sure to put some water on the O-ring to make it more slippery. Any damage or twist on the O-ring(7) may cause liquid leakage.
-

11.5. Inspection and replacement

- Wash disassembled parts that have touched liquid well.
- Clean disassembled parts that have not touched liquid well.
- Replace parts at their time for replacement.

Sensor holder A,B:

Replace the damaged or cracked holder.

O-ring:

Replace the damaged O-rings.

Others:

Modify or replace the deformed, damaged or defective parts.

11.6. Confirmation procedure

- Confirm that there is no liquid leakage in operation.
- Check the displayed concentration and the actual concentration.

When concentration is stabilized in the state of preparation, take sample liquid out of the sample port to measure the actual concentration by tools like FPD, or flame photometric detector.

11.7. Adjustment procedure

- (1) Adjust the displayed concentration so that it is within $\pm 1 \text{mEq/L}$ of the actual concentration.

12. Flow Sensor Assy



Please refer to 2.3.9 and 2.3.10 of the Maintenance parts list.

12.1. Removal and installation

<Illustration>



<Removal>

1. Open the right side cover.
2. Remove 2 bolts that fix the FS1.
3. Block the tubes at the IN and OUT sides with forceps.
4. Then remove the tubes and the flow sensor.

<FS1>



1. Remove FS2 as the same step as FS1.

CAUTION

- Put waste cloths so that liquids are not spilled in the machine.

<Installation>

1. Install the parts in the reverse order of the removal procedure.

12.2. Disassembly and assembly

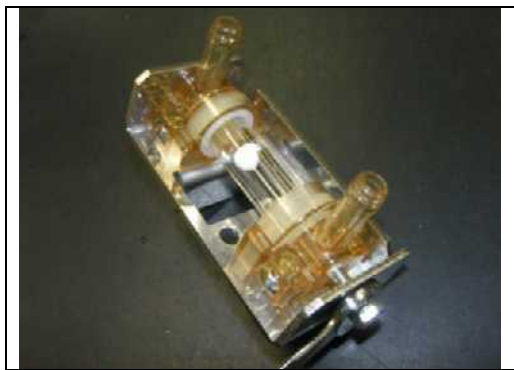
<Illustration>



<Disassembly>

1. Remove the screw (10), and take out the photo sensor (5).

<Illustration>



<Disassembly>

2. Loosen the nut (6) and hexagon bolt (7).



3. Remove the screw(9)



4. Disassembled.

<Illustration>

<Assembly>

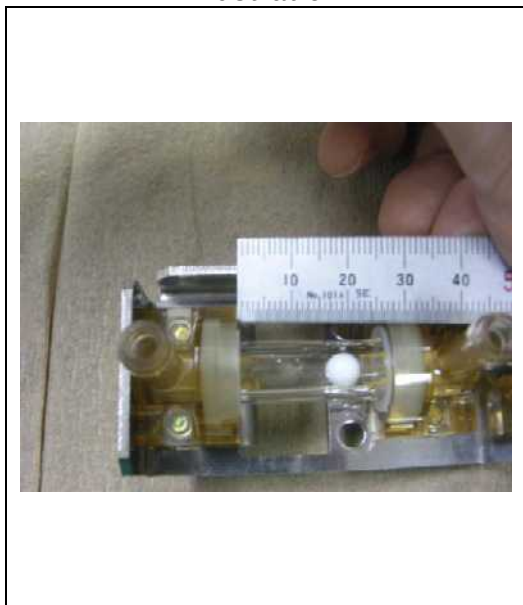
1. Set the nut (8) in the block (2) and (11).

2. Set the packing (3) in the block (2) and (11).

3. Put the pipe (6) or taper pipe (6) on the one side, and then install the teflon ball (4).

4. Set the block (2) and (11) in the both side of the pipe (6) or the taper pipe (6).

<Illustration>



<Assembly>

5. Temporarily fix the pipe which has been set to FS bracket (1).

⚠ CAUTION

- When assembling FS2, attach the taper pipe (6) so that its line mark faces to the opposite of the hexagon bolt side.
 - Attach the block (11) to the opposite of the hexagon bolt side.
-

6. Fix the block (2) or (11) of the opposite side of the hexagon bolt.

7. Tighten the hexagon bolt (7) so that the gap between blocks is 26 ± 0.2 mm.

8. Fix the block (2) of the side of the hexagon bolt.

9. Install the photo sensor (6) or (5) to the FS bracket (1).

The parts have been completely assembled.

Note: The parts number in this procedure corresponds to those for FS2.

12.3. Inspection and replacement

Packing

Replace a deformed, discolored, hardened, or impressed packing.

Photo sensor:

Replace a defective sensor.

Others:

Modify or replace defective parts.

12.4. Confirmation procedure

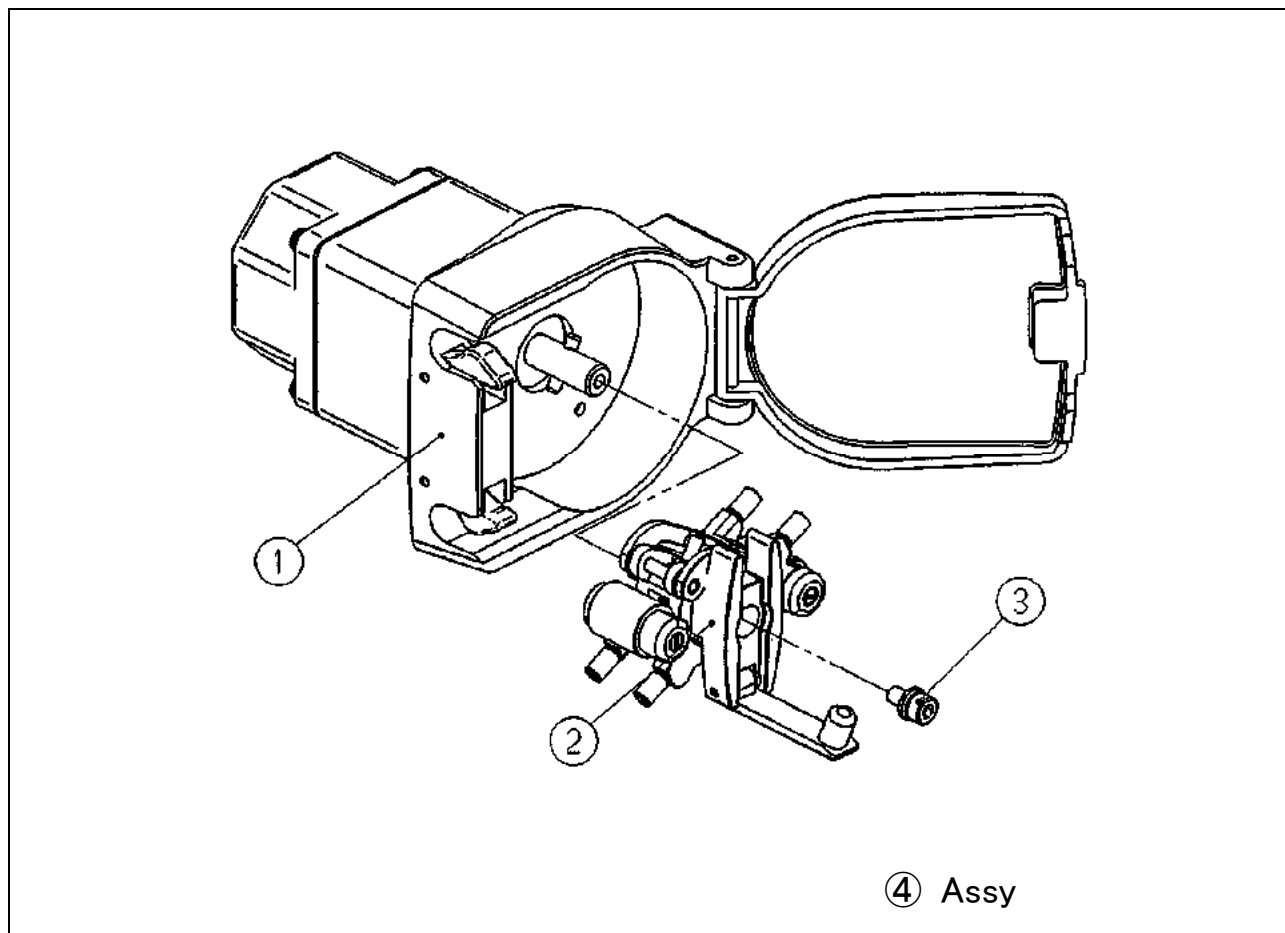
On the Flowchart screen check that On is displayed in the flow sensor normal status (with no liquid) and OFF in the reversal status (with liquid).

Adjust the photo sensor position if it does not normally operate.

13. Blood Pump



13.1. Overall configuration



13.2. List of components

No	Name	Model No.	Torque	Note
1	Housing Assy	—		
2	Rotor Assy	—		With a hand crank
3	Hexagon bolt with washer	M5x10	2. 9N·m	
4	Blood pump Assy	MRP22-011		

13.3. Removal and installation

<Illustration>



<Removal>

⚠ Caution

-
- Be sure to turn OFF the breaker.
-

Remove the right and left side face cover at the upper part.

Remove the left side face electric cover.

Remove the three screws fixing the blood pump from the front of the device.

Remove the power connector of the blood pump.

Remove the BP cover harness CN113 from the power board.
Remove all the banding bands then.

Remove the grounding cable.

Remove the blood pump from the device.

<Installation>

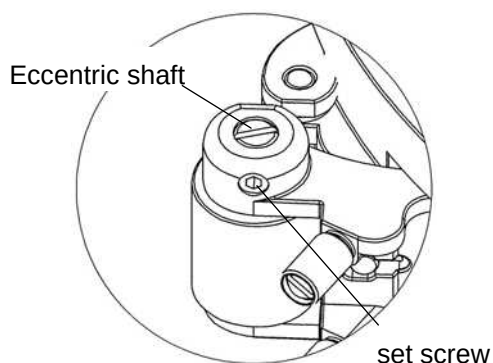
Install the parts in the reversal order of the removal procedure.

13.4. Adjust the gap for Small diameter tube



⚠ Caution

- Do not use a hexagon wrench whose head is round. It may damage the screw.
- The standard of the blood circuit is as follows. Using any other circuits is not guaranteed.
- Large diameter : $\phi 8.00 \times \phi 12.00 \pm 0.15\text{mm}$, L: 210mm or more, $t=2.00 \pm 0.1\text{mm}$
- Small diameter : $\phi 6.35 \times \phi 9.75 \pm 0.15\text{mm}$, L: 210mm or more, $t=1.70 \pm 0.1\text{mm}$



1. Lift the lever of the rotor and remove the screw.
2. Loosen the set screw using a hexagon wrench. (2mm width across flats).
3. Rotate the eccentric shaft using a flat head screwdriver to adjust the roller radius to 37.3mm (74.6mm in diameter)
4. Fasten the set screw maintaining the position of the eccentric shaft.



⚠ Caution

- Decenter the both rollers to the adjusting spring side.
- In case of large diameter tube, the radius is 37.1mm (74.2mm in diameter).

13.5. Adjustment procedure

Adjust in the following order

- (1) Maintenance Manual

9.4 Blood pump: Switching to the line with a small diameter tube

- (2) Confirmation of the blood pump shut-off status

BHAJ**-014

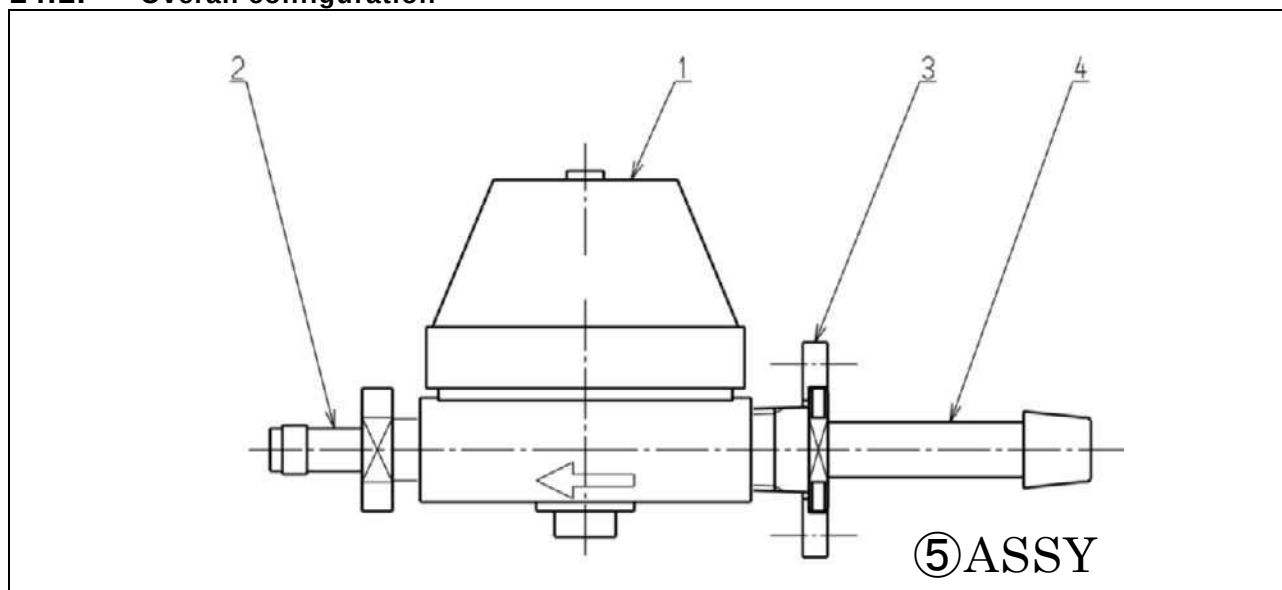
- (3) Blood pump flow adjustment

BHAJ**-015

14. Reducing Valve



14.1. Overall configuration



14.2. List of components

No	Name	Model No.	Torque	Note
1	Reducing valve body	ER0884ZBS8		
2	Hose opening	93312116129		
3	Flange	93442113002		
4	PR Hose opening	93282116113	2N·m	
5	PRV Assy	93892100		

14.3. Removal and installation

<Illustration>



<Removal>

Remove the lower right side face cover.

Shut off the tube on the IN side of the reducing valve by a pean pincer, and then pull out the tube.

Remove the screws on the reducing valve at the back of the device.

Shut off the tube on the OUT side of the reducing valve by a pean pincer, and then pull out the silicon tube.

⚠Caution:

- Put waste cloths so that liquids are not spilled in the device.。

<Installation>

Install the parts in the reversal order of the removal procedure.

14.4. Disassembly and assembly

<Disassembly>

Remove the hose opening ② and PR hose opening ④ from the reducing valve body ①.

<Assembly>

Wind up the seal tape (width 13mm) to Hose opening ② and PR hose opening ④ 2.5 rounds.

Silicone seal paste must be applied to the seal tape when Hose opening ② and PR hose opening ④ are assembled.

Make sure that the tightening torque of the hose opening ② is $2\text{N} \cdot \text{m}$ (20kgf·cm).

14.5. Inspection and confirmation

- Cleanse disassembled parts very carefully.
- Confirm that there are no foreign objects inside.
- Replace Assy if there is rust taking place inside.
- Replace those with modification, damage, or malfunction.
- Confirm that there is no fluid leakage in operation.

14.6. Adjustment procedure

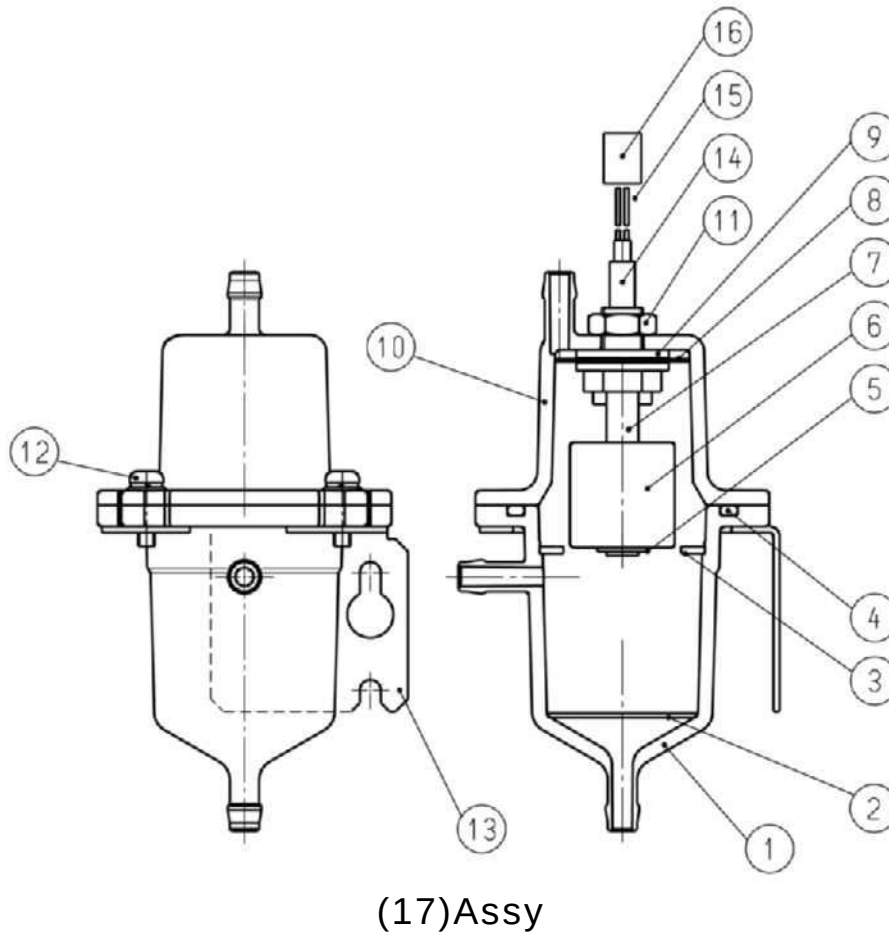
Implement adjustment in the following order:

- | | |
|--|------------|
| (1) Adjustment of Supplying Flow
and Fill-up Pressure | CXAJ**-009 |
| (2) Balance test | CXAJ**-016 |
| (3) UF precision test 1 | BHAJ**-018 |
| (4) UF precision test 2 | BHAJ**-019 |

15. Air Elimination Tank



15.1. Overall configuration



(17) Assy

15.2. List of components

No.	Name	Model No.	Code No.	Note
1	AS Case	-		
2	Wire net	91933015002	0-201-013	
3	AS Plate	-	-	
4	O-ring	G50	9-001-121	
5	Ring	For OLV-2P(Special)		
6	Float	For OLV-2P(Special)		
7	Float switch shaft	For OLV-2P(Special)		
8	Mesh	01930200089	0-501-030	
9	Packing	01930200088	0-201-016	
10	AS2 Case	-		
11	Nut	-		Tighten by hand, and then give a 1/4revolution by tool.
12	Screw	M4x15	-	With a flat washer and a spring washer, 1.18N·m(12kgf·cm)
13	AS Bracket			
14	Tube	-		
15	Pin	-		
16	Connector	-		
17	Air elimination tank Assy	90572400	-	
18	Float switch Assy	OLV-2P(Special)		(5),(6),(7),(11)

15.3. Removal and installation

<Illustration>



<Removal>

Open the left side cover.

Loosen two screws that fix the air elimination tank assembly, and take out the assembly.

Block the tubes connected to the hose connectors at the IN, OUT and AIR OUT sides with forceps.

When removing the tube, remove liquid from the air elimination tank to the container.

After removing liquid from the tank, remove the tube at the IN side, and remove the connector of the FSW harness. Then take out the assembly.

⚠ CAUTION

- Put waste cloths so that fluids are not spilled in the machine.
-

<Installation>

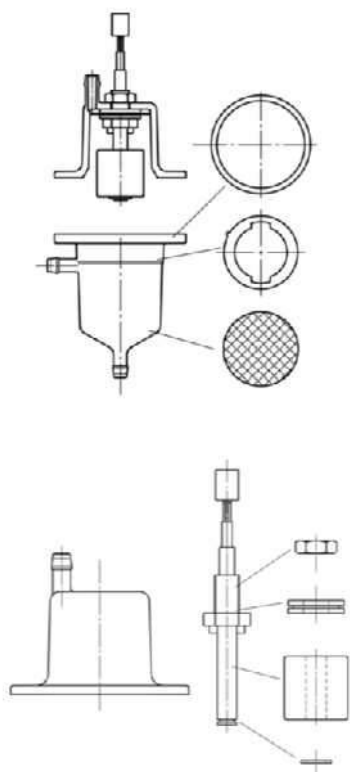
1. Install the parts in the reverse order of the removal procedure.

⚠ CAUTION

- Make sure to bundle the tubes using tie wraps.
-

15.4. Disassembly and Assembly

<Illustration>



<Disassembly>

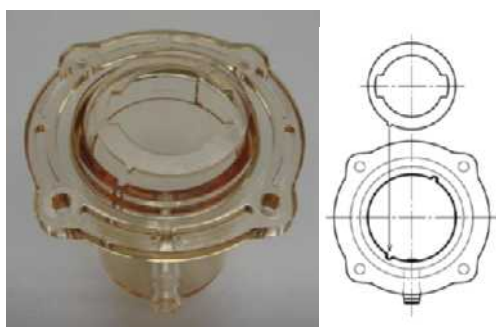
Remove screw(12),and disassemble the tank into

AS case (1),
AS2 case (10),
AS Plate (4),
and O-ring (4).

Remove Ring(5) and pull out float(6).

By using a special tool (if there is none, use tweezers, etc.), draw out the connector pin.

Remove float switch shaft(7), and then remove mesh(8) and packing(9).



Remove the AS Plate(4)



Remove wire net (2) from the port part of AS Case (1) using a screwdriver.

Wash the wire net (2).

<Assembly>

Assemble the parts in the reverse order of the disassembly procedure.

Direction of the float

Put the "CLOSED" surface to the side o ring (5)

**CAUTION**

-
- Fasten the tightening nut (11) tightly by hand, and then give a 1/4 revolution with a tool.
-

15.5. Inspection and replacement

- Wash disassembled parts well.
- Replace deformed or damaged parts.
- Replace parts at their time for replacement.

Float:

Replace those over 8 grams.

Packing:

Replace impressed parts.

O-ring:

Replace scratched parts.

Float switch:

Check conductivity by a tester. Replace the defective one.

Wire net & mesh:

Wash with water and confirm there is no clogging.

Others:

Modify or replace deformed, damaged or defective parts.

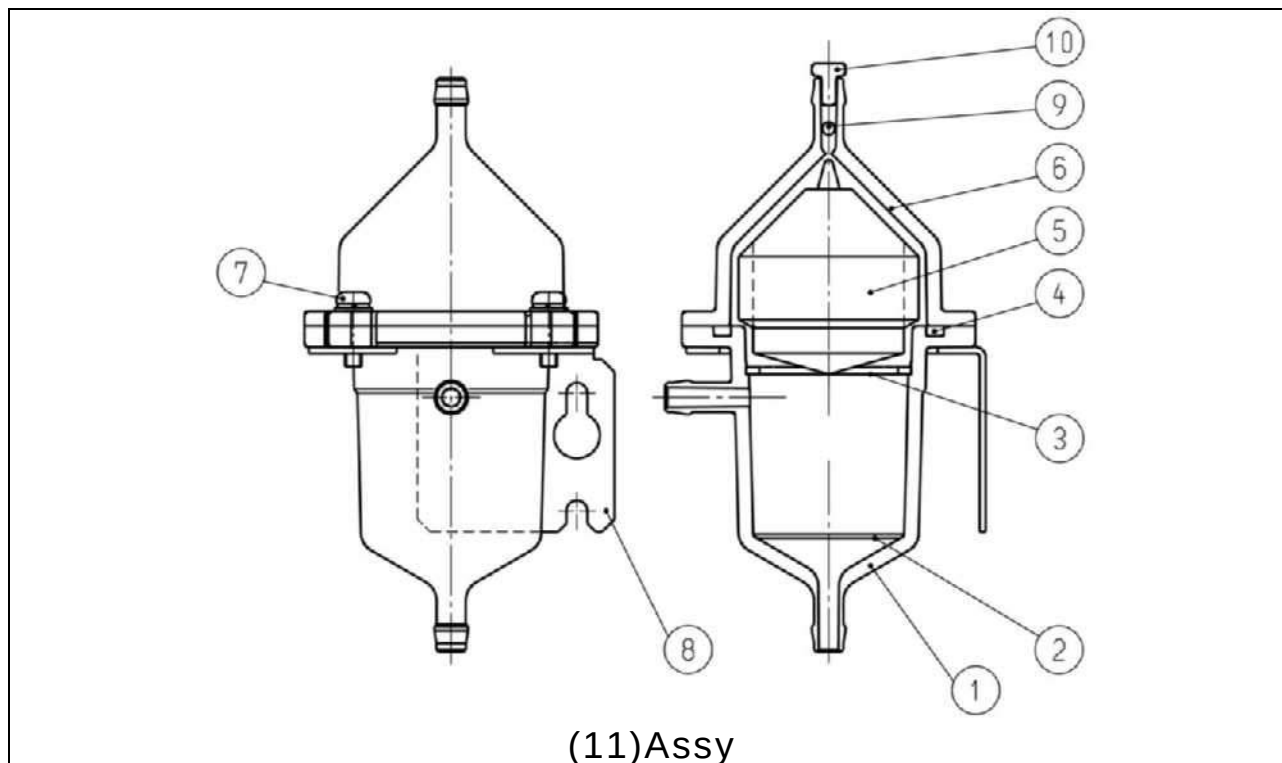
15.6. Confirmation procedure

- Confirm that there is no liquid leakage in operation.
- Infuse air in the preparation process and confirm that automatic gas purge is implemented.

16. Degassing tank



16.1. Overall configuration



16.2. List of components

No.	Name	Model No.	Code No.	Note
1	AS Case	90352311100	-	
2	Wire net	91933015002	0-201-013	
3	AS plate	90352314100	-	
4	O-ring	G50	9-001-121	Viton
5	Float	93898213100	0-501-143	
6	AS1 Case	90352312100	-	
7	Screw	M4x15		SUS304 1.18N·m(12kgf·cm)
8	AS Bracket	90612311001	-	
9	Steel ball	Grade28 spare1/8		SUS316
10	Stopper	90402316001		
11	Degassing tank Assy	90572300		

16.3. Removal and installation

<Illustration>



<Removal>

Open the left side cover of the machine.

Loosen the screws that fix the degassing tank.

Pull out the degassing tank from the machine, block the silicon tubes on the IN side, the OUT side, and the air OUT side with Pean forceps, and then pull out the silicon tube on the OUT side.

Turn the OUT side of the degassing tank to the container, and pull out the silicon tube on the air OUT side. Move liquid in the degassing tank to the container at that time.

⚠ CAUTION

- Do not lose the steel ball (10) when removing the silicone tube at the AIR OUT side.
 - Put waste cloths so that liquid is not spilled in the machine.
-

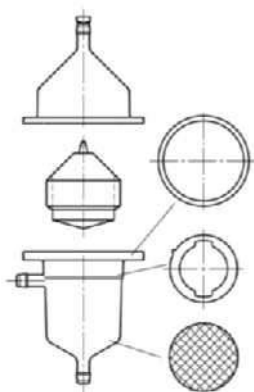
Pull out the silicon tube on the IN side, and remove the degassing tank from the machine body.

<Installation>

Install the parts in the reverse order of the removal procedure.

16.4. Disassembly and assembly

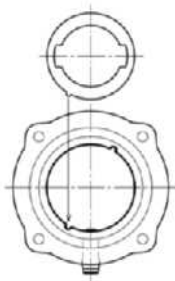
<Illustration>



<Disassembly>

Remove screws (7) and disassemble the assembly into the followings:

- AS case (1)
- AS1 case (6)
- AS Plate (3)
- Float (5)
- Wire net (2)
- O-ring (4)



Remove the AS Plate



Remove the wire net (2) from the port of AS case (1) with a screw driver.

Wash the wire net (2).



Pull the stopper (10), and take out the steel ball (9).

- The steel ball is small. Take care not lose it.
- Do not lose the steel ball when removing the tube since its stopper remains in the tube.

Stopper
Steel ball

<Assembly>

Assemble the parts in the reverse order of the disassembly procedure.

16.5. Inspection and replacement

- Wash the disassembled parts well.
- Replace parts at their time for replacement

Float:

Replace the deformed or damaged parts

Packing:

Replace those with impressed parts.

O-ring:

Replace scratched parts.

Wire net & mesh:

Wash with water and check for clogging.

Others:

Modify or replace deformed, damaged or defective parts.

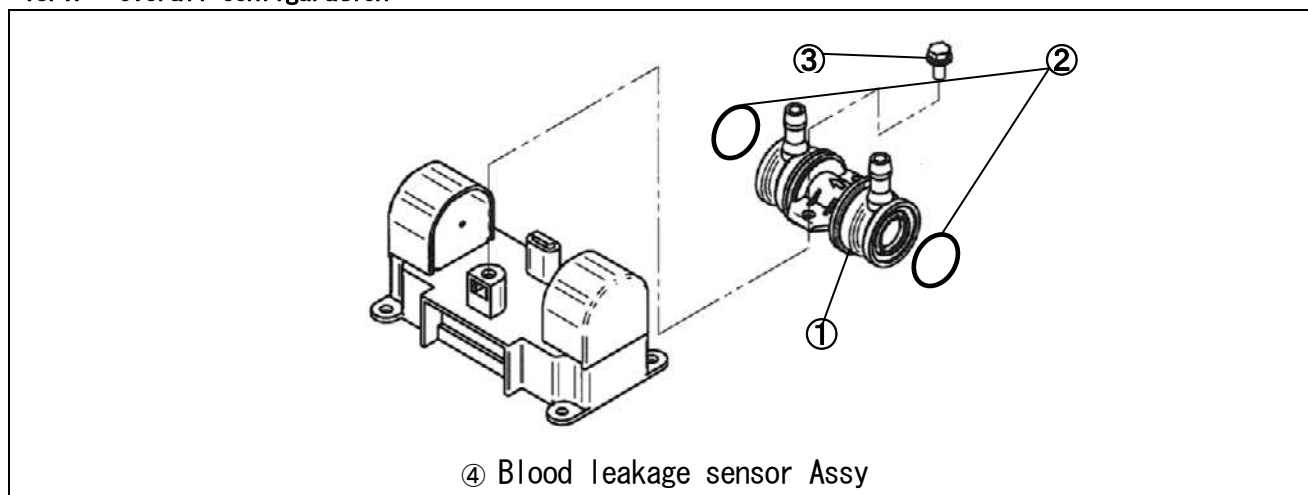
16.6. Confirmation procedure

- Confirm that there is no liquid leakage in operation.
- Confirm that the air in the degassing tank will be emitted to the outside when the machine is in operation.

18. Blood Leakage Sensor Assy



18.1. Overall configuration

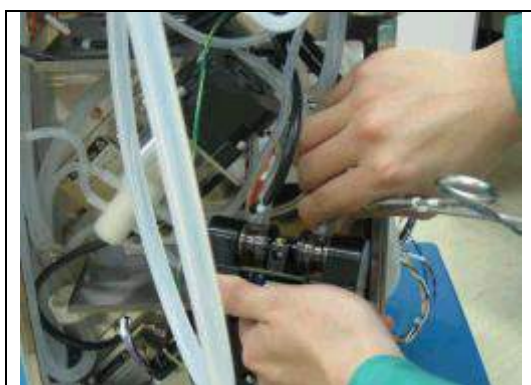
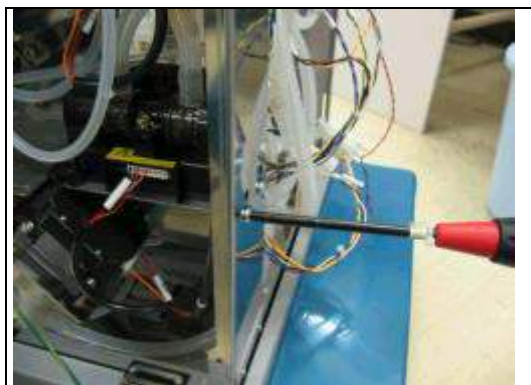


18.2. List of components

No	Name	Model No.	Torque	Note
1	Cartridge			
2	O-ring	P16		
3	Screw	M4x10	0.8N·m	
4	Blood leakage sensor Assy	BLD-01		

18.3. Removal and installation

<Illustration>



<Removal>

⚠ Caution:

-
- Be sure to turn OFF the breaker.
-

Remove the front cover and the lower right side face cover.

Remove the connector connected to the blood leakage sensor.
Remove the blood leakage sensor off together with the bracket.

Shut off silicon tubes on both sides of the blood leakage sensor by a pean pincer, and then pull out the silicon tubes.

Remove the blood leakage sensor from the bracket.

⚠ Caution:

-
- Put waste cloths so that liquids are not spilled in the device.
-

<Installation>

Install the parts in the reverse order of the removal procedure.

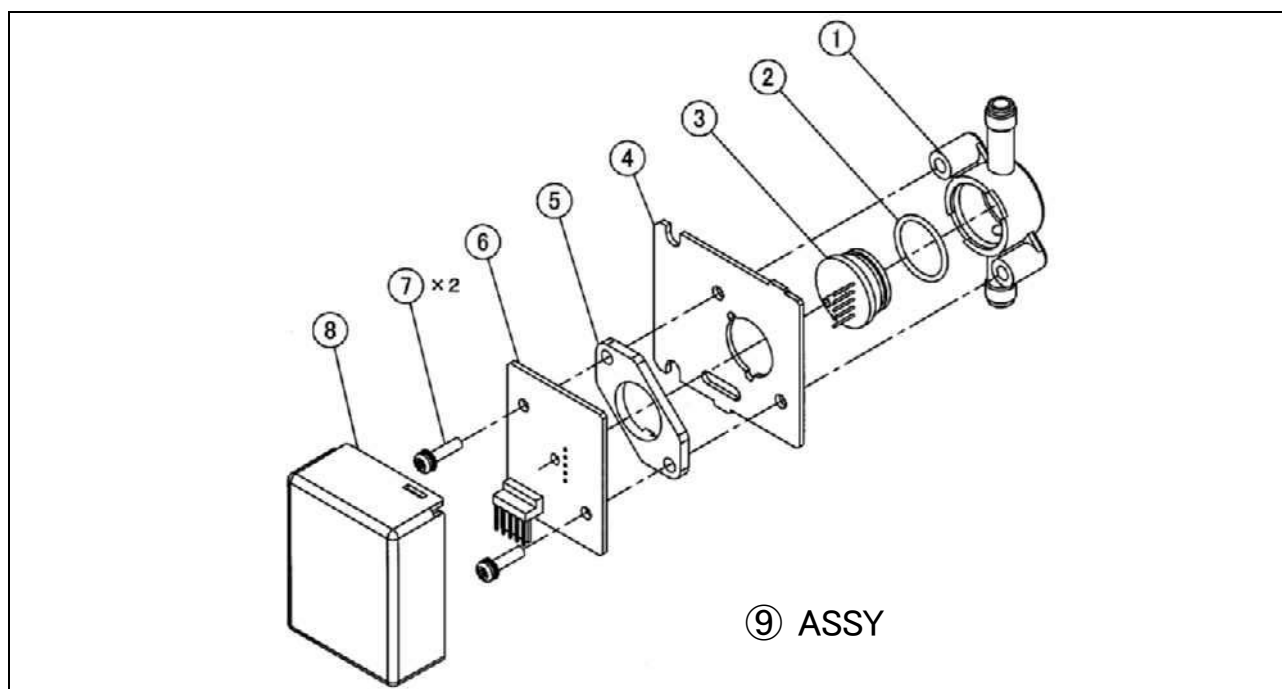
18.4. Adjustment procedure

- (1) Blood leakage sensor calibration BHJ**007

19. Dialysate Pressure Sensor



19.1. Overall configuration



19.2. List of components

No.	Name	Model No.	Torque	Note
1	Manifold	93882911100		
2	O-ring	S-16		
3	Pressure transducer	P-8505-102G-012		
4	Base plate	93882912003		
5	Spacer	93882913001		
6	Dialysate pressure sensor board	SSM-05025 (SBY0213)		
7	Screw	M3x12	0.8N·m	With a flat washer and a spring washer
8	Board cover	93882914100		
9	Dialysate pressure sensor Assy	PD-02		

19.3. Removal and installation

<Illustration>



<Removal>

⚠ Caution:

-
- Be sure to turn OFF the breaker.
-

Remove lower left-side cover.

Remove the screws that fix the dialysate pressure sensor connector and device (those of the other side are to be just loosened), and remove the dialysate pressure sensor.

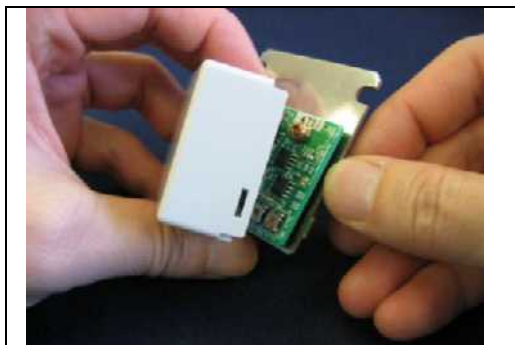
Shut off the silicon tubes on either end of the dialysate pressure sensor by a pean pincer, and then pull out the silicon tube.

<Installation>

Install the parts in the reverse order of the removal procedure.

19.4. Disassembly and assembly

<Illustration>



<Disassembly>

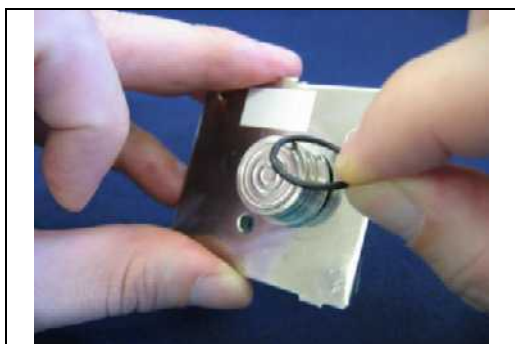
Remove the board cover ⑧.



Remove the screw ⑦.



Pull out the manifold ① from the pressure transducer ③.



Remove the O-ring ②.

⚠ Caution:

- Do not touch the pressure receiver surface of the dialysate pressure sensor. It may cause failure.

<Assembly>

Assemble the parts in the reverse order of the disassembly procedure.

⚠ Caution:

- Put some water on O-ring ② so as not to hurt it, and slowly insert it into the manifold ①.

19.5. Inspection and replacement

- Cleanse the disassembled parts very carefully.
- Replace the parts whose regular replacement time has come.

Manifold:

Replace those with injuries and cracks.

O-ring:

Replace those with injuries.

Others:

Modify or replace those with modification, damages, or malfunction.

19.6. Confirmation work

When you put the pressure of +700mmHg on the manifold, make sure that the leakage is within 9mmHg/20sec.

Furthermore, when you put the pressure of -700mmHg on the manifold, also make sure that the leakage is within 9mmHg/20sec.

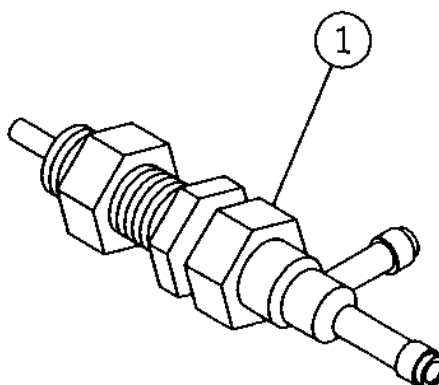
19.7. Adjustment procedure

- (1) Dialysate pressure sensor calibration BHAJ**-002

20. Needle Valve



20.1. Overall configuration



20.2. List of components

No.	Name	Model No.	Code No.	Note
1	Needle valve	-		NV1,NV4,NV5,NV6

20.3. Removal and installation

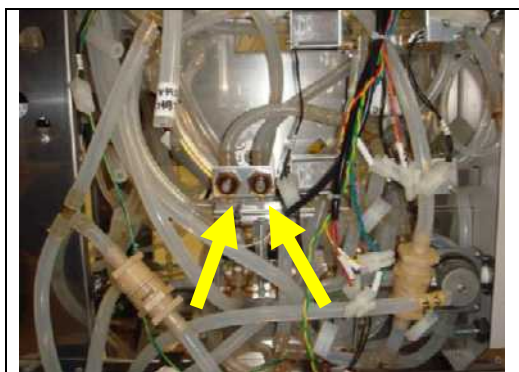
<Illustration>



<NV1>



<NV4>



<NV5>

<NV6>

<Removal>

1. Open the covers.
2. Block the silicon tubes on both sides of the needle valve with Pean forceps, and pull out the tubes.
3. Loosen the screws that fix the needle valve and take out the needle valve.
4. Take out the other needle vales by the same procedure.

⚠ CAUTION

- Put waste cloths so that liquids are not spilled in the machine.

<Installation>

1. Attach the parts in the reverse order of the removal procedure

⚠ CAUTION

- After replacing the needle valve, be sure to bundle tubes by tie wraps.

20.4. Confirmation work

- Confirm that there is no liquid leakage in operation

20.5. Adjustment procedure

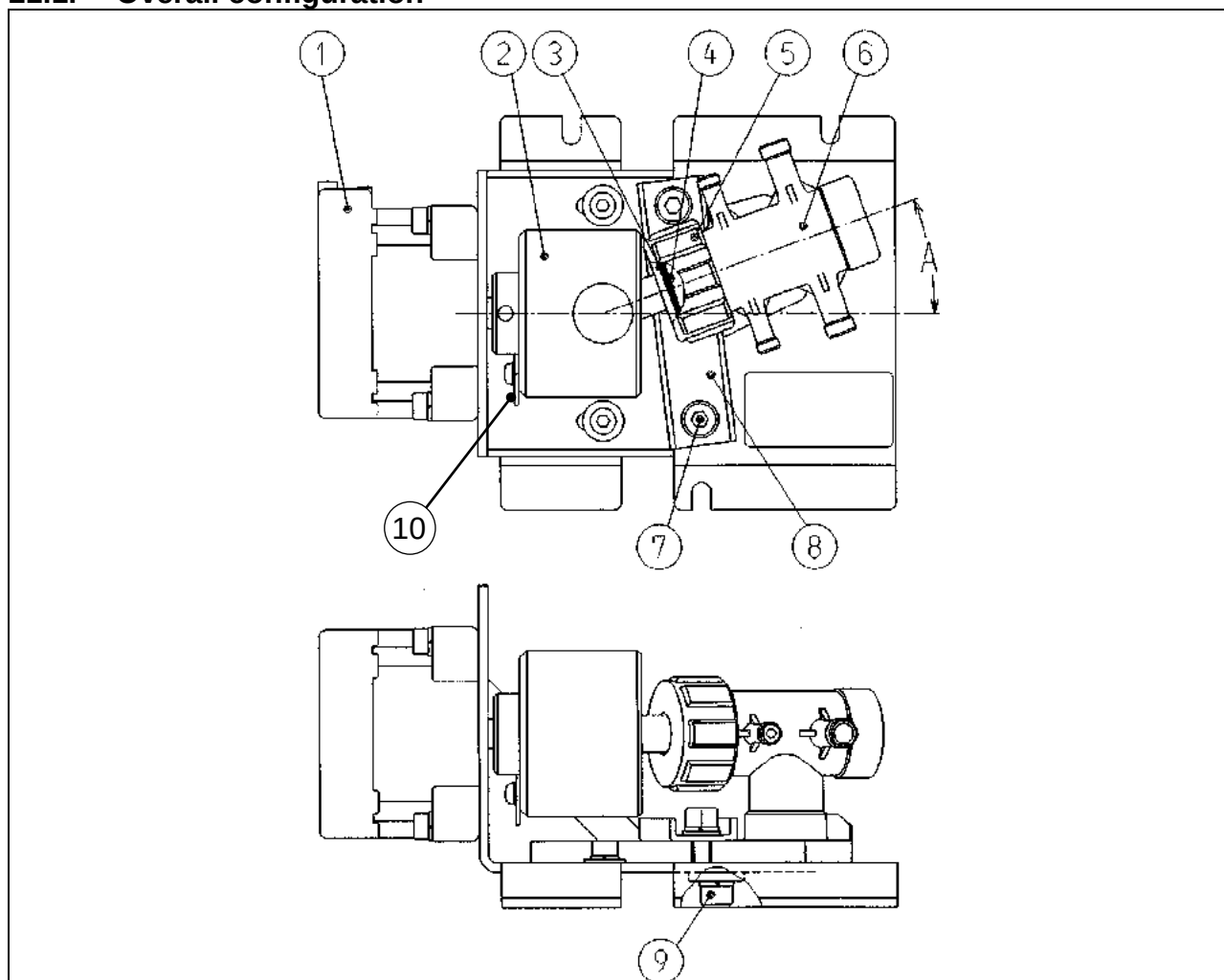
- (1) Adjustment of Supplying Flow and Fill-up Pressure

CXAJ**-009

21. Piston Pump



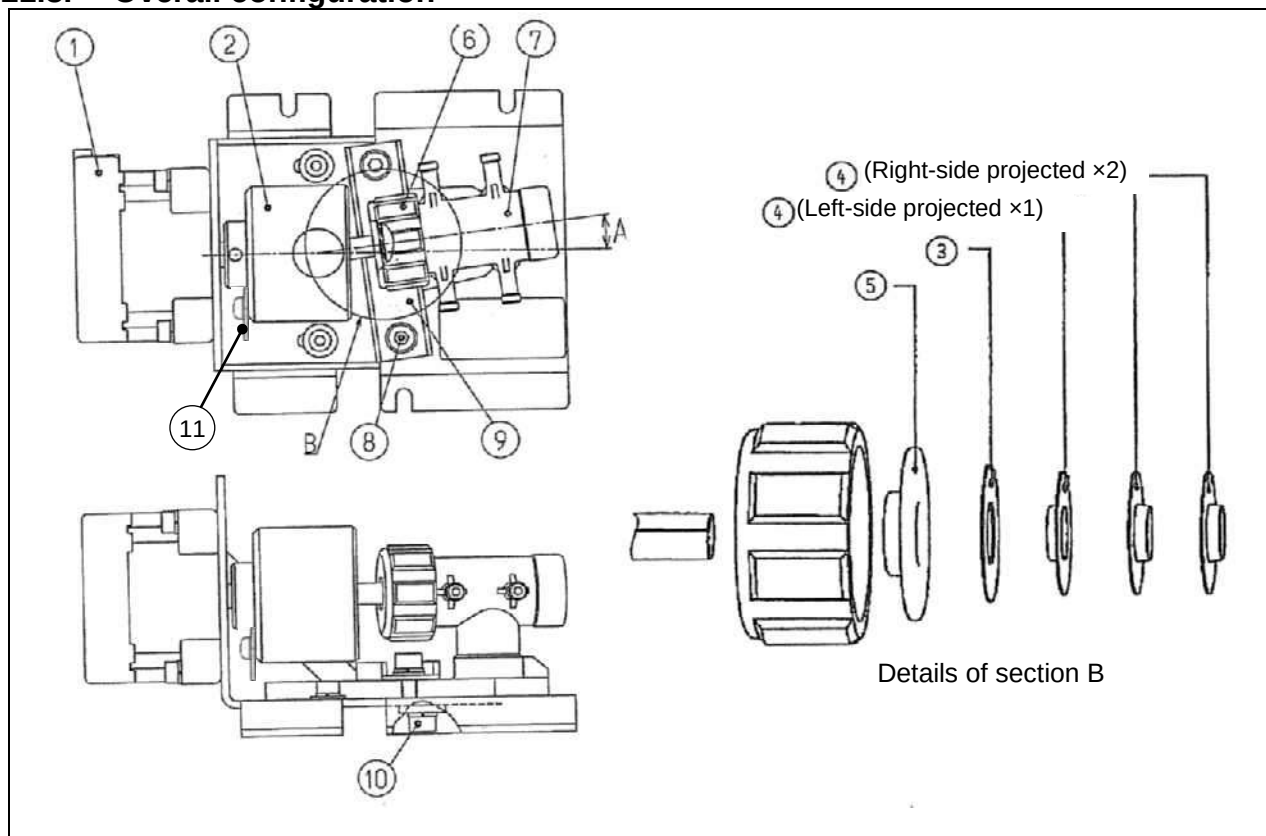
21.1. Overall configuration



21.2. List of components

No.	Name	Model No.	Code No.	Note
1	3-phase Stepping motor	KT56JM4-009		
2	Drive joint	-		
3	Back sheet	-	0-501-049	For $\phi 10$
4	Lip seal	-	0-501-050	For $\phi 10$
5	Hexagon cap nut	-		
6	Pump head Assy	2A03768		(3),(4),(5),(6) included
7	Fixed screw	-		2.9N·m
8	Fixed plate	-		
9	Set screw	-		2.9N·m
10	Dog	-		
11	Ultrafiltration pump Assy (Piston pump)	V-10AUM-P1H	0-501-168	SSM-08011

21.3. Overall configuration



21.4. List of components

No.	Name	Model No.	Code No.	Note
1	3-phase stepping motor	KT56JM4-009		
2	Drive joint	-		
3	Back sheet	-	0-501-049	For $\phi 10$
4	Lip seal	-	0-501-050	For $\phi 10$
5	Coned disk spring	-		
6	Hexagon cap nut	-		
7	Pump head Assy	-		(3),(4),(5),(6) included
8	Fixed screw	-		2.9N·m
9	Fixed plate	-		
10	Set screw	-		2.9N·m
11	Dog			
12	Pump for A concentrate Assy(Piston pump) (P3)	V-10AFP06-XP-43		SSM-12072 Infusion volume:0.3(mL/r)
	Pump for B concentrate Assy(Piston pump) (P4)	V-10AUM-P3BH	0-501-168	SSM-08014 Infusion volume:0.3(mL/r)

21.5. Removal and installation

□

<Illustration>



<Removal>

CAUTION

-
- Be sure to turn OFF the power breaker.
-

1. Open the left side cover.
2. Remove the silicon tubes connected to the pump head.
 - The illustration is in the case of P3.

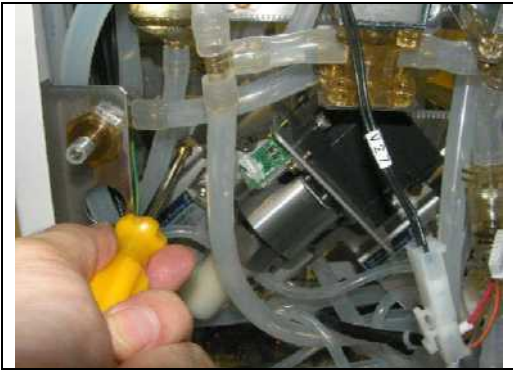


3. Remove the harness attached to the three-phase stepping motor.

Note: Remove the harness attached to the UFP sensor board.



4. Remove the V46 and V45 assy.



5. Remove three screws that fix the piston pump.



6. Take out the piston pump from the machine.

<Installation>

1. Install the parts in the reverse order of the removal procedure.

21.6. Disassembly and assembly

<Illustration>



Disassembly

1. Remove the positioning screw by the 5mm hexagon wrench.



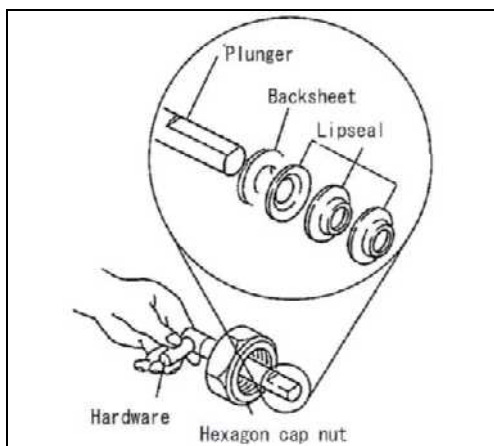
2. Remove the pump head assembly from the drive joint.



3. Remove the hexagon cap nut, the conical spring, the back sheet and the lip seals.

⚠ CAUTION

- Make sure to remove them very carefully by turning around one by one.



4. Being careful of the direction of the lip seals, attach the parts as shown in the left illustration.

⚠ CAUTION

- Be careful not to fold or injure the lip seals.
- After attaching the lip seals, wipe them off with alcohol to make sure that there is no hand oil or dust, etc., set onto the plunger.

<Assembly>

1. Assemble the parts in the reverse order of the disassembly procedure.

<Additional tightening>

The hexagon cap nut may loosen due to hot water. In this case, additionally tighten the nut in the Hot rinse process.

⚠ CAUTION

- Hot water approximately 90°C is running, and so make sure not to get burned.

<Angle A when pump head is fixed>

Name	Model No.	Angle A
UF Pump Assy(UFP)	V-10AUM-P1H	Approximately 20°
concentration solution pump Assy (P3)	V-10AFP06-XP-43	Approximately 7.9°
concentration solution pump Assy (P4)	V-10AUM-P3BH	Approximately 7.9°

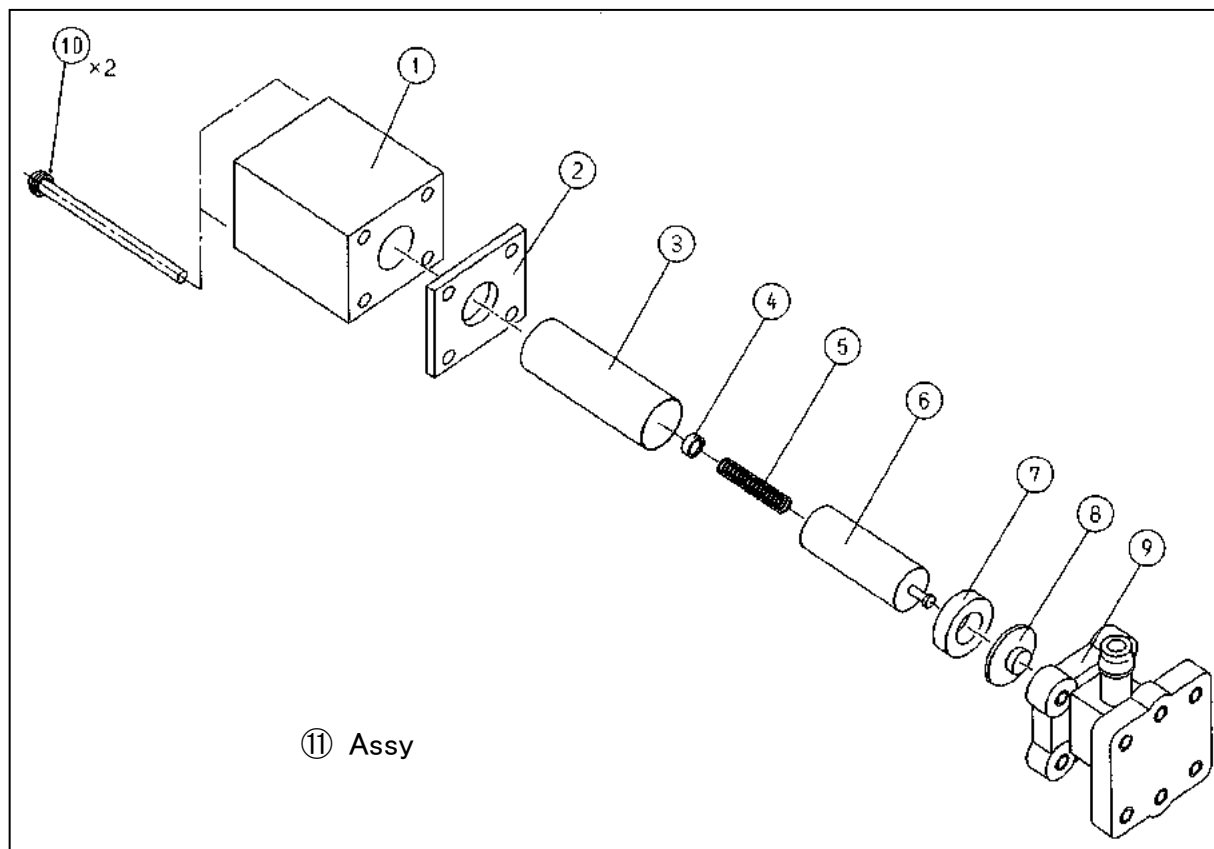
21.7. Adjustment procedure

- (1) Confirmation of Flow Volume of P3 and P4 CXAJ**-011
- (2) UFP flow measurement BHAJ**-017

22. Relief Solenoid Valve



22.1. Overall configuration



22.2. List of components

No.	Name	Model No.	Torque	Note
1	Actuator Assy	D3Y610		
2	Core	D3E370		
3	Packless pipe	D3I2600		
4	Sound deadening sheet	D3L490		
5	Spring	D2K130		
6	Plunger	D3I2430		
7	Diaphragm presser	D3H370		
8	Diaphragm	D3A540		
9	Valve body	D3B1810		
10	Screw	M3x45P2	0. 69N・m	
11	Relief solenoid valve	SV2-322-SRHW		V33

22.3. Removal and installation

<Removal>

⚠ Caution:

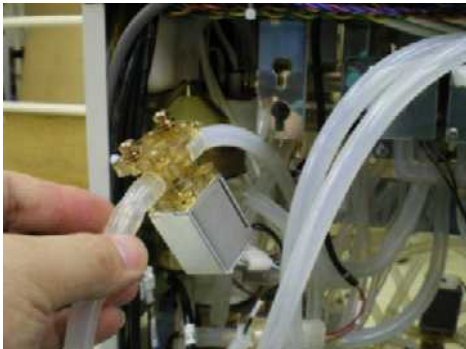
-
- Be sure to turn OFF the breaker.
-

Remove the lower left side face cover.

Remove the connector.



Loosen screws that fix the valve, and take out the relief solenoid valve.



Shut off the silicon tubes on both ends by a pean pincer, and then pull out the silicon tube.

⚠ Caution:

-
- Put waste cloths so that liquids are not spilled in the device.
-

<Installation>

Install the parts in the reverse order of the removal procedure.

22.4. Disassembly and assembly

<Illustration>



<Disassembly>

Remove the screws.



Actuator Assy is dissembled into:

Core
Packless pipe
Spring
Plunger
Diaphragm presser
Diaphragm
Valve body

Remove diaphragm from the plunger.

<Assembly>

Install the parts in the reverse order of the disassembly procedure.

⚠ Caution:

- When assembling the diaphragm, make sure that it has been tightly put into the plunger by turning it around two or three times. Also make sure that the tightening torque of the screws is $0.69\text{N}\cdot\text{m}$ ($7\text{kgf}\cdot\text{cm}$).

22.5. Inspection and replacement

- Replace the parts whose regular replacement time has come.
- Confirm that no foreign object is inside.
- Replace the parts with any modification, damage, and malfunction.

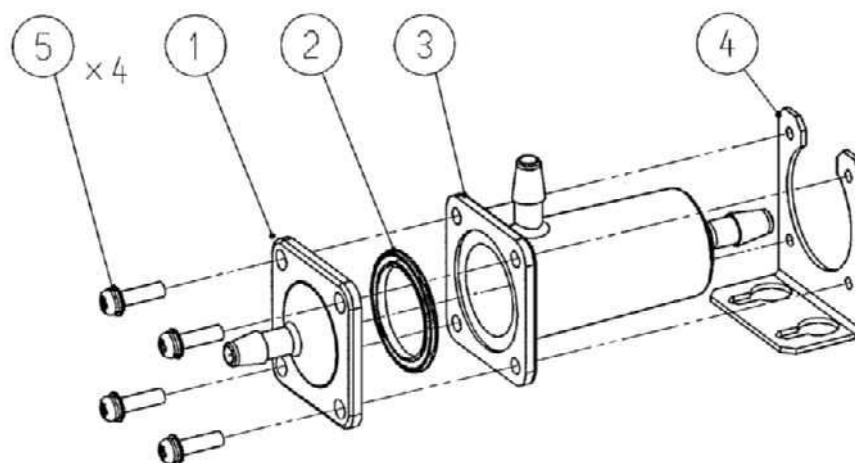
22.6. Confirmation work

- At Maintenance 4 "Operation status," manually turn on and off the two-way solenoid valve, and make sure that it is properly working by sound.
- Install a pressure gauge onto the OUT side of the relief solenoid valve, and make sure that pressure is released when the pressure of 0.15MPa (1.5kgf/cm²) $\pm 15\%$ is applied from the OUT side of the solenoid valve by syringe with no electricity conducted.
- Make sure that there is no fluid leakage in operation.



23. AS3

23.1. Overall configuration



⑥ Assy

23.2. List of components

No.	name	Model No.	torque	note
1	AS3 lid	90369712001		
2	Packing	90364413001		
3	Case	90364411001		
4	AS3 Bracket	90579714100		
5	screw	M4 × 15	1. 2N·m	SUS304
6	AS3	90369700		AS3

23.5. Removal and installation

<Removal>

Prepare a container to receive fluid from B Powder concentrate solution air elimination tank.

Remove the screws that fix the AS3.

Pull out the AS3 from the device, then shut off the silicon tubes on the IN side, the OUT side, and air OUT side by pean pincer, and pull out the silicon tube on the OUT side.

Put the OUT side of the AS3 to the container, and pull out the silicon tube on the air OUT side. Then, receive fluid from the AS3 by the container.

Pull out the silicon tube on the IN side, and remove the AS3 off the device body.

⚠Caution:

- Put waste cloths so that liquids are not spilled in the device.
-

<Installation>

Install the parts in the reverse of the removal procedure.

23.6. Disassembly and assembly

<Disassembly>

Remove the screw (5) and disassemble the AS3.

<Assembly>

Attach the parts in the reverse order of the disassembly procedure.

⚠ Caution

-
- Do not apply silicon or grease, etc., onto the fluid.
-

23.7. Inspection and replacement

Cleanse the disassembled parts very carefully.

- Replace the parts whose regular replacement time has come.
- Remove any foreign objects if they are attached.

O-ring:

Replace those with injuries.

Others:

Modify or replace parts with any modification, damage, and malfunction.

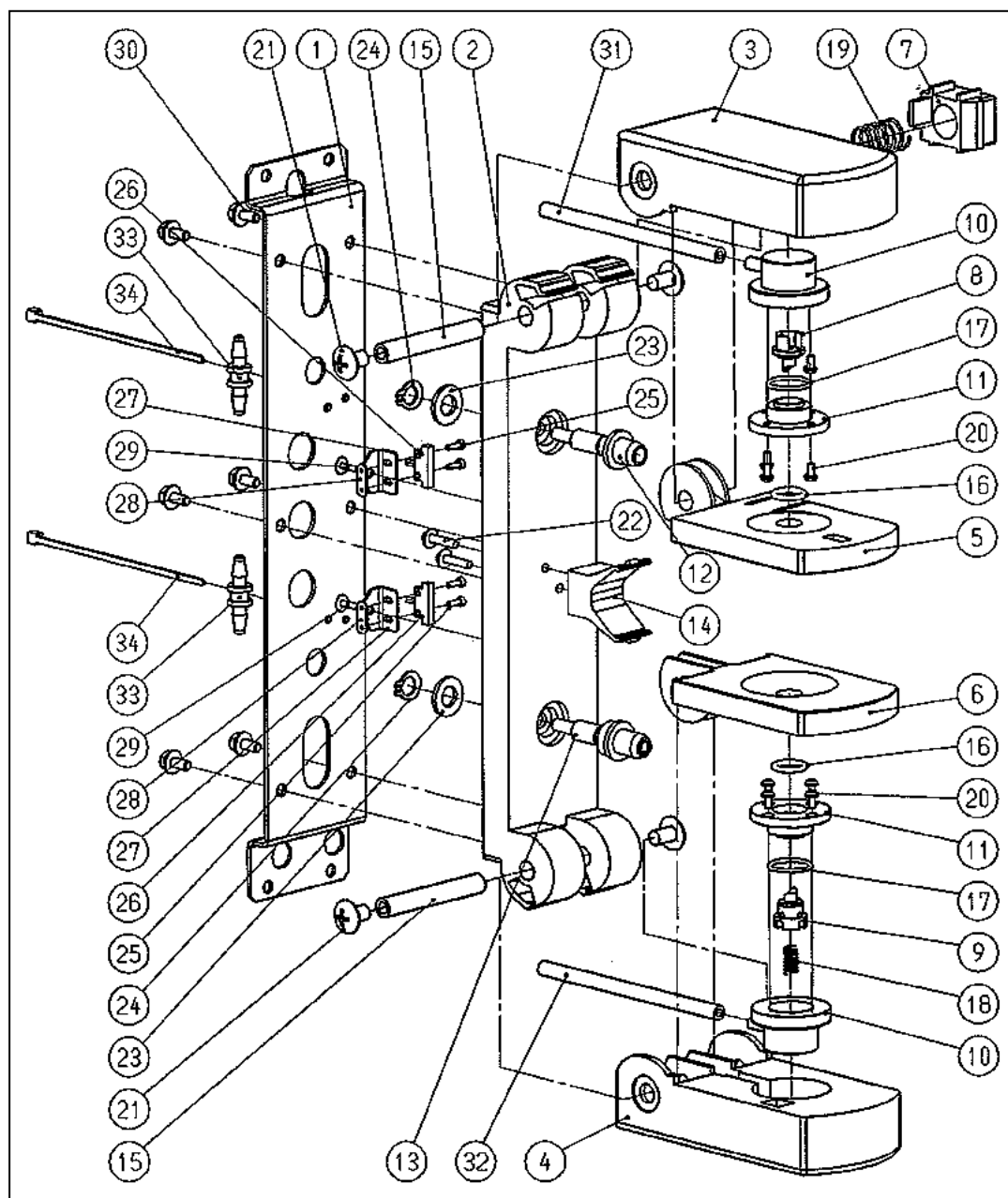
23.8. Confirmation work

- Confirm that there is no fluid leakage in operation.
- Confirm that the air in the B Powder concentrate solution air elimination tank generated due to work like part replacement will be discharged at operation.

24. B Powder Cartridge Holder



24.1. Overall configuration



24.2. List of components

No.	Name	Model no.	Torque	Note
1	Base BKT	90369311001		
2	Base	90369312001		
3	Arm A	90369313001		
4	Arm B	90369314001		
5	Arm A cover	90369315001		
6	Arm B cover	90369316001		
7	Button	90369317001		
8	Nozzle A	90369319001		
9	Nozzle B	90369320001		
10	Case	90369321001		
11	Seal ring	90369322001		
12	Nozzle receiver	90369325001		
13	Check valve	90369326001		
14	Stopper	90369318001		
15	Shaft	90369327001		
16	Nozzle packing	90369324001		
17	Packing	90369323001		
18	Spring	5172		
19	Spring	5626		
20	Screw	M2.5x4	0.32N·m	SUS
21	Truss screw	M6x10	4.9N·m	SUS
22	Screw	M3x10	0.8N·m	
23	Flat washer	For M8		
24	Grip hold ring	Spare 8		
25	Hexagon hole bolt	M2x6	0.2N·m	SUS
26	Proximity switch	RS-901S-500		
27	Sensor bracket	90369328001		
28	Nut plate	90333016001		
29	P tight	M3x8	Not removable	
30	Screw	M4x8	1.2N·m	
31	Silicone tube	Φ 3xΦ 6x135mm		
32	Silicone tube	Φ 3xΦ 6x140mm		
33	Hose opening	01930800124		
34	Insulation lock	T18R		L = 100mm, W = 2.5mm
35	B Powder Assy	90369300		

24.3. Removal and installation

<Removal>

⚠ Caution:

- Be sure to turn OFF the breaker.

Remove the front cover and the lower right side face cover.

Remove the connectors to the proximity switch ②⑥.

Remove the setscrews of the B Powder cartridge holder, and then remove it from the device.

Pull out the silicon tube on the device side from the hose opening ③③.

<Installation>

Install the parts in the reverse order of the removal procedure.

24.4. Disassembly and assembly

<Illustration>



<Disassembly>

Cut the insulation lock ③④.

Remove the hose opening ③③ from the silicon tubes ③①③②.

Remove the screws ②①, pull out the shaft ①⑤, and remove the arm A ③ and arm B ④.

Remove the screws ③①, and then remove the base ② from the base BKT ①.

Remove the screw ②⑨ and then take out the sensor bracket ②⑦ from the base ②.

Remove the hexagon hole bolt ②⑤, and take out the proximity switch ②⑥ from the sensor bracket ②⑦.

Remove the grip hold ring ②④ and take out the nozzle receiver ①②.

<Illustration>



[Arm disassembly]

Remove the arm cover from each arm.

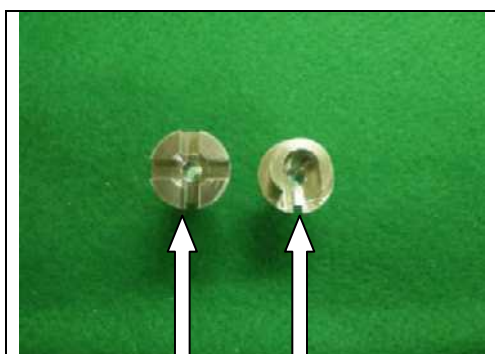


Remove the screws ⑳ and disassemble the parts.

<Assembly>

Assemble the parts in the reverse order of the disassembly procedure.

Make sure not to confuse the nozzle A ⑧ with nozzle B ⑨.



Nozzle A ⑧

Nozzle B ⑨

24.5. Inspection and replacement

Replacement

- Cleanse the disassembled parts very carefully.
- Replace the parts whose regular replacement time has come.

Inspection

- In maintenance 4 "Operation status," confirm the reaction of "LS7" .
 - Arm open: OFF
 - Arm close: ON

Service Manual List (Electric Parts) (for Type 2)

No.	Identification No.	Document Name	Option
1	CXSM00-100	Main CPU Board	
2	CXSM01-101	I/O Board	
3	BHSM02-102	Bridge Board	
4	BHSM00-103	Motor Drive Board	
5	BHSM01-104	PV/PA Board	
6	BHSM00-105	Power Board	
7	CXSM01-106	Operation Section	
8	BHSM00-107	Battery Assy	
9	BHSM01-108	Blood Pressure Gauge Assy	○
10	BHSM00-109	UFP Sensor Board	
11	CXSM00-110	Relay, SSR	
12	CXSM00-111	Power Section	
13	CXSM00-112	Electric Block Diagram	
14	CXSM00-113	LAN module Board	○

⚠ Caution

Use the following tools or equal tools for maintenance.

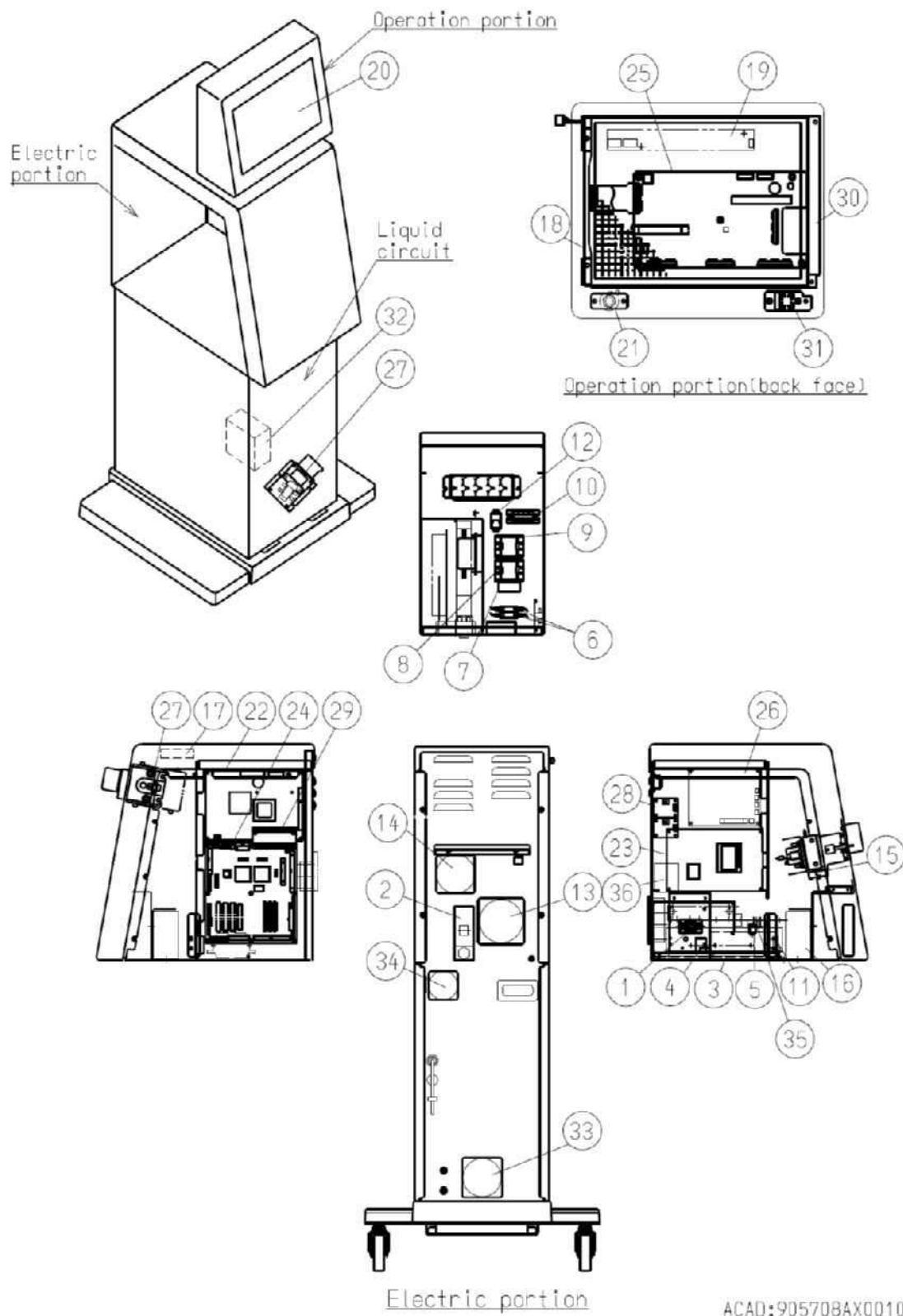
Each part or a part of each part may be changed for improvement without notice.

Be forewarned that a part of using product may be differed from this document.

- Remove static electricity, using an earth-band when a board is maintained or charged.
- Do not touch any point other than the mentioned point by a tester when a board is maintained, connecting probes or IC clips of tester. It could be a cause of board breakdown, depend on the touched place.
- Do not touch any board, electric wiring or terminal except for a maintenance.
- Do not touch any electric wiring or terminal which is not isolated.
- Do not touch any board with wet hand.
- Pull out a power plug from an outlet when the harness is removed from a Terminal block.
- Use a tester which is "True RMS TYPE" to measure precise alternating current value.

* RMS : Root Mean Square

Electric part diagram



ACAD:905708AX0010

No.	Name	Model No.	Identification No.	Remark
1	Terminal block	PTUL2-30-2P		
2	Earth leakage breaker	BAME2152C		
		BAME2202C		110V specification
3	Noise filter	SUP-EX20-R-6		
		SUP-EX30-R-6		110V specification
4	Noise suppression capacitor	XE474-Z		
5	Noise suppression capacitor	XYE-224102BE		
6	Fuse holder	F-64AD		
	Fuse	218 06.3P	CXSM**-110	
7	Relay	AHE2212	CXSM**-110	
8	SSR	AQA411VL	CXSM**-110	
9	SSR	AQA411VL	CXSM**-110	Only 230V specification
10	Terminal board	UF1005-20S-4P		230V specification
		UF1005-20S-2P		110V specification
11	SW electric	CME240P-24		
12	Relay	AHN36024		
13	FAN	D0925C24B4AS-00		
14	FAN	PUDC24D4		
15	FAN	EUDC24B8		
16	Nicd battery	20KR-CH (3AH)	BHSM**-107	
17	Speaker	F66A52-6		
18	LCD module	AA104VC10	BHSM**-106	Simultaneous order with No. 20
		AA104VH01		
19	Inverter board	CXA-0549-M	BHSM**-106	
	DC/DC Converter board	ALD-214012PM132		
20	Touch panel	FT-AS768-027H	BHSM**-106	Simultaneous order with No. 20
21	Encoder	RECW20D50-201-1	BHSM**-106	9036V351100
22	CPU board	SSM-3048	CXSM**-100	
		SSM-10027		
23	Electric board	SBY0261	BHSM**-105	
24	I/O board	SBY0324	CXSM**-101	ROM attachment
25	Panel CPU board	SBY0257	BHSM**-106	
26	Drive board	SBY0202	BHSM**-103	
27	UFP sensor board	SBY0200	BHSM**-109	
28	PV/PA sensor board	SBY0232	BHSM**-104	
29	Bridge board	SBY0238	BHSM**-102	
30	LCD connect board	SSM-3046		
31	I/O switch board	SBY0225		
32	Dialysate sensor board	SBY0213		
33	FAN	TUDC24D4		
34	FAN	TUDC24D4		
35	FAN	EUDC24B8		
36	LAN module board	SBY0245*	CXSM**-113	

100. Main CPU Board



100.1. Overall configuration

SSM-3048



SSM-10027



100.2. List of components

No.	Name	Model No.	Note
1-1	Main CPU board	SSM-3048	
1-2	Compact flash	TS512MCFL834014-9801632	Writing needed For 1-1 Main CPU board
2-1	Main CPU board	SSM-10027	
2-2	Compact flash	IPC17-001GW	Writing needed For 2-1 Main CPU board
3	Clock battery	CR-2032	

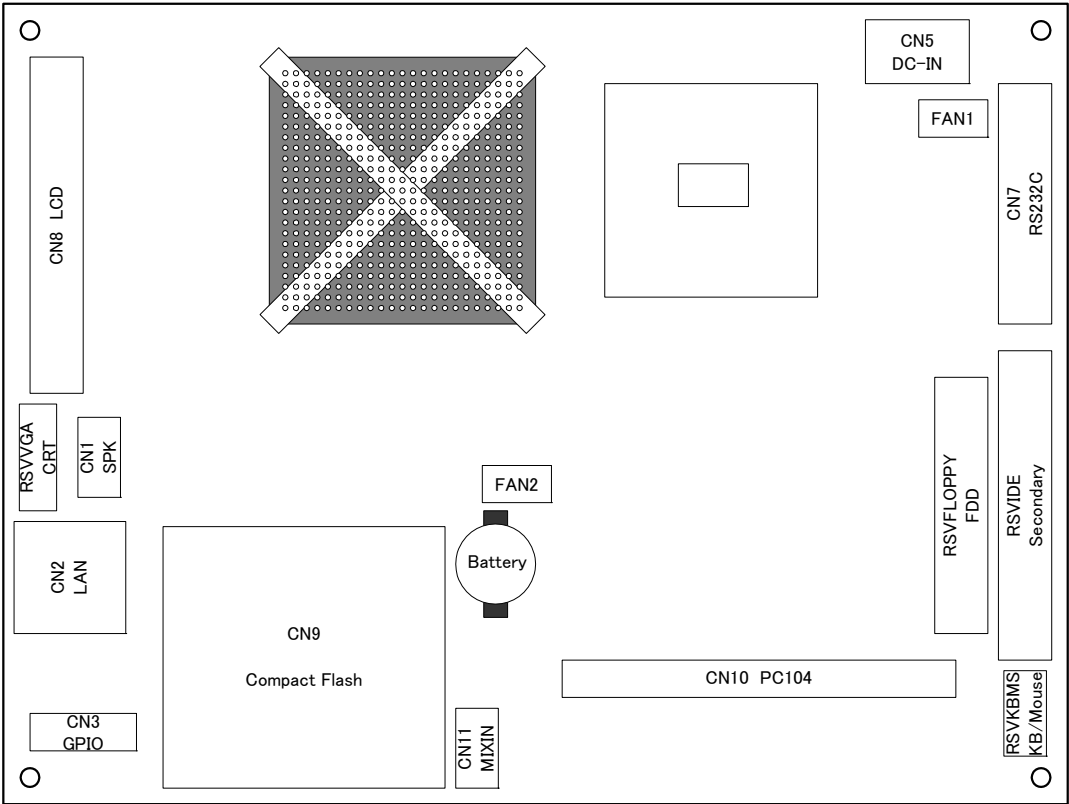
100.3. Summary and functions of the board

- It is attached onto the upper right side of the device.
- Software versions can be changed by replacing compact flashes.
- Main CPU board and compact flash have determined combination. (refer to pictures below)

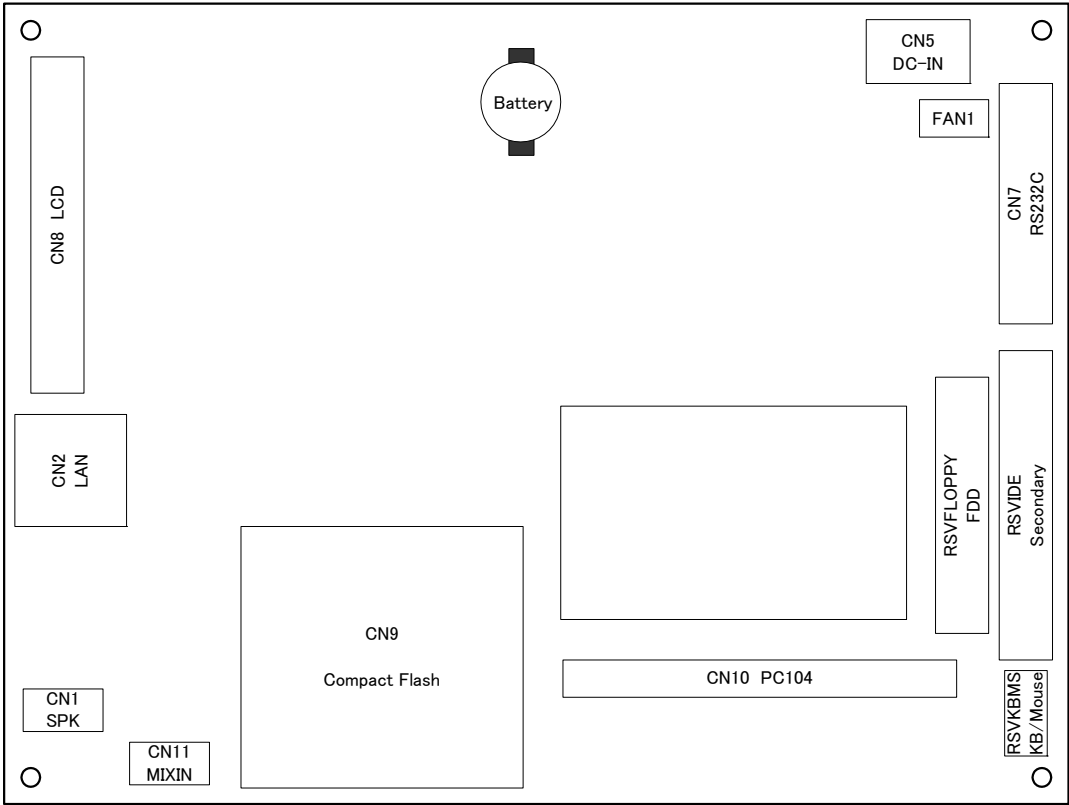
Compact flash for SSM-3048 	Compact flash for SSM-10027 
--	--

100.4. Configuration of test points and connectors

SSM-3048



SSM-10027



100.5. Connection of each connector

- CN1: Speaker connected
- CN7: Connected to I/O board using harness
- CN9: Connected to compact flash
- CN10: Connected to the bridge board connected to the I/O board

100.6. Sub alarm volume

For SSM-10027 (Main CPU board), turn VR2 clockwise to lower the volume and counterclockwise to raise it (default: maximum volume)

Refer to 101.6 for SSM-3048

100.7. Removal and installation

<Illustration>

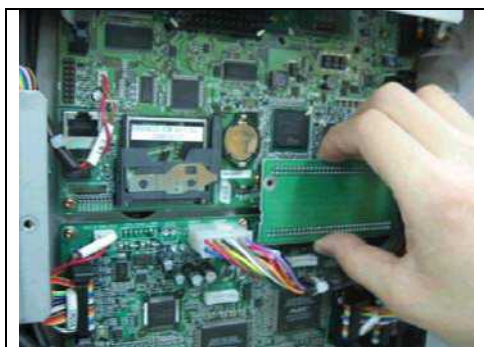


<Removal>

⚠ Caution:

- Be sure to turn OFF the breaker.

Remove the cover of the upper right side face and the electric cover of the right side face.



Remove the bridge board. Refer to BHS**--102.

Remove all the connectors on the main CPU board.

Remove the eight screws that fix the main CPU board.

<Installation>

Install the parts in the reverse order of the removal procedure.

⚠ Caution:

- Make sure not to damage the harness with the parts.

100.8. CF version change

<Illustration>

<Replacement>

⚠ Caution:

-
- Be sure to turn OFF the breaker.
-

Remove the cover of the upper right side face and electric cover of the right side face.



Push down the compact flash socket lever, and take out the compact flash.



Insert the compact flash into the socket.

<Panel CPU and Sub-CPU version change>

If panel CPU and sub-CPU versions that CF has at the time of the device start-up are different from that of the base board after CF versions are changed, versions will be automatically changed.

memo

-
- As the CF's version is prioritized, even if the panel CPU or sub-CPU versions are raised, they go down to the version of the CF.
 - Never turn off I/O-SW during the version change. The device will not start up.
 - As for data transmission, refer to BHSM**-101.
-

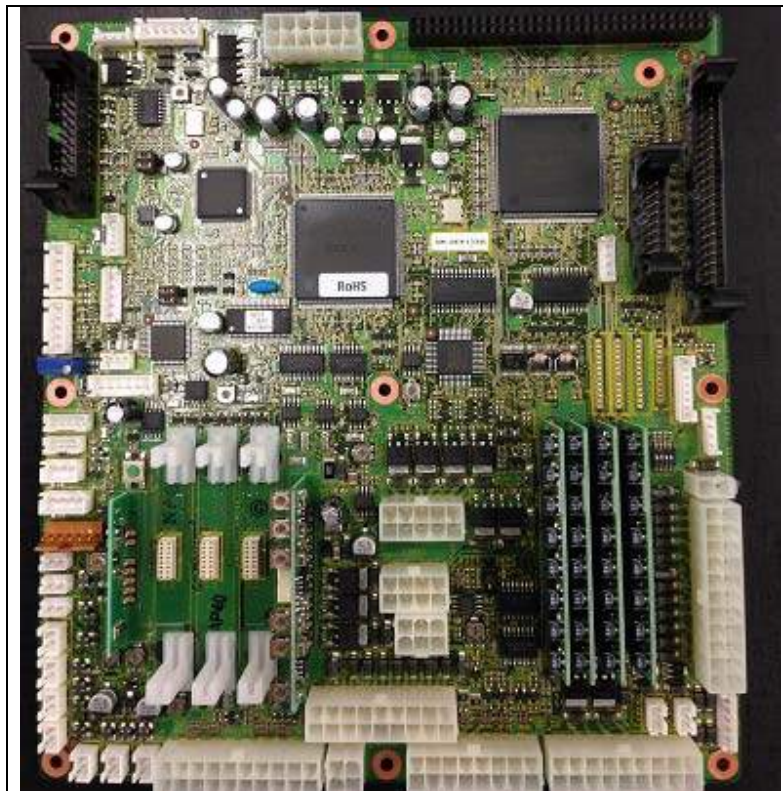
<Confirmation work>

Confirm that the CF version on the display matches the version inscribed on the CF seal.

101. I/O Board



101.1. Overall configuration



101.2. List of components

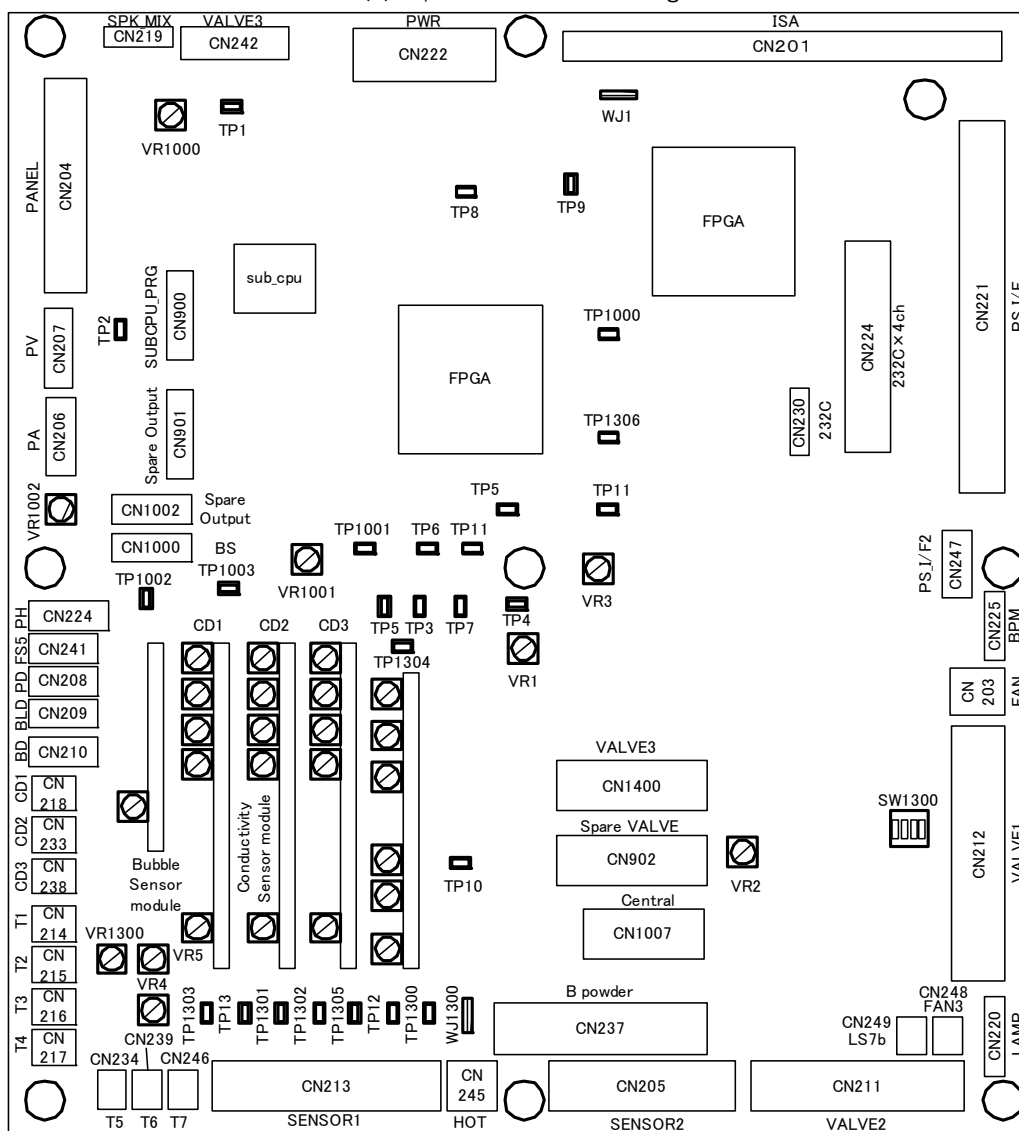
No.	Name	Model No.	Remark
1	I/O board	SBY0324	
2	Conductivity sensor module base board	SBY0300	

101.3. Summary and functions of the board

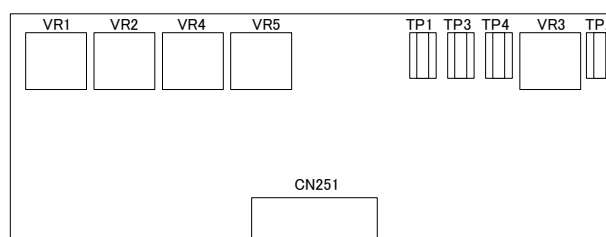
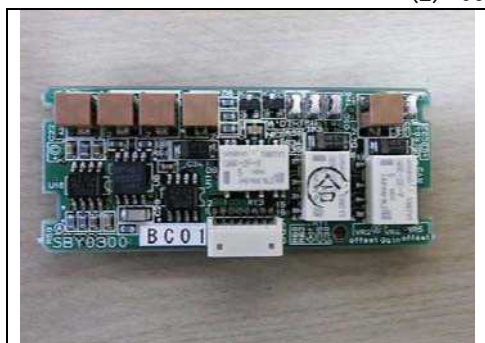
- It is attached to the upper right side of the device.
- Memorizes, by FRAM, information such as device eigenvalue, device set value, etc.
- CD1, CD2, and CD3 module boards are attached.
- Bubble sensor module board is attached.
- Temperature sensor module board is attached.
- Drives various kinds of valves.
- Inputs and outputs measurement values of various kinds of sensors.

101.4. Test points, adjustment volume, and connectors layout drawing

(1) I/O base board design



(2) Conductivity sensor module board



101.5. Connection of each connector

- CN201 : Using bridge board, connect the main CPU board.
- CN203 : Connect to FAN2.
- CN204 : Using harness, connect to the panel CPU board.
- CN205 : Using harness, connect to LS3, LS5, LS6, PH1, PH7, and PH8.
- CN206 : Using harness, connect to the arterial pressure sensor board.
- CN207 : Using harness, connect to the venous pressure sensor board.
- CN208 : Using harness, connect to the dialysate pressure sensor.
- CN209 : Using harness, connect to the blood leakage sensor.
- CN210 : Using harness, connect to bubble detector.
- CN211 : Using harness, connect to V9a, V9b, V10, V11, V12, V17, V19, V21, and V24.
- CN212 : Using harness, connect to V1, V4, V5a, V5b, V6a, V6b, V7a, V7b, V8a, and V8b.
- CN213 : Using harness, connect to FS1, FS2, FS3, FSW, LS2, LS7a, PH5, and PH6.
- CN214 : Connected ThermistorT1.
- CN215 : Connected ThermistorT2.
- CN216 : Connected ThermistorT3.
- CN217 : Connected ThermistorT4.
- CN218 : Using harness, connect to the CD1.
- CN219 : Connect to buzzer.
- CN220 : Using harness, connect to four consecutive representative light.
- CN221 : Using harness, connect to electric power board.
- CN222 : Using harness, connect to electric power board.
- CN224 : Using harness, connect to main CPU board.
- CN225 : Using harness, connect to blood pressure gauge.
- CN230 : Using harness, connect to LAN module board. (Use LAN option)
- CN233 : Using harness, connect to the CD2.
- CN234 : Connected to thermistorT5.
- CN237 : Using harness, connect to V27, V28, V29, V30, and V31. (At the time of B powder)
- CN238 : Using harness, connect to CD3. (At the time of B powder)
- CN239 : Using harness, connect to thermistorT6. (At the time of B powder)
- CN242 : Using harness, connect to V33, and FAN3b.
- CN245 : Using harness, connect to V34, and V35.
- CN246 : Using harness, connect to thermistorT7.
- CN247 : Using harness, connect to electric power board.
- CN248 : Connect to FAN3.
- CN249 : Connect to FAN4.
- CN1000 : Using harness, connect to Blood sensor.
- CN1007 : Using harness, connect to V45, V46, and V47. (At the time of Central option)
- CN1400 : Using harness, connect to V54, V55, V56, and V57.

101.6. Test points

(1) I/O board

TP	Signal name	Voltage value	Adjustment VR
TP1	B5V power source	DC4.95~5.20V	—
TP2	B12V power source	DC11.9~12.5V	—
TP3	Analog reference voltage	DC4.99~5.01V	VR1
TP4	Blood leakage sensor output voltage	—	—
TP5	Venous pressure output voltage	—	—
TP6	Arterial pressure output voltage	—	—
TP7	Dialysate pressure sensor output voltage	—	—
TP8	FPGA driven voltage ①	DC3.3V	
TP9	FPGA driven voltage ②	DC2.5V	
TP10	Analog reference voltage (Sub)	DC4.99~5.01V	VR2
TP12	T1 output voltage	DC1.24~1.26V	VR4
		DC1.99~2.01V	VR5
TP13	T1 reference voltage	DC2.23~2.27V	VR5
TP1002	Blood sensor output voltage	—	—
TP1303	Bubble sensor output voltage	—	—
—	Sub alarm volume	—	VR1000※

※For SSM-3048 (Main CPU board), clockwise to lower the volume and counterclockwise to raise it (default: maximum volume)

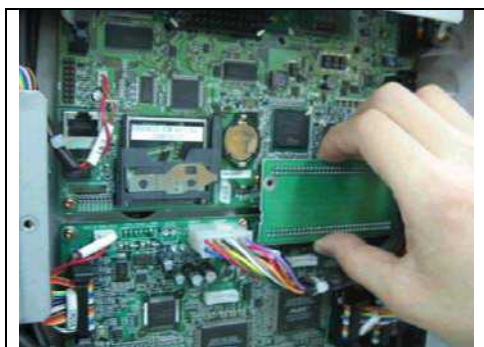
Refer to 100.6 for SSM-10027

(2) Conductivity sensor module board

TP	Signal name	Voltage value	Adjustment VR
TP1	Conductivity sensor output voltage	DC5.00V (GAIN)、2.50V (OFFSET)	VR1、VR2
TP2	Conductivity sensor reference voltage	AC100.0mV	VR3
TP3	Temperature sensor output voltage	DC1.25V (GAIN、25°C)	VR4
TP4	Temperature sensor output voltage	DC2.00V (OFFSET、40°C)	VR5

101.7. Removal and installation

<Illustration>



<Removal>

⚠ Caution

-
- Be sure to turn OFF the breaker.
-

Remove the upper right side face cover and electric cover of the right side face.

Remove the bridge board. Refer to BHSM**-102.

Remove all the connectors on the I/O board.

Remove nine the screws that fix the I/O board.

<Installation>

Install the parts in the reverse order of the removal procedure.

⚠ Caution

-
- Make sure not to damage the harness with the parts.
-

101.8. Analog reference voltage adjustment

<Removal>

⚠ Caution

-
- Be sure to turn OFF the breaker.
-

Remove the upper right side face cover and electric cover of the right side face.

Set the tester in the DC range, and then connect (+) probe to WJ1 and (-) probe to TP3.

⚠ Caution

-
- Be careful not to let the tester probe contact the other terminals. If the breaker is turned on while they have contact, the board may be damaged.
-

Turn ON the breaker, and adjust VR1 so that TP3 voltage becomes 5.00V+/-0.01V.

101.9. Adjustment of analog reference voltage (Sub)

<Removal>

⚠ Caution

-
- Be sure to turn OFF the breaker.
-

Remove the upper right side face cover and electric cover of the right side face.

Set the tester in the DC range, and then connect (+) probe to WJ1 and (-) probe to TP10.

⚠ Caution

-
- Be careful not to let the tester probe contact the other terminals. If the breaker is turned on while they have contact, the board may be damaged.
-

Turn ON the breaker, and adjust VR2 so that TP10 voltage becomes 5.00V+/-0.01V.

101.10. Data transfer

<Illustration>

<Data transfer from RAM to CF>

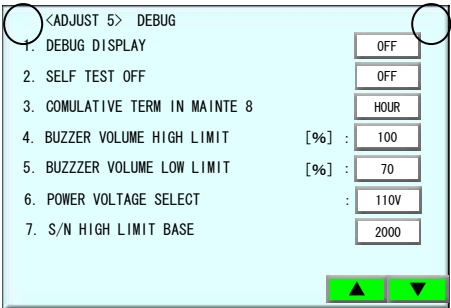
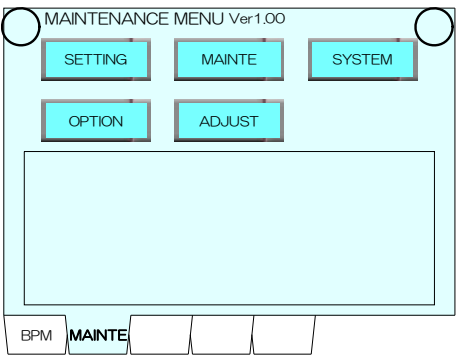
By turning ON “RAM->CF data transfer” of System 1 “Function setting” and turning OFF/ON on the I/O switch, setting values, device eigenvalue, device history, etc. can be stored in the compact flash.

memo

- When replacing the I/O board for a new one, setting values, device eigenvalue, device history, etc. will be initialized.

<Data transfer from CF to RAM>

It is necessary to have adjustment display shown when transferring data from CF to RAM. The “ADJUST” switch will be shown as in the left figure, when the two areas shown by the two circles are simultaneously pushed down for three seconds as shown in the left figure.

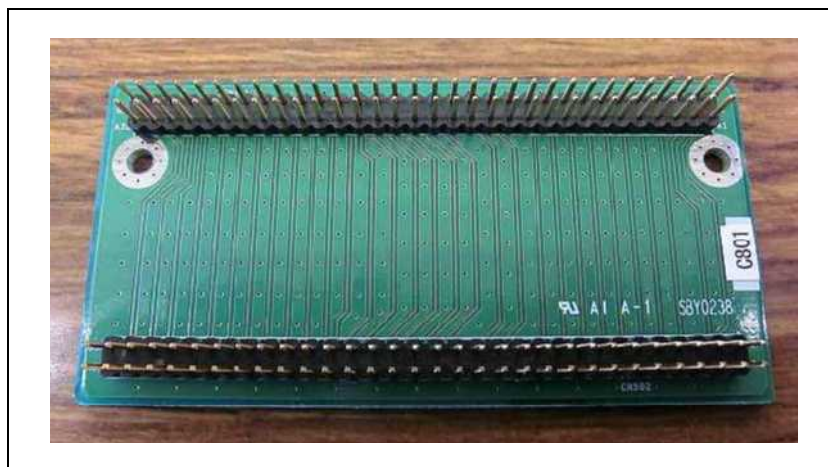


Open <ADJUST 5> display, and simultaneously push down the two areas shown by the two circles in the left figure for three seconds, and data will be transferred from CF to RAM.

102. Bridge Board



102.1. Overall configuration



102.2. List of components

No.	Name	Model No.	Remark
1	Bridge board	SBY0238	

102.3. Summary and functions of the board

- This is a board that connects the I/O board and the main CPU board.

102.4. Removal and installation

<Illustration>

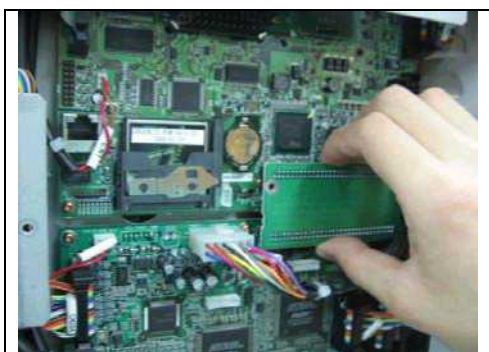


<Removal>

⚠ Caution

-
- Be sure to turn OFF the breaker.
-

Remove the upper right side face cover and electric cover of the right side face.



Pull out the connectivity terminals (pins) of the bridge board in an even manner so that they will not be folded and bent.

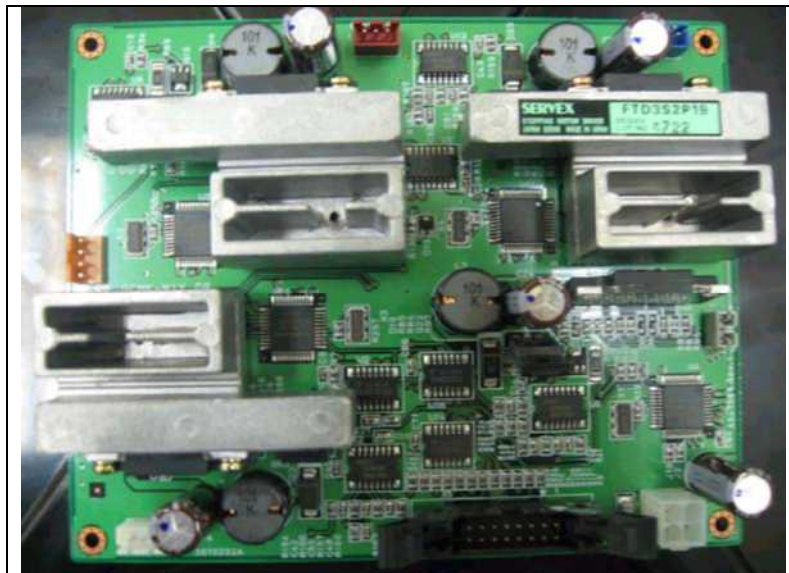
<Installation>

Confirm that connectivity terminals (pins) of the bridge board have not been folded or bent, and insert them one by one in upper, lower, left, and right positions in an even manner.

103. Motor Drive Board



103.1. Overall configuration



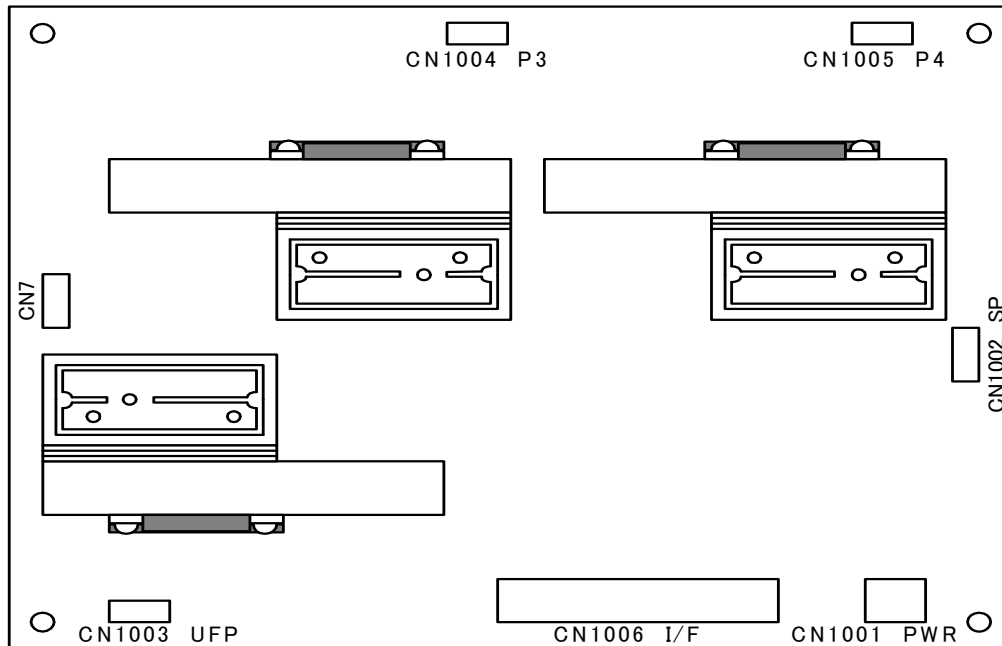
103.2. List of components

No.	Name	Model No.	Note
1	Motor drive board	FTD3S2P19 (SBY0202)	

103.3. Summary and functions of the board

- It is installed to the upper left side of the device.
- Drivers to drive HP, UFP, P3, and P4 are mounted.

103.4. Connector layout drawing



103.5. To which connectors are connected

- CN1001: Use harness and connect it to power source board.
- CN1002: Use harness and connect HP to it.
- CN1003: Use harness and connect UFP to it.
- CN1004: Use harness and connect P3 to it.
- CN1005: Use harness and connect P4 to it.
- CN1006: Use harness and connect it to power source board.

103.6. Removal and installation

<Illustration>



<Removal>

⚠ Caution:

-
- Be sure to turn OFF the breaker.
-

Remove the upper left side face cover and electric cover of the left side face.

Remove all the connectors on the motor drive board.

Remove all the four screws that fix the motor drive board.

<Installation>

Install the parts in the reverse order of the removal procedure.

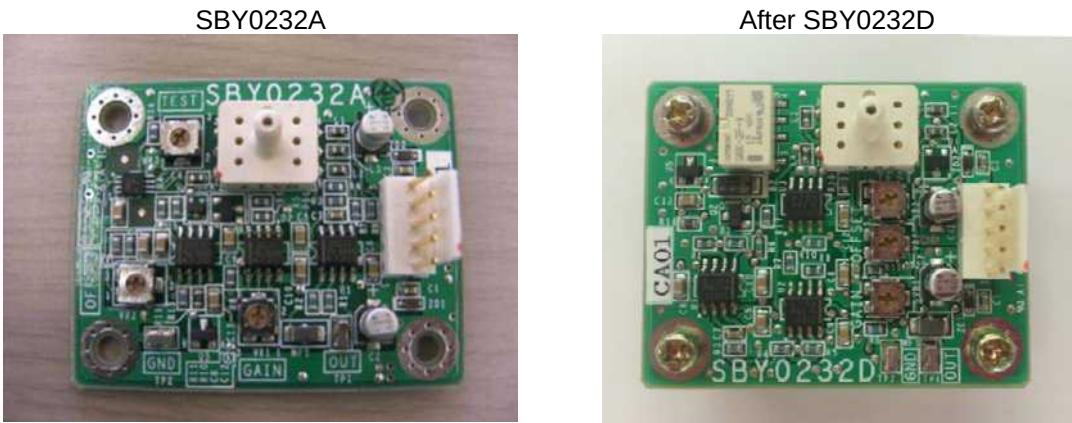
⚠ Caution:

-
- Make sure not to damage the harness with the parts.
-



104. PV/PA Board

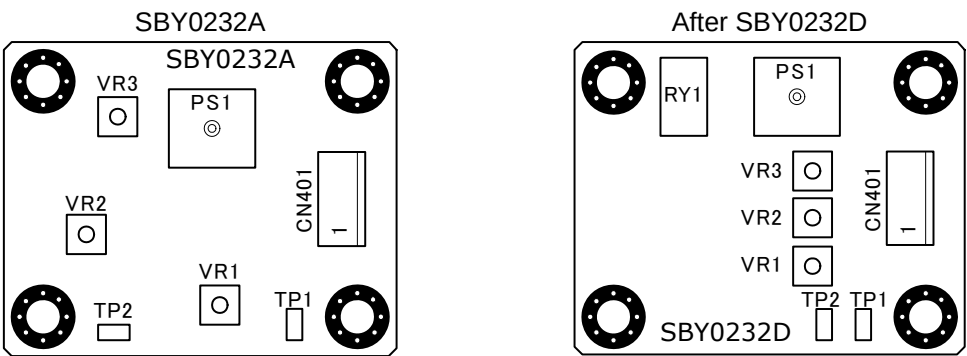
104.1. Overall configuration



104.2. List of components

No.	Name	Model No.	Note
1	PV/PA sensor board	SSM-07035 (SBY0232)	

104.3. Layout drawing of the connector test points



104.4. Summary and function of the board

- Measure the venous or arterial pressure in the blood circuit with the pressure sensor.
- The same board is used for both venous and arterial pressure.

104.5. Test point

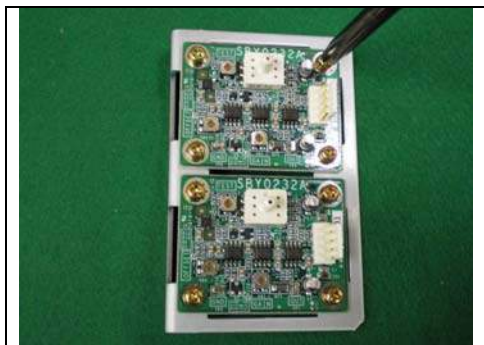
TP	Signal name	Voltage value	Adjustment VR, remark
TP1	Sensor output	0mmHg : $DC2.50V \pm 0.02V$ +300mmHg : $DC4.00V \pm 0.01V$ At self-diagnosis : $DC4.70V \pm 0.10V$	0mmHg : VR2 +300mmHg : VR1 At self-diagnosis : VR3
TP2	GND		

104.6. Connection of each connector

- CN401 : Using harness, connect to CN206 of the I/O board (arterial pressure sensor).
- CN401 : Using harness, connect to CN207 of the I/O board (venous pressure sensor).

104.7. Removal and installation

<Illustration>



<Removal>

⚠ Caution

-
- Be sure to turn OFF the breaker.
-

Remove the upper left side face cover.

Remove the connector and tube on the board.

Remove the board bracket fixing screws from the back of the device.

Remove the fixing screws.

<Installation>

Install in the reverse order of the removal procedure.
Venous pressure sensor (PV) is located on the upper part.
Arterial pressure sensor (PA) is located on the lower part.

105. Power Board



105.1. Overall configuration



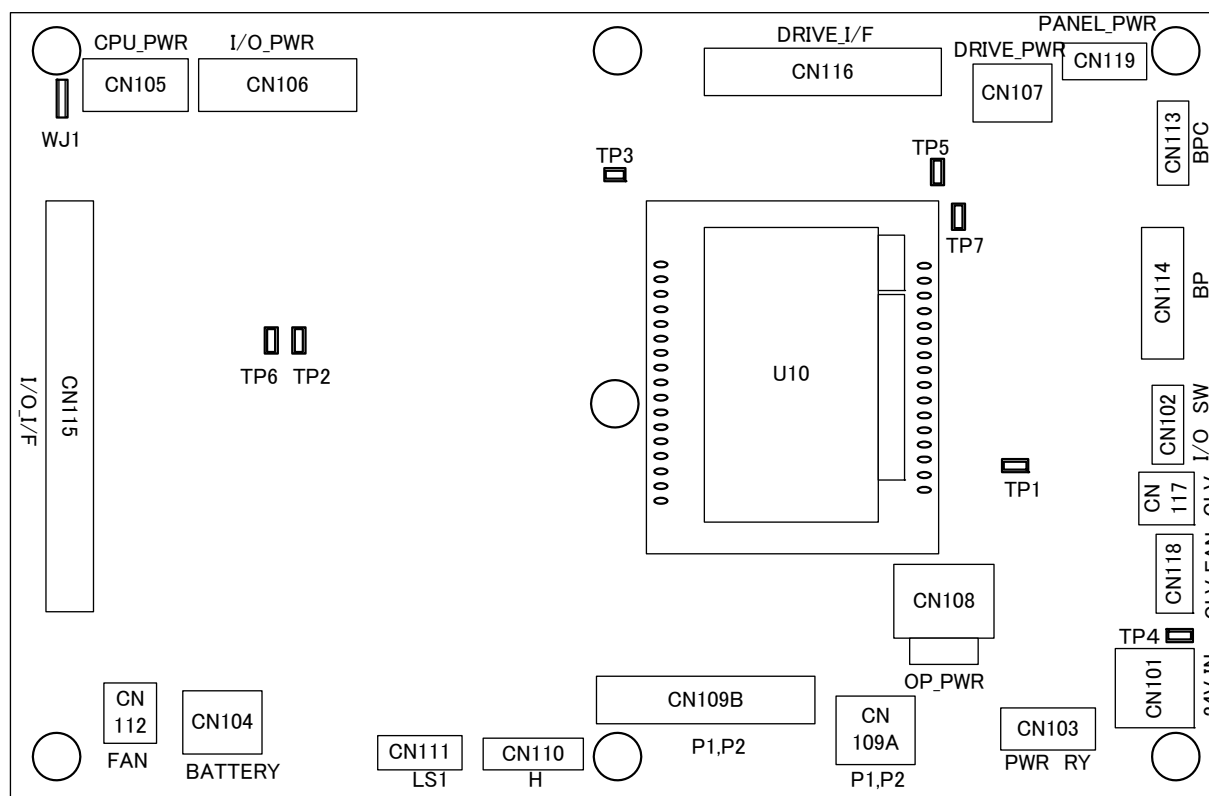
105.2. List of components

No.	Name	Model	Note
1	Power source board	SSM-08003 (SBY0261)	

105.3. Summary and function of the board

- Using the DC-DC converter, create 36V, +12V, -12V, +9V, +5V and -5VD from C24V.
- Drive P1, P2.
- Drive BP.
- Drive H.
- Drive the clamp.
- Perform charge and discharge of the battery.

105.4. Layout drawing of terminal block and connector



105.5. Test point

TP	Signal name	Voltage value	Adjustment VR, remark
TP1	I/OSW input circuit 5V power source	+5V	
TP2	24V	+24V	
TP3	Battery discharge voltage	+24V	
TP4	Battery voltage (at electric power failure)	+24V	At electric power failure, same value as TP3
TP5	24V power source for BP	+24V	
TP6	BP speed command voltage	—	
TP7	Battery charge 36V power source	+36V	
WJ1	GND		

105.6. To which connectors are connected

- CN102 : Using harness, connect I/O switch.
- CN103 : Using harness, connect the power relay.
- CN104 : Connect the battery.
- CN105 : Using harness, supply power to the CPU board.
- CN106 : Using harness, supply power to the I/O board.
- CN107 : Using harness, supply power to the motor drive board.
- CN108 : Connect the power harness of the internal organ blood pressure gauge (option).
- CN109 : Connect P1 and P2.
- CN110 : Connect the heater.
- CN111 : Using harness, connect the heater limit switch.
- CN112 : Connect FAN2.
- CN113 : Connect the cover switch of the blood pump
- CN114 : Using harness, connect the blood pump.
- CN115 : Using harness, connect to the I/O board.
- CN116 : Using harness, connect to the I/O board.
- CN117 : Connect the clamp.
- CN118 : Connect the clamp fan.
- CN119 : Using harness, supply power to the panel CPU board.

105.7. Removal and installation

<Illustration>



<Removal>

⚠ Caution

- Be sure to turn OFF the breaker.

Remove the upper left side face cover and electric cover of the left side face.

Disconnect all the connector on the power board.

Remove the seven screws fixing the power board.

<Installation>

Install in the reverse order of the removal procedure.

⚠ Caution

- Make sure not to damage the harness with the parts.
- Fix the screw on the terminal block securely. If loosen, the board may be burned out.

106. Operation Section



106.1. Overall configuration



106.2. List of components

No.	Name	Model No.	Remark
1	Panel CPU board	SSM-07015 (SBY0257)	
2-1	Inverter board	CXA-P1212B-WJL	For 2-2 LCD module
2-2	LCD module	LQ104V1DG51	
3-1	DC/DC converter board	ALD-214012PM132	For 3-2 LCD module
3-2	LCD module	AA104VH01	
4	Encoder	RECW20D50-201-1	
5	Touch panel	FT-AS768-027H	Attached to the panel plate
6	Operation section	—	

106.3. Summary and function of the board

◎ Panel CPU board

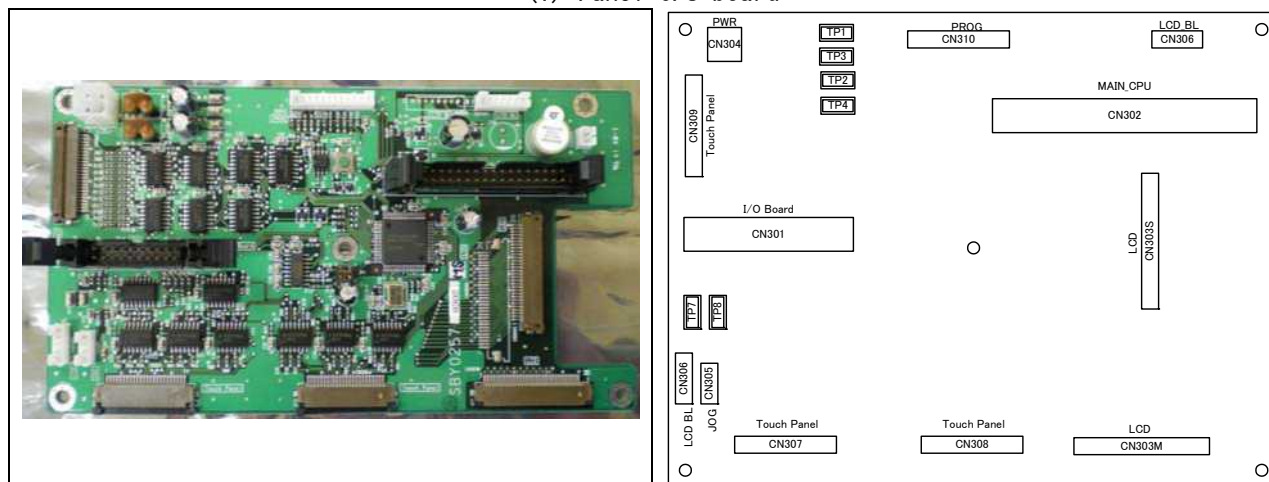
- Acknowledges the touch panel switch.
- Perform input/output of BP volume.

◎ Inverter board

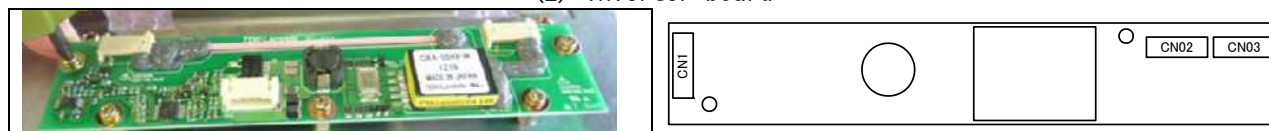
- Light up the back light of liquid crystal.

106.4. Base board drawing

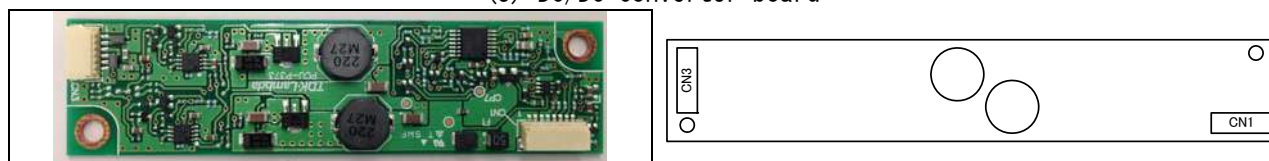
(1) Panel CPU board



(2) Inverter board



(3) DC/DC Converter board



106.5. Connection of each connector

◎ Panel CPU board

- CN301 : Using harness, connect to the I/O board.
- CN302 : Using harness, connect to the main CPU board.
- CN303 : Using LCD connector board, connect to the LC module.
- CN304 : Using harness, connect to the power board.
- CN305 : Connect to the encoder.
- CN306 : Using harness, connect to the LCD connector board.
- CN307 : Connect to the touch panel.
- CN308 : Connect to the touch panel.
- CN309 : Connect to the touch panel.

◎ Inverter board

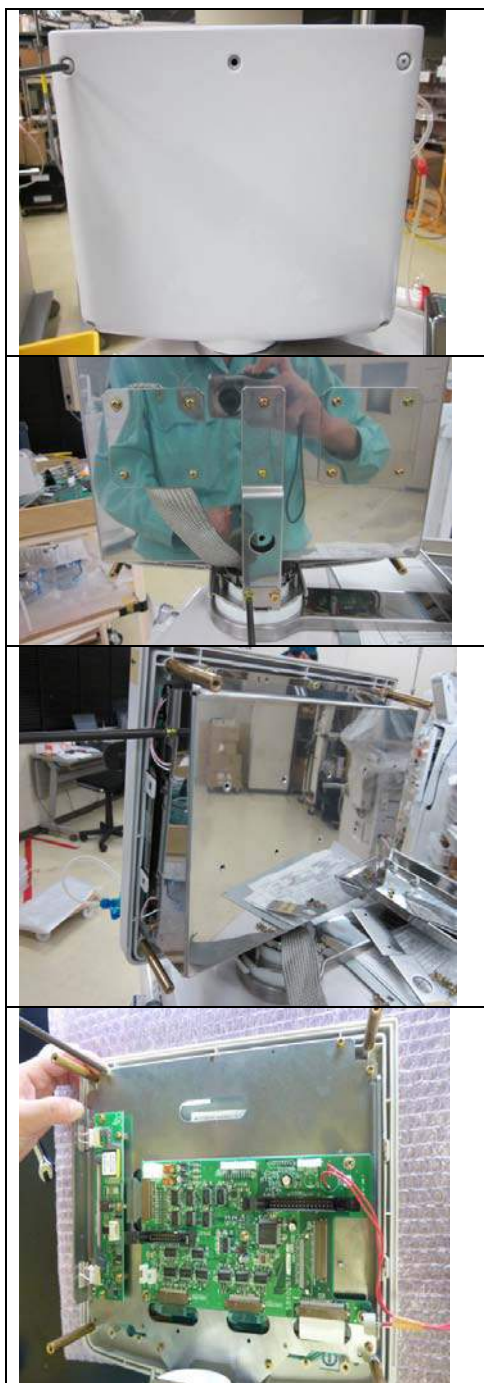
- CN1 : Using harness, connect to the panel CPU board.
- CN2, CN3 : Using harness, connect to the LCD module.

◎ DC/DC Converter board

- CN1 : Using harness, connect to the panel CPU board.
- CN3 : Using harness, connect to the LCD module.

106.6. Removal and installation

<Illustration>



<Removal>

⚠ Caution

- Be sure to turn OFF the breaker.

Remove the cover fixing screw of the operation panel section to remove the cover of the operation panel section.

Pull out CN301, CN302, CN304, and CN901. Remove the earth harness at the lower right.

Remove the five screws fixing the operation section to remove the front panel.

<Installation>

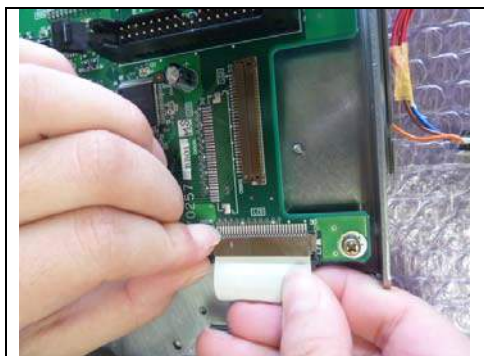
Install in the reverse order of the removal procedure.

⚠ Caution

- Make sure not to damage the harness with the parts. (Especially, take caution for handling the flexible cable of the panel seat.)

106.7. Disassembly and assembly

<Illustration>



<Removal>

(1) Panel CPU board

Remove all connectors connected to the panel CPU board.

⚠ Caution

- Remove the harness with the connector stopper free.



Remove the six screws fixing the panel CPU board to remove the panel CPU board.



(2) Inverter board(DC/DC Converter board)

Remove all the connectors connected to the inverter board (DC/DC converter board).

Remove the two screws fixing the inverter board (DC/DC converter board) to remove the inverter board from the operation section.

<Illustration>



<Removal>

(3) Encoder

Remove the cap with a flathead screwdriver.

Remove the fixing nut with long nose pliers.

Remove the blood pump flow setting volume.



Remove the screws fixing the blood pump flow setting volume to remove the volume from the operation section.

<Assembly>

Install in the reverse order of the removal procedure.

⚠ Caution

- Make sure not to damage the harness with the parts.
- Insert the flexible cable fully to apply a lock.



107. Battery Assy



107.1. Overall configuration

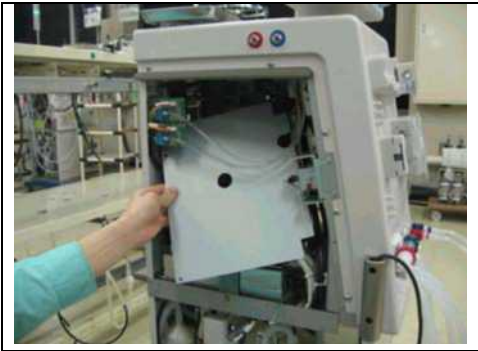


107.2. List of components

No.	Name	Model No.	Note
1	NICD battery	20KR-CH (3AH)	
2	Battery bracket	93897213101	

107.3. Removal and installation

<Illustration>



<Removal>

⚠ Caution

-
- Be sure to turn OFF the breaker.
-

Remove the right and left side face cover at the upper part.

Remove the electric cover of the left side face.



Remove the battery connector (CN104) on the power board.

Remove the battery bracket fixing screws.

Cut the banding band that bands the harness to remove the battery together with the bracket.

Cut the banding band to remove the battery from the bracket.

<Installation>

Install in the reverse order of the removal procedure.

107.4. Charging of battery

Battery will be charged by 48 hours with the power breaker on.

108. Blood Pressure Gauge Assy



108.1. Overall configuration

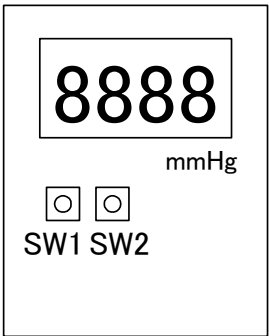


108.2. List of components

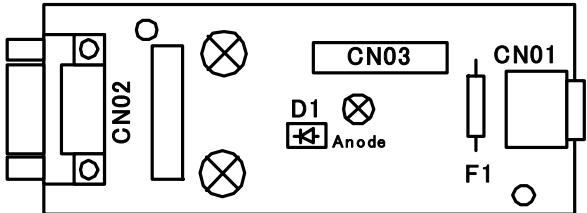
No.	Name	Model No.	Note
1	Blood pressure gauge module	M2100	
2	Blood pressure gauge I/F board	SBY0301	
3	BPM Bracket	93898012	
4	Blood pressure gauge Assy	—	

108.3. Switch/adjustment volume/connector layout drawing

(1) Blood pressure gauge module



(2) Blood pressure gauge I/F board



108.4. Switch application and test pin

<Blood pressure gauge module>

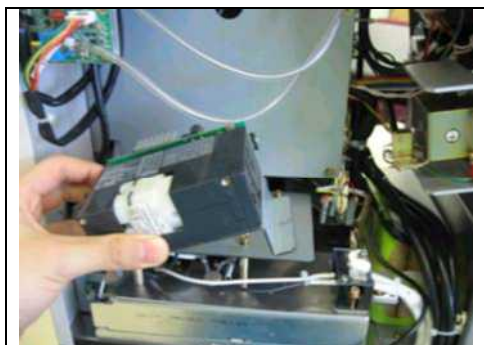
Switch	Name	Blood pressure gauge status by switch operation
SW1	Measurement mode setting switch	When pressing one time, measurement will start in the adult mode. If pressing the switch again during measurement, the measurement will stop.
SW2	Opening and closing switch of exhaust valve	When pressing one time, the exhaust valve will be shut off to be the standby state. If pressing the switch again, the test will stop.

<Blood pressure gauge I/F board>

Measure point	Name	Adjusted value	Remark
D1 anode	+5V power source	4.8~5.2V	
D1 cathode	+12V power source	11.8~12.2V	
F1 lead	+24V power source	23.2~24.8V	
CN3 16 pin	Power ground	GND terminal	

108.5. Removal and installation

<Illustration>



<Removal>

⚠ Caution

- Be sure to turn OFF the breaker.

Remove the left and right side face cover at the upper part.

Remove the power cable and communication cable.

Remove the silicon tube.

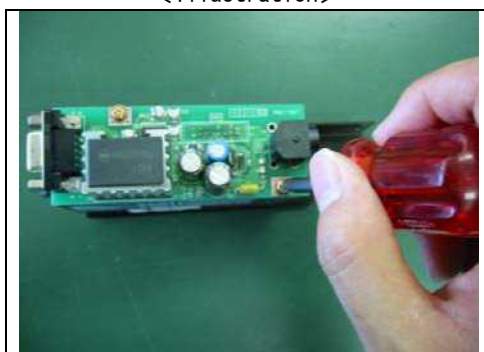
Remove the two screws fixing the blood pressure gauge Assy.

<Installation>

Install in the reverse order of the removal procedure.

108.6. Disassembly and assembly

<Illustration>



<Disassembly>

Remove the two screws fixing the blood pressure gauge I/F board to remove the board.

Remove the module from the bracket.

<Assembly>

Assembled in the reverse order of the disassembly procedure.

109. UFP Sensor Board



109.1. Overall configuration



109.2. List of components

No.	Name	Model No.	Note
1	UFP sensor board	SSM-3045 (SBY0200)	

109.3. Summary and function of board

- Acknowledges the dog of UFP and HP and outputs them.
- Monitors rotation direction and speed of the dog.

109.4. Connection of each connector

- CN701 : Using harness, connect to the I/O board.

109.5. Removal and installation

<Illustration>

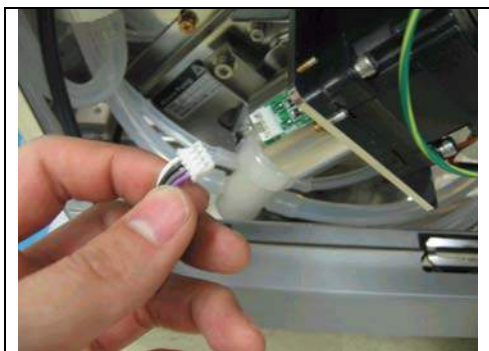
<Removal>

⚠ Caution

-
- Be sure to turn OFF the breaker.
-

Remove the front cover.

Remove CN701 connected to the UFP sensor board.



Remove the UFP sensor board.



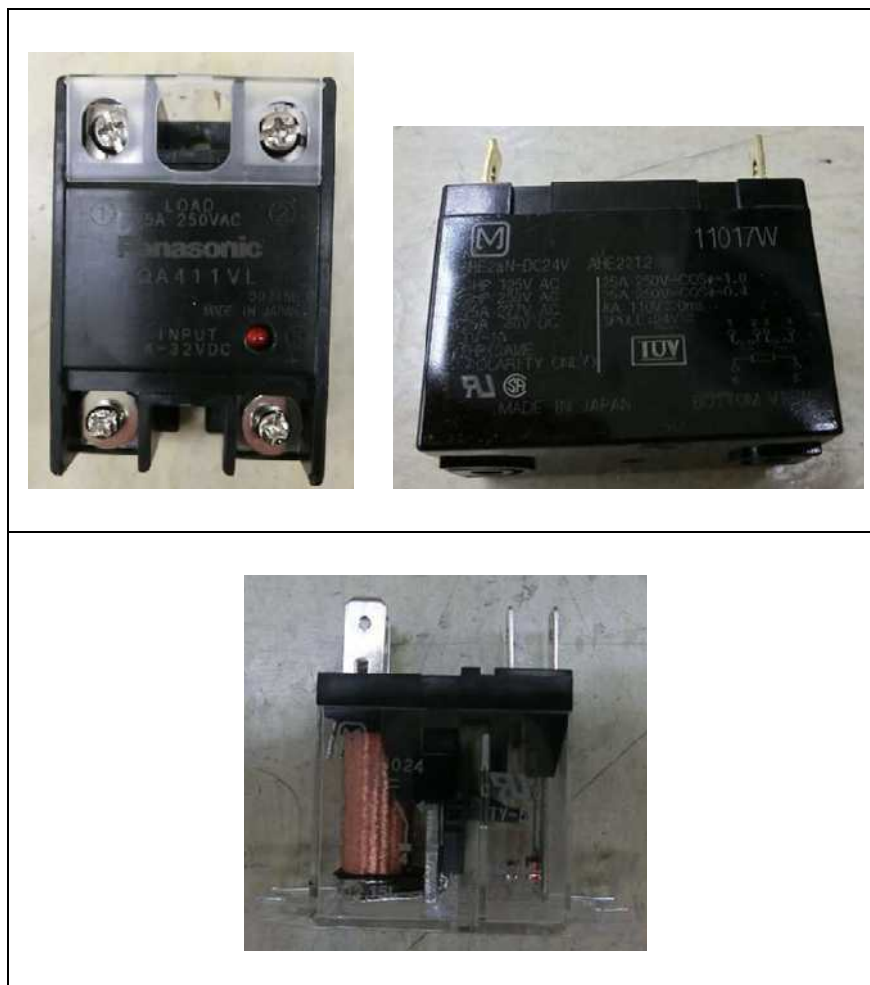
<Installation>

Install in the reverse order of the removal procedure.

110. Relay, SSR



110.1. Overall configuration



110.2. List of components

No.	Name	Model No.	Remark
1	Relay	AHE2212	
		AHN36024	
2	SSR	AQA411VL	
3	Fuse 6.3A	021306.3	

110.3. Removal and installation of SSR

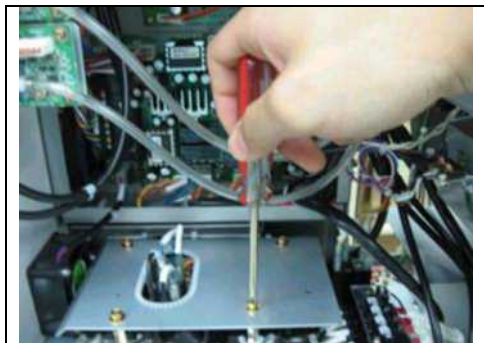
<Illustration>

<Removal>

⚠ Caution

- Be sure to turn OFF the breaker.

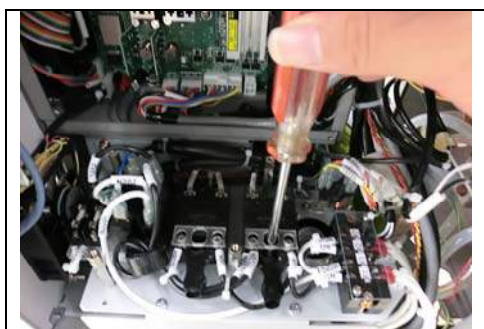
Remove the upper left side face cover.



Remove the four screws fixing the cover.



Remove the harness fixed to SSR.



Remove the two screws fixing SSR.



Remove SSR. At the back of SSR, cooling silicon is applied.

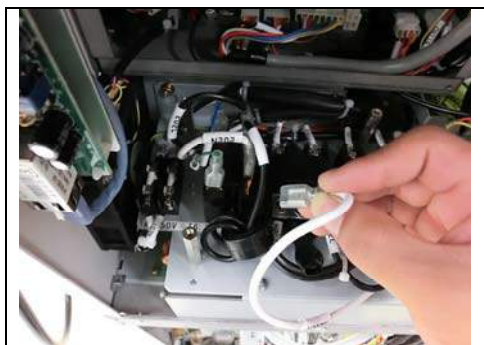
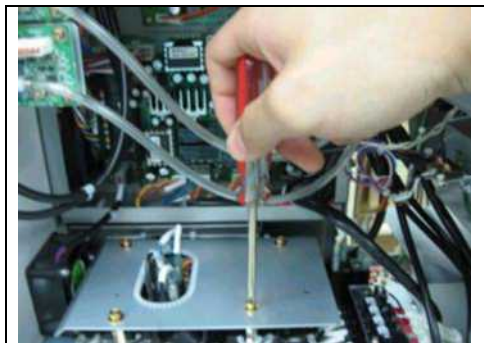
For installation, apply the cooling silicon uniformly to the metal part at the back of SSR.

<Installation>

Install in the reverse order of the removal procedure.

110.4. Removal and installation of Relay

<Illustration>



<Removal>

⚠ Caution

- Be sure to turn OFF the breaker.

Remove the upper left side face cover.

Remove the four screws fixing the cover.

Remove the fast-on terminal.

Remove the screws fixing the relay to remove the relay.

⚠ Caution

- Do not apply cooling silicon to the relay.

<Installation>

Install in the reverse order of the removal procedure.

110.5. Fuse replacement

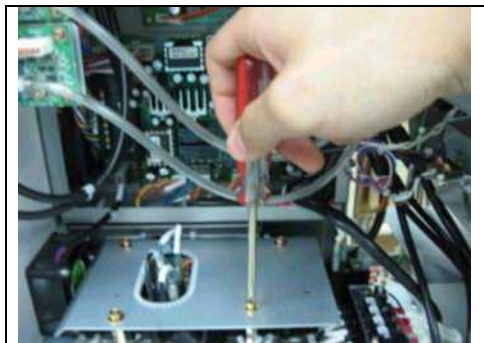
<Illustration>

<Removal>

⚠ Caution

-
- Be sure to turn OFF the breaker.
-

Remove the upper left face cover.



Remove the four screws fixing the cover.



Remove the fuse.

Install a new fuse.

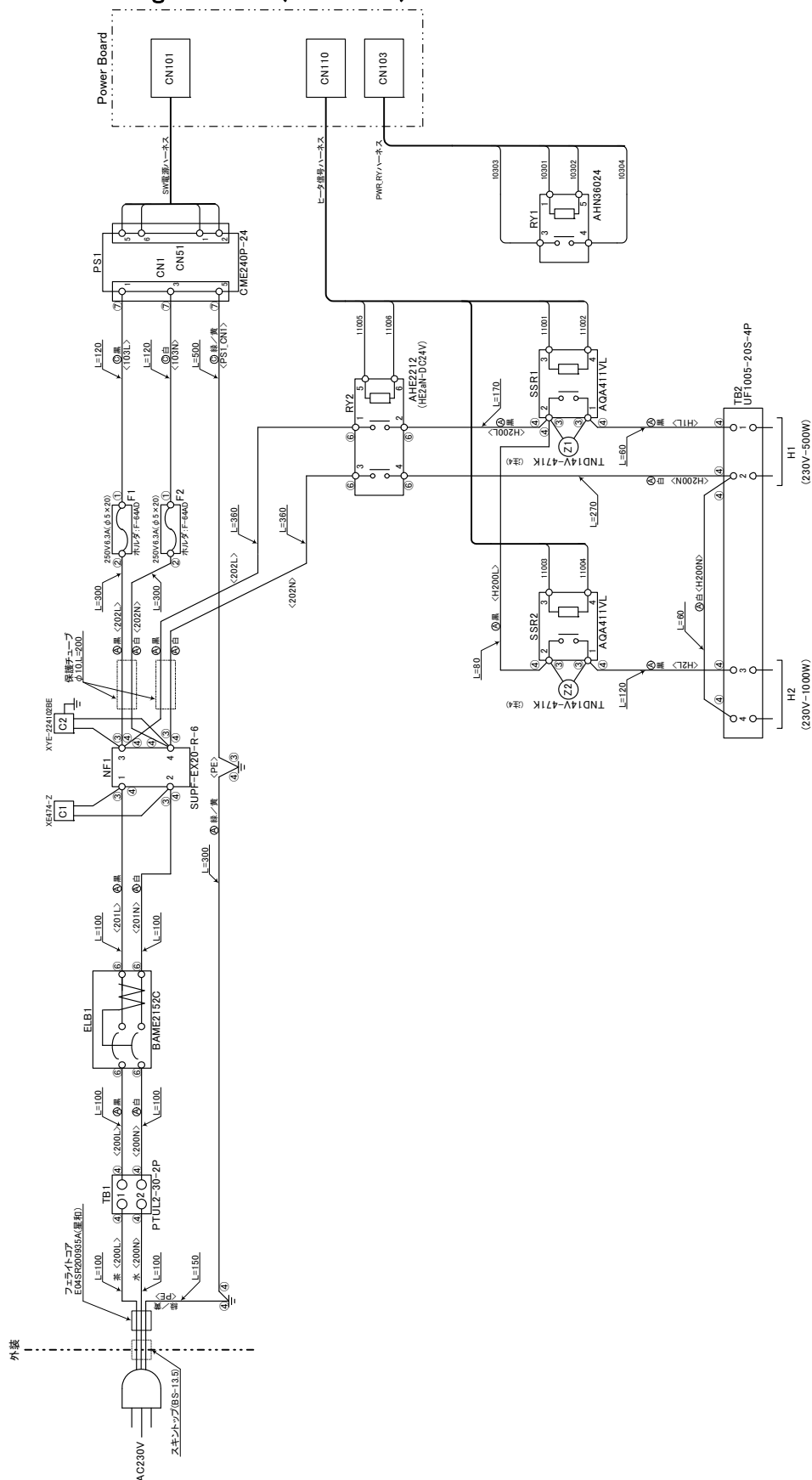
* Function of fuse

Protects AC power: 2 pcs. (250V, 6.3A)

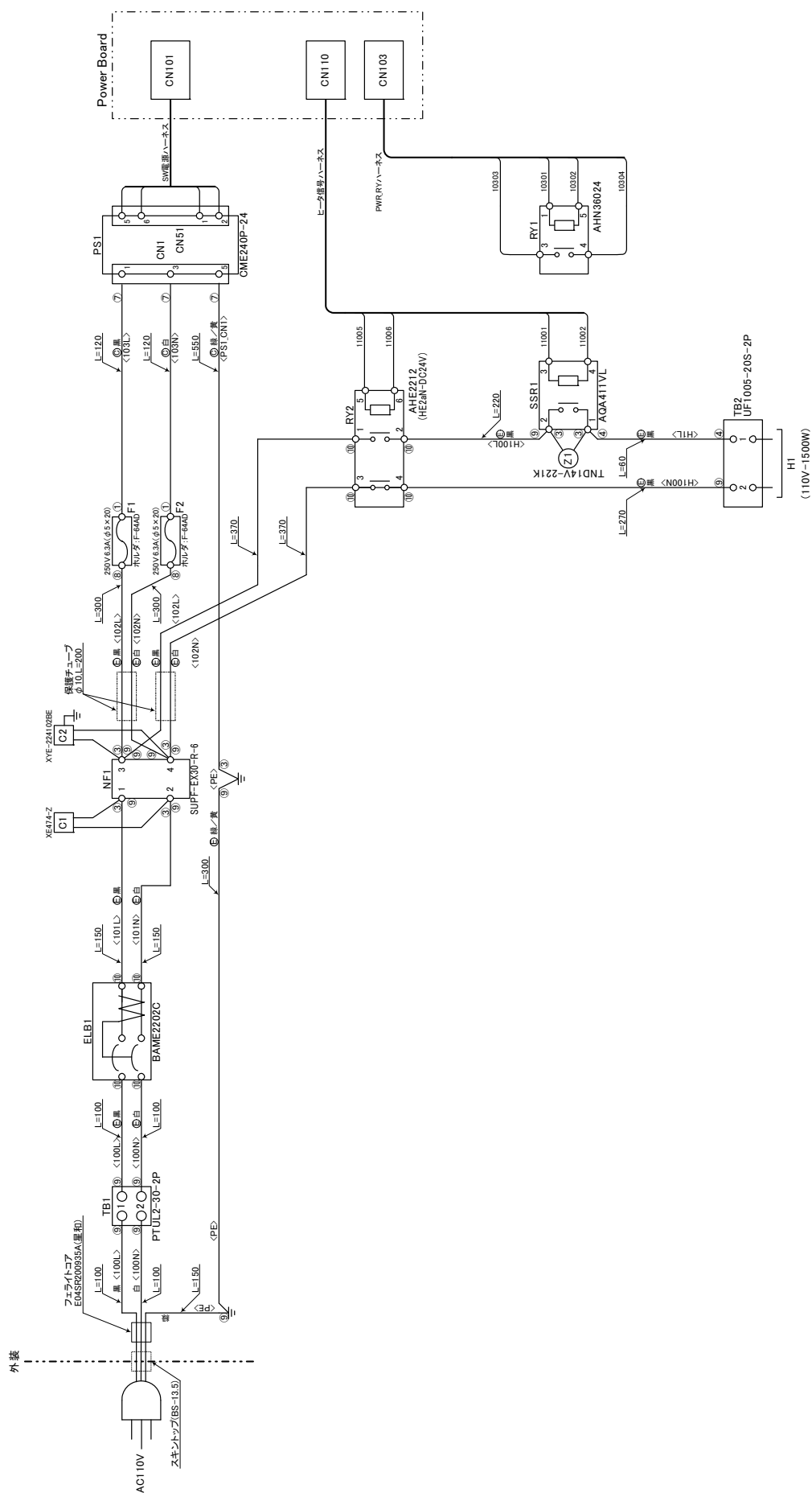
111. Power Section



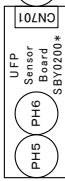
111.1. Overall configuration (230V)



111.2. Overall configuration (110V)



112. Electric Block Diagram



113. LAN Module Board



113.1. Overall configuration



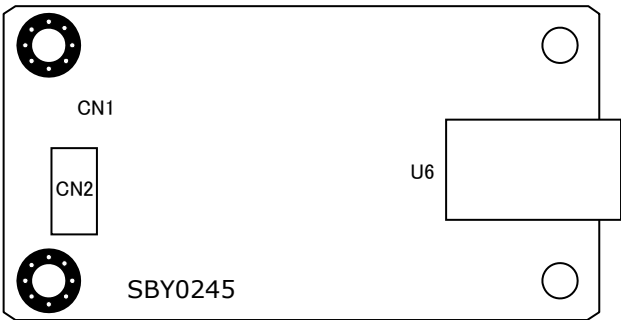
113.2. List of components

No.	Name	Model No.	Note
1	LAN Module Board	SBY0245*	

113.3. Summary and function of the board

- Performs LAN–serial conversion and communicates with the machine.

113.4. Layout drawing of the connector



113.5. Connection of each connector

- U6 : Using harness, connect to LAN cable.
- CN1 : Using harness, connect to CN120 of the Power Board.
- CN2 : Using harness, connect to CN230 of the I/O Board.

113.6. Removal and installation

<Illustration>



<Removal>

⚠ Caution

-
- Be sure to turn OFF the breaker.
-

Remove the upper left side face cover.

Remove the board bracket fixing screws from the back of the device.

Remove the connector on the LAN module board.

Remove the four screws fixing the LAN module board.

<Installation>

Install in the reverse order of the removal procedure.

4. Adjustment procedure

Adjustment Procedure List (for Type 2)

No.	Identification No.	Document name	Option
1	BHAJ04-001	PV/PA Board Calibration	
2	BHAJ04-002	Dialysate Pressure Sensor Calibration	
3	BHAJ03-022	Temperature Sensor Calibration ③ (T1)	
4	BHAJ04-003	Temperature Sensor Calibration ① (T2/T3/T7)	
5	BHAJ03-004	Temperature Sensor Calibration ② (T4/T5/T7)	
6	BHAJ04-005	Conductivity Sensor Calibration (CD1/CD2/CD3)	
7	CXAJ03-006	Bubble Detector Calibration	
8	BHAJ04-007	Blood Leakage Sensor Calibration	
9	CXAJ03-008	Adjustment of Reducing Valve	
10	CXAJ02-009	Adjustment of Supplying Flow and Fill-up Pressure	
11	CXAJ03-010	Adjustment of Closed Circuit Pressure	
12	CXAJ02-011	Confirmation of Flow Volume of P3 and P4	
13	BHAJ02-012	Clamp Leak Test	
14	BHAJ04-013	Syringe Test	
15	BHAJ04-014	Blood Pump Shutoff Status Check	
16	BHAJ03-015	Blood Pump Flow Adjustment	
17	CXAJ03-016	Balance Test	
18	BHAJ04-017	UFP Flow Measurement	
19	BHAJ04-018	UF Precision Test 1	
20	BHAJ04-019	UF Precision Test 2 (during SQHD)	
21	CXAJ02-020	Gas Purge Pressure Adjustment	
22	CXAJ02-021	Chamber Volumetric Measurement	
23	CXAJ03-023	Blood Sensor Calibration	

⚠ Caution

- Remove static electricity, using an earth-band when a board is maintained or changed.
- Do not touch any point other than the mentioned point by a tester when a board is maintained, connecting probes or IC clips of tester. It could be a cause of board breakdown, depending on the touched place.
- Do not touch any board, electric wiring or terminal except for a maintenance.
- Do not touch any electric wiring or terminal which is not isolated.
- Do not touch any board with wet hand.
- Pull out a power plug from an outlet when the harness is removed from a Terminal block.
- Use a tester which is “True RMS TYPE” to measure precise alternating current value.
* RMS: Root Mean Square.

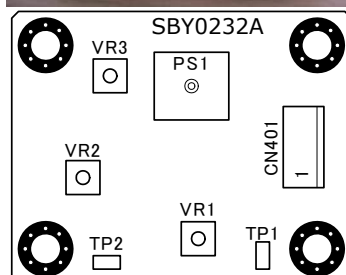
1. PV/PA Board Calibration



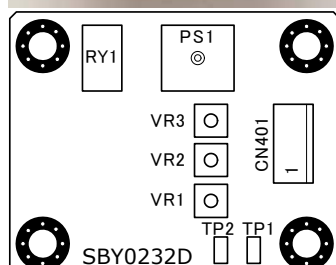
Standard	During air release : $0 \pm 2 \text{ mmHg}$ During pressurization with 300mmHg : $300 \pm 5 \text{ mmHg}$ (reference value: $-300 \pm 30 \text{ mmHg}$) During startup-test : $440 \pm 20 \text{ mmHg}$
Required jig	Pressure gauge, piping components, pean forceps x 1, ceramic driver for adjustment, syringe, tester
Purpose	To calibrate 2 points of offset and gain if the error is out of range, comparing device indication and manometer value.

< Illustration >

SBY0232A



After SBY0232D



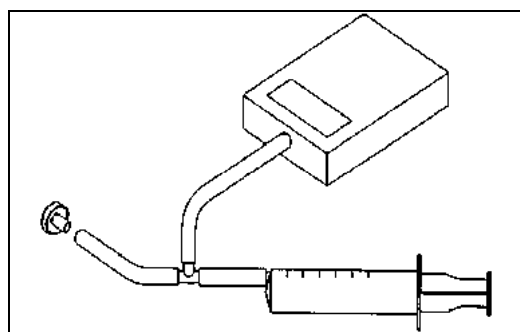
< Procedure >

Remove the cover of the upper left side.

For the two PV/PA placed in a line, the upper one is the venous pressure sensor board (PV) and the lower is the arterial pressure sensor board (PA).

Open the screen of MAINT - MAINT -
"1.DATE" "PRESSURE"

<Illustration>



Connection of pressure gauge

<Procedure>

【During air-release (offset adjustment)】

Set the pressure port to air-release and adjust the VR2 of the PV/PA board so that the displayed value becomes $0 \pm 2 \text{ mmHg}$.

【During pressurization with 300mmHg (gain adjustment)】

Pressurize by syringe, and when the pressure gauge value becomes approximately 300mmHg, shut off the silicon tube with pean forceps to fix the pressure.

NOTE

-
- If the pressure tends to be lowered even after shutting off the silicon tube with pean forceps, fix at the pressure slightly higher than 300mmHg and wait until the pressure becomes approximately 300mmHg.
-

Adjust the VR1 of the PV/PA board so that the displayed value of pressure gauge indicates $300 \pm 5 \text{ mmHg}$ when its actual value is 300mmHg.

Repeat the offset adjustment and gain adjustment until entering the standard value range.

【Check during pressurization with -300mmHg】

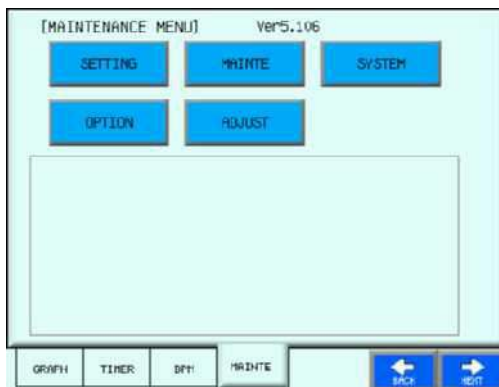
To check the linearity of the sensor output value, when 0mmHg and +300mmHg enter the standard value range, add -300mmHg with the syringe to the venous (arterial) pressure and check that the difference between the indication and displayed value is approximately ± 30 .

Change the board when the value can not be in this range.

NOTE

-
- This is only a check, so adjustment of the difference between indication and pressure gauge is unnecessary.
-

<Illustration>



<Procedure>

【Adjustment of startup-tested value】

● How to display the “Adjustment” menu

To adjust the startup-tested value, the “Adjustment” menu needs to be displayed. Simultaneously press the two parts indicated with ○ in the left figure for 3 seconds using the maintenance tag and the “ADJUST” switch is displayed as shown left.

Set the device to the rinse waiting process.

Press the PV (PA) switch at MAINT - ADJUST- “ 4.ROM ver” to simulatigly obtain the output voltage for startup-test.

NOTE

- During dialysis waiting process, output voltage for startup-tested cannot be obtained.

Adjust VR3 so that the displayed value becomes $440 \pm 20 \text{ mmHg}$.

CAUTION

- The “Adjustment” menu is for the manufacturer to perform adjustment and contains important items related to device control. After using it, be sure to hide it to prevent uninformed personnel from changing the contents.
- How to hide the menu is the same as how to display it.

Replace the following or make an adjustment if the value is not adjusted to the standard value.

Replace

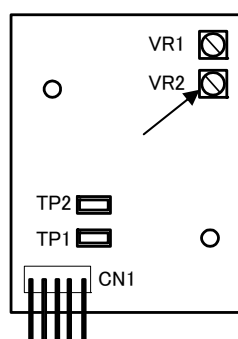
- PV/PA board

2. Dialysate Pressure Sensor Calibration



Standard	During air release : $0 \pm 2 \text{ mmHg}$ During pressurization with 300mmHg : $300 \pm 5 \text{ mmHg}$ (reference value: $-300 \pm 30 \text{ mmHg}$)
Required jig	Pressure gauge, piping components, pean forceps x 3, ceramic driver for adjustment, syringe
Purpose	To calibrate 2 points of offset and gain if the error is out of range, comparing device indication and manometer value.

<Illustration>



<Procedure>

【Draining fluid in the dialysate pressure sensor】

CAUTION

- Ensure that the breaker and I/O switch is turned off.

Remove the cover for the lower right side.

Remove the silicon tube from the dialysate pressure sensor to drain out the fluid.

【During air release (offset adjustment)】

Remove the cover of the dialysate pressure sensor.

Turn on the breaker and I/O switch.

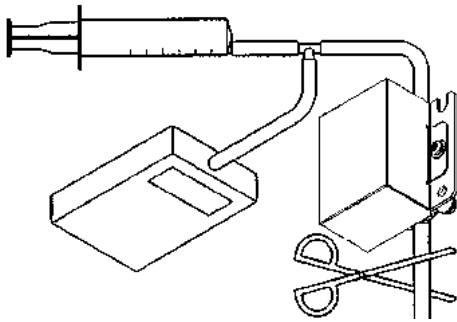
With the third page of MAINTENANCE -MAINTENANCE-

“6.CHARACTERISTICS” screen, set the dialysate pressure “display offset” to 0. If the dialysate pressure display offset is input intentionally, write down the input value for inputting again after adjustment.

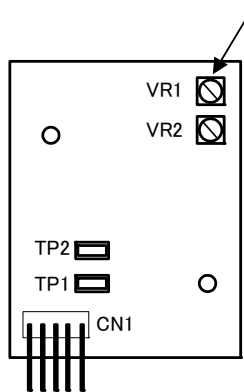
Open the “PRESSURE” screen of MAINTENANCE -MAINTENANCE- “1.DATA”.

Adjust the VR2 of the dialysate pressure sensor board so that the dialysis pressure displayed value becomes $0 \pm 2 \text{ mmHg}$ during air release.

<Illustration>



Connecting the pressure gauge



<Procedure>

【During pressurization with 300mmHg (gain adjustment)】

Connect the piping components to the dialysate pressure sensor to pressurize with the syringe. At the point where the pressure gauge value indicates approximately 300mmHg, shut off the silicon tube with the pean forceps to fix the pressure.

NOTE

- If the pressure tends to be lowered even after shutting off the silicon tube with the pean forceps, fix at the pressure slightly higher than 300mmHg and wait until the pressure becomes approximately 300mmHg.

Adjust the VR1 of the dialysate pressure sensor board so that the displayed value of pressure gauge indicates 300 ± 5 mmHg when its actual value is 300mmHg.

Repeat the offset adjustment and gain adjustment until entering the standard value range.

【Check during pressurization with -300mmHg】

To check the linearity of the sensor output value, when 0mmHg and +300mmHg enter the standard value range, add -300mmHg with the syringe and check that the difference between the indication and pressure gauge value is within ± 30 .

NOTE

- This is only a check, so adjustment of the difference between indication and pressure gauge is unnecessary.

Return the dialysate pressure display offset to the original input value if it is input intentionally.

Replace the following or make an adjustment if the value is not adjusted to the standard value.

Replace

- Dialysate pressure sensor Assy

3. Temperature Sensor Calibration① (T2/T3/T7)



Standard	T2/T3/T7 : 25°C±0.2°C, 40°C±0.2°C
Required jig	Temperature calibration device (Code NO.0-506-001), ceramic driver for adjustment, Phillips screwdriver
Purpose	To calibrate 2 points of offset and gain if the error is out of range, comparing device indication and manometer value.

<Illustration>



<Procedure>

CAUTION

- Ensure the breaker and I/O switch is turned off.

Remove the cover of the upper right side and the electric cover.

Remove the connector for the each temperature sensor, which is on the I/O board, and install the temperature calibration device.

NOTE

- Refer to CXSM**-101 for the connector position.

* Select based on the temperature sensor to be adjusted.

Temperature sensor		Connector name
T2	Heater outlet temperature	CN215
T3	Dialysate temperature	CN216
T7	Coupler temperature	CN246

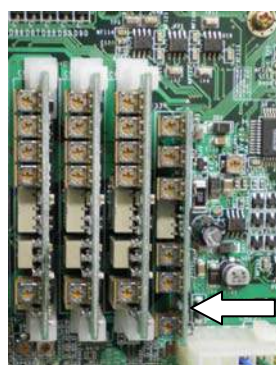
Temperature calibration device dummy resistance value

Temperature	Resistance value
25°C	8.494±0.024 kΩ
40°C	4.904±0.024 kΩ

【Configuration of temperature sensor module board】

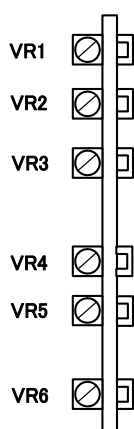


Right side of the device



Temperature sensor module board

<Illustration>



VR configuration of temperature sensor module board

<Procedure>

Turn on the breaker and I/O switch to confirm the indicated temperature with MAINTÉ – MAINTÉ - “1.DATA” - “TEM/CON (Temperature/concentration).”

【(Gain adjustment) at 25°C】

Set the switch of temperature calibration device is set to 25°C.

Adjust the adjustment VR on the temperature sensor module board so that the indicated temperature becomes $25.0 \pm 0.2^\circ\text{C}$.

【(Offset adjustment) at 40°C】

Set the switch of the temperature calibration device is set to 40°C.

Adjust the adjustment VR on the temperature sensor module board so that the indicated temperature becomes $40.0 \pm 0.2^\circ\text{C}$ when

Repeat 【offset adjustment】 and 【gain adjustment】 until entering the standard value range.

Temperature sensor	VR at 25°C	VR at 40°C
T2	VR3	VR4
T3	VR5	VR6
T7	VR1	VR2

- * Select based on the temperature sensor to be adjusted.
- * VR is on the temperature sensor module.

Replace the following or make an adjustment if the value is not adjusted to the standard value.

Replace

- I/O board

4. Temperature Sensor Calibration② (T4/T5/T6)



Standard	T4/T5/T6: 25°C±0.2°C、40°C±0.2°C
Required jig	Temperature calibration device (Code No. 0-506-001), ceramic driver for adjustment, Phillips screwdriver
Purpose	To calibrate 2 points of offset and gain if the error is out of range, comparing device indication and manometer value.

<Illustration>



<Procedure>

CAUTION

- Ensure that the breaker is turned off.

Remove the cover of the upper right side and the electric cover.

Remove the connector for the each temperature sensor, which is on the I/O board, install the temperature calibration device.

NOTE

- Refer to CXSM**-101 for the connector position.

* Select based on the temperature sensor to be adjusted.

Temperature sensor		Connector name
T4	For CD1 temperature compensation	CN217
T5	For CD2 temperature compensation	CN234
T6	For CD3 temperature compensation	CN239

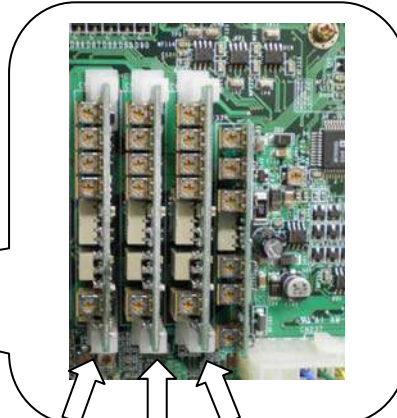
Temperature calibration device dummy resistance value

Temperature	Resistance value
25°C	8.494±0.024 kΩ
40°C	4.904±0.024 kΩ

【Configuration of conductivity sensor module board】



Right side of the device



CD1 CD2 CD3

<Illustration>

<Procedure>

NOTE

-
- Adjustment VR for T4/T5/T6 is on the conductivity sensor module board, not on the temperature sensor module board.
-

Turn on the breaker and I/O switch to confirm the indicated temperature with MAINTÉ – MAINTÉ - “1.DATA” - “TEM/CON (Temperature/concentration).”

【(Gain adjustment) at 25°C】

Set the switch of the temperature calibration device is set to 25°C.

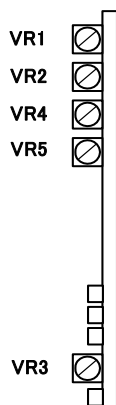
Adjust the adjustment VR on the conductivity sensor module board so that the indicated temperature becomes $25.0 \pm 0.2^\circ\text{C}$.

【(Offset adjustment) at 40°C】

Set the switch of the temperature calibration device is set to 40°C.

Adjust the adjustment VR on the conductivity sensor module board so that the indicated temperature becomes $40.0 \pm 0.2^\circ\text{C}$.

Repeat 【offset adjustment】 and 【gain adjustment】 until entering the standard value range.



VR configuration of conductivity sensor module board

Temperature sensor	VR at 25°C	VR at 40°C
T4 (CD1)	VR4	VR5
T5 (CD2)	VR4	VR5
T6 (CD3)	VR4	VR5

- * Select based on the temperature sensor to be adjusted.
- * VR is on the temperature sensor module.

Replace the following or make an adjustment if the value is not adjusted to the standard value.

Replace

- I/O board

5. Conductivity Sensor Calibration (CD1/CD2/CD3)



Standard	Standard voltage waveform :100±1.0mVAC	
	CD1	:2.50±0.02VDC (offset), 5.00±0.02VDC(gain)
	CD2/CD3	:0.20±0.02VDC (offset), 1.50±0.02VDC (gain)
Required jig	Concentration calibration device (Code No. 0-506-007), ceramic driver for adjustment, Phillips screwdriver, tester	
Purpose	To calibrate 2 points of offset and gain if the error is out of range, comparing device indication and manometer value.	

<Illustration>



<Procedure>

CAUTION

- Ensure that the breaker is turned off.

Remove the cover of the upper right face and the electric cover for the right face. Remove the connector for the conductivity sensor to be worked on, which is on the I/O board, and install the concentration calibration device.

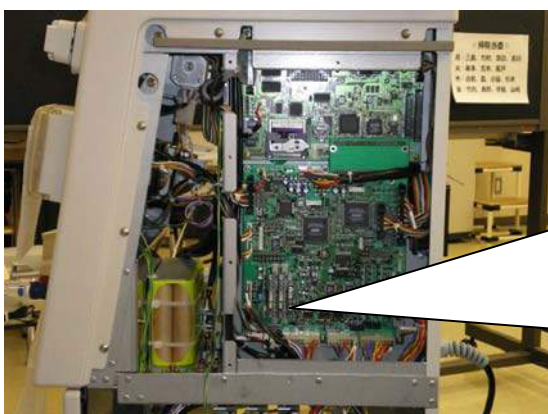
*Select based on the conductivity sensor to be adjusted.

Conductivity sensor		Connector name
CD1	A+B conductivity	CN218
CD2	B conductivity (moni)	CN233
CD3	B conductivity (cont)	CN238

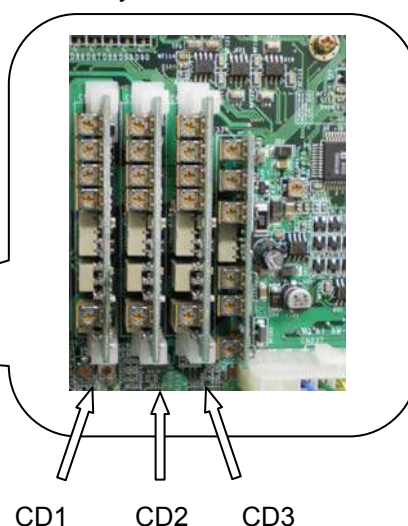
Concentration calibration device dummy resistance value

Cond. sensor	Conductivity	Resistance value
CD1	30mS	133.3±0.1Ω
	15mS	266.6±0.1Ω
CD2/CD3	9mS	444.4±0.1Ω
	1.2mS	3333.3±1.0Ω

【Configuration of conductivity sensor module board】



Right face of the device

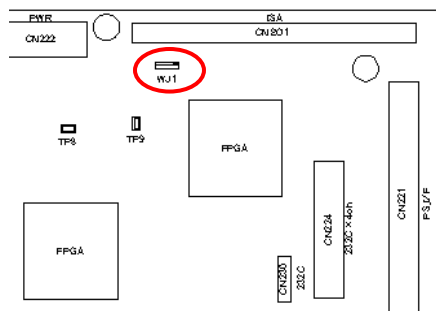


CD1

CD2

CD3

<Illustration>



WJ1 on the I/O board

<Procedure>

Turn on the breaker and I/O switch.

【Adjustment of standard voltage waveform】

Adjust the conductivity sensor to be worked on under the following conditions.

Set the tester range to “VAC.”

Tester (+) probe	Conductivity sensor module board TP2
Tester (-) probe	I/O board WJ1
Standard	100±1.0mVAC
Adjustment VR	Conductivity sensor module board VR3

【Gain adjustment】

Adjust the adjustment VR on the conductivity sensor module board so that the tester voltage becomes standard when the knob of the concentration calibration device is aligned.

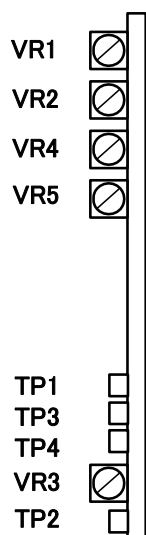
【Offset adjustment】

Adjust the adjustment VR on the conductivity sensor module board so that the tester voltage becomes standard when the knob of the concentration calibration device is aligned.

NOTE

●Do not refer to the indicated conductivity of screen as it's compensated value by 25 degrees.

Repeat 【offset adjustment】 and 【gain adjustment】 until entering the standard value range.



Configuration of conductivity sensor module board

Tester (+) probe	Conductivity sensor module board TP1
Tester (-) probe	I/O board WJ1
Gain adjustment VR	Conductivity sensor module board VR1
Offset adjustment VR	Conductivity sensor module board VR2

* Set the tester range to “VDC.”

* Select based on the conductivity sensor to be adjusted.

Standard

Conductivity sensor	Concentration calibration device Knob	Standard (VDC)
CD1	30mS (gain)	5.00±0.02
	15mS (offset)	2.50±0.02
CD2/CD3	9mS (gain)	1.50±0.02
	1.2mS (offset)	0.20±0.02

Replace the following or make an adjustment if the value is not adjusted to the standard value.

Replace

- I/O board

6. Bubble Detector Calibration



Standard	3.75±0.25V with the tube installed, 0.2V or less without the tube Single bubble of 0.03mL^{+0.005} is to be detected. 1.25V±0.1V (Self-Test voltage)
Required jig	Blood circuit, tester, ceramic driver for adjustment, micro syringe
Purpose	• To calibration output voltage of bubble detector. • Confirmation of bubble detector detection. • To calibration Self-Test voltage of bubble detector.

<Illustration>



<Procedure>

CAUTION

- Ensure that the breaker is turned off.

Remove the cover of the upper right side and the electric cover.

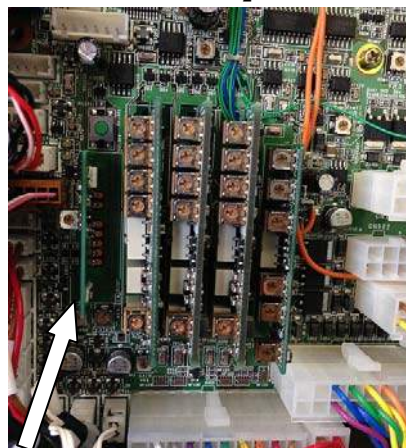
Turn on the breaker and I/O switch.

Set the bubble detection condition setting of the third page of MAINTÉ – MAINTÉ - “2.ALARM SETTING” screen to “SG10.”

【Configuration of Ultrasonic transceiver module board】

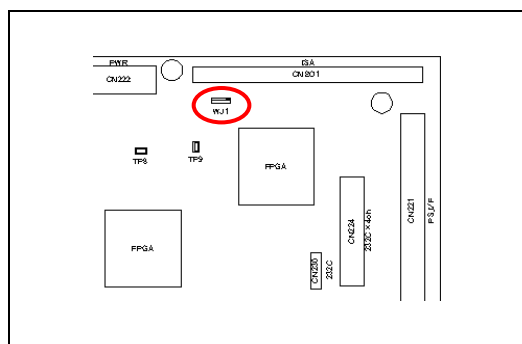


Right side of the device

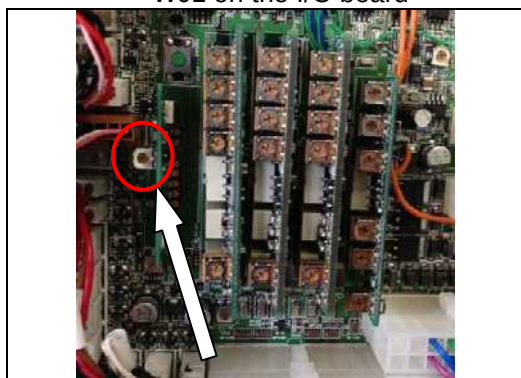
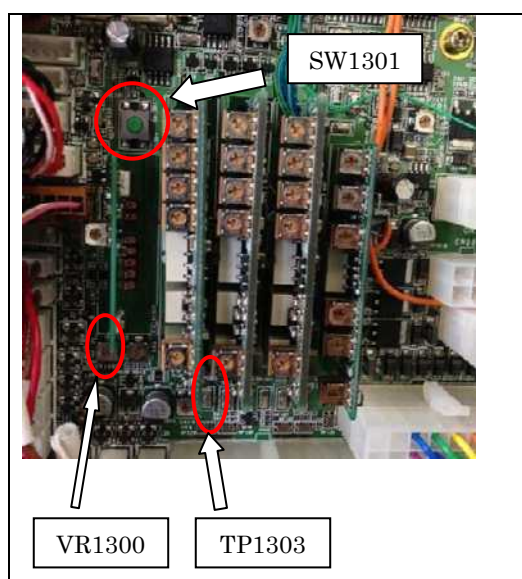


Ultrasonic transceiver module board

<Illustration>



WJ1 on the I/O board

Ultrasonic transceiver module board
VR1 locationBubble sensor adjustment
Volume, test point location

<Procedure>

【Voltage adjustment】

Set the standard tube containing the tepid water of approximately 37°C to the bubble detector.
Turn VR1300 on I/O board to the left end.

Set the tester range to “VDC” .

Connect the tester (+) probe to TP1303 on the I/O board.

Connect the tester (-) probe to WJ1 on the I/O board.

Turn VR1 on the ultrasonic transceiver module board to the left end and confirm 4.0V or more.

After confirmation, turn VR1 to the right end.

Turn VR1 slowly to the left and set to $3.75 \pm 0.25V$ by lowering the voltage gradually. .

*If the voltage lowers in the middle of this process, turn VR1 to the right end and start again.

Adjust TP1303 voltage to 1.15 ~ 1.35V with VR1300, pressing SW1301.

【Detection confirmation】

Set the device to the dialysis waiting process.
(To rotate blood pump)

Rotate the blood pump by 200mL/min. to circulate the tepid water of approximately 37°C in the blood circuit.

Turn on the bubble detector operation switch and confirm that the clamp is closed, blood pump stops and “E17(E18) AIR BUBBLES ARE DETECTED” is detected when bubble of 0.03mL is infused with a micro syringe.

Confirm that the TP1303 voltage value becomes 0.2V or less and “E19(E20) TUBE NON-EQUIPPED OR LARGE BUBBLE” is detected when removing the tube from the bubble sensor.

Set the bubble detection condition setting of the third page of MAINT – MAINT - “2.ALARM SETTING” screen to the original state.

Replace the following or make an adjustment if the value is not adjusted to the standard value.

Replace

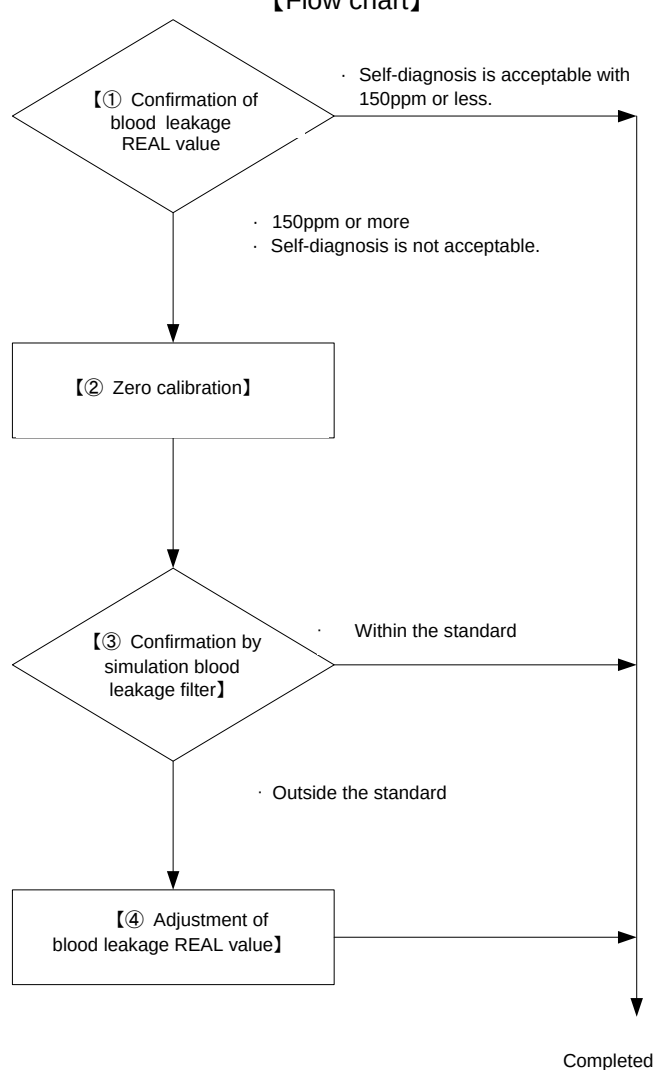
- I/O board
- Bubble detector ASSY

7. Blood Leakage Sensor Calibration



Standard	【① Confirmation of blood leakage REAL value】 : Self-diagnosis should be acceptable with 150ppm or less. 【③ Confirmation with simulation blood leakage filter】 : Blood leakage REAL value after calculation should be not be more than the standard value $\pm 80\text{ppm}$ described on the simulation blood leakage filter.
Required jig	Phillips screw driver, ceramic driver for adjustment, BLD-01 calibration cartridge
Purpose	To confirm whether blood leak is detected properly or not. Calibrate the sensor to detect blood leak properly.

【Flow chart】



<Illustration>

<Procedure>

【① Confirmation of blood leakage REAL value】

1. Display MAINTe -MAINTe- "1.DATA" - "BLD" screen."
2. Confirm the "BLD REAL VALUE" with no dirt or bubbles on the blood leakage cartridge window.
3. Procedure is completed if self-diagnosis is acceptable with 150ppm or less.
 - If self-diagnosis is not acceptable, proceed to 【②Zero calibration】 .
 - If 150ppm or more, proceed to 【②Zero calibration】 .

【②Zero calibration】

CAUTION

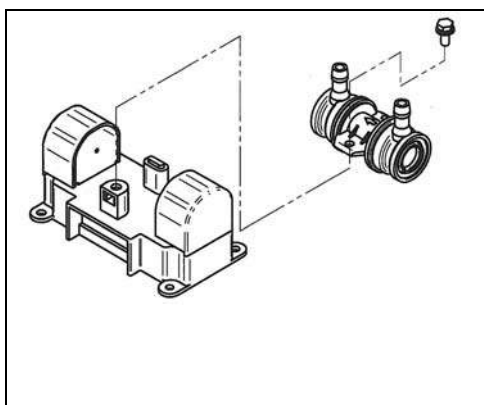
- Make adjustment after making sure that no dirt is attached by water rinse, chemical disinfection and acid rinse. Compensation with the dirt attached may cause the blood leakage detection value to deviate.

<Purpose>

Set the "BLD REAL VALUE" to 30 to 40ppm with no dirt or bubbles on the blood leakage cartridge window when the "BLD REAL VALUE" is 150ppm or more.

<Reason why the blood leakage real value should be adjusted to 30 to 40ppm, not 0>

No "BLD REAL VALUE" that are 0ppm or less shall be indicated, so if it is set to 0ppm, there is no judging whether it is 0 or under 0. Therefore, 30 to 40ppm, a little higher, shall be used for adjustment.



1. Operate the device for more than 5 minutes by T WTR rinse operation.

CAUTION

- Set T WTR rinse temperature to 37°C.

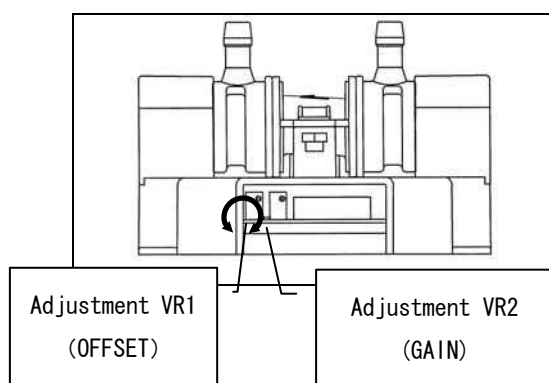
2. Remove the front cover.
3. Remove the blood leakage sensor cartridge, confirm there is no bubble within the cartridge, then return the cartridge to the original position.

CAUTION

- When removing the blood leakage sensor cartridge, do not lose O-rings at the both ends. These O-rings prevent the dialysate from entering the board when there is a leak on the cartridge.

4. Display MAINTe -MAINTe- "1.DATA" - "BLD" screen.

<Illustration>



<Procedure>

- Adjust the adjustment VR1 so that the "BLD REAL VALUE" becomes 30 to 40ppm.

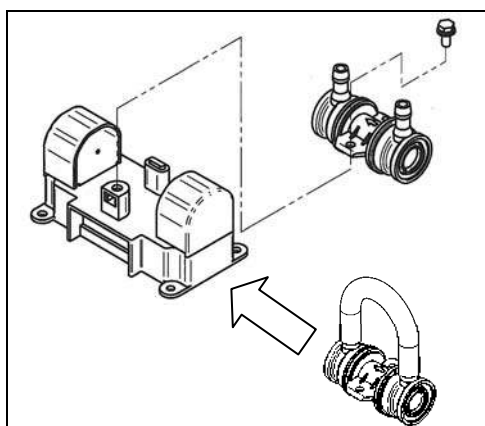
CAUTION

- Rotate the adjustment VR1 counterclockwise when the "BLD REAL VALUE" is 41ppm or more.
- Rotate the adjustment VR1 clockwise when the "BLD REAL VALUE" is 29ppm or less.

- Proceed to 【③ Confirmation by the simulation blood leakage filter】.

【③ Confirmation by the simulation blood leakage filter】<Purpose>

Make a blood leakage state by the simulation blood leakage filter, confirm the "BLD REAL VALUE" rises to the standard (refer to the followings) in that state, then confirm the blood leakage sensor detects the leakage normally.

**CAUTION**

- If the bubbles remain in the zero compensation cartridge, remove them by patting or shaking.

- Remove the cartridge of the blood leakage sensor and install the zero compensation cartridge.

* The "BLD REAL VALUE" at this time is to be (A).

CAUTION

- If the bubbles remain in the cartridge with simulation blood leakage filter, remove them by patting or shaking.

- Remove the zero compensation cartridge and install the cartridge with the simulation blood leakage filter.

* The "BLD REAL VALUE" at this time is to be (B).

<Illustration>

<Procedure>

3. Confirm the value obtained by $(B) - (A)$ is in the range of $\pm 80\text{ppm}$ from the standard value indicated on the simulation blood leakage filter.

Sample) Standard value: 360ppm

$$(B) - (A) = 360 \pm 80\text{ppm}$$

4. The procedure is completed if within the standard range.
 - If out of the standard range, proceed to 【④ Adjustment of the blood leakage REAL value】.

【④ Confirmation of blood leakage REAL value】

<Purpose>

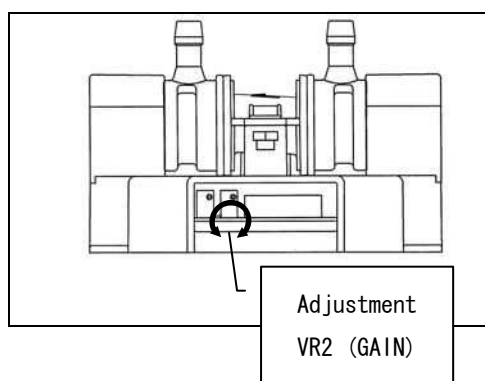
Adjust the blood leakage sensor so that it detects the blood leakage state normally.

“Zero point adjustment”

1. Replace the zero compensation cartridge with the same procedure as 【③ Confirmation by the simulation blood leakage filter】
2. Adjust the adjustment VR2 so that the "BLD REAL VALUE" becomes $80 \pm 10\text{ppm}$.

CAUTION

- Rotate the volume clockwise when the "BLD REAL VALUE" is over the standard value. Rotate the volume counterclockwise when the "BLD REAL VALUE" is under the standard value.



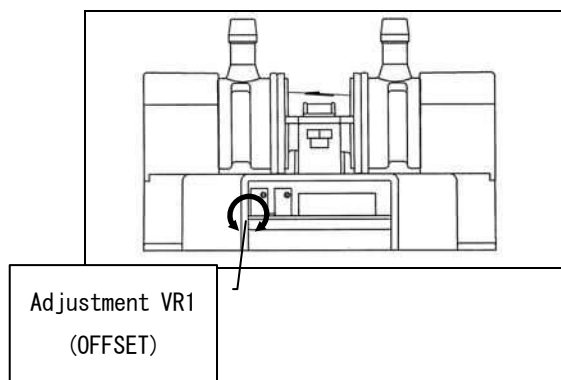
“Adjustment by the simulation blood leakage filter”

3. Replace on the cartridge with the simulation blood leakage filter with the same procedure as 【③ Confirmation by the simulation blood leakage filter】

4. Adjust the adjustment VR1 so that the "BLD REAL VALUE" becomes the standard value indicated on the simulation blood leakage filter $+80 \pm 10\text{ppm}$.

Sample) Standard value: 360ppm

$$(360 + 80) \pm 10\text{ppm}$$



<Illustration>

<Procedure>

5. Repeat the zero point adjustment and the simulation blood leakage filter adjustment until the "BLD REAL VALUE" enters in the range.
6. Remove the cartridge with the simulation blood leakage filter.
7. Install the original cartridge, operates for 5 minutes by T WTR rinse operation, then adjust VR1 so that the "BLD REAL VALUE" becomes 30 to 40ppm.

CAUTION

- Ensure that the both ends of the cartridge are installed with O-rings.
 - Set T WTR rinse temperature to 37°C.
-

Replace the following or make an adjustment if the value is not adjusted to the standard value.

Replace

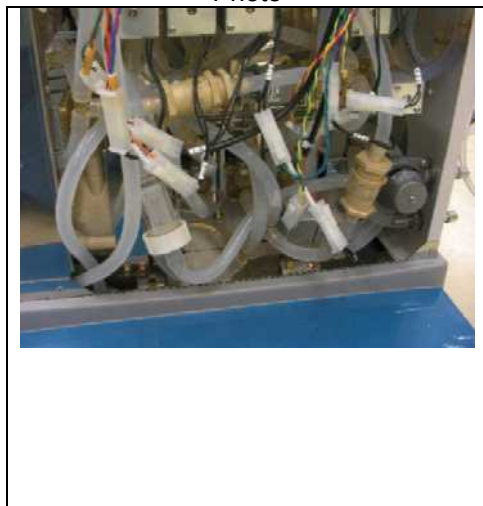
- Blood leak sensor Assy

8. Adjustment of Reducing Valve



Standard	0.034 to 0.038 MPa (257 to 287 mmHg)
Required jig	Pressure gauge, piping components, pean pincer x2, hexagon wrench (3 mm)
Purpose	Adjustment reducing valve to make proper secondary pressure.

<Photo>



<Operation procedure>

【Reducing valve check】

1. Set the machine to the Dialysis waiting process.
2. Turn V1, V55, V10, V4 and V35 to ON in MAINT-MAINT- "4.FLOW CHART" screen.
3. Turn V4 OFF and confirm the displayed PD pressure value in MAINT-AJUST- "1.DEBUG" screen is within the range of 257 to 287 mmHg (0.034 to 0.038 MPa). If not, perform the following adjustment. (Perform after adjusting the dialysate pressure sensor.)

CAUTION

- Confirm a numerical value in [] for PD value on the screen. Offset value is added in the value in [] on the left.

【Reducing valve adjustment】

1. Open the side cover in the right of machine.
2. Set the machine to the Dialysis waiting process.
3. Adjust the displayed PD pressure value in MAINT-AJUST- "1.DEBUG" screen to be within 257 to 287mmHg (0.034 to 0.038 MPa) by the reducing valve with a 3 mm hexagon wrench. (Secondary pressure can be measured by 【Reducing valve check】)
4. Repeat the above procedure to confirm the repeatability of the pressure value.
5. After adjustment, open V4 and V35 manually for approximately 2 seconds to release the pressure inside the machine.
6. Set all the valves to OFF

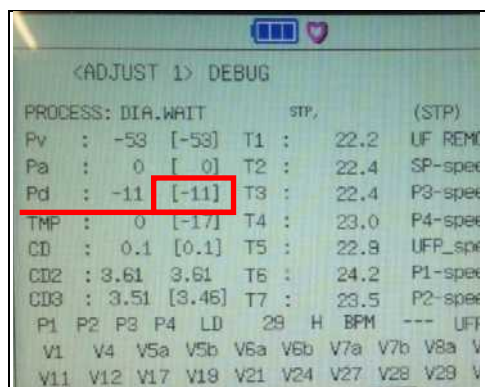
Replace the following or make an adjustment if the value is not adjusted to the standard value.

Replace

- Reducing valve

Adjustment

- Dialysate Pressure Sensor Calibration

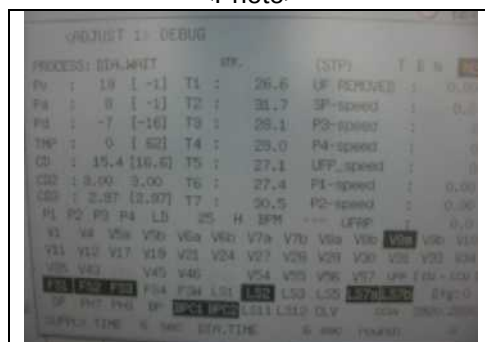


9. Adjustment of Supplying Flow and Fill-up Pressure



Standard	Supplying flow : Supply time must be 5.8 to 6.2 seconds Fill-up pressure : 0.098 to 0.108 MPa (1.0 to 1.1 kgf/cm ²) Degassing pressure : up to -650 mmHg
Required jig	Flat head screwdriver, pressure gauge, piping components, Clamping forceps x 2, monkey wrench
Purpose	Adjustment of supplying flow: Adjustment of flow so that supply completes before the liquid on the closed side runs out. Adjustment of fill-up pressure: Adjustment of fill-up pressure to make the lower adjustable UF volume within the criteria. Confirmation of degassing pressure: Confirmation of pressure to ensure the efficient dialysis with a dialyzer.

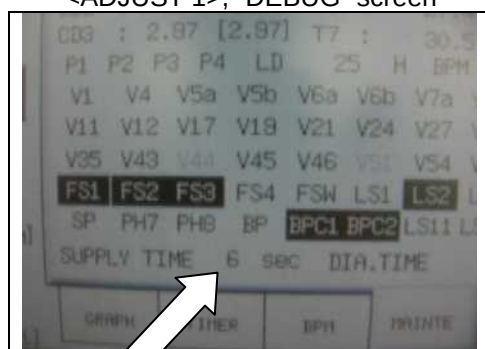
<Photo>



<Operation procedure>

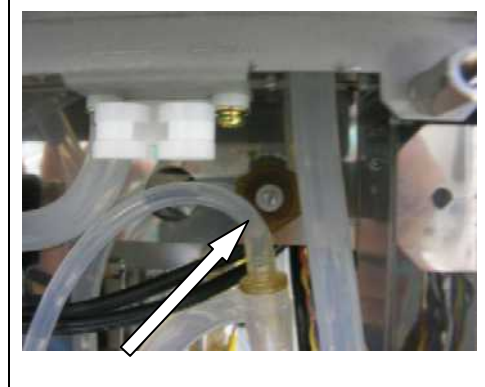
1. Start the machine operation at the Standby process.
2. Wait until the machine is in the Preparation complete (Bypass release) process.
3. Open the screen of MAINT-ADJUST- "1.DEBUG".

<ADJUST 1>, "DEBUG" screen



【Fill-up rate (time) adjustment】

1. Open the front cover of machine.
2. Check the "SUPPLY TIME" display in the screen of MAINT-ADJUST- "1.DEBUG".
3. Adjust the needle valve (NV1) so that the time displayed is 5.8 to 6.2 seconds.

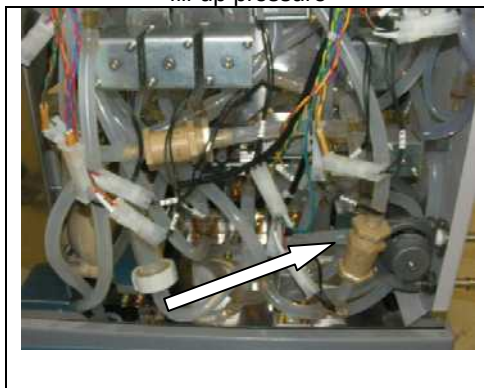


SUPPLY TIME display and NV1

<Photo>



Install position of pressure gauge for
fill-up pressure



Position of RV1



Install position of pressure gauge for
Degassing pressure

<Operation procedure>

1. Turn the breaker to OFF.
2. Install the pressure gauge for fill-up pressure just before "IN side of V5a and V7a".
3. Turn the breaker to ON.

【Fill-up pressure adjustment】

1. To keep the fill-up time interval (approx. 20 sec.), set the dialysate flow volume to 300 mL/min.
2. Adjust the relief valve (RV1) so that the pressure gauge indicates 0.098 to 0.108 MPa (1.0 to 1.1 kgf/cm²) at fill-up.
3. After completing RV1 adjustment, fix the lock nut with a monkey wrench.

1. Turn the breaker to OFF.
2. Install the pressure gauge for degassing pressure at before "P1 IN".
3. Turn the breaker to ON.

【Degassing pressure check】

1. Check that the degassing pressure is -650 mmHg or below at the above state.

Remove the pressure gauge.

Replace the following or make an adjustment if the value is not adjusted to the standard value.

Adjustment

- Adjustment of reducing pressure valve

10. Adjustment of Closed Circuit Pressure



Standard	Adjustment value without CF Flow rate at 1.2 V: 240 to 260 mL/min Pressure at 3.3 V: 0.098 to 0.108 MPa (1.0 to 1.1 kgf/cm ²) Adjustment value with CF Flow rate at 1.5V: 290 to 340 mL/min Flow rate at 4.2 V: 820 to 840 mL/min
Required jig	Flat head screwdriver, flow meter, pressure gauge, Pean forceps, piping components, blood line, dialyzer, monkey wrench, container of water
Purpose	Flow rate adjustment at 1.2 V(1.5V): Preliminary adjustment for RV2 adjustment Pressure adjustment at 3.3 V(4.2V): Adjustment to obtain the maximum dialysate flow rate

<Photo>

<Operation procedure>

1. Open the cover at both sides.

CAUTION

- Adjustment will be different in with CF and without CF. Make sure the procedure and adjust the pressure.

<Adjustment without CF>

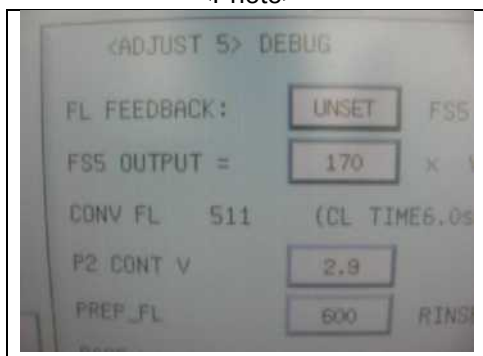
【Flow rate adjustment at 1.2 V】

1. Operate the device and wait until it is in the Preparation complete (Bypass release) process.
2. Set the blood circuit and the dialyzer. (Prepare water in a container to circulate as an alternative to blood.)
3. Prime the blood line and dialyzer and connect a coupler to the dialyzer.
4. Insert a needle to the blood circuit of the venous side.
5. Press the DIALYSE SW to start the dialysis operation.

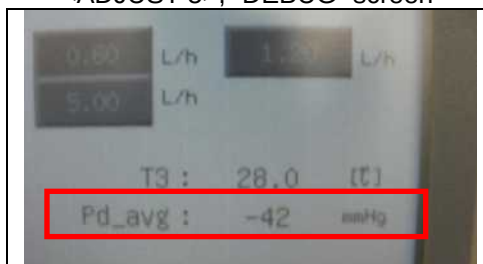
NOTE

- The pressure does not rise until the needle is inserted.

<Photo>



<ADJUST 5>, "DEBUG" screen



<ADJUST 2>, "UF CONTROL" screen



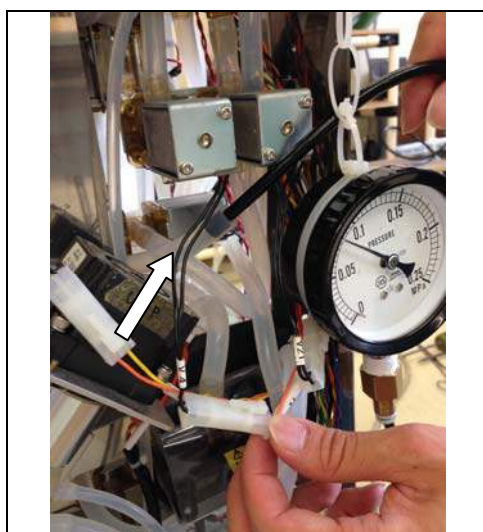
Right side of machine
Left: NV5; Right: NV6

<Operation procedure>

6. Open the fourth page of MAINT -ADJUST- "5.DEBUG SET" screen.
7. Turn "FL FEEDBACK" at the upper section to "UNSET" (OFF).
8. Change "P2 CONT V" to 1.2V.
9. Open the MAINT -ADJUST- "2.UF CONTROL" screen.
10. Turn the blood pump to ON and adjust the blood pump flow rate to set the Pd_avg to 250 ± 10 mmHg.
11. Open the fourth page of MAINT -ADJUST- "5.DEBUG SET" screen.
12. Adjust NV5 so that the CONV FL indicates the flow rate of 240 to 260mL/min.

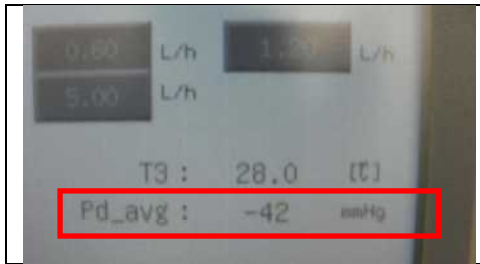
【Pressure adjustment at 3.3V】

1. Install the pressure gauge to "IN side of V4".
2. Turn the breaker to ON.
3. Open the fourth page of MAINT -ADJUST- "5.DEBUG SET" screen.
4. Change "P2 CONT V" to 3.3 V.
5. Operate the machine and wait until Preparation complete (Bypass release).

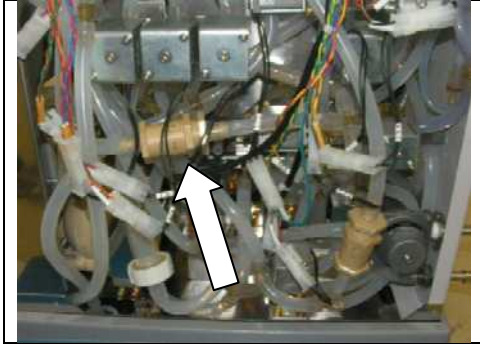


Installing position of pressure gauge for
close circuit pressure

<Photo>



<ADJUST 2>, "UF CONTROL" screen



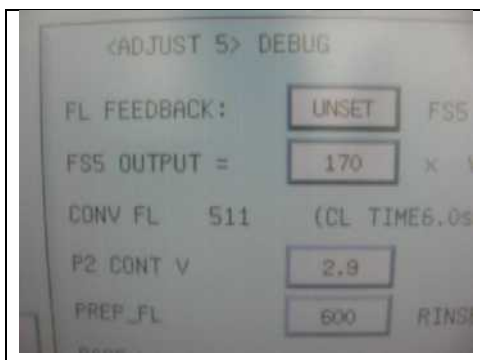
Position of RV2

<Operation procedure>

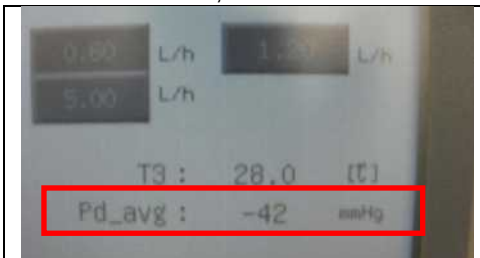
6. Open the MAINTÉ -ADJUST- "2.UF CONTROL" screen.
7. Turn the blood pump to ON and adjust the blood pump flow rate to set the Pd_avg to 250 ± 50 mmHg.
8. Adjust RV2 so that the closed circuit pressure immediately before chamber switching is 0.098 to 0.108 MPa (1.0 to 1.1 kgf/cm²).
9. After completing RV2 adjustment, fix the lock nut with a monkey wrench.
10. Open the fourth page of MAINTÉ -ADJUST- "5.DEBUG SET" screen.
11. Turn "FL FEEDBACK" at the upper section to "SET" (ON).
12. Remove the pressure gauge

【Confirm of adjustment】

1. Open the fourth page of MAINTÉ -ADJUST- "5.DEBUG SET" screen.
2. Turn "FL FEEDBACK" at the upper section to "UNSET" (OFF).
3. Change "P2 CONT V" to 1.2V.
4. Open the MAINTÉ -ADJUST- "2.UF CONTROL" screen.
5. Turn the blood pump to ON and adjust the blood pump flow rate to set the Pd_avg to 250 ± 10 mmHg.



<ADJUST 5>, "DEBUG" screen



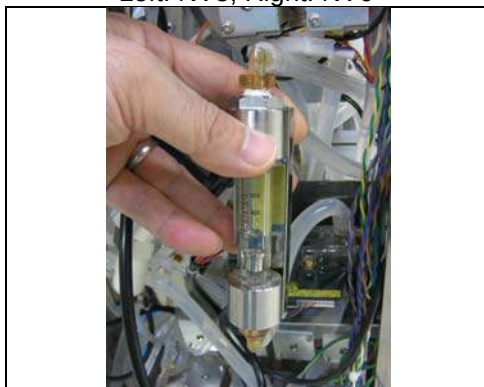
<ADJUST 2>, "UF CONTROL" screen

<Photo>



Right side of machine

Left: NV5; Right: NV6



Flow meter

<Operation procedure>

6. Open the fourth page of MAINTENANCE -ADJUST- "5.DEBUG SET" screen.

7. Confirm the CONV FL indicates the flow rate of 240 to 260mL/min.

1. Change the "P2 CONT V" to 1.5 V.

2. Turn the breaker to OFF.

3. Install Flow meter to the tube at P2 return line (BLD SENSOR line)

4. Turn the breaker to ON.

5. Confirm the flow meter is 150mL/min or above when the Pd_avg is 250 ± 10 mmHg.

6. Turn the breaker to OFF.

7. Remove Flow meter

<Photo>

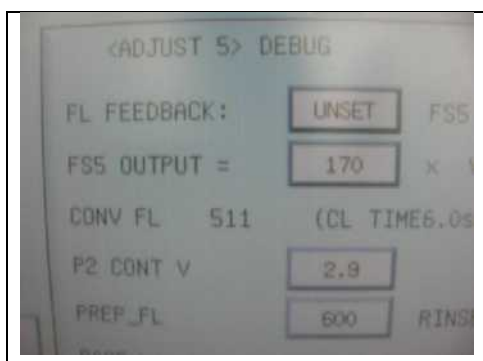
<Operation procedure>

<Adjustment with CF>**【Flow rate adjustment at 1.5 V】**

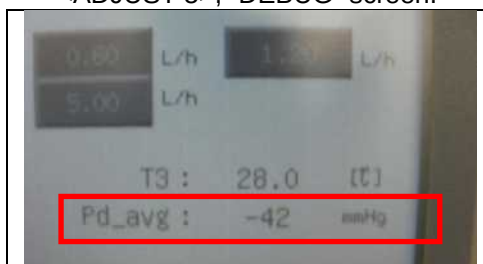
1. Operate the machine and wait until it is in the Preparation complete (Bypass release) process.
2. When the preparation complete process is ready, place the blood circuit and the dialyzer and connect a coupler to the dialyzer. (Pour water in a container to circulate as an alternative to blood.)
3. Perform the priming.
4. Insert a needle to the blood circuit of the venous side.

NOTE

- The pressure does not rise until the needle is inserted.



<ADJUST 5>, "DEBUG" screen.



<ADJUST 2>, "UF CONTROL" screen



Right side of machine

Left: NV5; Right: NV6

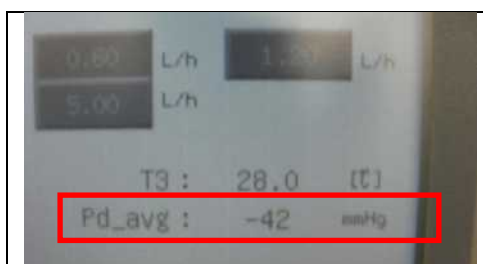
5. Open the fourth page of MAINTe -ADJUST- "5.DEBUG SET" screen.
6. Turn "FL FEEDBACK" at the upper section to "UNSET" (OFF).
7. Change "P2 CONT V" to 1.5V.
8. Open the MAINTe -ADJUST- "2.UF CONTROL" screen.
9. Turn the blood pump to ON and adjust the blood pump flow rate to set the Pd_avg to 250 ± 10 mmHg.
10. Open the fourth page of MAINTe -ADJUST- "5.DEBUG SET" screen.
11. Adjust NV5 so that the CONV FL indicates the flow rate of 290 to 340mL/min.

<Photo>

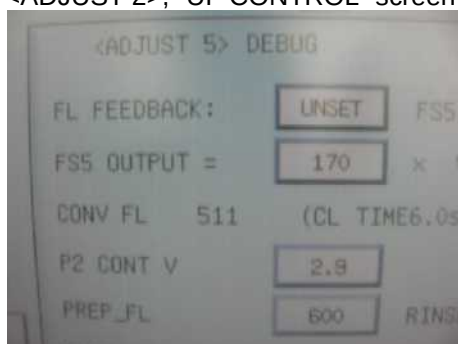
<Operation procedure>

【Pressure adjustment at 4.2V】

1. Turn the breaker to ON.
2. Open the fourth page of MAINT-ADJUST- “5.DEBUG SET” screen.
3. Change “P2 CONT V” to 4.2 V.
4. Operate the machine and wait until Preparation complete (Bypass release).
5. Open the MAINT-ADJUST- “2.UF CONTROL” screen.
6. Turn the blood pump to ON and adjust the blood pump flow rate to set the Pd_avg to 250 ± 10 mmHg.



<ADJUST 2>, “UF CONTROL” screen



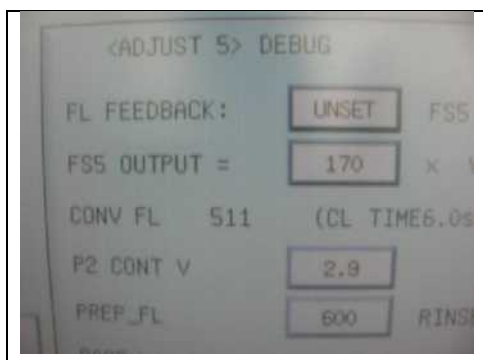
<ADJUST 5>, “DEBUG” screen.



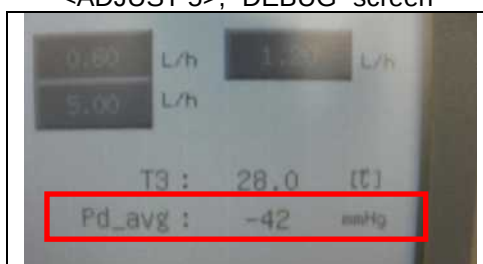
Position of RV2

7. Open the fourth page of MAINT-ADJUST- “5.DEBUG SET” screen.
8. Adjust RV2 so that the CONV FL indicates the flow rate of 800~820mL/min (target: 820mL/min).
9. After completing RV2 adjustment, fix the lock nut with a monkey wrench.
10. Turn “FL FEEDBACK” at the upper section to “SET” (ON).

<Photo>



<ADJUST 5>, "DEBUG" screen

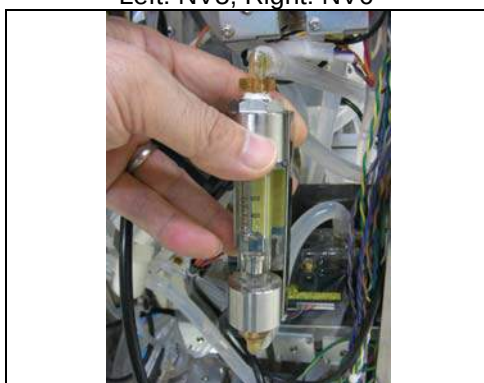


<ADJUST 2>, "UF CONTROL" screen



Right side of machine

Left: NV5; Right: NV6



Flow meter

<Operation procedure>

【Confirm of adjustment】

1. Open the fourth page of MAINT-ADJUST- "5.DEBUG SET" screen.
2. Turn "FL FEEDBACK" at the upper section to "UNSET" (OFF).
3. Change "P2 CONT V" to 1.5V.
4. Open the MAINT-ADJUST- "2.UF CONTROL" screen.
5. Turn the blood pump to ON and adjust the blood pump flow rate to set the Pd_avg to 250 ± 10 mmHg.
6. Open the fourth page of MAINT-ADJUST- "5.DEBUG SET" screen.
7. Confirm the CONV FL indicates the flow rate of 290 to 340mL/min.
8. Turn the breaker to OFF.
9. Install Flow meter to the tube at P2 return line (BLD SENSOR line)
10. Turn the breaker to ON.
11. Confirm the flow meter is 150mL/min or above when the Pd_avg is 250 ± 10 mmHg.
12. Turn the breaker to OFF.
13. Remove Flow meter

11. Confirmation of Flow Volume of P3 and P4



Standard	P3 Flow volume: 0.285 to 0.315mL/rev P4 Flow volume: 0.285 to 0.315mL/rev
Required jig	Electric scale, container (approximately 2L)
Purpose	Weighing of pump discharging volume and input of it for the correct infusion volume of concentrate.

<Illustration>



<Procedure>

[Confirmation of Flow Volume of P3]

Open the first page of MAINT—ADJUST—5.DEBUG SET screen when the machine is in Dialysis waiting mode, and turn 1.DEBUG DISPLAY “ON”.

1. Set the container containing water to the electric scale and perform zero reset.
2. Place A concentration (P3) suction tube into the container and set P3 of MAINT—MAINT— 4.FLOWCHART screen to “ON”.
3. Remove the A concentrate (P3) suction tube when P3 auto stops (100 revolutions = about 40 sec), and read the display value of the scale. (Caution: Dripping water from the tube)
4. Open the first page of MAINT—MAINT— 6.CHARACTERISTICS, and enter the value obtained by dividing the scale value by 100 in 1.P3 FLOW.
5. Return the A concentrate suction tube

<Illustration>



<Procedure>

[Confirmation of Flow Volume of P4]

1. Set the container containing water to the electric scale and perform zero reset.
2. Place B concentration (P4) suction tube into the container and set P4 of MAINT—MAINT— 4.FLOWCHART screen to "ON".
3. Remove the B concentrate (P4) suction tube when P4 auto stops (100 revolutions = about 40 sec), and read the display value of the scale. (Caution: Dripping water from the tube)
4. Open the first page of MAINT—MAINT— 6.CHARACTERISTICS, and enter the value obtained by dividing the scale value by 100 in 2.P4 FLOW.
5. Return the B concentrate suction tube
6. Open the first page of MAINT—ADJUST—5.DEBUG SET, and return 1.DEBUG DISPLAY to "OFF".

Replace the following or make an adjustment if the value is not adjusted to the standard value.

Replace

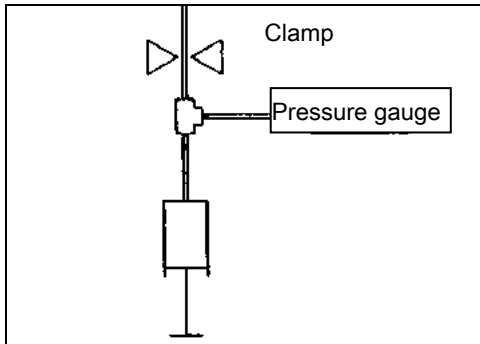
- Pump for A concentrate Assy (P3)
- Pump for B concentrate Assy (P4)

12. Clamp Leak Test



Standard	1mmHg or less/ 2 minutes
Required jig	Pressure gauge, piping components, pean forceps x 2, syringe, stopwatch
Purpose	Leak test in clamped state

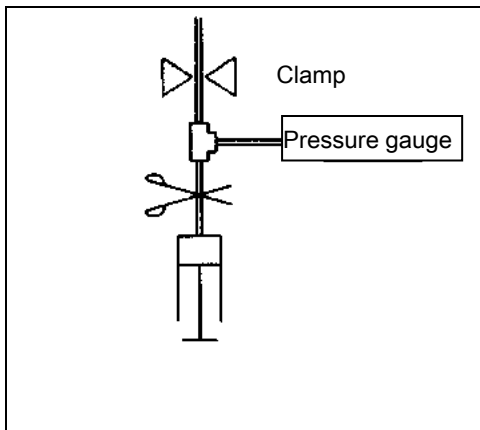
<Illustration>



<Procedure>

Set the pressure gauge, piping components (with no water) and syringe to the clamp.

With bubbles, turn on the bubble detector operation switch to clamp.

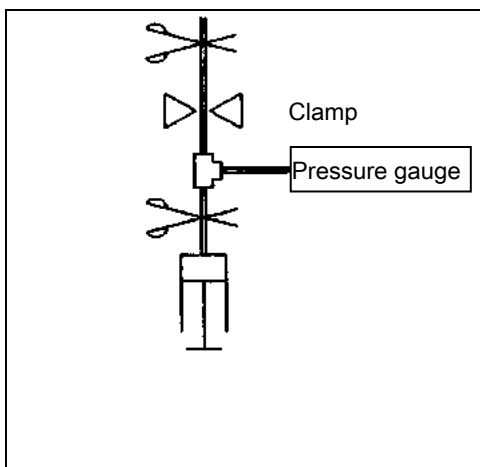


Pressurize with a syringe and fix the pressure gauge value to 150mmHg with pean forceps.

Pressure can be easily pulled out immediately after fixation, so leave it for approximately 90 seconds.

The pressure gauge value after leaving for 90 seconds is to be A.

The pressure gauge value after leaving for another 2 minutes is to be B.



Shut off the adjusting tool for leakage test with a forceps as in the left figure.

The pressure gauge value after leaving for another 2 minutes is to be C.

The value obtained by (A-B) is to be D.

The value obtained by (B-C) is to be E.

NOTE

- This value E indicates the leakage amount from the piping components.

<Illustration>

<Procedure>

The value obtained by (D-E) is to be F.

Confirm that the value F is 1mmHg or less.

Remove the adjusting tool for the leakage test to the original state.

Replace the following or make an adjustment if the value is not adjusted to the standard value.

Replace

- Clamp Assy

13. Syringe Test



Standard	Stroke: 0.494 to 0.546mm/min (Syringe ratio : 100%), overload : 500 to 800mmHg
Required jig	Stopwatch, caliper square, calculator, pressure gauge, syringe (20mL), piping components
Purpose	Stroke: For syringe injection rate verification. Overload: To verify whether the syringe is stopped at the specified pressure or not.

<Illustration>



<Procedure>

【Stroke test】

11.SYRINGE RATIO of the second page of MAINTENANCE- "6.CHARACTERISTICS" is to be 100%.

Move the pusher of heparin pump to the right end and measure the position with the caliper square. "Distance A"
Press the heparin pump flow display section and set the heparin pump flow to 9.9mL/hour.

Make sure the blood pump is running. (Without blood pump revolution, the syringe pump does not work.)

Start up time measurement by a stopwatch simultaneously with turning on the blood pump operation switch.

NOTE

- The heparin pump operates in conjunction with the blood pump.
- Measurement with the actual injection amount is impossible due to individual difference of syringe. Therefore, normality of the heparin pump injection amount should be confirmed by obtaining the pusher travel distance per 1 minute in the stroke test and confirming the obtained distance is within the standard range.



Turn OFF the heparin pump operation switch after operation for 30 minutes then measure the pusher move distance with the caliper square. "Distance B"

Move distance per 1 minute =

Calculate (Distance A [mm]—Distance B[mm])÷ operation time [min].

Confirm the travel distance per 1 minute is 0.494 to 0.546 mm/min.

If out of the standard range, refer to BHSM**-003.

CAUTION

- After the test, return the syringe ratio to the original value.

<Illustration>



<Procedure>

【Overload adjustment】

Remove the cover for the upper left face.

Set the syringe with the pressure gauge installed at the end.

Turn ON the heparin pump operation switch.

Adjust the nut ④⑦ so that the “E63 syringe overload or injection completion” alarm is detected when long-pressing the heparin pump flow fast-forward switch and adding pressure of approximately 0.086Mpa (500 to 800mmHg). (For the nut ④⑦ position, refer to BHSM**-003.)

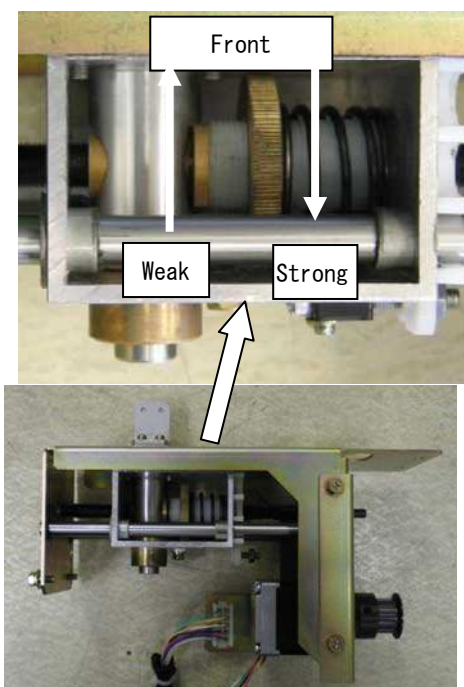
CAUTION

- Overload detection pressure fluctuates due to sliding resistance of the syringe gasket.

Rotating in the compressing direction of O.L. spring ④⑨ rises the overload detection pressure. Rotating in the extending direction lowers the pressure.

NOTE

- Overload detection value changes approximately 0.022Mpa (160mmHg) per one turn of nut. (Pressure change is the reference value when the Nipro 20mL is used.)



* No need to remove the heparin pump for adjustment.

Replace the following or make an adjustment if the value is not adjusted to the standard value.

Replace

- Heparin pump Assy

14. Blood Pump Shutoff Status Check



Standard	Discharge pressure to be 0.1 - 0.2Mpa (1.0 - 2.0kgf/cm ²).
Required jig	Test tools, pressure gauge, blood circuit
Purpose	To adjust occlusion.

<Illustration>



<Procedure>

【Large diameter】

Set the test tools to the blood circuit and operate the blood pump at 350mL/min.

Discharge pressure to be 0.10 - 0.20Mpa (1.0 - 2.0kgf/cm²).

CAUTION

- Specification of the blood circuit set here is as follows. If a blood circuit not complying with the specification, it will not be covered by the warranty.
- $\phi 8.00 \times \phi 12.00 \pm 0.15\text{mm}$, L : 210mm or more, t=2.00±0.1mm

【Small diameter】

Set the test tools to the blood circuit and operate the blood pump at 300mL/min.

Discharge pressure to be 0.10 - 0.20Mpa (1.0 - 2.0kgf/cm²).

CAUTION

- Specification of the blood circuit set here is as follows. If a blood circuit not complying with the specification, it will not be covered by the warranty.
- $\phi 6.35 \times \phi 9.75 \pm 0.15\text{mm}$, L : 210mm or more, t=1.70±0.1mm

Refer to the following for the gap adjustment.

- Blood pump

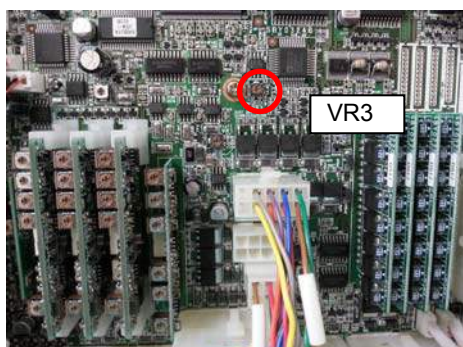
BHSM**-013

15. Blood Pump Flow Adjustment



Standard	Difference between the set flow and actual flow: within 10%
Required jig	Blood circuit, electric scale, stopwatch container
Purpose	Adjustment of difference between the set flow and actual flow

<Illustration>



<Procedure>

Set the blood circuit.

Use the electric scale to measure the actual flow per one minute by the given flow setting.

Adjust VR3 on the I/O board so that the difference between the set flow rate and the actual flow rate becomes within 10%.

On the screen of MAINT -MAINT- "4.FLOW CHART", the blood pump rotation number "BP REAL value" is displayed. Take it as a guide to rotate VR3.

After adjustment, use the electric scale again to measure the actual flow per one minute by the given flow setting and confirm that it is within the standard.

CAUTION

- Specification of the blood circuit set here is as follows. If a blood circuit not complying with the specification, it will not be covered by the warranty.
- $\phi 8.00 \times \phi 12.00 \pm 0.15\text{mm}$, L : 210mm or more, $t = 2.00 \pm 0.1\text{mm}$
- $\phi 6.35 \times \phi 9.75 \pm 0.15\text{mm}$, L : 210mm or more, $t = 1.70 \pm 0.1\text{mm}$

Replace the following or make an adjustment If the value is not adjusted to the standard value. Refer to the following for the gap adjustment.

- Blood pump BHSM**-013

Replace the following.

- I/O board

16. Balance Test



Standard	-10 to 60g/h
Required jig	Blood line, dialyzer, electric scale, container (approximately 2L), stopwatch
Purpose	Balance check of closed circuit

<Photo>



<Operation procedure>

1. Operate the device and wait until it is in the Preparation complete (Bypass release) process.
2. Set the blood circuit and the dialyzer. (Prepare water in a container to circulate as an alternative to blood.)
3. Prime the blood line and dialyzer and connect a coupler to the dialyzer.
4. Insert a needle to the blood circuit of the venous side.
5. Press the DIALYSE SW to start the dialysis operation.

NOTE

- Pressure will not rise unless a needle is attached.

6. Set the dialysate flow rate to 500 mL/min.

7. Open the MAINTENANCE-ADJUST- "2.UF CONTROL" screen.

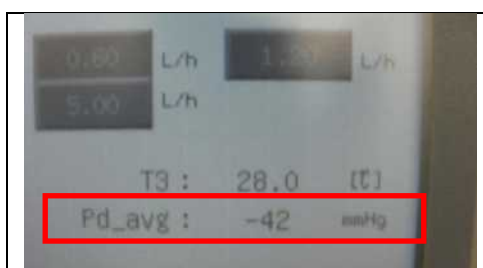
8. Turn the blood pump to ON and adjust the blood pump flow rate to set the Pd_avg to 200 ± 20 mmHg.

NOTE

- Do not change the blood pump flow rate setting onwards.

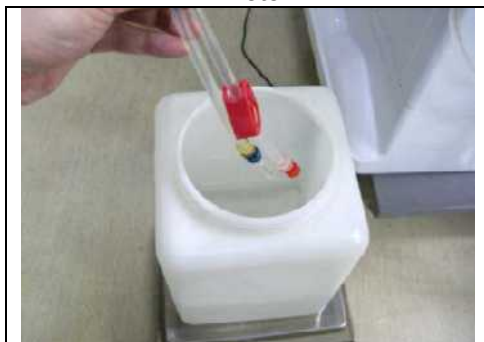
9. Wait until the dialysate temperature is within $36.5 \pm 0.3^\circ\text{C}$ with the temperature setting of 36.5°C .

10. When dialysate pressure is stabilized at 200 ± 20 mmHg, turn off the blood pump operation switch to stop the blood pump.



<ADJUST 2>, "UF CONTROL" screen

<Photo>



<Operation procedure>

11. Pull out the arterial/venous terminal of the blood circuit from the container and wait for the last drop.

CAUTION

- Ensure to collect the dripping inside the container.
-



12. Place the container on the electric scale and reset the weight to 0.

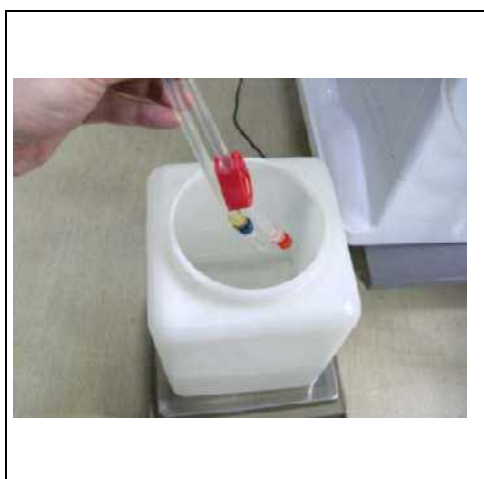
13. Put in the blood circuit to the container, and turn on the blood pump operation switch. Start the stop watch at the same time.

14. After 30 minutes, turn off the blood pump operation switch.

15. Extract the arterial/venous terminal of the blood circuit from the container and wait until the last drop.

CAUTION

- Ensure to collect dripping inside the container.
-



16. Read the value on the electric scale and calculate the UF volume per hour.

UF volume [g/h] = (decrease rate of water from the scale reading) [g/30 min.] × 2

If the UF volume does not reach -10 to 60 g/h, perform the adjustment below or the leakage inspection for closed circuit.

- Fill-up rate/fill-up pressure
- Closed circuit

17. UFP Flow Measurement



Standard	0.700 to 0.800mL/STROKE
Required jig	Blood circuit, dialyzer, electric scale, container (approximately 5L), stopwatch
Purpose	Measure discharge volume of pump and calculate the coefficient value, to execute precise ultra-filtration.

<Illustration>



<Procedure>

1. Operate the device and wait until it is in the Preparation complete (Bypass release) process.
2. Set the blood circuit and the dialyzer. (Prepare water in a container to circulate as an alternative to blood.)
3. Prime the blood line and dialyzer and connect a coupler to the dialyzer.
4. Insert a needle to the blood circuit of the venous side.

CAUTION

- Ensure that the venous side of the blood circuit is installed with a needle.

Set the UF RATE to 5.00L/h and set the UF GOAL to 2.50L/h or more.

Set the "26.UF RATE ALARM MAX" of the fourth page of MAINTENANCE - MAINTENANCE - "2.ALARM SETTING" to 100%. (Default value: 30%)

Change the "3.UFP FLOW" set value of the first page of the MAINTENANCE - MAINTENANCE - "6.CHARACTERISTICS" to 0.750mL.

Wait until the dialysate temperature enters the range of $36.5 \pm 0.5^{\circ}\text{C}$ with the setting at 36.5°C .

Pull out the arterial and venous end of the blood circuit from the container and wait until there is no drop.

CAUTION

- Ensure that the drops are put in the container.

Place the container on the electrical scale and reset the weight display to zero.



<Illustration>



<Procedure>

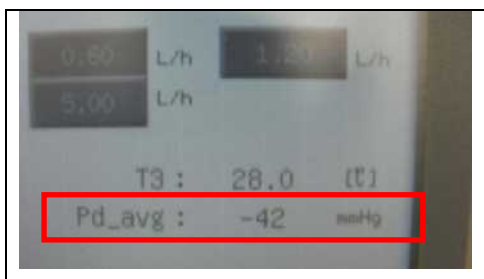
Put the blood circuit in the container and operate the blood pump.

Press the dialysis SW and start up the dialysis operation.

Start up the stopwatch simultaneously with the UF start.

NOTE

- When the UF speed dot display changes from lighting to blinking, the UF starts.



Open the MAINTENANCE -ADJUST- "2.UF CONTROL" screen.

Turn the blood pump to ON and adjust the blood pump flow rate to set the Pd_avg to 0 ± 10 mmHg.

After 30 minutes have passed, press the RETRANS switch. And turn OFF the blood pump operation switch.

<ADJUST 2>, "UF CONTROL" screen



Pull out the arterial and venous end of the blood circuit from the container and wait until there is no drop.

CAUTION

- Ensure that the drops are put in the container.

Read the value on the electric scale and calculate the UF volume per one hour.

$$\text{UF volume [g/hour]} = \frac{(\text{reduced water volume read}) [\text{g/30 min.}] \times 2}{}$$

Calculate the UFP flow from the UF volume per hour and input it to the device.

$$\text{UFP flow rate [ml/STROKE]} = \frac{\text{UF volume [g/h]} \times 1.5 \div 10000}{(0.750:5000 = \text{UFP flow rate: from [UF volume per hour]})}$$

Example: When UF volume is 5072.6g/h
 $5072 \times 1.5 \div 10000 = 0.761$ [mL/STROKE]

Set "26.UF RATE ALARM MAX" of the forth page of MAINTENANCE -MAINTENANCE- "2.ALARM SETTING" to the original state.

Replace the following or make an adjustment if the value is not adjusted to the standard value.

Adjustment

- Balance test

18. UF Precision Test 1



Standard	Set value $\pm$ 30g/h or less
Required jig	Blood circuit, dialyzer, electric scale, container (approximately 5L), stopwatch
Purpose	Verification of UF precision

<Illustration>



<Procedure>

1. Operate the device and wait until it is in the Preparation complete (Bypass release) process.
2. Set the blood circuit and the dialyzer. (Prepare water in a container to circulate as an alternative to blood.)
3. Prime the blood line and dialyzer and connect a coupler to the dialyzer.
4. Insert a needle to the blood circuit of the venous side.

CAUTION

- Ensure that the venous side of the blood circuit is installed with a needle.

Set the UF speed to the given value.

NOTE

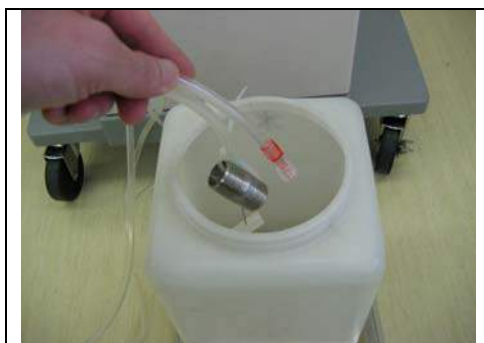
- Change the "26.UF RATE ALARM MAX" of the fourth page of MAINTENANCE - MAINTENANCE - "2.ALARM SETTING" when performing with high UF.

Wait until the dialysate temperature enters the range of $36.5\pm 0.5^{\circ}\text{C}$ with the setting at 36.5°C .

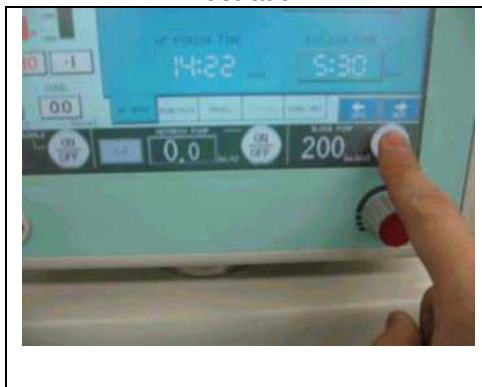
Pull out the arterial and venous end of the blood circuit from the container and wait until there is no drop.

CAUTION

- Ensure that the drops are put in the container.



<Illustration>



<Procedure>

Place the container on the electrical scale and reset the weight display to zero.

Put the blood circuit in the container and operate the blood pump.

Press the DIALYSE SW to start the dialysis operation.

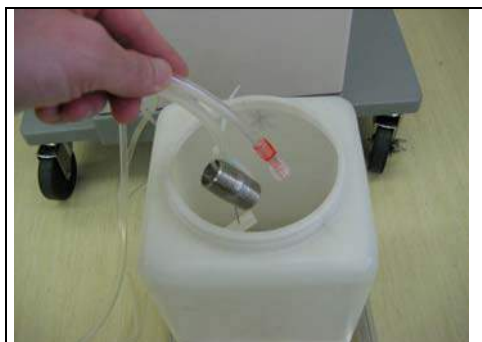
Start up the stopwatch simultaneously with the UF start.

NOTE

-
- When the UF speed dot display changes from lighting to blinking, the UF starts.
-

The blood pump flow is to be the given value.

After 30 minutes have passed, press the RETRANS switch. And turn OFF the blood pump operation switch.



Pull out the arterial and venous end of the blood circuit from the container and wait until there is no drop.

CAUTION

-
- Ensure that the drops are put in the container.
-

Read the value on the electric scale and calculate the UF volume per one hour.

UF volume [g/hour] = (reduced water volume read) [g/30 min.] x2

If within the error UF volume per hour ± 30 g, it is regarded as passed.

Set "26.UF RATE ALARM MAX" of the fourth page of MAINTENANCE - "2.ALARM SETTING" to the original state.

Replace the following or make an adjustment if the value is not adjusted to the standard value.

Adjustment

- Balance test
- UFP Flow Measurement

19. UF Precision Test 2 (during SQHD)



Standard	Within set value $\pm$ 30g/h
Required jig	Blood circuit, dialyzer, electric scale, container (approximately 5L), stopwatch
Purpose	Verification of UF precision during SQHD

<Illustration>



<Procedure>

1. Operate the device and wait until it is in the Preparation complete (Bypass release) process.
2. Set the blood circuit and the dialyzer. (Prepare water in a container to circulate as an alternative to blood.)
3. Prime the blood line and dialyzer and connect a coupler to the dialyzer.
4. Insert a needle to the blood circuit of the venous side.

CAUTION

- Ensure that the venous side of the blood circuit is installed with a needle.

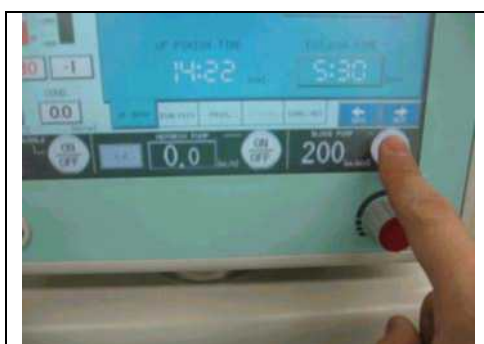
Set the UF speed to the given value.

Operate the device in SQHD.

Pull out the arterial and venous end of the blood circuit from the container and wait until there is no drop.

CAUTION

- Ensure that the drops are put in the container.



Place the container on the electrical scale and reset the weight display to zero.

Put the blood circuit in the container and operate the blood pump.

<Illustration>

<Procedure>

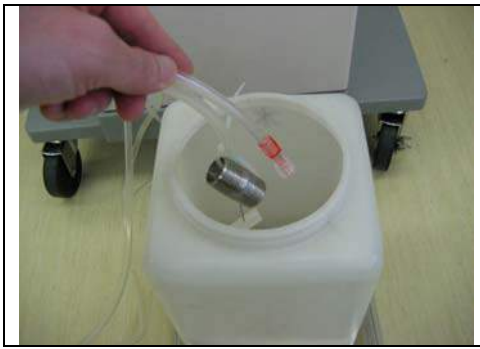
Press the DIALYSE SW to start the dialysis operation.
Start up the stopwatch simultaneously with the UF start.

NOTE

-
- When the UF speed dot display changes from lighting to blinking, the UF starts.
-

The blood pump flow is to be the given value.

After 30 minutes have passed, press the RETRANS switch.
And turn OFF the blood pump operation switch.



Take out the arterial and venous end of the blood circuit from the container and wait until there is no drop.

CAUTION

-
- Ensure that the drops are put in the container.
-

Example: Input the coefficient b'
When increasing UF by 10g/h
Current value +10
When decreasing UF by 10g/h
Current value -10

Read the value on the electric scale and calculate the UF volume per one hour.

UF volume [g/hour] = (reduced water volume read) [g/30 min.] x2

Display the screen of MAINTe -ADJUST- "2.UF CONTROL"

Input the coefficient b' so that UF volume per hour is within the set value $\pm$ 30g/h and test again.

Replace the following or make an adjustment if the value is not adjusted to the standard value.

Adjustment

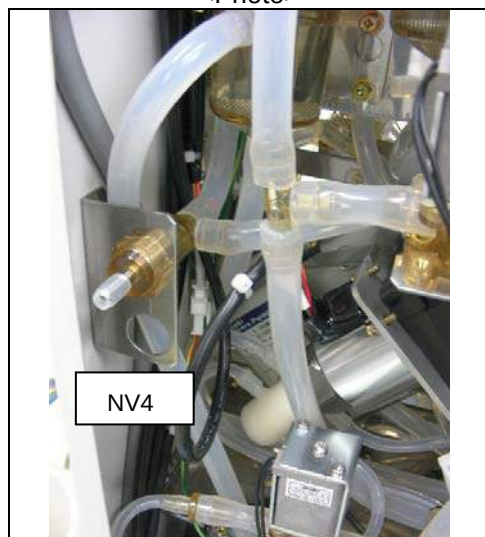
- UFP Flow Measurement
- UF Precision Test 1

20. Gas Purge Pressure Adjustment



Standard	30±20 mmHg
Required jig	Flat head screwdriver, Tweezers
Purpose	Adjustment of purge pressure for normal air discharge in the closed circuit

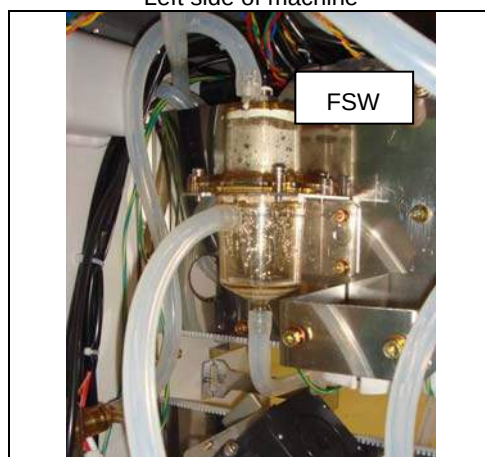
<Photo>



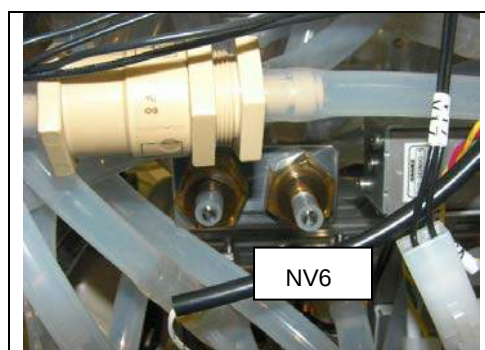
<Operation procedure>

1. Operate the machine, and wait until it is in the Preparation Complete process (Bypass released).
Do not connect to dialyzer.
2. Open the right and left side covers.
3. NV4 is fully open.
4. Press GAS PURGE switch.
Adjust NV6 at the right side of machine so that the dialysate pressure during gas purge is 30±20 mmHg.

Left side of machine



Left side of machine



Right side of machine

At adjustment, after fully open NV4, perform the following operation:

To increase gas purge pressure: loose NV6 (CCW)

To decrease gas purge pressure: Tighten NV6 (CW)

21. Chamber Volumetric Measurement



Standard	95 to 105mL
Required jig	Pean forceps, container (approximately 2L), electric scale
Purpose	Chamber volume measurement and input of the value for proper infusion of concentrate.

<Photo>



Left side of machine

<Operation procedure>

Remove the left side cover at the lower part of machine.
Operate the machine with the Preparation Complete process.
Remove the V21 connector (*1).

Using pean forceps, close the silicon tube at the air discharge side of AS1.

Wait for approximately 2 minutes until no air is discharged from the drain port any more.

To check the time interval from drain discharge completion to the chamber switch (should be approximately 20 sec), set the dialysate flow rate to 300 mL/min.

Sequence step



With the screen of MAINTENANCE -MAINTENANCE- "4.FLOW CHART", measure the drain volume by an electric scale.

Volume drained with the sequence step of JB(JKB): Volume of the Chamber A side

Volume drained with the sequence step of JA(JKA): Volume of the Chamber B side



Drain discharge completion → Read the electric scale indication and record it → Zero reset of the electric scale indication → Wait until drain discharge completes

Repeat the above step three times for both chambers A and B, measure the drain volume, and check that the average volume is within the range: 95 to 105 mL.

Input the average measurement value to the chamber volume in MAINTENANCE -MAINTENANCE- "5.CONC.ADJUST" screen.

<Photo>

<Operation procedure>

Connect the V21 connector, and remove the pean forceps from the air discharge port A21.

Replace the following or make an adjustment if the value is not adjusted to the standard value.

Replace

- Chamber Assy

22. Temperature Sensor Calibration ③ (T1)



Standard	25°C±0.2°C, 40°C±0.2°C
Required jig	Temperature calibration device (Code No. 0-506-001), ceramic driver for adjustment, Phillips screwdriver
Purpose	To calibrate 2 points of offset and gain if the error is out of range, comparing device indication and dummy resistance value.

<Illustration>



<Procedure>

CAUTION

- Ensure that the breaker is turned off.

Remove the cover of the upper right side and the electric cover at the right side.

Remove the connector CN214 for the temperature sensor, which is on the I/O board, and install the temperature calibration device.

NOTE

- Refer to CXSM**-101 for the connector position.

Temperature calibration device dummy resistance value

Temperature	Resistance value
25°C	8.494±0.024 kΩ
40°C	4.904±0.024 kΩ

Turn on the breaker and I/O switch and confirm the displayed temperature with MAINTÉ – MAINTÉ - “1.DATA” - “TEM/CON (Temperature/concentration).”

For the position of the adjustment volume VR4 and VR5 on the I/O board, confirm with the BHSM**-101 layout drawing.

【At 25°C (gain adjustment)】

Set the switch of the temperature calibration device is set to 25°C.

Adjust the VR4 on the I/O board so that the displayed temperature becomes 25.0±0.2°C .

【At 40°C (offset adjustment)】

Set the swtch of the temperature calibration device is set to 40°C.

Adjust the VR5 on the I/O board so that the displayed temperature becomes 40.0±0.2°C.

<Illustration>

<Procedure>

Repeat the offset adjustment and gain adjustment until entering the standard range.

Replace the following or make an adjustment if the value is not adjusted to the standard value.

Replace

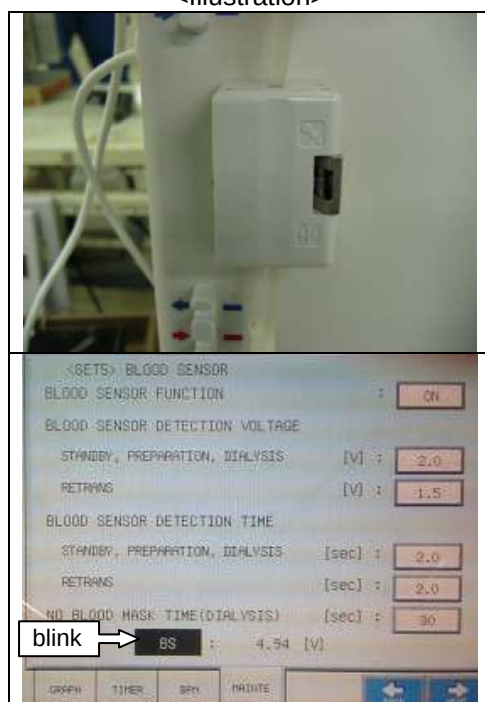
- I/O board

23. Blood Sensor Calibration

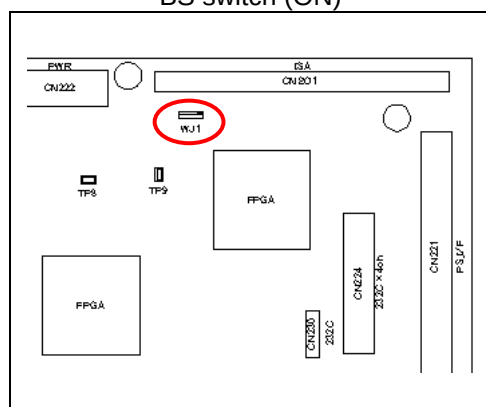


Standard	Receiver GAIN voltage : $9.00 \pm 0.05 \text{VDC}$ Voltage during light shielding : 0.3VDC or less
Required jig	Blood circuit, tester, ceramic driver for adjustment
Purpose	Blood sensor sensitivity calibration

<Illustration>



BS switch (ON)



WJ1 on the I/O board

<Procedure>

CAUTION

- Ensure that the breaker is turned off.

Remove the cover of the upper right face and the electric cover.

Turn on the breaker and I/O switch.

Open MAINTENANCE -OPTION- "1.OPTION" screen and turn BLOOD SENSOR ON

Open MAINTENANCE -SETTING- "5.BLOOD SENSER" screen and press "BS" to blink it.

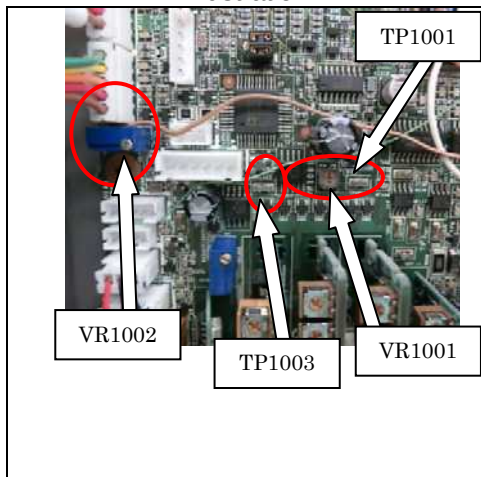
Set the tester range to "VDC."

Connect the tester (+) probe to following measurement point on the I/O board.

Connect the tester (-) probe to WJ1 on the I/O board.

Procedure	Adjustment point	Measurement point	Adjustment Voltage	Lighting status
No.1	VR1001	TP1003	$9 \text{V} \pm 0.2 \text{V}$	Lighting
No.2	—	TP1001	2V or less	Lighting
No.3	VR1002	TP1003	$9.00 \text{V} \pm 0.05 \text{V}$	Lighting
No.4	—	TP1003	0.3V or less	Shielding
No.5	—	Setting 5 "Blood sensor"	More than 4.0V	Blinking

<Illustration>



Blood sensor adjustment Volume, test point location

<Procedure>

【NO.1】

Adjust TP1003 voltage to $9.00V \pm 0.2$ using the adjust VR1001 on the board, without blood circuit.

【NO.2】

Confirm TP1001 voltage is 2.0V or less.

【NO.3】

Adjust TP1003 voltage to $9.00V \pm 0.05$ using the adjust VR1002 on the board.

【NO.4】

Shield the blood sensor by paper completely and confirm TP1003 voltage is 0.3V or less.

【NO.5】

Open MAINT -SETTING-“5.BLOOD SENSER” screen and press “BS” not to blink it.

Confirm BS LED is blinking in standby process.

Place an empty blood circuit to BS section.

Open MAINT -SETTING-“5.BLOOD SENSER” screen and confirm the display voltage is 4.0V or higher.

Replace the following or make an adjustment if the value is not adjusted to the standard value.

Replace

- I/O board
- Blood sensor Assy

Appendix

Appendix

Maintenance tool List

⚠ CAUTION

Use the following tools or equal tools for maintenance.

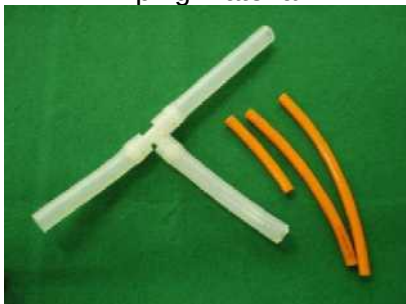
Each part or a part of each part may be changed for improvement without notice.

Be forewarned that a part of using product may be differed from this document.

Name of product		Gauge
Flow meter  CodeNo. 0-201-345		Manufacturer : RYUKI Model: SEMF-600 Measure range: 100 to 600mL/min
Circuit tester  CodeNo.0-506-022		Manufacturer : FLUKE Model: 87V(FLUKE-87-5) *This could be sourced locally.
SPEC		
Function	Range	Reference accuracy
DC	600.0mV, 6.000V, 60.00V, 600.0V, 1000V	0.05%
AC	600.0mV, 6.000V, 60.00V, 600.0V, 1000V	0.7% (True RMS value)
DC	600.0μA, 6000μA, 60.00mA, 600.0mA, 6.000A, 10.00A	0.2%
AC	600.0μA, 6000μA, 60.00mA, 600.0mA, 6.000A, 10.00A	1.0% (True RMS value)
Resistance	600.0Ω, 6.000kΩ, 60.00kΩ, 600.0kΩ, 6.000MΩ, 50.00MΩ	0.2%
Use a tester which is "True RMS TYPE" to measure precise alternating current value. * RMS: Root Mean Square		

Name of product	Gauge
Probe  CodeNo.0-506-023(Red) CodeNo.0-506-024(Black)	Name of product: DIY SMD Test clip Manufacturer: Tokiwa Model: 5243-Red, 5243-Black Name of product: Chip Jack Manufacturer: SATO PARTS Model: TJ-2167 (Red) TJ-2167(Black)
Stopwatch  CodeNo.0-506-025	Manufacturer: CASIO Model: HS-3C-8AJH *This could be sourced locally.
Electronic balance  CodeNo.0-506-026 (110V only)	Measure range : 0 to 5000g -5000 to 0g Unit of measurement: 1g *This should be sourced locally.
Container  CodeNo.0-506-027(2L)	SPEC: About 2L *This could be sourced locally.

Name of product	Gauge
Container  CodeNo.0-506-028(5L)	SPEC: About 5L *This could be sourced locally.
Slide gauge  CodeNo.0-506-029	Manufacturer: Mitutoyo Model: N15 Measure range: 0 to 150mm
Calculator  CodeNo.0-506-030	*This could be sourced locally.
Torque screwdriver  CodeNo.0-706-002 CodeNo.0-706-003 CodeNo.0-706-004	Manufacturer: KANON Model: N20LTDK SPEC: 0.4 to 2 N.m Manufacturer: KANON Model: N6LTDK SPEC: 0.05 to 0.6 N.m Name of product: Bit+ Manufacturer: VESSEL Model: B35-2050G

Name of product	Gauge
Torque wrench  CodeNo.0-706-005 CodeNo.0-706-006 CodeNo.0-706-007	Manufacturer: KANON Model: N60LCK SPEC: 2 to 6 N.m Manufacturer: KANON Model: N60QLK SPEC: 2 to 6 N.m Open end head Manufacturer: KANON Model: 230SCK 19 SPEC: 19mm
Digital manometer  CodeNo.0-506-032	Manufacturer: Copal Electronics Model: PG-100 102GH Measure range: 0 to 735mmHg *This could be sourced locally.
Analog manometer  CodeNo.0-506-033	Manufacturer: KOGANEI Measure range: 0 to 1MPa Piping
Piping material  CodeNo.0-506-034	Name of product of product: T-tube Manufacturer: SHIBUYA Model:93613911100* Name of product of product: Silicon tube SPEC:d5×D11×L150 Name of product of product: TE-6-100 SPEC:d4×D6×L100

Name of product	Gauge
Syringe  CodeNo.0-506-035	Manufacturer: NIPRO Model: 20ml SPEC: 20ml
Wrist strap  CodeNo.0-506-038	Manufacturer: HOZAN Model: F-153
Ceramic driver  CodeNo.0-506-039	Manufacturer: ENGINEER Model: DA-55
(+)Driver  CodeNo.0-506-040	Manufacturer: VESSEL Model: 600P.2-100

Name of product	Gauge
(+)Driver  CodeNo.0-506-041	Manufacturer: VESSEL Model: 600P.1-75
(-)Driver  CodeNo.0-506-042	Manufacturer: VESSEL Model: 600 6-100
Replacement Driver  CodeNo.0-506-043	Manufacturer: HOZAN Model: D-2
Power socket driver  CodeNo.0-506-044	Manufacturer: VESSEL Model: 4100 7-125

Name of product	Gauge
Power socket driver  CodeNo.0-506-045	Manufacturer: VESSEL Model: 4100 8-125
Pean forceps  CodeNo.0-506-046	Manufacturer: REDA Model: 13710-18
Needle-nose pliers  CodeNo.0-506-047	Manufacturer: HOZAN Model: P-15-125
Nipper  CodeNo.0-506-048	Manufacturer: HOZAN Model: N-4-125

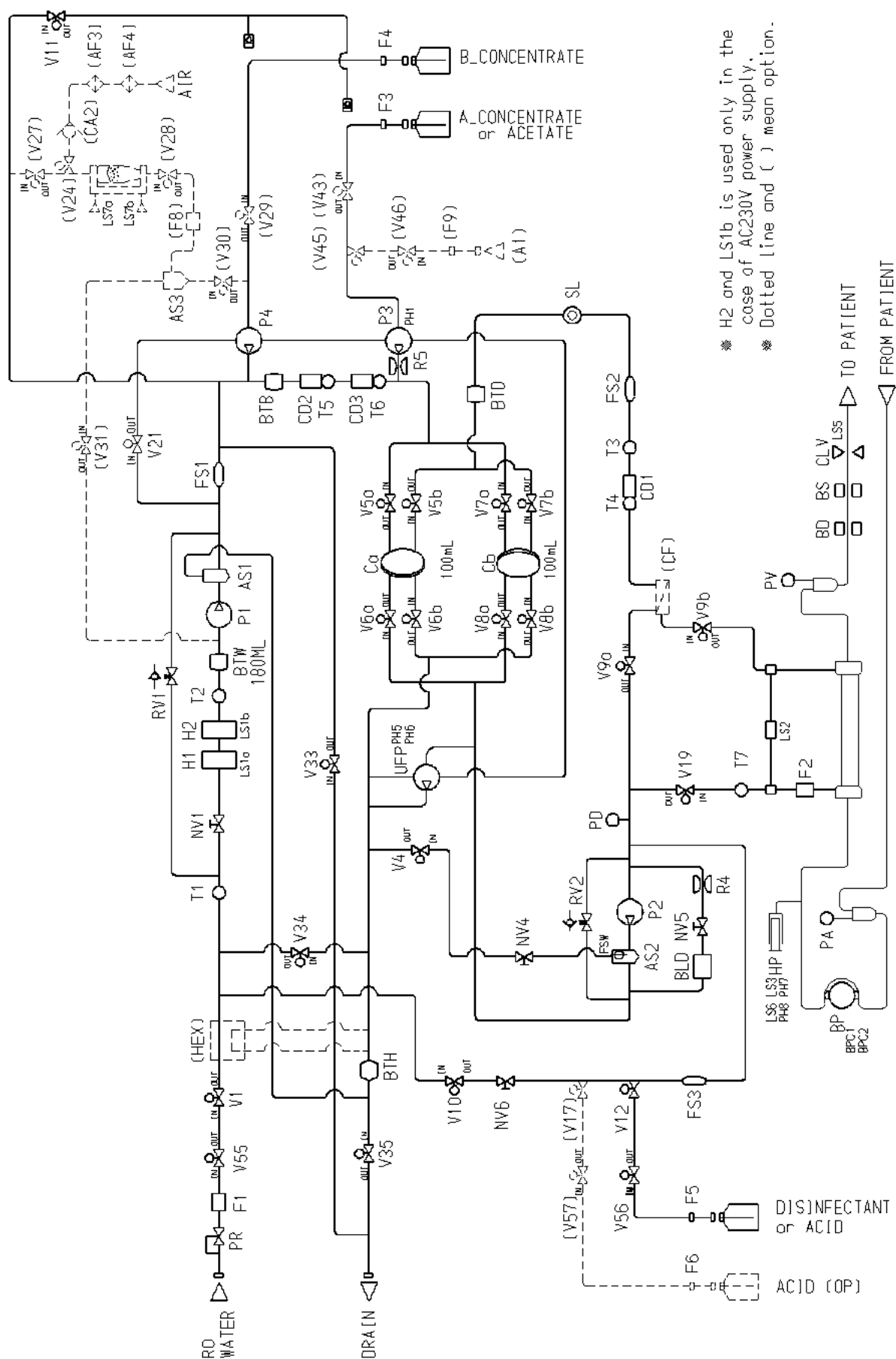
Name of product	Gauge
Tweezers  CodeNo.0-506-049	Manufacturer: HOZAN Model: P-887
Spanner  CodeNo.0-506-050	Manufacturer: HOZAN Model: W-521
Hexagonal wrench  CodeNo.0-506-051	Manufacturer: PB Model: 211HC-6
Monkey wrench  CodeNo.0-506-052	SPEC: 200mm

Name of product	Gauge
Pliers  CodeNo.0-506-053	SPEC: 200mm
Resistance switch BOX (For temperature sensor)  CodeNo.0-506-001	Manufacturer : SHIBUYA
Resistance switch BOX (For conductivity sensor)  CodeNo.0-506-007	Manufacturer : SHIBUYA
Resistance switch BOX (For temperature sensor and conductivity sensor)  CodeNo.0-506-066	Manufacturer : SHIBUYA

Name of product	Gauge
BLD-01Cartridge for calibration  CodeNo.4-030-077	Manufacturer : SHIBUYA CodeNo.4-30-077
MOLYKOTE grease  CodeNo.0-506-054	Manufacturer: MOLYKOTE Model: BR2-PLUS
Bond  CodeNo.0-506-055	Manufacturer: Three Bond Model: 1401C
Silicon sealant  CodeNo.0-506-056	Manufacturer: GE TOSHIBA Silicon Model: TSE382-W

Name of product	Gauge
Seal tape  CodeNo.9-001-131	Gauge: 0.1mm×8mm
Seal tape  CodeNo.9-001-056	Gauge: 0.1mm×13mm

Flow Chart





Manufacturer

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